

Addition of vectors

When adding vectors, the horizontal movements and the vertical movements are added separately:

Thus, if $a = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $b = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ then,

$$\begin{aligned} a + b &= \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \\ &= \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} \end{aligned}$$

Questions on Transformations & Vectors

Q1) $P(-1, 2)$ has image $Q(-2, -3)$ under a translation. Find the column vector of the translation.

Q2) $\underline{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$. Work out

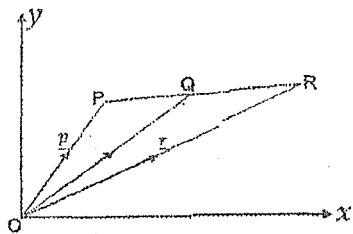
- $4\underline{a}$
- $3\underline{a} - 2\underline{b}$
- $3(\underline{a} - \underline{b})$

Q3) $OPQR$ is a parallelogram with vertices $O(0,0)$, $P(5,2)$ and $R(-2, -6)$. Find the coordinates of Q .

Q4) Find the coordinates of point T , given that $\overrightarrow{OT} = \overrightarrow{OA} + \overrightarrow{OB}$ and $A(3, 5)$ and $B(5, -2)$

Q5) Use vectors to find the coordinates of the mid-point, R , of PQ where $P = (3, 7)$ and $Q = (5, -1)$

Q6) $\overrightarrow{OT} = \underline{p}$, $\overrightarrow{OR} = \underline{r}$ and Q is a point such that $3\overrightarrow{PQ} = \overrightarrow{QR}$. Find $\overrightarrow{OQ}$ in terms of $\underline{p}$ and $\underline{r}$.



Q7) Find the lengths of these vectors.

- $\begin{pmatrix} 5 \\ 12 \end{pmatrix}$
- $\begin{pmatrix} x \\ y \end{pmatrix}$

Q8) If $\underline{s} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and $\underline{t} = \begin{pmatrix} -8 \\ 3 \end{pmatrix}$

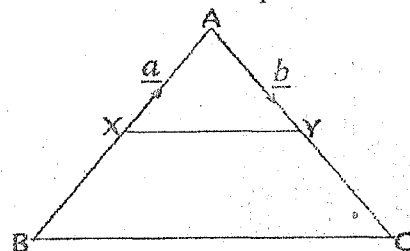
Find:

- $|\underline{s}|$
- $|\underline{s} - \underline{t}|$

Q9) The coordinates of A , B , C and D are $(2, 6)$, $(5, 5)$, $(2, 2)$ and $(-1, 3)$ respectively. Show that $ABCD$ is a parallelogram.

Q10) The coordinates of P , Q , R and S are $(0, 1)$, $(7, 8)$, $((1, -1)$ and $(9, 7)$ respectively. Show that $\overrightarrow{PQ}$ is parallel to $\overrightarrow{RS}$.

Q11) In triangle ABC , X is the mid-point of AB and Y is the mid-point



Show that XY is parallel to BC and half

its length.

Q12) Show that $(-4, 4)$, $Y(2, 7)$ and $Z(6, 9)$ are collinear.

Q13) Show that each of the following pairs of vectors are parallel.

- $\underline{u} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 9 \\ 6 \end{pmatrix}$
- $\underline{w} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$ and $\underline{x} = \begin{pmatrix} 2 \\ 4 \\ 10 \end{pmatrix}$

Q14) Find the magnitude of the following vector. $\underline{g} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

Q15) Show that $A(1, 2, 3)$, $B(2, 4, 6)$ and $C(6, 12, 18)$ are collinear.

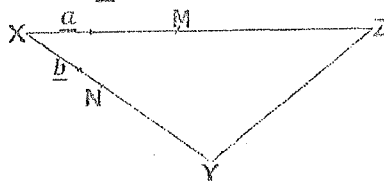
Q16) The points $P(-1, -4)$, $Q(1, 2)$, $R(4, 4)$ and $S(2, -2)$ form a quadrilateral. Show by vector method, that $PQRS$ is a parallelogram.

Q17) A , B , C and D are four points on the Cartesian plane such that $A(-6, 4)$, $C(6, 2)$, $\overrightarrow{AB} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}$ and $\overrightarrow{AD} = \begin{pmatrix} 8 \\ -8 \end{pmatrix}$. Calculate the coordinates of points B and D .

Q18) A point P has been translated by the translation vector $T = \begin{pmatrix} -3 \\ -11 \end{pmatrix}$ to $P' = \begin{pmatrix} -1 \\ -4 \end{pmatrix}$. Find the coordinates of the point P .

Q19) Given that $\underline{a} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$, calculate $|\underline{a} + \underline{b}|$

Q20) In figure below, M and N are mid-points of $\overline{XZ}$ and $\overline{XY}$ respectively.
 $\overline{XM} = \underline{a}$ and $\overline{XN} = \underline{b}$



Q21) Show that $\overline{MN}$ and $\overline{ZY}$ are parallel.
 Given that $P(-1, 4)$, $Q(3, 0)$ and $R(2, a)$ are collinear, find the value of a .

Q22) Given that $\begin{pmatrix} a-1 \\ 2 \end{pmatrix} + \underline{b} = \begin{pmatrix} 16 \\ 7 \end{pmatrix}$,
 Find $\underline{b}$

Q23) Points P and Q have position vectors $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 9 \\ 15 \end{pmatrix}$ respectively. Find
 (i). $\overrightarrow{PQ}$
 (ii). $|\overrightarrow{PQ}|$

Q24) Given that vector $A = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is translated by the vector $B = \begin{pmatrix} a \\ b \end{pmatrix}$ to a vector $A' = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$. Calculate the values of a and b .

Q25) Given $\underline{a} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$.
 Find $\frac{1}{2}(\underline{a} - \underline{b})$.

Q26) Given that $\overrightarrow{AB} = \begin{pmatrix} -9 \\ 4 \end{pmatrix}$, calculate the length of $\overline{AB}$, leaving your answer correct to 3 significant figures.

Q27) A point $P(-2, 4)$ is translated to a point P' . If P' is 5 units down and 3 units to the right of point P , find the coordinates of P' .

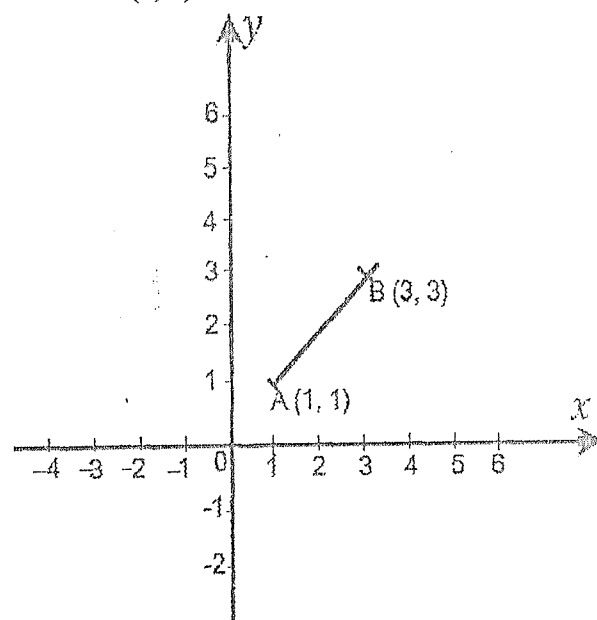
Q28) Let $\overrightarrow{AB} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. Calculate the mid-point of $\overline{BC}$

Q29) Given that $C' = (5, -5)$ is the image of $C = (1, 1)$ after translation, find the translation vector that maps C into C' .

Q30) Solve for x and y in the vector equation

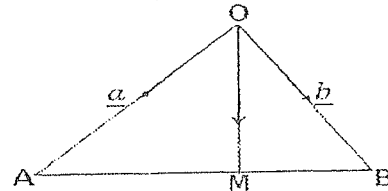
$$\begin{pmatrix} x \\ 3 \\ y \\ 3 \end{pmatrix} = 4 \begin{pmatrix} -2 \\ 1 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}$$

Q31) In figure below, AB is the straight line joining the points $A(1, 1)$ and $B(3, 3)$



- Draw the image of AB under reflection in the y -axis.
- Find the coordinates of A' and B'

Q32) In figure below, $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and M is the mid-point of $\overline{AB}$.



Find $\overrightarrow{OM}$ in terms of $\underline{a}$ and $\underline{b}$

Transformations & Vectors (Model Solutions)

Q1) Solution

In this case, if the column vector is $\begin{pmatrix} x \\ y \end{pmatrix}$ then

$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

So, $x - 1 = -2$, hence $x = -1$
and $y + 2 = -3$, hence $y = -5$

The column vector is $\begin{pmatrix} -1 \\ -5 \end{pmatrix}$

Alternative method

We can work this out more quickly without an equation by noting that:

$$\overrightarrow{AB} = \begin{pmatrix} -2 - (-1) \\ -3 - 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$$

Q2) Solution

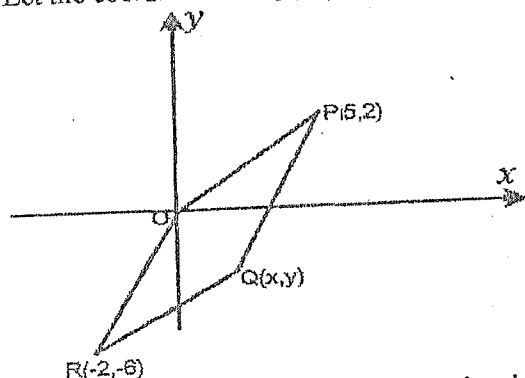
$$(i). \quad 4\underline{a} = 4 \times \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$$

$$(ii). \quad 3\underline{a} - 2\underline{b} = 3 \times \begin{pmatrix} -2 \\ -1 \end{pmatrix} - 2 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix} \\ = \begin{pmatrix} -14 \\ 3 \end{pmatrix}$$

$$(iii). \quad 3(\underline{a} - \underline{b}) = 3 \begin{pmatrix} -2 - 4 \\ -1 - (-3) \end{pmatrix} = \begin{pmatrix} -18 \\ 6 \end{pmatrix}$$

Q3) Solution

Let the coordinates of $Q(x, y)$.



Since OPQR is a parallelogram, opposite sides OP and RQ are parallel and equal in length.

$$\text{So, } \overrightarrow{OP} = \overrightarrow{RQ}$$

$$\text{But } \overrightarrow{OP} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 - 0 \\ 2 - 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$$\text{and } \overrightarrow{RQ} = \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} -2 \\ -6 \end{pmatrix} = \begin{pmatrix} x - (-2) \\ y - (-6) \end{pmatrix} \\ = \begin{pmatrix} x + 2 \\ y + 6 \end{pmatrix}$$

$$\text{So, } \begin{pmatrix} x + 2 \\ y + 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$$x = 5 - 2 = 3$$

$$y = 2 - 6 = -4 \text{ and } Q = (3, -4)$$

Q4) Solution

Since the position vectors of points A and B are given as the vectors $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ respectively

Then,

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$\therefore \overrightarrow{OT} = \overrightarrow{OA} + \overrightarrow{OB}$$

$$= \begin{pmatrix} 3 \\ 5 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 + 5 \\ 5 + (-2) \end{pmatrix}$$

$$= \begin{pmatrix} 8 \\ 3 \end{pmatrix}$$

$\therefore$ The coordinates of T are (8,3)

Q5) Solution

$$\overrightarrow{OP} = \underline{p} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \overrightarrow{OQ} = \underline{q} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OR} = \frac{1}{2}(\underline{p} + \underline{q})$$

$$= \frac{1}{2} \begin{pmatrix} 3 + 5 \\ 7 + (-1) \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

So, $R = (4,3)$

Q6) Solution

$$\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$$

$$= \underline{p} + \overrightarrow{PQ}$$

$$\text{and } \overrightarrow{PR} = -\underline{p} + \underline{r}$$

$$= \underline{r} - \underline{p}$$

Now, $3\overrightarrow{PQ} = \overrightarrow{QR}$, and $\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR}$

$$\text{So, } \overrightarrow{PR} = \overrightarrow{PQ} + 3\overrightarrow{PQ}$$

$$= 4\overrightarrow{PQ}$$

$$\text{and } \overrightarrow{PQ} = \frac{1}{4}\overrightarrow{PR}$$

$$\text{So, } \overrightarrow{OQ} = \underline{p} + \frac{1}{4}\overrightarrow{PR}$$

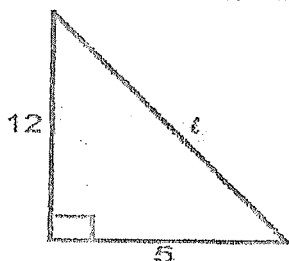
$$= \underline{p} + \frac{1}{4}(\underline{r} - \underline{p})$$

$$= \frac{1}{4}(\underline{r} + 3\underline{p})$$

Q7) Solution

- (i). The vector represents a journey of 5 along, and 12 up.

The vector is the hypotenuse of a right-angled triangle with the other two sides 5 and 12.



$$l^2 = 5^2 + 12^2$$

$$= 25 + 144 = 169$$

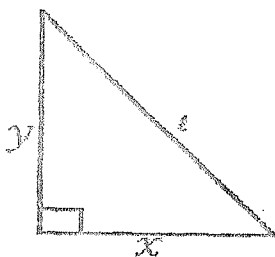
$$l = \sqrt{169}$$

$$l = 13$$

- (ii). $\begin{pmatrix} x \\ y \end{pmatrix}$

Solution

Similarly, this vector has the hypotenuse of a right-angled triangle with the other two sides x and y .



$$l^2 = x^2 + y^2$$

$$l = \sqrt{x^2 + y^2}$$

Q8) Solution

a) $|\underline{s}| = \sqrt{(3^2 + (-5)^2)}$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$= 5.83 \text{ (to 3 s.f.)}$$

b) $|\underline{s} - \underline{t}| = \begin{pmatrix} 3 \\ -5 \end{pmatrix} - \begin{pmatrix} -8 \\ 3 \end{pmatrix}$

$$= \begin{pmatrix} 3 - (-8) \\ -5 - 3 \end{pmatrix}$$

$$= \begin{pmatrix} 3 + 8 \\ -5 - 3 \end{pmatrix}$$

$$= \begin{pmatrix} 11 \\ -8 \end{pmatrix}$$

Q9) Solution

$$\overrightarrow{AB} = \begin{pmatrix} 5 - 2 \\ 5 - 6 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

and $\overrightarrow{DC} = \begin{pmatrix} 2 - (-1) \\ 2 - 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

$\Rightarrow \overrightarrow{AB} = \overrightarrow{DC}$, so the opposite sides of ABCD are equal and parallel.

Or Similarly,

$$\overrightarrow{AD} = \begin{pmatrix} -1 - 2 \\ 3 - 6 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

and $\overrightarrow{BC} = \begin{pmatrix} 2 - 5 \\ 2 - 5 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$

$\Rightarrow \overrightarrow{AD} = \overrightarrow{BC}$, so the opposite sides of ABCD are also equal and parallel.

Note: When we know one pair of sides is parallel and equal. This is sufficient information to make a conclusion.

Q10) Solution

$$\overrightarrow{PQ} = \begin{pmatrix} 7 - 0 \\ 8 - 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \end{pmatrix} = 7 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and $\overrightarrow{RS} = \begin{pmatrix} 9 - 1 \\ 7 - (-1) \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \end{pmatrix} = 8 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\therefore \overrightarrow{PQ}$ is parallel to $\overrightarrow{RS}$

Q11) Solution

We need to show that $2 \times \overrightarrow{XY} = \overrightarrow{BC}$

Let $\overrightarrow{XA} = \underline{a}$ and $\overrightarrow{AY} = \underline{b}$.

Then, $\overrightarrow{XY} = \underline{a} + \underline{b}$.

Since X is the mid-point of AB, $\overrightarrow{BA} = 2\underline{a}$

Since Y is the mid-point of AC, $\overrightarrow{AC} = 2\underline{b}$

$$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC}$$

$$= 2\underline{a} + 2\underline{b}$$

$$= 2(\underline{a} + \underline{b})$$

$$= 2\overrightarrow{XY}$$

So, XY is parallel to BC and half its length.

Q12) Solution

$$\overrightarrow{XY} = \begin{pmatrix} 2 + 4 \\ 7 - 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

$$\overrightarrow{YZ} = \begin{pmatrix} 6 - 2 \\ 9 - 7 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

So, $\overrightarrow{XY} = 3 \times \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\overrightarrow{YZ} = 2 \times \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

So, $\overrightarrow{XY}$ and $\overrightarrow{YZ}$ are parallel, as they have a common point they must be collinear.

Q13) Solution

a) Given $u = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and

$$v = \begin{pmatrix} 9 \\ 6 \end{pmatrix} \Rightarrow v = 3 \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Thus, $v = 3u$

$\therefore u/v$

b) Given $w = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$ and

$$x = \begin{pmatrix} 2 \\ 4 \\ 10 \end{pmatrix} \Rightarrow x = 2 \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$$

Thus, $x = 2w$.

Hence, w/x

Q14) Solution

$$\text{Given } g = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow |g| &= \sqrt{(2^2 + (-3)^2 + 5^2)} \\ &= \sqrt{(4 + 9 + 25)} \\ &= \sqrt{38} \\ &= 6.164 \text{ (to 3 d.p.)} \end{aligned}$$

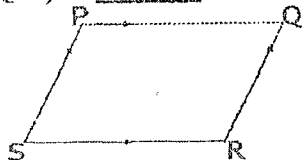
Q15) Solution

$$AB = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$BC = \begin{pmatrix} 6 \\ 12 \\ 18 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 12 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Thus, $BC = 4 AB$. BC is a multiple of AB .
Therefore, $BC \parallel AB$. But, B is a common point.
Therefore, A, B and C are collinear.

Q16) Solution



Given $P(-1, -4), Q(1, 2), R(4, 4)$ and $S(2, -2)$

$$\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ -4 \end{pmatrix}$$

$$\begin{aligned} &= \begin{pmatrix} 1 - (-1) \\ 2 - (-4) \end{pmatrix} \\ &= \begin{pmatrix} 1 + 1 \\ 2 + 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 6 \end{pmatrix} \\ \overrightarrow{SR} &= \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} 4 - 2 \\ 4 + 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 6 \end{pmatrix} \end{aligned}$$

Q17) Solution

Since $A(-6, 4)$ and $C(6, 2)$

In this case, let;

$B(x, y)$ and $D(x, y)$

$$\text{So, } \overrightarrow{AB} = B - A = \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} -6 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 8 \\ 2 \end{pmatrix} + \begin{pmatrix} -6 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 8 - 6 \\ 2 + 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 6 \end{pmatrix} \end{aligned}$$

$B(2, 6)$

Also,

$$\overrightarrow{AD} = D - A = \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} -6 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ -8 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 8 \\ -8 \end{pmatrix} + \begin{pmatrix} -6 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 8 - 6 \\ -8 + 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -4 \end{pmatrix} \end{aligned}$$

$D(2, -4)$

Q18) Solution

Translation

Object + Translation = Image

$$\Rightarrow P + T = P'$$

$$\text{But, } P(x, y), T = \begin{pmatrix} -3 \\ -11 \end{pmatrix} \text{ and } P' = \begin{pmatrix} -1 \\ -4 \end{pmatrix}$$

$$\begin{aligned} P + T &= P' \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -3 \\ -11 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} -1 \\ -4 \end{pmatrix} - \begin{pmatrix} -3 \\ -11 \end{pmatrix} \\ &= \begin{pmatrix} -1 - (-3) \\ -4 - (-11) \end{pmatrix} \\ &= \begin{pmatrix} -1 + 3 \\ -4 + 11 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 7 \end{pmatrix} \end{aligned}$$

Q19) Solution

Given $\underline{a} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$

Need to find $|\underline{a} + \underline{b}|$

$$\begin{aligned} \Rightarrow \underline{a} + \underline{b} &= \begin{pmatrix} 3 \\ 5 \end{pmatrix} + \begin{pmatrix} 9 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 3+9 \\ 5+4 \end{pmatrix} \\ &= \begin{pmatrix} 12 \\ 9 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} |\underline{a} + \underline{b}| &= \sqrt{12^2 + 9^2} \\ &= \sqrt{144 + 81} \\ &= \sqrt{225} \\ &= 15 \end{aligned}$$

Q20) Solution

Given M and N as mid-points of XZ and XY respectively

$$\Rightarrow XM = MZ$$

$$XN = NY$$

Also, given that $\overrightarrow{XM} = \underline{a}$, $\overrightarrow{XN} = \underline{b}$

$$\Rightarrow \overrightarrow{MX} = -\underline{a}$$

$$\overrightarrow{MX} + \overrightarrow{XN} = \overrightarrow{MN}$$

$$-\underline{a} + \underline{b} = \overrightarrow{MN}$$

$$\overrightarrow{ZX} + \overrightarrow{XY} = \overrightarrow{ZY}$$

$$-2\underline{a} + 2\underline{b} = \overrightarrow{ZY}$$

$$2(-\underline{a} + \underline{b}) = \overrightarrow{ZY}$$

$$\text{Since } \overrightarrow{MN} = \frac{1}{2} \overrightarrow{ZY}$$

$\therefore \overrightarrow{MN}$ and $\overrightarrow{ZY}$ are parallel

Q21) Solution

Given that P (-1, 4), Q (3, 0) and R (2, a) are collinear

Need to find a.

$$\begin{aligned} \text{Now, } \overrightarrow{PQ} &= \begin{pmatrix} 3 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 - (-1) \\ 0 - 4 \end{pmatrix} \\ &= \begin{pmatrix} 3+1 \\ 0-4 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ -4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{QR} &= \begin{pmatrix} 2 \\ a \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 2-3 \\ a-0 \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ a \end{pmatrix} \end{aligned}$$

Since the points are collinear, then;
 $\overrightarrow{QR} = k\overrightarrow{PQ}$, for k a scalar.

$$\begin{pmatrix} -1 \\ a \end{pmatrix} = k \begin{pmatrix} 4 \\ -4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 \\ a \end{pmatrix} = \begin{pmatrix} k4 \\ -k4 \end{pmatrix}$$

From above:

$$\Rightarrow -1 = k4 \Rightarrow k = \frac{-1}{4}$$

Also, $a = -k4$

$$\Rightarrow a = -\left(\frac{-1}{4}\right) \times 4 \text{ (Substitute } \frac{-1}{4} \text{ for } k)$$

$$a = \frac{1}{4} \times 4 = 1$$

$\therefore$ The value of a is 1.

Q22) Solution

Given that $\begin{pmatrix} a-1 \\ 2 \end{pmatrix} + \underline{b} = \begin{pmatrix} 16 \\ 7 \end{pmatrix}$

Need to find $\underline{b}$.

$$\begin{pmatrix} a-1 \\ 2 \end{pmatrix} + \underline{b} = \begin{pmatrix} 16 \\ 7 \end{pmatrix}$$

$$\underline{b} = \begin{pmatrix} 16 \\ 7 \end{pmatrix} - \begin{pmatrix} a-1 \\ 2 \end{pmatrix}$$

$$\underline{b} = \begin{pmatrix} 16-a+1 \\ 7-2 \end{pmatrix}$$

$$\therefore \underline{b} = \begin{pmatrix} 17-a \\ 5 \end{pmatrix}$$

Q23) Solution

(i). Given the position vectors P (3, 7) and Q (9, 15)

$$\overrightarrow{PQ} = \begin{pmatrix} 9 \\ 15 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$= \begin{pmatrix} 9-3 \\ 15-7 \end{pmatrix}$$

$$\therefore \overrightarrow{PQ} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$$

$$(ii). |\overrightarrow{PQ}| = \sqrt{x^2 + y^2}$$

$$= \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$\therefore |\overrightarrow{PQ}| = 10$$

Q24) Solution

Given $A = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, translated by $\begin{pmatrix} a \\ b \end{pmatrix}$

$$A' = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

Since, $\text{Object} + \text{Translation} = \text{Image}$

$$\Rightarrow A + B = A'$$

$$\Rightarrow \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

Equating corresponding terms we get

$$2 + a = 5 \Rightarrow a = 5 - 2 = 3$$

$$3 + b = 7 \Rightarrow b = 7 - 3 = 4$$

$\therefore$ The value of a is 3 and the value of b is 4

Q25) Solution

Given $\underline{a} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$.

$$\text{Now, } \frac{1}{2}(\underline{b} - \underline{a}) = \frac{1}{2} \left\{ \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} -2 \\ -4 \end{pmatrix} \right\}$$

$$= \frac{1}{2} \begin{pmatrix} -4 - (-2) \\ 0 - (-4) \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} -4 + 2 \\ 0 + 4 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} \times -2 \\ \frac{1}{2} \times 4 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Q26) Solution

Given $\overrightarrow{AB} = \begin{pmatrix} -9 \\ 4 \end{pmatrix}$

Length or modulus of $\overrightarrow{AB} = |\overrightarrow{AB}|$

$$\Rightarrow |\overrightarrow{AB}| = \sqrt{x^2 + y^2}$$

$$= \sqrt{(-9)^2 + (4)^2}$$

$$= \sqrt{81 + 16}$$

$$= \sqrt{97}$$

$$= 9.85 \text{ (to 3 s.f.)}$$

$\therefore$ The length of $\overrightarrow{AB}$ is 9.85 units

Q27) Solution

Given $P(-2, 4)$ translated to P'

Since P' is 5 units down and 3 units to the right of point P ,

Then translation vector, $T = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$

Object + Translation = Image

Thus, $P + T = P'$

$$\Rightarrow P' = \begin{pmatrix} -2 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

$$= \begin{pmatrix} -2 + 3 \\ 4 - 5 \end{pmatrix}$$

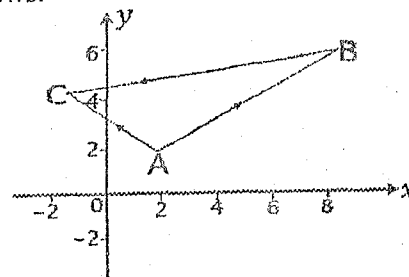
$$= \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\therefore$ The coordinates of P' are $(1, -1)$

Q28) Solution

Given $\overrightarrow{AB} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$

Let's have a sketch for easy work out
As follows:



$$\begin{aligned} \overrightarrow{AB} + \overrightarrow{BC} &= \overrightarrow{AC} \\ \Rightarrow \overrightarrow{BC} &= \overrightarrow{AC} - \overrightarrow{AB} \\ &= \begin{pmatrix} -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 6 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} -4 - 6 \\ 2 - 4 \end{pmatrix} \\ &= \begin{pmatrix} -10 \\ -2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{Mid-point of } \overrightarrow{BC} &= \frac{1}{2} \begin{pmatrix} -10 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} \times (-10) \\ \frac{1}{2} \times (-2) \end{pmatrix} \\ &= \begin{pmatrix} -5 \\ -1 \end{pmatrix} \end{aligned}$$

$\therefore$ The coordinates of the mid-point of $\overrightarrow{BC}$ is $(-5, -1)$.

Q29) Solution

Given $C = (1, 1)$ translated to $C' = (5, -5)$

In this, case, let $\begin{pmatrix} x \\ y \end{pmatrix}$ be the translation vector

Then, $\Rightarrow C + \begin{pmatrix} x \\ y \end{pmatrix} = C'$

$$\Rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 + x \\ 1 + y \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \end{pmatrix}$$

Equating corresponding values:

$$1 + x = 5 \Rightarrow x = 5 - 1 = 4$$

$$1 + y = -5 \Rightarrow y = -5 - 1 = -6$$

$\therefore$ The translation vector that maps C into C' is $\begin{pmatrix} 4 \\ -6 \end{pmatrix}$

Q30) Solution

$$\begin{pmatrix} x \\ 3 \end{pmatrix} = 4 \begin{pmatrix} -2 \\ 1 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \times (-2) \\ 4 \times (1) \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x \\ 3 \end{pmatrix} = \begin{pmatrix} -8 \\ 4 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x \\ 3 \end{pmatrix} = \begin{pmatrix} -8 - x \\ 4 - y \end{pmatrix}$$

Equating the corresponding terms of the vectors:

$$\frac{x}{3} = -8 - x$$

$$\frac{x}{3} \times (3) = -8 \times (3) - x \times (3)$$

$$x = -24 - 3$$

$$3x + x = -24$$

$$\frac{4x}{4} = \frac{-24}{4}$$

$$x = -6$$

Also,

$$\frac{y}{3} = 4 - y$$

$$\frac{y}{3} \times (3) = (3) \times 4 - y \times (3)$$

$$y = 12 - 3y$$

$$3y + y = 12$$

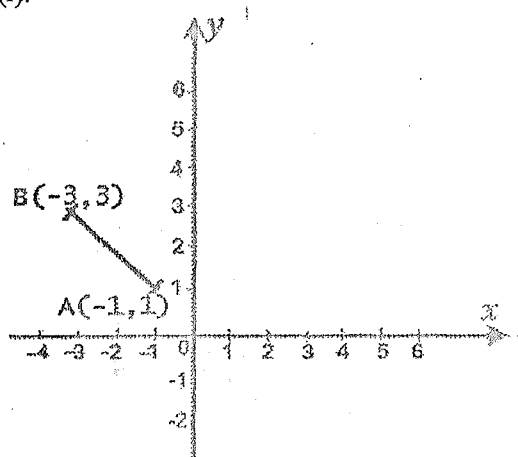
$$\frac{4y}{4} = \frac{12}{4}$$

$$y = 3$$

$\therefore$ The value of x is -6 and the value of y is 3

Q31) Solution

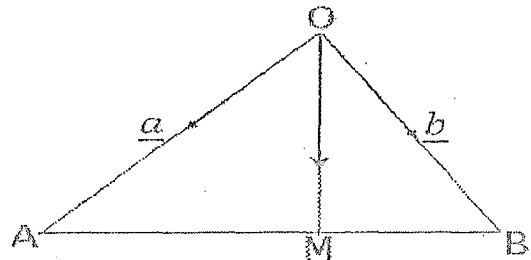
(i).



(ii). Coordinates of $A' = (-1, 1)$

Coordinates of $B' = (-3, 3)$

Q32) Solution



Need to find OM in terms of $\underline{a}$ and $\underline{b}$

$$\overrightarrow{OA} + \overrightarrow{AM} = \overrightarrow{OM}$$

$$\underline{a} + \overrightarrow{AM} = \overrightarrow{OM} \dots \dots \dots (\alpha)$$

$$\overrightarrow{OB} + \overrightarrow{BM} = \overrightarrow{OM}$$

$$\underline{b} + \overrightarrow{BM} = \overrightarrow{OM} \dots \dots \dots (\beta)$$

But $\overrightarrow{AM} = \overrightarrow{MB}$ (since M is the mid-point)
 $= -\overrightarrow{BM}$

Substitute $-\overrightarrow{BM}$ for $\overrightarrow{AM}$ in (α)

$$\underline{a} - \overrightarrow{BM} = \overrightarrow{OM} \dots \dots \dots (@)$$

Then $(@) + (\beta)$:

$$\underline{a} - \overrightarrow{BM} + \underline{b} + \overrightarrow{BM} = \overrightarrow{OM} + \overrightarrow{OM}$$

$$\underline{a} + \underline{b} - \overrightarrow{BM} + \overrightarrow{BM} = \overrightarrow{OM} + \overrightarrow{OM}$$

$$\underline{a} + \underline{b} = 2\overrightarrow{OM}$$

$$\therefore \frac{\underline{a} + \underline{b}}{2} = \overrightarrow{OM}$$

Topic 7

Exponential & Logarithmic functions

Introduction

An algebraic term such as 2^x is called an exponential function. The number is raised to power which is x ; such a variable is called an *exponent*. The number 2 is called the *base* of the function.

The base of an exponential can be a variable: $f(x) = y^x$ is an *exponential function*.

An equation such as $2^x = 32$ is an *exponential equation*. Such equations can be solved by expressing both sides of the equation in terms of the same base if possible. If not, a general method using logarithms can be applied.

The following laws of indices are true for all non-zero values of a , b , n , and x .

Laws of indices

- a) $x^a \times x^b = x^{a+b}$
- b) $x^a / x^b = x^{a-b}$
- c) $(x^a)^b = x^{ab}$
- d) $x^{1/n} = \sqrt[n]{x}$
- e) $a^0 = 1 (a \neq 0)$
- f) $a^1 = a$
- g) $a^{-1} = \frac{1}{a} (a \neq 0)$

Laws of logarithms

The following laws of logarithms are true for all non-zero values of a , p , m , x and y .

- a) $\log(xy) = \log x + \log y$
- b) $\log\left(\frac{x}{y}\right) = \log x - \log y$
- c) $\log x^p = p \log x$
- d) $\log \sqrt[m]{x} = \frac{1}{m} \log x$
- e) $\log_a 1 = 0$
- f) $\log_a a = 1$

Questions on Exponential & Logarithmic functions

- Q1) Solve for x $2^x = \frac{1}{32}$
- Q2) Find the value of n , if $27^n = 9$ $2^6 = 2^6$
- Q3) Without using tables, evaluate $\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 180$ $x = c$
- Q4) Solve the following equations
- a) $\log_4(x^2 - 3x) = 1$
 - b) $(3^x)^2 - 4(3^x) + 3 = 0$
- Q5) Solve for a
 $\log_a 45 + \log_a 5 = 2$
- Q6) Solve for x $2^x + 2^4 = 17$
- Q7) Given that $\frac{a^7}{a^{-3} \times a^2} = a^y$
- Q8) Solve the equation $\log_{10}(2m + 6) = 1 + \log_{10}(m - 1)$
- Q9) Solve for x if $(2^x) - 9(2^x) + 8 = 0$
- Q10) Given that $\log_m 27 = 3$, find m
- Q11) Given that $\log_9 27^k = k + 1$
- Q12) Given that $\log_{10} n - \log_{10} m = 2 \log_{10} h$, show that $n = mh^2$
- Q13) Evaluate $2 \log_a a + 3 \log_a 1 + \log_2 \sqrt{2}$
- Q14) Solve the equation $\log_7 343 = 2x - 5$
- Q15) Given that $\log_x(1/2) + \log_x 16 = 3$
- Q16) Solve for y $xy^3 = 5 - 2y^3$
- Q17) Solve the equation $3^y = \frac{9y}{81}$
- Q18) Given that $\frac{1}{2} \log_3 x = \log_3(6 - x)^{1/2}$, find the value of x .
- Q19) Given that $\log_{10} 5 = 0.6990$, without using a calculator or tables, evaluate $\log_{10} 2$
- Q20) Given that $\log_5 x + \log_5 y = 3 \log_5 q$. Show that $x = \frac{q^3}{y}$
- Q21) Given that $\log_y 864 - \log_y 6 = 2$, find the value of y .
- Q22) Given that $\log_x 6 \frac{1}{4} = 2$. Solve for x .
- Q23) Given that $2^{2a} - 5(2^a) + 4 = 0$, find the values of a .
- Q24) Without using four-figure tables or a calculator evaluate $\log_a 2 - \log_a 6 + \log_a 3$

Q25) Given that $\log_a c = 0.4475$, evaluate $\log_a \left(\frac{a^2}{c}\right)$

Exponential & Logarithmic Functions (Model Solutions)

Q1) Solution

$$2^x = \frac{1}{32} \quad (\text{Express 32 in terms of base 2})$$

$$\Rightarrow 2^x = \frac{1}{2^5}$$

$$2^x = 2^{-5}$$

Equating the indices

$$\therefore x = -5$$

Q2) Solution

$$27^n = 9 \quad (\text{Expressing 27 in base 3})$$

$$\Rightarrow 3^{3n} = 3^2$$

Equating the indices (Since bases are equal)

$$3n = 2$$

$$\therefore n = \frac{2}{3}$$

Q3) Solution

$$\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 180$$

$$= \log_{10} 25^{1/2} - \log_{10} 3^2 + \log_{10} 180$$

$$= \log_{10} 5 - \log_{10} 9 + \log_{10} 180 \quad \{25^{1/2} = \sqrt{25} = 5\}$$

$$= \log_{10} 100$$

$$= 2$$

Q4) Solution

a) $\log_4(x^2 - 3x) = 1$

$$\Rightarrow 4^1 = (x^2 - 3x)$$

$$4 = x^2 - 3x$$

$$x^2 - 3x = 4$$

$$x^2 - 3x - 4 = 0$$

$$x^2 + x - 4x - 4 = 0$$

$$x(x+1) - 4(x+1) = 0$$

$$(x+1)(x-4) = 0$$

Either $(x+1) = 0 \Rightarrow x = -1$

or $(x-4) = 0 \Rightarrow x = 4$

$$\therefore x = -1 \text{ or } x = 4$$

b) **Solution**

$$(3^x)^2 - 4(3^x) + 3 = 0$$

In this case, let $3^x = y$

Then substituting y for 3^x in the equation

$$y^2 - 4y + 3 = 0$$

$$y^2 - 3y - y + 3 = 0$$

$$y(y-3) - 1(y-3) = 0$$

$$(y-3)(y-1) = 0$$

$$\text{Either } (y-3) = 0 \Rightarrow y = 3$$

$$\text{or } (y-1) = 0 \Rightarrow y = 1$$

Substitute 3^x for y to get

$$3^x = 3 \Rightarrow 3^x = 3^1$$

$$\text{or } 3^x = 1 \Rightarrow 3^x = 3^0$$

Equating the powers

$$\Rightarrow x = 1 \text{ or } x = 0$$

Q5) Solution

$$\log_a 45 + \log_a 5 = 2$$

$$\log_a (45 \times 5) = 2$$

$$\log_a 225 = 2$$

$$a^2 = 225$$

$$a^2 = 15^2$$

Since the powers are equal the bases should also be equal.

$$\therefore a = 15$$

Q6) Solution

$$2^x + 2^4 = 17$$

$$2^x = 17 - 16$$

$$2^x = 1$$

$$2^x = 2^0$$

Since the bases are equal, the powers should also be equal

$$\therefore x = 0$$

Q7) Solution

$$\frac{a^7}{a^{-3} \times a^2} = a^y$$

$$\frac{a^7}{a^{-3+2}} = a^y$$

$$\frac{a^7}{a^{-1}} = a^y \quad (\text{Multiply both sides by } a^{-1})$$

$$a^7 = a^{-1} a^y$$

$$a^y = \frac{a^7}{a^{-1}}$$

$$a^y = a^{7-(-1)}$$

$$a^y = a^{7+1=8}$$

$$a^y = a^8$$

Since the bases are equal, the powers must also be equal

$$\therefore y = 8$$

Q8) Solution

$$\log_x 125^{-1} = -3$$

$$-1 \log_x 125 = -3 \quad (\text{Multiply both sides by } (-1))$$

$$\log_x 125 = 3$$

$$x^3 = 125$$

$$x^3 = 5^3$$

Since the powers are equal, the bases must be equal
 $\therefore x = 5$

Q9) Solution

$$\log_{10}(2m+6) = 1 + \log_{10}(m-1)$$

Subtract $\log_{10}(m-1)$ both sides to get

$$\log_{10}(2m+6) - \log_{10}(m-1) = 1$$

$$\text{Then, } \log_{10} \frac{(2m+6)}{(m-1)} = 1$$

$$\frac{(2m+6)}{(m-1)} = 10^1$$

(Multiply both sides by $(m-1)$)

$$(m-1) \times \frac{(2m+6)}{(m-1)} = 10 \times (m-1)$$

$$2m+6 = 10m-10$$

$$2m-10m = -10-6$$

$$-8m = -16$$

$$\frac{-8m}{-8} = \frac{-16}{-8}$$

$$m = 2$$

$\therefore$ The value of m is 2

Q10) Solution

$$\log_m 27 = 3$$

$$m^3 = 27 \text{ (Express 27 in base 3)}$$

$$m^3 = 3^3$$

Since the powers are equal, the bases should also be equal

$$m = 3$$

$\therefore$ The value of m is 3

Q11) Solution

$$\log_9 27^K = K + 1$$

$$\Rightarrow 9^{K+1} = 27^K$$

$$3^{2(K+1)} = 3^{3K}$$

$$3^{2K+2} = 3^{3K}$$

Since the bases are equal, the powers should be equal

$$\text{Thus, } 2K+2 = 3K$$

$$3K-2K = 2$$

$$\therefore K = 2$$

Q12) Solution

$$\log_{10} n - \log_{10} m = 2 \log_{10} h$$

$$\log_{10} \frac{n}{m} = \log_{10} h^2$$

$$\text{antilog}_{10}(\log_{10} \frac{n}{m}) = \text{antilog}_{10}(\log_{10} h^2)$$

$$\therefore \frac{n}{m} = h^2$$

$$\Rightarrow n = mh^2$$

Q13) Solution

$$2 \log_a a + 3 \log_a 1 + \log_2 \sqrt{2}$$

$$2 \log_a a + 3 \log_a 1 + \log_2 2^{1/2}$$

$$\text{Since } \log_a a = 1$$

$$\log_a 1 = 0$$

$$\text{and } \log_2 2^{1/2} = \frac{1}{2} \log_2 2$$

Then,

$$2 \log_a a + 3 \log_a 1 + \log_2 2^{1/2}$$

$$= 2(1) + 3(0) + \frac{1}{2} \log_2 2$$

$$= 2 + 0 + \frac{1}{2} (1)$$

$$= 2 + \frac{1}{2} = 2\frac{1}{2}$$

Q14) Solution

$$\log_7 343 = 2x - 5$$

$$\Rightarrow 7^{2x-5} = 343$$

$$7^{2x-5} = 7^3$$

Since the bases are equal, then the powers should be equal

$$\therefore 2x - 5 = 3$$

$$2x = 3 + 5$$

$$2x = 8$$

$$\frac{2x}{2} = \frac{8}{2}$$

$$x = 4$$

$\therefore$ The value of x is 4

Q15) Solution

$$\log_x \left(\frac{1}{2}\right) + \log_x 16 = 3$$

$$\Rightarrow \log_x \left(\frac{1}{2} \times 16\right) = 3$$

$$\log_x 8 = 3$$

$$x^3 = 8$$

$$x^3 = 2^3$$

Since the powers are equal, then the bases should be equal

$$x = 2$$

$\therefore$ The value of x is 2

Q16) Solution

$$xy^3 = 5 - 2y^3$$

$$\Rightarrow xy^3 + 2y^3 = 5$$

$$y^3(x+2) = 5$$

$$\frac{y^3(x+2)}{(x+2)} = \frac{5}{(x+2)}$$

$$y^3 = \frac{5}{(x+2)}$$

$$y = \sqrt[3]{\frac{5}{(x+2)}}$$

Q17) Solution

$$\begin{aligned} 3^y &= \frac{9^y}{81} \\ \Rightarrow 3^y \times 81 &= 9^y \\ 3^y \times 3^4 &= 3^{2y} \\ 3^{y+4} &= 3^{2y} \end{aligned}$$

Since the bases are equal, then the powers must be equal

$$\begin{aligned} \therefore y + 4 &= 2y \\ 2y - y &= 4 \\ y &= 4 \end{aligned}$$

Hence the value of y is 4

Q18) Solution

$$\begin{aligned} \frac{1}{2} \log_3 x &= \log_3 (6-x)^{1/2} \\ \frac{1}{2} \log_3 x &= \frac{1}{2} \log_3 (6-x) \\ \frac{1}{2} \log_3 x - \frac{1}{2} \log_3 (6-x) &= 0 \\ \frac{1}{2} \{\log_3 x - \log_3 (6-x)\} &= 0 \end{aligned}$$

Multiply both sides by (2) to get

$$\begin{aligned} \log_3 x - \log_3 (6-x) &= 0 \\ \log_3 \frac{x}{6-x} &= 0 \\ 3^0 &= \frac{x}{6-x} \\ 1 &= \frac{x}{6-x} \\ 1(6-x) &= x \\ 6-x &= x \\ x+x &= 6 \\ \frac{2x}{2} &= \frac{6}{2} \\ x &= 3 \end{aligned}$$

$\therefore$ The value of x is 3

Q19) Solution

$$\begin{aligned} \log_{10} 0.6990, \\ \log_{10} 2 &= \log_{10} \frac{10}{5} \\ &= \log_{10} 10 - \log_{10} 5 \\ &= 1 - 0.6990 \\ &= 0.301 \\ \therefore \log_{10} 2 &= 0.301 \end{aligned}$$

Q20) Solution

$$\text{Given } \log_5 x + \log_5 y = 3 \log_5 q.$$

We need to show that $x = \frac{q^3}{y}$

$$\begin{aligned} \log_5 x + \log_5 y &= 3 \log_5 q \\ \Rightarrow \log_5 (x \times y) &= \log_5 q^3 \\ \log_5 xy &= \log_5 q^3 \end{aligned}$$

$$\text{antilog}(\log_5 xy) = \text{antilog}(\log_5 q^3)$$

$$\begin{aligned} xy &= q^3 \\ \therefore x &= \frac{q^3}{y} \end{aligned}$$

Q21) Solution

$$\begin{aligned} \log_y 864 - \log_y 6 &= 2 \\ \log_y \left(\frac{864}{6} \right) &= 2 \text{ (Rule of division of logs)} \\ y^2 &= 144 \\ y &= \sqrt{144} \\ y &= 12 \end{aligned}$$

Q22) Solution

$$\begin{aligned} \log_x 6 \frac{1}{4} &= 2 \\ \Rightarrow \log_x \frac{25}{4} &= 2 \\ x^2 &= \frac{25}{4} \\ x &= \sqrt{\frac{25}{4}} \\ x &= \frac{5}{2} \end{aligned}$$

Q23) Solution

$$\begin{aligned} \text{Given } 2^{2a} - 5(2^a) + 4 &= 0 \\ \text{Let } 2^a &= x \\ \text{Then} \\ x^2 - 5x + 4 &= 0 \\ x^2 - 4x - x + 4 &= 0 \\ x(x-4) - 1(x-4) &= 0 \\ (x-4)(x-1) &= 0 \\ \text{Either } (x-4) &= 0 \Rightarrow x = 4 \\ \text{or } (x-1) &= 0 \Rightarrow x = 1 \\ 2^a &= 4 \text{ or } 2^a = 1 \text{ (substituting } 2^a \text{ for } x) \\ 2^a &= 2^2 \text{ or } 2^a = 2^0 \\ \therefore a &= 2 \text{ or } a = 0 \end{aligned}$$

Q24) Solution

$$\begin{aligned} \log_a 2 - \log_a 6 + \log_a 3 \\ &= \log_a 2 + \log_a 3 - \log_a 6 \\ &= \log_a \frac{(2 \times 3)}{6} \\ &= \log_a \frac{6}{6} \\ &= \log_a 1 \\ &= 0 \end{aligned}$$

Q25) Solution

$$\text{Since } \log_a c = 0.4475 \text{ (Given)}$$

$$\begin{aligned} \text{Then, } \log_a \left(\frac{a^2}{c} \right) &= \log_a a^2 - \log_a c \\ &= 2 \log_a a - \log_a c \text{ (But } \log_a a = 1) \\ &= 2(1) - 0.4475 \\ &= 2 - 0.4475 \\ &= 1.5525 \end{aligned}$$

Topic 8

Working with formulae (Subject of the formula)

- Q1) Make a the subject of the formula
 $m = t - a^2$
- Q2) Make x the subject of the formula
 $c = \frac{x^2 - y}{y}$
- Q3) Make c the subject of the formula
 $\frac{a}{b} = \sqrt{\frac{c-1}{c+1}}$
- Q4) Make d the subject of the formula
 $S = \frac{n}{2}[2a + (n-1)d]$
- Q5) Make the b the subject of the formula
 $c = ab - \frac{b}{a}$
- Q6) Make t the subject of the formula
 $y = \sqrt{a - \frac{t}{t+b}}$
- Q7) Make m the subject of the formula
 $y = \frac{m}{1+m}$
- Q8) Make r the subject of the formula
 $d = (t - r/r)^2$
- Q9) Make x the subject of the formula
 $4y = x^2$
- Q10) Make r the subject of the formula
 $S = \pi(2r)^2$
- Q11) Make y the subject of the formula
 $y^{1/2} = \frac{w + w(xy)^{1/2}}{x^{1/2}}$
- Q12) Make t the subject of the formula
 $M = K + \frac{3y^2}{t}$
- Q13) Make r the subject of the formula
 $P = m(r/x)^3$
- Q14) Make a the subject of the formula
 $x = \frac{\sqrt{a}}{y} + b$
- Q15) Make y the subject of the formula

- Q16) Make y the subject of the formula
 $xy^3 = 5 - 2y^3$
 $p = \sqrt{y - a}/y + 1$
- Q17) Make k the subject of the formula
 $r = (\sqrt{k+1})^2$
- Q18) Make m the subject of the formula
 $r = \frac{n}{\sqrt[3]{k-m}}$
- Q19) Make q the subject of the formula
 $a - \frac{3}{q^2} = p$
- Q20) Make y the subject of the formula
 $p = \frac{1}{4}xy^3h$
- Q21) Make R the subject of the formula
 $A = P\left(1 + \frac{R}{100}\right)^T$
- Q22) Make n the subject of the formula
 $an + 2b^2 = bn + 2a^2$
- Q23) Make t the subject of the formula
 $L = W\frac{(k-t)}{M} - t$
- Q24) Make x the subject of the formula
 $a^2 = b^2 - 2bcx + c^2$
- Q25) Make y the subject of the formula
 $\frac{y+d}{c} = \frac{3d}{y-d}$
- Q26) Make n the subject of the formula
 $T = ar^n$

Working with formulae Model Solutions

Q1) Solution

$$\begin{aligned}
 m &= t - a^2 \\
 \Rightarrow t - a^2 &= m \\
 -a^2 &= m - t \\
 \text{(Divide both sides by } -1) \\
 a^2 &= t - m \\
 \text{(Taking roots both sides)} \\
 \sqrt{a^2} &= \sqrt{t - m} \\
 \therefore a &= \sqrt{t - m}
 \end{aligned}$$

Q2) Solution

$$c = \frac{x^2 - y}{y} \text{ (Cross multiply to get)}$$

$$cy = x^2 - y$$

(Taking roots both sides)

$$\sqrt{x^2} = \sqrt{cy + y}$$

$$\therefore x = \pm \sqrt{cy + y}$$

Q3) Solution

$$\frac{a}{b} = \sqrt{\frac{c-1}{c+1}}$$

Square both sides

$$\frac{a^2}{b^2} = \frac{c-1}{c+1}$$

$$\frac{a^2}{b^2} = \frac{c-1}{c+1} \text{ (Cross multiply)}$$

$$b^2(c-1) = a^2(c+1)$$

Expand the brackets to get

$$b^2c - b^2 = a^2c + a^2$$

Collecting like terms together

$$b^2c - a^2c = a^2 + b^2$$

$$\frac{c(b^2 - a^2)}{b^2 - a^2} = \frac{a^2 + b^2}{b^2 - a^2} \text{ (Divide both sides by } b^2 - a^2)$$

$$\therefore c = \frac{a^2 + b^2}{b^2 - a^2}$$

Q4) Solution

$$S = \frac{n}{2} [2a + (n-1)d]$$

$$2S = n[2a + (n-1)d] \text{ (Multiply both sides by 2)}$$

$$2S = n(2a + dn - d)$$

$$2S = 2an + dn^2 - dn$$

$$2S - 2an = d(n^2 - n) \text{ (Factorising } dn^2 - dn)$$

$$d \frac{(n^2 - n)}{(n^2 - n)} = \frac{2S - 2an}{(n^2 - n)}$$

$$d = \frac{2S - 2an}{(n^2 - n)} \text{ [Dividing both sides by } (n^2 - n)]$$

$$d = \frac{2S - 2an}{n(n-1)}$$

Q5) Solution

$$c = ab - \frac{b}{a}$$

$$ab - \frac{b}{a} = c$$

$$(a) \times ab - \frac{b}{a} \times (a) = c \times (a)$$

$$a^2b - b = ac$$

$$b(a^2 - 1) = ac$$

$$b = \frac{ac}{a^2 - 1}$$

Q6) Solution

$$y^2 = \left(\sqrt{a - \frac{t}{t+b}} \right)^2 \text{ (Squaring both sides)}$$

(Cross-multiplying we get)

$$y^2 = a - \frac{t}{t+b}$$

$$y^2(t+b) = a - t$$

$$y^2t + y^2b = a - t$$

Rearranging like terms we get

$$y^2t + t = a - y^2b$$

$$t(y^2 + 1) = a - y^2b$$

$$t = \frac{a - y^2b}{y^2 + 1}$$

Q7) Solution

Multiply both sides by $(1+m)$

$$(1+m) \times y = \frac{m}{1+m} \times (1+m)$$

$$y + ym = m$$

$$ym - m = -y$$

$$m(y-1) = -y$$

$$m \frac{(y-1)}{(y-1)} = \frac{-y}{(y-1)} \text{ (Divide both sides by } (y-1))$$

$$m = \frac{-y}{(y-1)}$$

Q8) Solution

$$\sqrt{d} = \sqrt{\left(\frac{t-r}{r} \right)^2} \text{ (Taking roots both sides)}$$

$$\sqrt{d} = \left(\frac{t-r}{r} \right) \text{ (Cross-multiplying we get)}$$

$$r\sqrt{d} = (t-r)$$

$$r\sqrt{d} + r = t$$

$$r(\sqrt{d} + 1) = t$$

Dividing both sides by $(\sqrt{d} + 1)$, we get

$$r = \frac{t}{(\sqrt{d} + 1)}$$

Q9) Solution

$$\begin{aligned} 4y &= a^x \\ \Rightarrow \log a^x &= \log 4y \\ x \log_a a &= \log_a 4y \\ x \times 1 &= \log_a 4y \quad (\text{Since } \log_a a = 1) \\ \therefore x &= \log_a 4y \end{aligned}$$

Q10) Solution

$$\begin{aligned} S &= \pi(2r)^2 \Rightarrow S = \pi(4r^2) \\ \pi(4r^2) &= S \\ r^2 &= S/4\pi \\ \sqrt{r^2} &= \sqrt{S/4\pi} \quad (\text{taking roots both sides}) \\ r &= \sqrt{S/4\pi} \\ \therefore r &= \pm \frac{1}{2} \sqrt{\frac{S}{\pi}} \end{aligned}$$

Alternative method

$$\begin{aligned} S &= \pi(2r)^2 \\ \Rightarrow \pi(2r)^2 &= S \\ \frac{\pi(2r)^2}{\pi} &= \frac{S}{\pi} \\ (2r)^2 &= \frac{S}{\pi} \quad (\text{Taking square roots both sides}) \\ 2r &= \sqrt{\frac{S}{\pi}} \quad (\text{Dividing both sides by 2}) \\ \therefore r &= \pm \frac{1}{2} \sqrt{\frac{S}{\pi}} \end{aligned}$$

Q11) Solution

Note that $y^{1/2} = \sqrt{y}$

Therefore,

$$y^{1/2} = \frac{w+w(xy)^{1/2}}{x^{1/2}} \Rightarrow \sqrt{y} = \frac{w+w\sqrt{xy}}{\sqrt{x}}$$

Cross-multiplying we have

$$\begin{aligned} \sqrt{x}\sqrt{y} &= w + w\sqrt{xy} \\ \sqrt{x}\sqrt{y} - w\sqrt{xy} &= w \end{aligned}$$

$$\begin{aligned} \sqrt{x}\sqrt{y} - w\sqrt{x}\sqrt{y} &= w \\ \sqrt{x}\sqrt{y}(1-w) &= w \\ \frac{\sqrt{x}\sqrt{y}(1-w)}{\sqrt{x}(1-w)} &= \frac{w}{1-w} \end{aligned}$$

$$\sqrt{y} = \frac{w}{\sqrt{x}(1-w)}$$

$$\begin{aligned} (\sqrt{y})^2 &= \left\{ \frac{w}{\sqrt{x}(1-w)} \right\}^2 \\ y &= \frac{1}{x} \left(\frac{w}{1-w} \right)^2 \end{aligned}$$

Q12) Solution

$$\begin{aligned} K + \frac{3y^2}{t} &= M \\ \frac{3y^2}{t} &= M - K \quad (\text{Subtract both sides by } K) \\ t(M - K) &= 3y^2 \quad (\text{Dividing both sides by } M - K) \\ \therefore t &= \frac{3y^2}{M - K} \end{aligned}$$

Q13) Solution

$$P = m(r^3/x^3)$$

$$P = mr^3/x^3$$

$$mr^3 = Px^3$$

$$r^3 = Px^3/m$$

$$\sqrt[3]{r^3} = \sqrt[3]{Px^3/m} \quad (\text{taking cube roots both sides})$$

$$r = x \sqrt[3]{P/m}$$

Q14) Solution

$$\frac{\sqrt{a}}{y} = x - b \quad (\text{Cross-multiplying we get})$$

$$\sqrt{a} = y(x - b)$$

Squaring both sides to remove the root

$$\begin{aligned} (\sqrt{a})^2 &= \{y(x - b)\}^2 \\ \therefore a &= y^2(x - b)^2 \end{aligned}$$

Q15) Solution

$$xy^3 = 5 - 2y^3$$

$$xy^3 + 2y^3 = 5$$

$$y^3(x + 2) = 5 \text{ (Factorising)}$$

$$y^3 \frac{(x+2)}{(x+2)} = \frac{5}{(x+2)} \text{ (Dividing both sides by } (x+2) \text{)}$$

$$y^3 = \frac{5}{(x+2)} \text{ (Cubing both sides we get)}$$

$$y = \sqrt[3]{\frac{5}{(x+2)}}$$

Q16) Solution

$$p = \sqrt{y - a / y + 1} \text{ (Squaring both sides)}$$

$$(p)^2 = \left(\sqrt{y - a / y + 1} \right)^2$$

$$p^2 = \frac{y-a}{y+1} \text{ (Cross-multiplying we get)}$$

$$p^2(y+1) = y-a$$

$$p^2y + p^2 = y-a$$

$$p^2 + a = y - p^2y$$

$$p^2 + a = y(1 - p^2)$$

$$y(1 - p^2) = p^2 + a$$

$$\text{(Dividing both sides by } (1 - p^2) \text{)}$$

$$y = \frac{p^2 + a}{(1 - p^2)} \Rightarrow \frac{p^2 + a}{(1 - p)(1 + p)}$$

Q17) Solution

$$\sqrt{r} = \sqrt{(\sqrt{k} + 1)^2}$$

$$\text{(taking roots both sides we get)}$$

$$\sqrt{r} = \sqrt{k} + 1$$

$$(\sqrt{r})^2 = (\sqrt{k})^2 + (1)^2$$

$$r = k + 1$$

$$\text{Hence, } k = r - 1$$

Alternative method

$$\sqrt{r} = \sqrt{(\sqrt{k} + 1)^2}$$

$$\text{(taking roots both sides we get)}$$

$$\sqrt{r} = \sqrt{k} + 1$$

$$r^{1/2} - 1 = \sqrt{k}$$

$$(r^{1/2} - 1)^2 = (\sqrt{k})^2 \text{ (Squaring both sides)}$$

$$\text{Therefore, } k = (r^{1/2} - 1)^2$$

Q18) Solution

$$r = \frac{n}{\sqrt[3]{k-m}} \text{ (Cross-multiplying we get)}$$

$$(\sqrt[3]{k-m})r = n$$

$$\text{(Cubing both sides we get)}$$

$$(r)^3 \times (\sqrt[3]{k-m})^3 = (n)^3$$

$$r^3 \times (k-m) = n^3$$

$$r^3k - r^3m = n^3$$

$$r^3m = r^3k - n^3$$

$$m = \frac{r^3k - n^3}{r^3}$$

$$= \frac{r^3k}{r^3} - \frac{n^3}{r^3}$$

$$= k - \frac{n^3}{r^3}$$

Alternative method (No. 18)

$$r = \frac{n}{\sqrt[3]{k-m}} \text{ (Multiplying both sides by } \sqrt[3]{k-m} \text{)}$$

$$(\sqrt[3]{k-m})r = n$$

$$\text{Dividing both sides with } r \text{ we get}$$

$$(\sqrt[3]{k-m}) = \frac{n}{r}$$

$$\text{(cubing both sides we get)}$$

$$(\sqrt[3]{k-m})^3 = \left(\frac{n}{r}\right)^3$$

$$k-m = \frac{n^3}{r^3}$$

$$-m = \frac{n^3}{r^3} - k$$

$$m = k - \frac{n^3}{r^3}$$

Q19) Solution

$$a - \frac{3}{q^2} = p$$

$$\text{Then,}$$

$$\Rightarrow \frac{3}{q^2} = a - p$$

$$\text{(Cross-multiplying we get)}$$

$$q^2(a-p) = 3$$

$$\text{Dividing both sides by } (a-p) \text{ we get}$$

$$\frac{q^2(a-p)}{(a-p)} = \frac{3}{(a-p)}$$

$$\text{Hence } \sqrt{q^2} = \sqrt{\frac{3}{(a-p)}}$$

$$\therefore q = \sqrt{\frac{3}{(a-p)}}$$

Q20) Solution

$$(4) \times p = \frac{1}{4} xy^3 h \times (4)$$

$$4p = xy^3 h$$

$$xy^3 h = 4p$$

$$y^3 = \frac{4p}{xh}$$

$$\sqrt[3]{y^3} = \sqrt[3]{\frac{4p}{xh}} \quad (\text{cubing both sides})$$

$$y^3 = \sqrt[3]{\frac{4p}{xh}}$$

Q21) Solution

$$\frac{P(1+\frac{R}{100})^T}{P} = \frac{A}{P}$$

$$(1+\frac{R}{100})^T = \frac{A}{P}$$

$$\sqrt[T]{(1+\frac{R}{100})^T} = \sqrt[T]{(\frac{A}{P})} \quad (\text{taking } T^{\text{th}} \text{ root both sides})$$

$$(1+\frac{R}{100}) = \sqrt[T]{(\frac{A}{P})} \quad (\text{subtracting 1 both sides to get})$$

$$\frac{R}{100} = \sqrt[T]{(\frac{A}{P})} - 1 \quad (\text{multiply both sides by 100})$$

$$R = [\sqrt[T]{(\frac{A}{P})} - 1] 100$$

$$\therefore R = 100[\sqrt[T]{(\frac{A}{P})} - 1]$$

Q22) Solution

$$an + 2b^2 = bn + 2a^2$$

$$\Rightarrow an + bn = 2a^2 - 2b^2$$

Factor out common factors to get

$$n(a+b) = 2(a^2 - b^2)$$

$$n(a+b) = 2[(a-b)(a+b)]$$

Divide both sides by $(a+b)$

$$n \frac{(a+b)}{(a+b)} = 2 \left[\frac{(a-b)(a+b)}{(a+b)} \right]$$

$$\therefore n = 2(a-b)$$

Q23) Solution

$$-L = W \frac{(k-t)}{M} - t$$

$$\Rightarrow L \times M = M \times W \frac{(k-t)}{M} - t \times M$$

$$LM = W(k-t) - Mt$$

$$LM = Wk - Wt - Mt$$

$$Wt + Mt = Wk - LM$$

$$t(W+M) = Wk - LM \quad (\text{Factorising})$$

Dividing both sides by $W+M$

$$\therefore t = \frac{Wk - LM}{W+M}$$

Q24) Solution

$$a^2 = b^2 - 2bcx + c^2$$

$$2bcx = b^2 + c^2 - a^2$$

$$\frac{2bcx}{2bc} = \frac{b^2 + c^2 - a^2}{2bc}$$

$$x = \frac{b^2 + c^2 - a^2}{2bc}$$

Q25) Solution

$$(y+d)(y-d) = c \times (3d) \quad [\text{Cross multiply}]$$

Expand using difference of two squares

$$y^2 - d^2 = 3cd$$

$$y^2 = 3cd + d^2$$

Taking square root both sides, gives

$$\therefore y = \sqrt{3cd + d^2}$$

Q26) Solution

$$T = ar^n \quad (\text{Dividing both sides by } a \text{ we get})$$

$$\frac{T}{a} = r^n$$

Expressing the exponential to a logarithm equation

$$\frac{T}{a} = r^n$$

$$\log_r \frac{T}{a} = n$$

$$\therefore n = \log_r \frac{T}{a}$$

Topic 9

Trigonometry (SOHCAHTOA)

$$\tan = \frac{\text{opp}}{\text{adj}} \quad \cos = \frac{\text{adj}}{\text{hyp}} \quad \sin = \frac{\text{opp}}{\text{hyp}}$$

Introduction

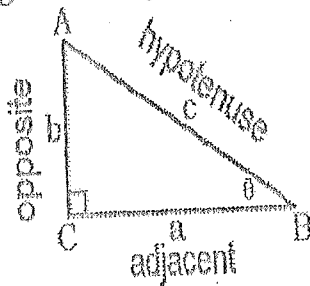
How are heights of mountains measured? Mountain climbers do not carry tape measures. Mathematics called **trigonometry** is used. The word "trigonometry" has three parts:

"tri" means "three",
"gon" means "shape", and
"metry" means "measure".

We will use rules and formulae to calculate the side lengths and the angles of right-angled triangles.

Trigonometrical formulae

Right-angled triangles



Right angle - In a right-angled triangle, there is always a right angle or 90° .

Angle θ - The angle in which we are interested; uses the Greek letter "theta" which is the equivalent of the English letter "T"

Hypotenuse - The side opposite the right angle

Opposite side - The side opposite angle; the side all the way across the triangle from angle θ

Adjacent side - The side adjacent angle θ ; the side next door to angle θ

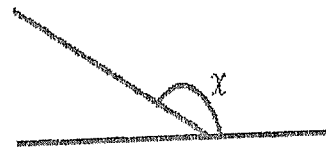
In the right-angled triangle ABC above

$$c^2 = a^2 + b^2 \quad (\text{Pythagoras' Theorem})$$

$$\sin A = \frac{a}{c} \quad \cos A = \frac{b}{c} \quad \tan A = \frac{a}{b}$$

$$\sin B = \frac{b}{c} \quad \cos B = \frac{a}{c} \quad \tan B = \frac{b}{a}$$

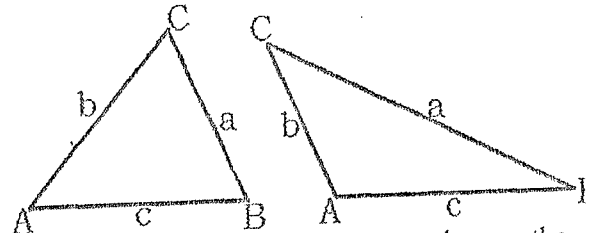
Obtuse angles



$$\sin x = \sin(180 - x)$$

$$\cos x = -\cos(180 - x)$$

$$\tan x = -\tan(180 - x)$$



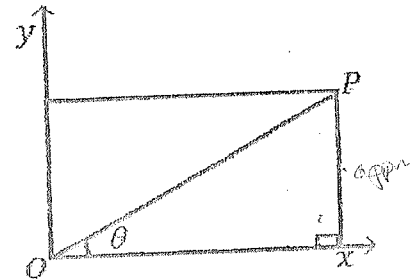
In both of the triangles shown above, the following formulae may be used to solve the triangle.

$$\text{Sine rule} \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\text{Cosine rule} \quad c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{Area of } \triangle ABC = \frac{1}{2}ab \sin C$$



The projection of OP on Ox is $OP \cos \theta$

The projection of OP on Oy is $OP \sin \theta$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Trigonometrical ratios of the angles

$0^\circ, 90^\circ, 30^\circ, 60^\circ, 45^\circ$

$$\sin 0^\circ = 0 \quad \cos 0^\circ = 1 \quad \tan 0^\circ = 0$$

$$\sin 90^\circ = 1 \quad \cos 90^\circ = 0 \quad \tan 90^\circ \text{ is undefined}$$

$$\sin 30^\circ = \frac{1}{2} \quad \cos 30^\circ = \frac{\sqrt{3}}{2} \quad \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2} \quad \cos 60^\circ = \frac{1}{2} \quad \tan 60^\circ = \sqrt{3}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}} \quad \cos 45^\circ = \frac{1}{\sqrt{2}} \quad \tan 45^\circ = 1$$

Angle of elevation

The angle of elevation is the angle between an imaginary horizontal line drawn from the place where the observation is taking place and a point or object above that place.

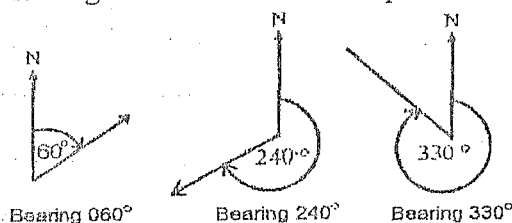
Angle of depression

The angle of depression is the angle between an imaginary horizontal line drawn from the place where the observation is taking place and a point or object below that place.

Bearing

- A bearing is an acute angle measured from a north-south line towards either the east or the west. In short, bearings are a measure of direction, with North taken as a reference
- If you are travelling North, your bearing is 000° , and this is usually represented as straight up on the page.
- If you are travelling in any other direction, your bearing is measure clockwise from North.

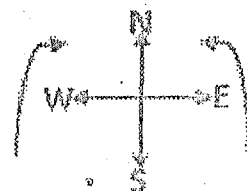
REMEMBER: Bearings are always measured clockwise from North and are given as 3 digits. Here are some examples:



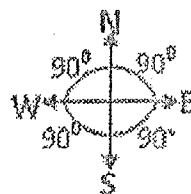
Rules for Solving Bearing and Distances

- Taking reading in bearing starts from the North Pole in clockwise direction and

ends also at the North pole. I.e. North – North Pole reading.



- All angles formed while taking reading in bearing is equal to 360 degrees.



- All questions in bearing leads into the formation of a triangle.



- In triangle PQR , $PQ = 7\text{ cm}$, angle $Q = 90^\circ$ and angle $P = 25^\circ$, find the length of PQ to 2 significant figures.
- A ladder whose base is 6 m from a wall reaches 10 m up the wall. What is the angle between the ladder and the ground?
- An isosceles triangle has equal sides of 8 cm and a third side of length 6 cm . calculate the equal angles.
- Find two values of x between 0° and 360° which satisfy the equation:
 - $\sin x = 0.7$
 - $5 \cos x = 2 \sin x$
- If $\cos x = -\frac{1}{\sqrt{5}}$ and $90^\circ \leq x \leq 180^\circ$ Calculate:
 - $\sin x$
 - $\tan x$
- Given that $x = \frac{5}{12}$, evaluate

$$\frac{\cos x - \sin x}{\cos x + \sin x}$$

Q7) A ship travels 10 km on a course heading 50° east of north. How far north and how far east has the ship travelled at this point?

Q8) A cable car at a ski resort carries skiers up to the top of the mountain. It is supported by a steel cable. The vertical height of the mountain is 3000 metres and the angle that the cable makes with the ground is 40° . What is the sloped length of the cable?

Q9) For a motorbike stunt, stuntman Frank Gareta must ride at high speed up a ramp with a vertical height of 20 metres and a sloped length of 30 metres. What is the angle with the ground?

Q10) Kite aerial photography uses remote control to move a kite and take camera shots from a bird's eye view. The length of the kite cable is 2000 metres. The angle that the cable makes with the ground is 36° . What is the distance in a horizontal direction that the kite is away from the photographer?

Q11) In Nicaragua in South America, volcano boarding is becoming a popular extreme sport. If the horizontal distance from the centre of the volcanic hill is 1280 metres and the angle of the slope with level ground is 50° , what is the length of the whole boarding slope?

Q12) If I stand 20 metres away from the base of the tree, the angle from my feet to the top of the tree is 60° . What is the height of the tree?

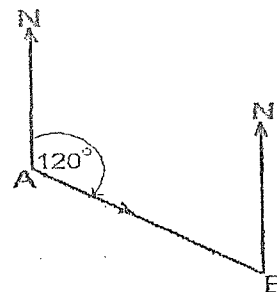
Q13) Village B is on the bearing of 135° and a distance of 40 km from village A. Village C is on a bearing of 225° and a distance of 62 km from village A.

- Show that A, B and C form a right-angled triangle.
- Calculate angle ACB to the nearest degree.

Q14) At a cinema, a wheelchair ramp must be built next to steps that are 3 metres high. For safety, the maximum angle of the ramp is

5° . What must be the horizontal length of the ramp?

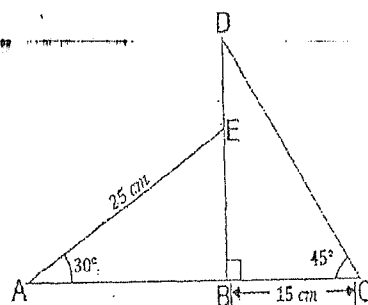
Q15) A ship sails from A to B on a bearing of 120° .



On what bearing will it have to sail to return from B to A?

Q16) Given that angle θ is acute and that $\log \cos \theta = \bar{1}.75$, evaluate $(\cos \theta)^2$

Q17) In Figure below, DB is perpendicular to the line ABC, AE = 25 cm, BC = 15 cm, angle EAB = 30° , and angle BCD = 45° .



Q18) Given that $\sin \theta = \frac{1}{3}$, find $\cos \theta$, leaving your answer in surd form

Q19) Without using a calculator or four-figure tables, find $\cos B$ if $\sin B = 0.8$

Q20) An aeroplane takes off at an angle of 23° to the horizontal ground and flies for 13 km. Calculate its height above the ground, leaving your answer correct to 1 decimal place.

Q21) Without using a calculator or four-figure tables, evaluate $\frac{\tan 60^\circ}{\tan 30^\circ}$

Q22) Find the values of: $\sin 25^\circ$, $\cos 25^\circ$ and $\tan 25^\circ$ and show that:

$$a) \frac{\sin 25^\circ}{\cos 25^\circ} = \tan 25^\circ$$

$$b) \sin^2 25^\circ + \cos^2 25^\circ = 1$$

Q23) From the top of a 200 meters high building, the angle of depression to the bottom of a second building is 20 degrees.

From the same point, the angle of elevation to the top of the second building is 10 degrees. Calculate the height of the second building.

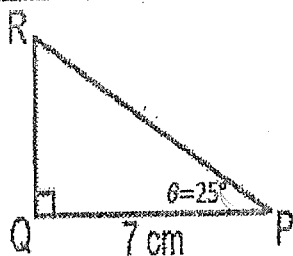
Q24) A boat sails 6km from a port X on a bearing of 065° and thereafter 13km on a bearing of 136°. What is the distance and bearing of the boat from X?

Q25) An aircraft takes off from an airstrip at an average speed of 20km/hr on a bearing of 052° for 3 hours. It then changes course and flies on a bearing of 028° at an average speed of 30km/hr for another 1½ hours. Find;

- its distance from the starting point,
- the bearing of the aircraft from the airstrip.

Trigonometry (Model Solutions)

Q1) Solution



$$\text{Since } \tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{RQ}{7}$$

$$\tan 25^\circ = \frac{RQ}{7} \text{ (Multiply both sides by 7)}$$

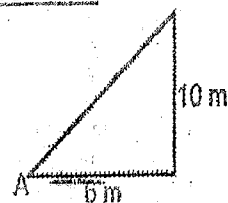
$$7 \times \tan 25^\circ = RQ$$

$$RQ = 7 \times 0.4663$$

$$RQ = 3.2641 \text{ cm}$$

$$RQ = 3.3 \text{ cm (to 2 s.f.)}$$

Q2) Solution

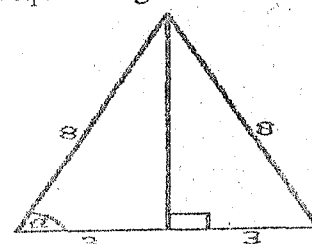


$$\tan A = \frac{10}{6} = 1.6667$$

$$\angle A = 59.0^\circ \text{ (to 3 s.f.)}$$

Q3) Solution

We divide the isosceles triangle into two right-angled triangles. In this case, the required angle is a .



$$\cos a = \frac{3}{8}$$

$$= 0.375$$

$$\angle a = 68.0^\circ \text{ (to 3 s.f.)}$$

Q4) Solution

$$(i) \sin x = 0.7$$

$$x = \sin^{-1} 0.7$$

$$x = 44.43^\circ$$

$$\sin x = (180^\circ - 44.3^\circ)$$

$$x = (180^\circ - 44.3^\circ)$$

$$x = (135.57^\circ)$$

$$x = 44.3^\circ \text{ or } 135.57^\circ$$

$$(ii) 5 \cos x = 2 \sin x$$

Divide both sides by $2 \sin x$

$$\frac{2}{5} = \frac{\sin x}{\cos x}$$

$$\Rightarrow \tan x = \frac{5}{2}$$

$$x = \tan^{-1} \left(\frac{5}{2} \right)$$

$$= 68.2^\circ$$

$$\begin{aligned}\tan x &= -\tan(180^\circ - x) \\ \therefore \tan x &= (180^\circ - 68.2^\circ) \\ \therefore \tan x &= (111.8^\circ) \\ \therefore x &= 68.2^\circ \text{ or } 111.8^\circ\end{aligned}$$

Q5) Solution

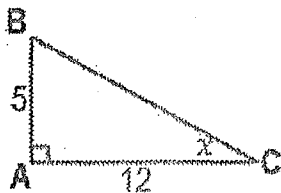
(ii). Starting from

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1 \\ \Rightarrow \sin^2 x &= 1 - \cos^2 x \\ \text{But } \cos x &= \frac{1}{\sqrt{5}} \text{ (given)} \\ \therefore \cos^2 x &= \frac{1}{5} \\ \sin^2 x &= 1 - \frac{1}{5} \\ &= \frac{4}{5} \\ \Rightarrow \sin x &= \pm \sqrt{\frac{4}{5}} = \pm \frac{2}{\sqrt{5}} \\ \sin x &= \pm \frac{2}{\sqrt{5}}\end{aligned}$$

(iii). $\tan x = \frac{\sin x}{\cos x}$

$$\begin{aligned}&= \frac{\frac{2}{\sqrt{5}}}{\frac{1}{\sqrt{5}}} \\ &= \frac{2}{\sqrt{5}} \div \frac{1}{\sqrt{5}} \\ &= \frac{2}{\sqrt{5}} \times \frac{\sqrt{5}}{1} = 2 \\ \therefore \tan x &= 2\end{aligned}$$

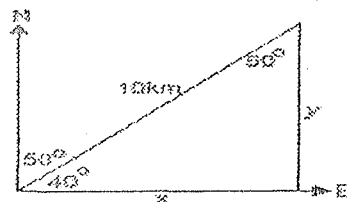
Q6) Solution



$$\begin{aligned}BC^2 &= AB^2 + AC^2 \text{ (Pythagoras Theorem)} \\ BC^2 &= 5^2 + 12^2 \\ BC^2 &= 25 + 144 \\ BC^2 &= 169 \\ BC &= \sqrt{169} \\ \therefore BC &= 13 \\ \cos x &= \frac{AC}{BC} = \frac{12}{13}\end{aligned}$$

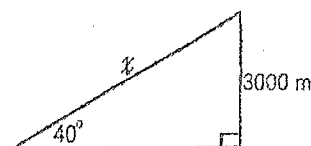
$$\begin{aligned}\sin x &= \frac{AB}{BC} = \frac{5}{13} \\ \frac{\cos x - \sin x}{\cos x + \sin x} &= \frac{\frac{12}{13} - \frac{5}{13}}{\frac{12}{13} + \frac{5}{13}} \\ &= \frac{\frac{12-5}{13}}{\frac{12+5}{13}} \\ &= \frac{7}{17} \div \frac{17}{13} \\ &= \frac{7}{17} \times \frac{13}{17} \\ &= \frac{91}{289}\end{aligned}$$

Q7) Solution



$$\begin{aligned}\sin 40^\circ &= \frac{\text{opp}}{\text{hyp}} \\ \therefore \sin 40^\circ &= \frac{y}{10} \\ 10 \times \sin 40^\circ &= y \\ y &= 10 \times 0.6428 \\ y &= 6.43 \text{ km} \\ \sin 50^\circ &= \frac{\text{opp}}{\text{hyp}} \\ \therefore \sin 50^\circ &= \frac{x}{10} \\ 10 \times \sin 50^\circ &= x \\ x &= 10 \times 0.7660 \\ x &= 7.66 \text{ km}\end{aligned}$$

Q8) Solution



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin 40^\circ = \frac{3000}{x}$$

$$(x) \times \sin 40^\circ = 3000$$

$$x \times 0.6428 = 3000$$

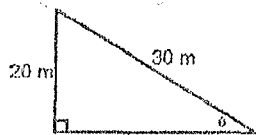
$$0.6428x = 3000$$

$$x = \frac{3000}{0.6428}$$

$$x = 4667.17 \text{ m}$$

The sloped length of the cable must be 4667.17 metres.

Q9) Solution



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin \theta = \frac{20}{30}$$

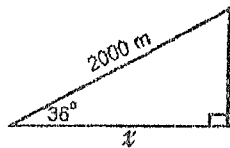
$$\theta = \sin^{-1}\left(\frac{20}{30}\right)$$

$$\theta = \sin^{-1}(0.6667)$$

$$\theta = 41.81^\circ$$

The angle of the stunt ramp is 41.81° .

Q10) Solution



$$\cos \theta = \frac{\text{adj}}{\text{hyp}} \Rightarrow \frac{x}{2000}$$

$$\cos 36^\circ = \frac{x}{2000}$$

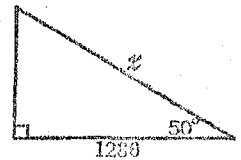
$$2000 \times \cos 36^\circ = x$$

$$x = 2000 \times 0.8090$$

$$\therefore x = 1618 \text{ m}$$

The horizontal distance is 1618 metres.

Q11) Solution



$$\cos \theta = \frac{\text{adj}}{\text{hyp}} \Rightarrow \frac{1280}{x}$$

$$\cos 50^\circ = \frac{1280}{x}$$

$$x \times \cos 50^\circ = 1280$$

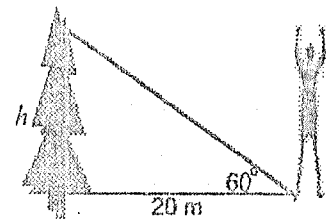
$$x = \frac{1280}{\cos 50^\circ}$$

$$x = \frac{1280}{0.6428}$$

$$x = 1991.33$$

The sloped length is 1991.33 metres.

Q12) Solution



$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{h}{20 \text{ m}}$$

$$\tan 60^\circ = \frac{h}{20 \text{ m}}$$

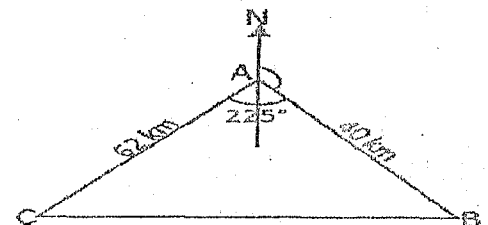
$$h = 20 \text{ m} \times \tan 60^\circ$$

$$h = 20 \text{ m} \times 1.7321$$

$$h = 34.64 \text{ m}$$

The height of the tree is 34.64 metres.

Q13) Solution



a) Need to show that one of the angles of $\triangle ABC$ is a right angle.

$$\text{Reflex } \angle NAC = 225^\circ$$

$$\text{Obtuse } \angle NAB = 135^\circ$$

$$\angle BAC = \text{Reflex } \angle NAC - \text{Obtuse } \angle NAB$$

$$= 225^\circ - 135^\circ$$

$$= 90^\circ$$

$\therefore \angle BAC = 90^\circ$ (which is a right-angle)

Hence A, B, A form a right-angled Δ

b) Need to calculate $\angle ACB$:

$$\text{Now, } \tan \angle ACB = \frac{AB}{AC}$$

$$\tan \angle ACB = \frac{40 \text{ km}}{62 \text{ km}}$$

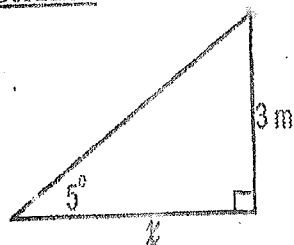
$$\tan \angle ACB = \frac{40}{62}$$

$$\angle ACB = \tan^{-1} \left(\frac{40}{62} \right)$$

$$\angle ACB = 33^\circ \text{ (to the nearest degree)}$$

Hence, the value of $\angle ACB$ is 33°

Q14) Solution



$$\tan 5^\circ = \frac{\text{opp}}{\text{adj}} = \frac{3}{x}$$

$$\tan 5^\circ = \frac{3}{x}$$

$$3 = x \times \tan 5^\circ$$

$$x \times \tan 5^\circ = 3$$

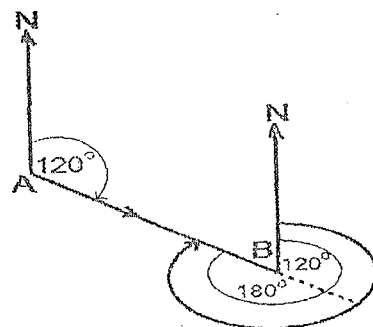
$$x = \frac{3}{\tan 5^\circ}$$

$$\therefore x = 34.29$$

The horizontal length of the ramp must be 34.29 metres.

Q15) Solution

We can extend the line from A to B, then rotate through 180° to head in the opposite direction.



We can see from the diagram above that the bearing from B to A is 300° , because $120^\circ + 180^\circ = 300^\circ$.

Q16) Solution

$$\cos^2 \theta = (\cos \theta)^2$$

$$\log(\cos^2 \theta) = \log(\cos \theta)^2$$

$$\text{RHS} \Rightarrow 2 \log(\cos \theta)$$

$$= 2(-1 + 0.75)$$

$$= 2(-0.25)$$

$$= -0.5$$

$$= -1 + 0.5$$

{Note: $(-1 + 0.5)$ is equal to -0.5 }

$$(\cos \theta)^2 = 10^{\log(\cos \theta)^2}$$

$$= 10^{-1+0.5}$$

$$= 10^{-1} \times 10^{0.5}$$

$$= \frac{1}{10} \times 3.16$$

$$= 0.316$$

$$\therefore \cos^2 \theta = 0.316$$

Q17) Solution

In ΔDBC ;

$$\tan \angle BCD = \frac{DB}{CB}$$

$$\tan 45^\circ = \frac{DB}{15 \text{ cm}}$$

$$DB = 15 \text{ cm} \times \tan 45^\circ$$

$$= 15 \text{ cm} \times 1$$

$$= 15 \text{ cm}$$

In ΔEBA ;

$$\sin \angle EBA = \frac{EB}{AE}$$

$$\sin 30^\circ = \frac{EB}{25 \text{ cm}}$$

$$EB = 25 \text{ cm} \times \sin 30^\circ$$

$$EB = 25 \text{ cm} \times \frac{1}{2} \quad (\text{since } \sin 30^\circ = \frac{1}{2})$$

$$EB = 12.5 \text{ cm}$$

$$\therefore DE = DB - EB$$

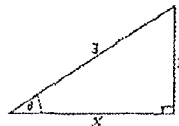
$$= 15 \text{ cm} - 12.5 \text{ cm}$$

$$= 2.5 \text{ cm}$$

$\therefore$ The length of DE is 2.5 cm

Q18) Solution

Given $\sin \theta = \frac{1}{3} = \frac{\text{opp}}{\text{hyp}}$



$$x^2 = 3^2 - 1^2$$

$$x^2 = 9 - 1$$

$$x^2 = 8$$

$$x = \pm\sqrt{8}$$

Since x is the length, the negative number is not considered.

$$\therefore x = \sqrt{8}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{x}{3}$$

$$\cos \theta = \frac{\sqrt{8}}{3}$$

$$= \frac{\sqrt{4 \times 2}}{3} = \frac{\sqrt{2} \times \sqrt{4}}{3}$$

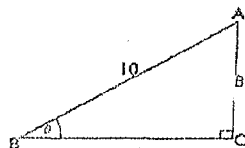
$$= \frac{2\sqrt{2}}{3}$$

Q19) Solution

$$\sin B = 0.8$$

$$= \frac{8}{10}$$

This can be shown diagrammatically as follows:



$$AB^2 = BC^2 + AC^2$$

$$10^2 = BC^2 + 8^2$$

$$100 = BC^2 + 64$$

$$BC^2 = 100 - 64$$

$$BC^2 = 36$$

$$BC = \sqrt{36}$$

$$\therefore BC = 6$$

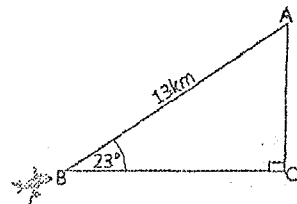
$$\cos B = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos B = \frac{BC}{AB}$$

$$\cos B = \frac{6}{10}$$

$$\therefore \cos B = 0.6$$

Q20) Solution



We need to find the length AC.

Now: $\sin B = \frac{AC}{AB}$

$$\sin 23^\circ = \frac{AC}{13 \text{ km}}$$

$$AC = 13 \text{ km} \times \sin 23^\circ$$

$$= 13 \text{ km} \times 0.3907$$

$$= 5.1 \text{ km (to 1 decimal place)}$$

$\therefore$ The height of the airplane above the ground is 5.1 km

Q21) Solution

Since $\tan 60^\circ = \sqrt{3}$ and $\tan 30^\circ = \frac{1}{\sqrt{3}}$

$$\frac{\tan 60^\circ}{\tan 30^\circ} \Rightarrow \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}}$$

$$= \sqrt{3} \div \frac{1}{\sqrt{3}}$$

$$= \sqrt{3} \times 3$$

$$= \sqrt{9}$$

$$= 3$$

Q22) Solution

a) $\sin 25^\circ = 0.4226$,
 $\cos 25^\circ = 0.9063$
 and $\tan 25^\circ = 0.4663$ (to 4 d.p)

$$\begin{aligned} \text{LHS} &\Rightarrow \frac{\sin 25^\circ}{\cos 25^\circ} \\ &= \frac{0.4226}{0.9063} \\ &= 0.4663 \\ \therefore \text{LHS is } 0.4663 \end{aligned}$$

Similarly,

$$\begin{aligned} \text{RHS} &\Rightarrow \tan 25^\circ \\ &= 0.4663 \end{aligned}$$

$\therefore \text{RHS is } 0.4663$

Now, since $\text{LHS} = \text{RHS}$

$$\Rightarrow \frac{\sin 25^\circ}{\cos 25^\circ} = \tan 25^\circ \text{ (as required)}$$

b) $\sin^2 25^\circ + \cos^2 25^\circ = 1$

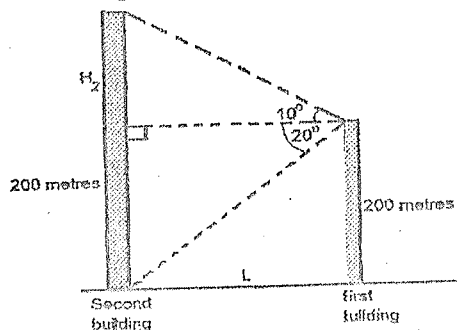
$$\begin{aligned} \text{LHS} &\Rightarrow \sin^2 25^\circ + \cos^2 25^\circ \\ &= (0.4226)^2 + (0.9063)^2 \\ &= 0.1786 + 0.8214 \\ &= 1 \end{aligned}$$

Also, $\text{RHS} = 1$

$$\therefore \text{LHS} = \text{RHS} \Rightarrow \sin^2 25^\circ + \cos^2 25^\circ = 1$$

Q23) Solution

The diagram below shows the two buildings and the angles of depression and elevation.



$$\tan 20^\circ = \frac{\text{opp}}{\text{adj}} = \frac{200}{L}$$

$$\tan 20^\circ = \frac{200}{L}$$

$$L \times \tan 20^\circ = 200$$

$$L = \frac{200}{\tan 20^\circ}$$

Similarly,

$$\tan 10^\circ = \frac{\text{opp}}{\text{adj}} = \frac{H_2}{L}$$

$$\tan 10^\circ = \frac{H_2}{L}$$

$$L \times \tan 10^\circ = H_2$$

Height of the second building

$$= 200 + \left(\frac{200 \times \tan 10^\circ}{\tan 20^\circ} \right)$$

$$= 258 \text{ (to 1 d.p)}$$

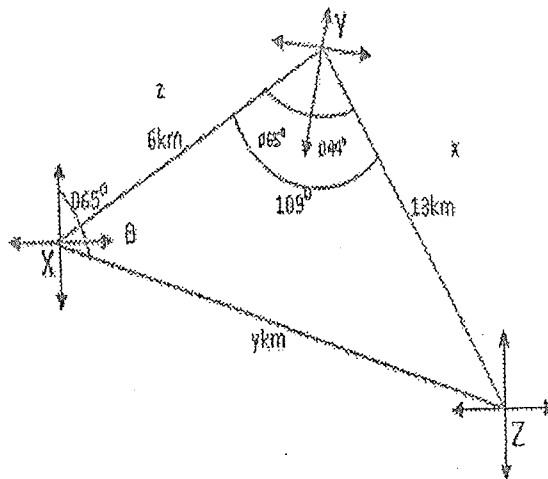
Q24) Solution

Step 1

You will need to represent the question on a triangle.

You will actually need a ruler and pencil for drawing along with a good eraser in case you make mistake.

Take your pencil and draw a point X, take ruler and rule a straight line to another point Y and rule another straight line to point Z and finally close it up at point X again. See below;



Step 2

Use Cosine Rule to solve for y km;

From Cosine Rule;

$$b^2 = a^2 + c^2 - 2ab \cos B$$

Step 3

Replace a, b, c and B with x, y, z and Y respectively;

$$y^2 = x^2 + z^2 - 2xz \cos Y$$

$$y^2 = 13^2 + 6^2 - 2(13)(6) \cos 109^\circ$$

$$y^2 = 169 + 36 - 156 \times (-0.3256)$$

$$y^2 = 205 - 156 + 0.3256$$

$$y^2 = 205 - 50.7886$$

$$y^2 = 255.7886$$

Step 4

Take square root of both sides;

$$y = \sqrt{255.7886}$$

$$y = 15.9934 \text{ km} \sim 16 \text{ km to the nearest km}$$

Step 5

To find the bearing of the boat from X, we will apply Sine Rule;

From Sine Rule;

$$\frac{\sin X}{x} = \frac{\sin Y}{y}$$

$$\frac{\sin \theta}{13 \text{ km}} = \frac{\sin 109^\circ}{16 \text{ km}}$$

Step 6

Cross Multiply;

$$16 \text{ km} \times \sin \theta = 13 \text{ km} \times \sin 109^\circ$$

$$16 \text{ km} \times \sin \theta = 13 \text{ km} \times 0.9455$$

$$16 \text{ km}(\sin \theta) = 12.2917$$

Step 7

Divide both sides by 16km;

$$\sin \theta = \frac{12.2917}{16 \text{ km}}$$

$$\sin \theta = 0.7682$$

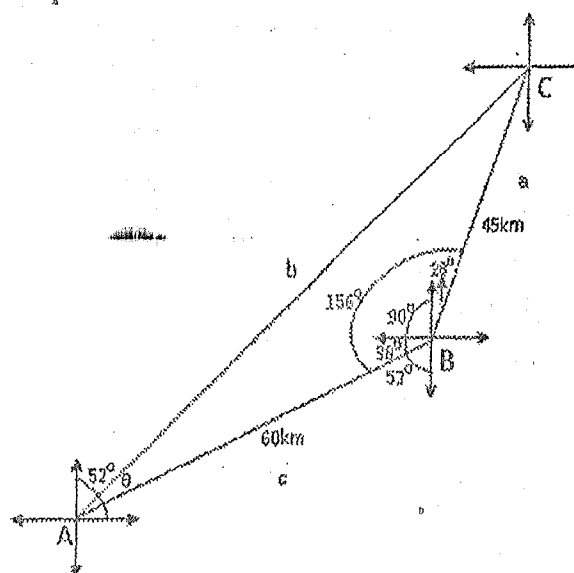
$$\theta = \sin^{-1} 0.7682$$

$$\theta = 50.1955^\circ \sim 50^\circ \text{ (to the nearest degree)}$$

Step 8

$\therefore$ The bearing of boat from the port will be $065^\circ + \theta$

Which is $065^\circ + 50^\circ = 115^\circ$ (answer).

Q25) Solution**Step 1****Step 2**

Use Cosine rule to solve for side b ;

From Cosine Rule;

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$b^2 = 45^2 + 60^2 - 2(45)(60) \cos 156^\circ$$

$$b^2 = 2025 + 3600 - 5400(-0.9135)$$

$$b^2 = 5625 + 5400 \times 0.9135$$

$$b^2 = 5625 + 4932.9$$

$$b^2 = 10557.9$$

$$b = \sqrt{10557.9}$$

$$b = 102.7516 \sim 103 \text{ (nearest whole number)}$$

$$\therefore b = 103 \text{ km}$$

Step 3

To find the bearing of the aircraft from the airstrip, you must first find θ ;

Find θ using Sine Rule below;

From Sine Rule;

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin \theta}{45 \text{ km}} = \frac{\sin 156^\circ}{103 \text{ km}}$$

Step 4

Cross Multiply;

$$103\text{km}(\sin \theta) = 45\text{km} \times \sin 156^\circ$$

Step 5

Divide both sides by 103km;

$$\frac{103\text{km}(\sin \theta)}{103\text{ km}} = \frac{45\text{km} \times \sin 156^\circ}{103\text{ km}}$$

$$\sin \theta = \frac{18.3015}{103\text{ km}}$$

$$\sin \theta = 0.1776$$

(Note; when taking Sin to the RHS, it becomes inverse)

$$\theta = \sin^{-1} 0.1776$$

$$\theta = 10.2299^\circ$$

To find the bearing will be $052^\circ - \theta$

$$\therefore \text{Bearing} = 052^\circ - 10.2299^\circ$$

$$\text{Bearing} = 41.77^\circ$$

Topic 10

Similar Figures

Area and similar triangles theorems

In fact *two* triangles are similar if any one of the following is true:

- (i). The sides of one triangle are proportional to the sides of the other triangle.
- (ii). The angles of one triangle are respectively equal to the angles of the other triangle.
- (iii). One angle of one triangle is equal to one angle of the other triangle, and the sides adjacent to the equal angles are proportional.

If any one of these statements is true, then the other two are true.

- (iv). The ratio of the areas of similar triangles is equal to the ratio of the squares of corresponding sides.

In general, *if the lengths of two similar figures are in the ratio $k:1$, the ratio of their areas is $k^2:1$*

We say that the *area factor* is $k^2:1$

- (v). The ratio of the volumes of similar solids is the cube of the scale factor of the two solids.
- (vi). If a perpendicular is drawn from the right angle of a right-angled triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other.
- (vii). Parallelograms on the same base and between the same parallels are equal in area.
- (viii). If two triangles are equiangular, their corresponding sides are proportional.
- (ix). If two triangles have their corresponding sides proportional, they are equiangular. (*This theorem is the converse of the above theorem*).
- (x). For any given triangle, if a parallel line is drawn from one side of the triangle to the

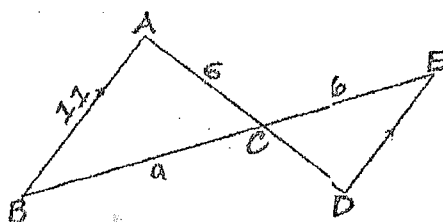
other side then the two triangles formed will be similar.

- (xi). It is also true that if a line divides two sides of a triangle proportionally, then it is parallel to the third side of the triangle.

(This is the converse of the above theorem)

Questions on Similar Figures

- Q1) In the Figure below, $AB \parallel ED$ and AD and BE intersect at C . $AB = 11 \text{ cm}$, $BC = 9 \text{ cm}$ and $AC = CE = 6 \text{ cm}$. Find the lengths of CD and DE .



- Q2) A polygon has an area of 28 cm^2 . What is the area of a similar polygon all of whose sides are 5 times as long.
- Q3) A piece of cloth 4 cm by 5 cm costs K800. How much will a piece of cloth cost that is twice as long and twice as broad? (Assume that the price is proportional to the area.)
- Q4) A woman uses 5 m^2 and 1.2 m^2 of cloth when making similar dresses for herself and her daughter respectively. If the woman is 165 cm tall, how tall is the daughter?
(Assume that the woman and daughter are similar in build)
- Q5) A map is drawn to a scale of $1:5000$. On the map, a village has an area of 6 cm^2 . Find the true area of the village in hectares. ($1 \text{ ha} = 10000 \text{ m}^2$)
- Q6) Two heaps of rice are of similar shape and contain 128 kg and 250 kg of rice respectively. If the height of the bigger

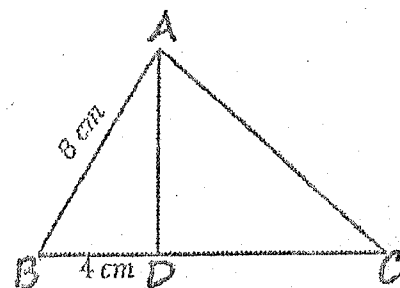
heap is 70 cm , find the height of the smaller one.

- Q7) Two similar tins contain 960 g and 405 g of margarine respectively. Find the area of the base of the smaller tin.
- Q8) Triangles ABC and XYZ are similar. The sides of triangle AEC are 6 cm , 7 cm and 8 cm . The shortest side of triangle XYZ is 2 cm . Given that the area of triangle XYZ is 4.5 cm^2 , calculate the area of triangle ABC .
- Q9) A cylindrical metal bar whose volume is 594 cm^3 is melted and cast into a sphere. Calculate the radius of the sphere, leaving your answer correct to 2 decimal places.

$$(\text{Volume of a sphere} = \frac{4}{3}\pi r^3,$$

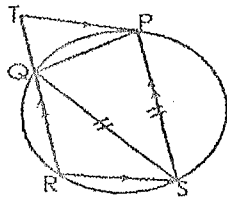
Take $\pi = 3.142$).

- Q10) In Figure below, triangle ABC is similar to triangle DBA . $AB = 8 \text{ cm}$, $BD = 4 \text{ cm}$ and area of triangle DBA is 24 cm^2 . Find the area of triangle ABC .



- Q11) A trapezium has a height of 3 cm and its area is 6 cm^2 . Calculate the area of a similar trapezium with a height of 12 cm .
- Q12) Two triangles LMN and ABC are similar. In triangle LMN , $LM = 4 \text{ cm}$, $MN = 5 \text{ cm}$ and $LN = 6 \text{ cm}$. The longest side of triangle ABC is 5 cm longer than its shorter side. Find the ratio of the areas of the two triangles.

Q13)

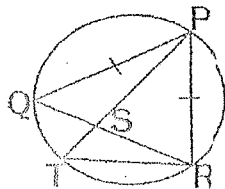


In the Figure above, TP is parallel to RS and TR is parallel to PS in the circle PQRS. TP is a tangent to the circle at P and $PS = SQ$;

Prove that:

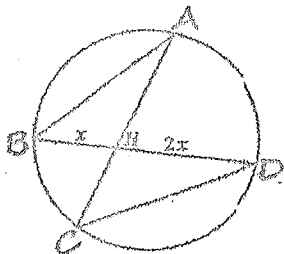
- (i). $TP = QP$
- (ii). triangles PTO and SPQ are similar and hence prove that $SR^2 = TO \cdot TR$

Q14) In Figure below, triangle, PQR is inscribed in a circle such that $PR = PQ$. S is a point on QR such that PS produced cuts the circle at T.



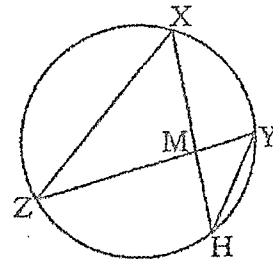
Prove that triangles PRS and PTR are similar hence show that $PR^2 = PT \cdot PS$

Q15) In Figure below, circle ABCD has chords AC and BD intersecting at H. Length of line HD is twice that of BH.



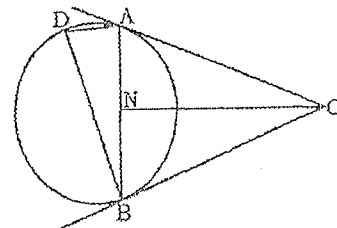
Find the ratio of areas of triangle ABH to triangle CDH.

Q16)



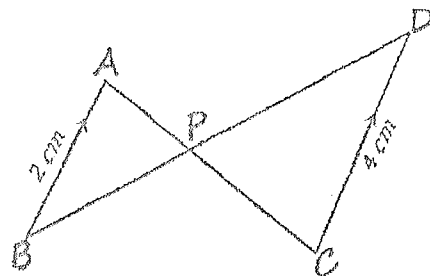
In figure above, prove that $XZ \cdot MY = XM \cdot YH$

Q17) In Figure below, CA and CB are tangents to the circle ABD in which BD is a diameter. CN meets the chord AB at right angles.



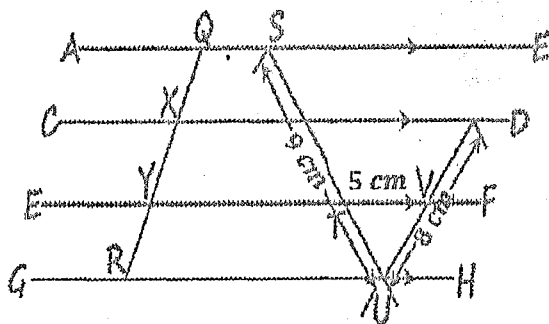
- (i). Prove that angle ACN = angle ABD.
- (ii). Prove that N is the mid-point of AB.
- (iii). Using similar triangles, show that $2AN^2 = NC \cdot DA$.

Q18) In Figure below, AB is parallel to DC, $AB = 2 \text{ cm}$, $CD = 4 \text{ cm}$ and the area of triangle ABP = 6 cm^2 .



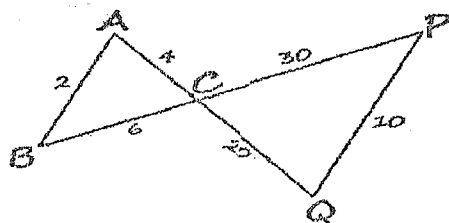
Calculate the area of triangle CDP.

Q19) In Figure below, AB is parallel to CD, EF and GH. The parallel lines AB, CD, EF and GH intersect QR such that $QX = XY = YR$. $SU = 9 \text{ cm}$, $DU = 8 \text{ cm}$ and $TV = 5 \text{ cm}$.



Prove that angle $TUV = 90^\circ$

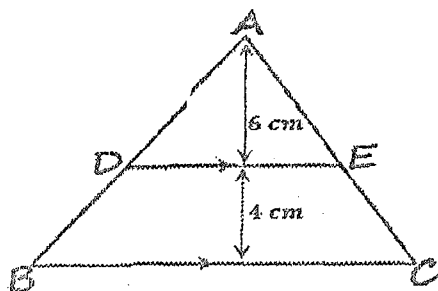
Q20) In Figure below ACQ and BCP are straight lines



Show that:

- triangle ABC is similar to triangle QPC
- AB is parallel to PQ.

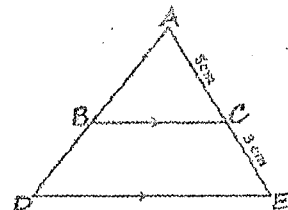
Q21) Figure below two similar triangles ADE and ABC in which DE is parallel to BC. The area of triangle ADE = 12 cm^2 .



If the heights of triangle ADE and trapezium DECB are 6 cm and 4 cm respectively, calculate the length of BC.

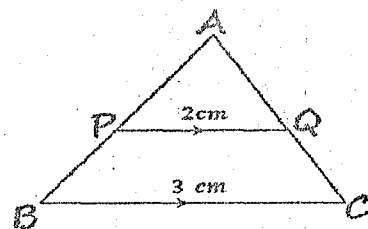
Q22) The areas of two similar triangles ABC and XYZ are 1: 16. If the height of the smaller triangle is 2 cm, calculate the height of the bigger triangle.

Q23) Figure below shows triangle ADE which $AC = 5 \text{ cm}$, $CE = 3 \text{ cm}$ and BC is parallel to DE. B is on AD and is on AE.



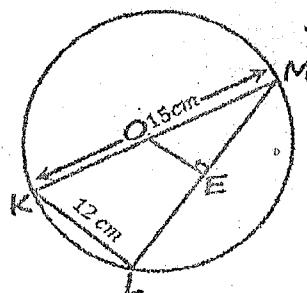
Calculate the ratio of the area of triangle ABC to the area of trapezium BCED.

Q24) In Figure below, $PQ = 2 \text{ cm}$, $BC = 3 \text{ cm}$ and the area of trapezium OQCB = 10 cm^2



If the area of triangle APQ = $x \text{ cm}^2$. Calculate the value of x.

Q25) Figure below, shows a circle KLM centre O. The diameter KM = 15 cm and chord KL = 12 cm. E is a point on LM.

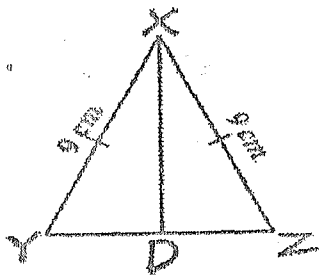


If OE is perpendicular to the chord LM, calculate the length of EM.

Q26) The area of a rectangle ABCD is 72 cm^2 and its width is 6 cm. Calculate the area of

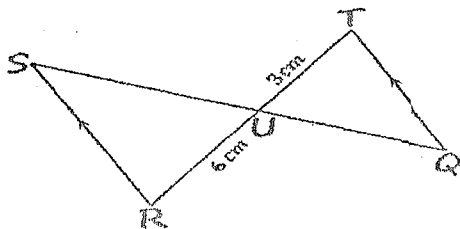
similar rectangle KLMN that has a width 9 cm.

- Q27) Figure below, shows an isosceles triangle XYZ with sides $XY = XZ = 9 \text{ cm}$ and $YZ = 12 \text{ cm}$. XD is a perpendicular bisector of YZ.



Calculate angle YXZ, giving your answer correct to the nearest degree.

- Q28) Figure shows two similar triangles TQU and RSU, $TU = 3 \text{ cm}$, $UR = 6 \text{ cm}$ and RS is parallel to QT.



Calculate the ratio of the area of triangle TQU to the area of triangle RSU, leaving your answer in its simplest form.

- Q29) The areas of two similar triangles ABC and HKL are 100 cm^2 and 256 cm^2 respectively. If the length of AB is 5 cm, calculate the length of HK.

Similar Figures (Model Solutions)

Q1) Solution

In Δs ABC and CDE

$$\angle BAC = \angle EDC \text{ (alt } \angle s, AB \parallel ED)$$

$$\angle ABC = \angle DEC \text{ (alt. } \angle s, AB \parallel ED)$$

$$\therefore \angle BCA = \angle EDC \text{ (}\angle \text{ sum of a } \Delta)$$

So, $\Delta ABC \parallel \Delta DEC$ (they are equiangular)

$$\text{Hence } \frac{CD}{CA} = \frac{DE}{CB}$$

$$\frac{CD}{6} = \frac{6}{9}$$

$$\therefore CD = \frac{6 \times 6}{9} = 4 \text{ cm}$$

Also,

$$\frac{DE}{AB} = \frac{CE}{CB}$$

$$\frac{DE}{11} = \frac{6}{9}$$

$$\text{Therefore, } DE = \frac{6 \times 11}{9} = 7\frac{1}{3} \text{ cm}$$

Q2) Solution

$$\begin{aligned} \text{The area} &= 28 \times \text{area factor} \\ &= 28 \times 5^2 \text{ cm}^2 \\ &= 700 \text{ cm}^2 \end{aligned}$$

Q3) Solution

Since the scale factor is 2:1, the area of factor is 4:1.

$$\text{The cloth will cost } 4 \times K800 = K3200$$

Q4) Solution

Area factor of daughter's dress to woman's dress

$$= \frac{3.2 \text{ m}^2}{5 \text{ m}^2} = \frac{32}{40} = \frac{16}{25} = \left(\frac{4}{5}\right)^2$$

$$\text{Scale factor} = \text{square root of area factor} = \frac{4}{5}$$

$$\text{Height of woman} = 165 \text{ cm}$$

$$\text{Height of daughter} = \frac{4}{5} \text{ of } 165 \text{ cm}$$

$$= \frac{4 \times 165}{5} \text{ cm}$$

$$= 4 \times 33 \text{ cm}$$

$$= 132 \text{ cm}$$

Q5) Solution

$$\text{Scale factor} = 5\,000$$

$$\text{Area factor} = (5\,000)^2$$

$$= 25\,000\,000$$

$$\text{Area of village} = 25\,000\,000 \times 6\text{ cm}^2$$

$$= \frac{25\,000\,000 \times 6}{100 \times 100} \text{ m}^2$$

$$= \frac{25\,000\,000 \times 6}{10\,000 \times 100 \times 100} \text{ ha}$$

$$= \frac{25 \times 6}{100} \text{ ha}$$

$$= 1.5 \text{ ha}$$

Q6) Solution

Mass is proportional to volume, thus, ratio of volumes = ratio of masses.

$$= \frac{128}{250} = \frac{64}{125} = \frac{4^3}{5^3} = \left(\frac{4}{5}\right)^3$$

$$\text{Scale factor} = \frac{4}{5}$$

$$\text{Height of bigger heap} = 70 \text{ cm}$$

$$\text{Height of smaller heap} = \frac{4}{5} \times 70 \text{ cm}$$

$$= 4 \times 14 \text{ cm}$$

$$= 56 \text{ cm}$$

Q7) Solution

Ratio of volumes = ratio of masses

$$= \frac{960}{405} = \frac{64}{27} = \frac{4^3}{3^3} = \left(\frac{4}{3}\right)^3$$

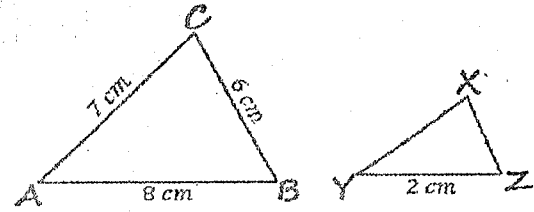
$$\text{Thus, scale factor} = \frac{4}{3}$$

$$\text{and the area factor} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$$

$$\text{Area of larger base} = 120 \text{ cm}^2$$

$$\text{Area of smaller base} = \frac{9}{16} \times 120 \text{ cm}^2$$

$$= 67\frac{1}{2} \text{ cm}^2$$

Q8) Solution

$$\text{Area of } \triangle XYZ = 4.5 \text{ cm}^2 \quad (\text{given})$$

Since $\triangle ABC \sim \triangle XYZ$, then

$$\text{Scale factor} = \frac{AB}{XY} = \frac{AC}{XZ} = \frac{BC}{YZ} \text{ and}$$

$$\text{Area factor} = (\text{Scale factor})^2$$

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle XYZ} = \left(\frac{BC}{YZ}\right)^2 \quad (\text{Area factor})$$

$$\Rightarrow \text{Area of } ABC = \text{Area of } \triangle XYZ \times \left(\frac{BC}{YZ}\right)^2$$

$$= 4.5 \text{ cm}^2 \times \left(\frac{6 \text{ cm}}{2 \text{ cm}}\right)^2$$

$$= 4.5 \text{ cm}^2 \times 3^2$$

$$= 4.5 \text{ cm}^2 \times 9$$

$$= 40.5 \text{ cm}^2$$

$$\therefore \text{The area of triangle } ABC \text{ is } 40.5 \text{ cm}^2$$

Q9) Solution

Given that metal bar whose volume is 594 cm^3 is melted and cast into a sphere.

In this case, volume of cylinder = Volume of sphere

$$\text{Now, volume of cylinder} = 594 \text{ cm}^3 \quad (\text{given})$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3 \quad (\text{given})$$

$$\Rightarrow 594 \text{ cm}^3 = \frac{4}{3}\pi r^3$$

$$594 \text{ cm}^3 = \frac{4}{3} \times 3.14 \times r^3$$

Multiply both sides by 3

$$(3) \times 594 \text{ cm}^3 = \frac{4 \times 3.14 \times r^3}{3} \times (3)$$

$$1782 \text{ cm}^3 = 12.568 r^3$$

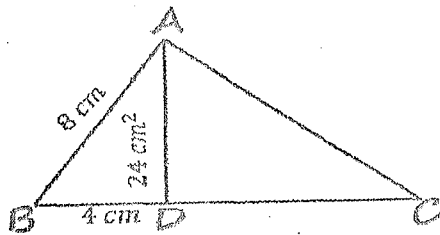
$$\Rightarrow \frac{1782 \text{ cm}^3}{1782} = \frac{12.568 r^3}{1782}$$

$$r^3 = \frac{1782}{12.568} \text{ cm}^3$$

$$r^3 = \sqrt[3]{\frac{1782}{12.568}} \text{ cm}^3$$

$$r = 5.21 \text{ cm (to 2 d.p.)}$$

Q10) Solution



Given: $\triangle ABC \sim \triangle DBA$

$$\text{Scale factor} = \frac{AB}{DB} = \frac{8}{4} = 2$$

$$\text{Area factor} = (\text{scale factor})^2$$

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DBA} = 2^2$$

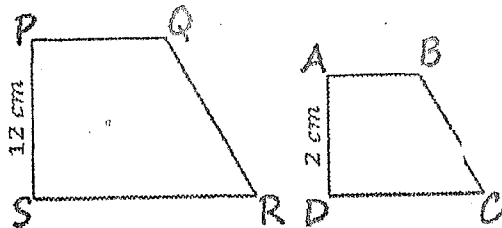
$$\frac{\text{Area of } \triangle ABC}{24 \text{ cm}^2} = 4$$

$$\text{Area of } \triangle ABC = 24 \text{ cm}^2 \times 4 = 96 \text{ cm}^2$$

$\therefore$ The area of triangle ABC is 96 cm^2

Q11) Solution

Sketch:



Since the trapezium are similar, then

$$\frac{\text{Area of PQRS}}{\text{Area of ABCD}}$$

$$= \left(\frac{\text{Height of PQRS}}{\text{Height of ABCD}} \right)^2 \text{ (area factor)}$$

$$\frac{\text{Area of PQRS}}{6} = \left(\frac{12}{2} \right)^2$$

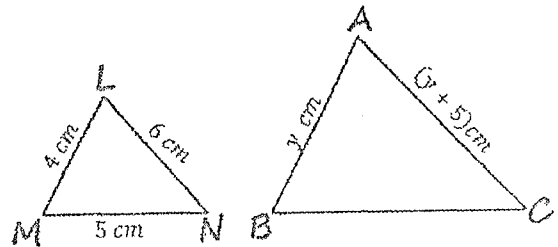
$$\frac{\text{Area of PQRS}}{6} = \frac{144}{4}$$

$$\text{Area of PQRS} = \frac{144}{4} \times 6$$

$$= 96 \text{ cm}^2$$

$\therefore$ The area of the trapezium with height 12 cm is 96 cm^2

Q12) Solution



Let the shortest side of $\triangle ABC = y \text{ cm}$

Then, the longest side $= (y + 5) \text{ cm}$

Since $\triangle LMN \sim \triangle ABC$

$$\Rightarrow \frac{LM}{AB} = \frac{MN}{BC} = \frac{LN}{AC}$$

$$\Rightarrow \frac{LM}{AB} = \frac{LN}{AC}$$

$$\frac{4}{y} = \frac{6}{y+5}$$

Cross multiply to get

$$4(y + 5) = 6y$$

$$4y + 20 = 6y$$

$$4y - 6y = -20$$

$$-2y = -20$$

$$\frac{-2y}{-2} = \frac{-20}{-2}$$

$$y = 10$$

$$\therefore AB = 10 \text{ cm}$$

$$\text{Now, } \frac{\text{Area of } \triangle LMN}{\text{Area of } \triangle ABC} = \left(\frac{LM}{AB} \right)^2 \text{ (Area factor)}$$

$$= \left(\frac{4}{10} \right)^2$$

$$= \frac{16}{100} = \frac{4}{25}$$

$$\therefore \triangle LMN : \triangle ABC = 4 : 25$$

$\therefore$ The ratio of the areas of the two triangles is 4:25

Q13) Solution

(i) PSRT is a parallelogram (opp. sides $\parallel$)

$\therefore \angle RSP = \angle RTP$ (opp. $\angle$ s of $\parallel$ gram)

$\angle RSP = \angle TQP$ (ext. $\angle$ of cyclic quad)

$\therefore \angle RTP = \angle TQP$

$\therefore \triangle TQP$ is isosc. (Base $\angle$ s equal)

$\therefore TP = QP$ (opp. sides of isosc. $\triangle$)

(ii) $\angle PTQ = \angle TQP$ (Proved)

$\therefore \angle QPS = \angle PQS = \angle FTQ = \angle TQP$

$$= \frac{180^\circ - \angle TPQ}{2} \quad (\angle \text{sum of a } \Delta)$$

$$\Delta PTQ \parallel \Delta SPQ \quad (AAA)$$

$$\therefore \frac{PT}{SP} = \frac{TQ}{PQ} \quad (\text{ratios of corr. sides of } \parallel \Delta)$$

Cross-multiplying we get

$$PT \cdot PQ = TQ \cdot SP \dots \dots \dots (i)$$

Further, $PT = SR$ (corr. sides of a //gram)
and $PT = PQ$ (proved)

$$\therefore PQ = SR$$

$$\therefore PT \cdot PQ = SR \cdot SR$$

$$= SR^2$$

$$PT = SR \quad (\text{opp. sides of a //gram})$$

$$\therefore TQ \cdot SR = TQ \cdot TR$$

$$\therefore \text{from (1)} \Rightarrow SR^2 = TQ \cdot TR$$

Q14) Solution

Given: PQR inscribed in a circle such that $PR = PQ$

Need to prove: Triangles PRS and PTR // and hence
 $PR^2 = PT \times PS$

Proof:

In Δ s PRS, PTR:
 $\angle RPS = \angle RPT$ (common)
 $PR = PQ$ (common)
 $\angle PRQ = \angle PQR$ (base $\angle$ s of isos. Δ PQR)
 $\angle PTR = \angle PQR$ ($\angle$ s in same seg)
 $\therefore \angle PRQ = \angle PTR$ (both = $\angle PQR$)
but $\angle PRQ = \angle PTR$
 $\therefore \angle PRS = \angle PTR$ (3^{rd} remaining $\angle$)

Hence

In Δ s PRS and PTR;
 $\angle RPS = \angle RPT$
 $\angle PRS = \angle PTR$
 $\angle PSR = \angle PRT$
 $\therefore \Delta PRS \parallel \Delta PTR$ (AAA)
 $\therefore \frac{PR}{PT} = \frac{PS}{PR}$ (ratio of corr. sides if $\parallel \Delta$ s)

Cross multiply to get

$$PR \times PR = PS \times PT$$

$$PR^2 = PT \times PS$$

Q15) Solution

$\angle BAH = \angle CDH$ ($\angle$ s in same seg)
 $\angle ABH = \angle DCH$ ($\angle$ s in same seg)
 $\angle BHA = \angle CHD$ (vert. opp. $\angle$ s)

$$\therefore \frac{\text{Area of } \Delta ABH}{\text{Area of } \Delta CDH} = \left(\frac{BH}{DH}\right)^2$$

$$= \left(\frac{x}{2x}\right)^2$$

$$= \left(\frac{x^2}{4x^2}\right)$$

$$= \frac{1}{4}$$

Ratio of areas of ΔABH to ΔDCH is 1:4

$\therefore \Delta ABH \parallel \Delta DCH$ (equiangular)

Q16) Solution

$\angle ZXM = \angle MYH$ ($\angle$ s in the same segment)
 $\angle XMZ = \angle YMH$ (vert. opp. $\angle$ s)
 $\therefore \angle XZM = \angle YHM$ (Third $\angle$ s)

$\therefore \Delta XZM \parallel \Delta YHM$ (AAA)

$$\therefore \frac{XZ}{YH} = \frac{XM}{YM} \quad (\text{corr. sides of } \parallel \Delta\text{s})$$

Cross - multiply

$$XZ \cdot YM = XM \cdot YH$$

Q17) Solution

(i) In ΔANC , $\angle ANC = 90^\circ$ (given)

In ΔABD , $\angle DAB = 90^\circ$ ($\angle$ in semi circle)

$$\angle NAE = \angle ANC + \angle ACN \quad (\text{ext. } \angle \text{ of a } \Delta)$$

$$= 90^\circ + \angle ACN$$

Also,

$$\angle NAE = \angle DAB + \angle DAE$$

$$= 90^\circ + \angle DAE$$

$$\therefore 90^\circ + \angle ACN = 90^\circ + \angle DAE$$

$$\therefore \angle ACN = \angle DAE$$

$$\angle DAE = \angle ABD \quad (\angle\text{s in alt. seg.})$$

$$\therefore \angle ABD = \angle ACN$$

(ii) In Δ s ANC and BNC,

$AC = BC$ (tangents from same ext. point)

$\angle CAN = \angle CBN$ (base $\angle$ s of an isosc. Δ ABC)

$\angle CNA = \angle CNB$ (rt. $\angle$ s)

$\Delta CAN = \Delta CBN$ (RHS)

$$\therefore AN = BN \quad (\text{corr. sides in } \equiv \Delta\text{s})$$

$\therefore N$ is the midpoint of AB

(iii) In $\Delta s ANC, ABD$

$$\angle ACN = \angle ABD \text{ (proved)}$$

$$\angle DAB = \angle ANC \text{ (rt. } \angle s)$$

$\therefore 3^{rd} \angle s$ are equal

$$\therefore \Delta ACN \sim \Delta DBA \text{ (AAA)}$$

$$\therefore \frac{CN}{BA} = \frac{AN}{DA} \text{ (Corr. sides in } \sim \Delta s)$$

But $BA = 2AN$ (proved, N midpoint of AB)

$$\therefore \frac{CN}{BA} = \frac{CN}{2AN}$$

$$\frac{CN}{2AN} = \frac{AN}{DA} \text{ (Cross-multiply)}$$

$$CN \cdot DA = 2AN^2$$

Q18) Solution

Let us consider triangles ABP and CDP

$$\angle ABP = \angle CDP \text{ (alt. } \angle s, AB \parallel CD)$$

$$\angle BAP = \angle DCP \text{ (alt. } \angle s, AB \parallel CD)$$

$$\angle APB = \angle DPC \text{ (vert. opp. } \angle s)$$

$$\therefore \Delta ABP \sim \Delta DCP \text{ (AAA)}$$

$$\therefore \frac{\text{Area of } \Delta CDP}{\text{Area of } \Delta ABP} = \left(\frac{DC}{AB}\right)^2 \text{ (Area factor)}$$

$$\frac{\text{Area of } \Delta CDP}{6 \text{ cm}^2} = \left(\frac{4}{2}\right)^2$$

$$\frac{\text{Area of } \Delta CDP}{6 \text{ cm}^2} = 4$$

$$\text{Area of } \Delta CDP = 4 \times 6 \text{ cm}^2$$

$$= 24 \text{ cm}^2$$

$\therefore$ The area of triangle CDP is 24 cm^2

Q19) Solution

Need to prove that angle $TUV = 90^\circ$

Since $AB \parallel CD, EF, GH$, then

$$UT = \frac{US}{3} = \frac{9 \text{ cm}}{3} = 3 \text{ cm}$$

$$UV = \frac{UD}{2} = \frac{8 \text{ cm}}{2} = 4 \text{ cm}$$

$$TV = 5 \text{ cm (given)}$$

Using Pythagoras theorem as a test for the presence of right-angle

$$TV^2 = UT^2 + UV^2$$

$$= (3 \text{ cm})^2 + (4 \text{ cm})^2$$

$$= 9 \text{ cm}^2 + 16 \text{ cm}^2$$

$$= 25 \text{ cm}^2$$

$$TV = \sqrt{25 \text{ cm}^2}$$

$$TV = 5 \text{ cm}$$

Hence, the sides of UTV satisfies the Pythagoras theorem with TV hypotenuse

$$\therefore \text{Angle } TUV = 90^\circ$$

Q20) Solution

(i). To prove that $\Delta ABC \sim \Delta QPC$

$$\frac{AB}{PQ} = \frac{2}{10} = \frac{1}{5}$$

$$\frac{BC}{CP} = \frac{6}{30} = \frac{1}{5}$$

$$\frac{AC}{CQ} = \frac{4}{20} = \frac{1}{5}$$

$\therefore \Delta ABC \sim \Delta QPC$ (three sides equal in proportion)

(ii). To prove that $AB \parallel PQ$

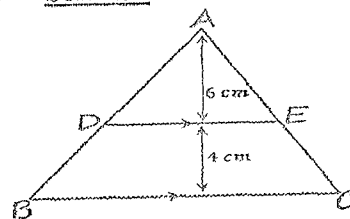
Since $\Delta ABC \sim \Delta QPC$,

Either

$$\begin{cases} \angle B = \angle P \\ \text{or } \angle A = \angle Q \end{cases} \text{ But these are alt. } \angle s$$

$$\therefore AB \parallel PQ$$

Q21) Solution



$$\text{Area of } \Delta ADE = 12 \text{ cm}^2 \text{ (given)}$$

$$\Rightarrow \frac{1}{2} \times b \times h = 12 \text{ cm}^2$$

$$\frac{1}{2} \times DE \times 6 \text{ cm} = 12 \text{ cm}^2$$

$$3 \text{ cm } DE = 12 \text{ cm}^2$$

$$DE = \frac{12 \text{ cm}^2}{3 \text{ cm}} = 4 \text{ cm}$$

Let $BC = k$

Since $\Delta ADE \sim \Delta ABC$, then

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ADE} = \left(\frac{BC}{DE}\right)^2 \text{ (area factor)}$$

$$\frac{\frac{1}{2} \times k \times 10}{12} = \left(\frac{k}{4}\right)^2$$

$$\frac{5k}{12} = \frac{k^2}{16}$$

Cross multiply to get

$$5k \times 16 = 12 \times k^2$$

$$80k = 12k^2$$

$$12k^2 - 80k = 0$$

$$4k(3k - 20) = 0$$

$$\text{Either } 4k = 0 \Rightarrow k = 0$$

$$\text{or } 3k - 20 = 0 \Rightarrow 3k = 20$$

$$\Rightarrow k = \frac{20}{3} = 6.67 \text{ (to 2 d.p.)}$$

But, from the diagram, $k > 0$

$\therefore$ The length of BC is 6.67 cm.

Q22) Solution

Given: $\triangle ABC \parallel \triangle XYZ$ with ratio of their areas as 1:16

Let's denote h_s to be the height of smaller $\triangle$ and h_b to be the height of bigger $\triangle$.

Since $\triangle ABC \parallel \triangle XYZ$, then,

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle XYZ} = (\text{scale factor})^2$$

$$\Rightarrow \frac{1}{16} = \left(\frac{h_s}{h_b}\right)^2$$

$$\frac{1}{16} = \left(\frac{2}{h_b}\right)^2$$

$$\frac{1}{16} = \frac{4}{h_b^2}$$

$$-h_b^2 = 4 \times 16$$

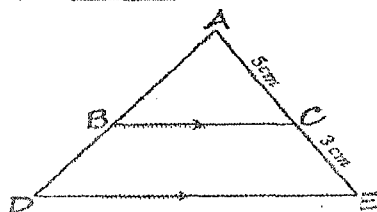
$$h_b^2 = 64$$

$$h_b = \sqrt{64}$$

$$\therefore h_b = 8$$

$\therefore$ The height of the bigger triangle is 8 cm.

Q23) Solution



In $\triangle s$ ABC and ADE:

$\angle DAE$ is common

$\angle ABC = \angle ADE$ (corr. $\angle s$, $BC \parallel DE$)

$\angle ACB = \angle AED$ (corr. $\angle s$, $BC \parallel DE$)

$\therefore \triangle ABC \parallel \triangle ADE$ (AAA)

$$\therefore \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ADE} = \left(\frac{AC}{AE}\right)^2 \text{ (area factor)}$$

$$= \left(\frac{AC}{AC + CE}\right)^2$$

$$= \left(\frac{5}{5 + 3}\right)^2$$

$$= \left(\frac{5}{8}\right)^2$$

$$= \frac{25}{64}$$

But $\triangle ADE = \triangle ABC + \text{trapezium BCED}$

$$\Rightarrow \frac{\triangle ABC}{\triangle ABC + \text{trapezium BCED}} = \frac{25}{64}$$

$$64 \times \triangle ABC = 25(\triangle ABC + \text{trapezium BCED})$$

$$64\triangle ABC = 25\triangle ABC + 25 \times \text{trapezium BCED}$$

$$64\triangle ABC - 25\triangle ABC = 25 \times \text{trapezium BCED}$$

$$39\triangle ABC = 25 \times \text{trapezium BCED}$$

$$\Rightarrow \frac{\triangle ABC}{\text{trapezium BCED}} = \frac{25}{39}$$

$\therefore$ The ratio of area of triangle ABC to the area of trapezium BCED is 25:39.

Q24) Solution

In $\triangle APQ$ and $\triangle ABC$

$\angle APQ = \angle ABC$ (corr. $\angle s$, $PQ \parallel BC$)

$\angle AQP = \angle ACB$ (corr. $\angle s$, $PQ \parallel BC$)

$\angle A$ is common

$\therefore \triangle APQ \parallel \triangle ABC$ (AAA)

$\Rightarrow$ Ratio $\triangle ABC : \triangle APQ$

$$\text{Scale factor} = 3:2 = \frac{3}{2}$$

$$\text{Area factor} = 9:4 = \frac{9}{4}$$

$$\text{Area of } \triangle APQ = x \text{ cm}^2$$

$$\text{Area of } \triangle ABC = x + 10 \text{ cm}^2$$

$$\text{Ratio of areas} = \frac{ABC}{APQ} = \frac{x+10}{x} = \frac{9}{4}$$

$$4(x + 10) = 9x$$

$$4x + 40 = 9x$$

$$9x - 4x = 40$$

$$5x = 40$$

$$x = 8$$

$\therefore$ Area of $\triangle APQ = 8 \text{ cm}^2$

Q25) Solution

Given: circle KLM, diameter $KL = 15 \text{ cm}$

$$KL = 12 \text{ cm}$$

$$OM = \frac{1}{2} KL$$

$$OM = 7.5$$

$\therefore \triangle MOE \parallel \triangle MKL$ (equal angles)

$$\therefore \text{Scale factor} = 1:2 = \frac{1}{2}$$

$$\therefore \frac{OM}{MK} = \frac{OE}{KL} = \frac{1}{2}$$

$$\therefore \frac{OE}{KL} = \frac{1}{2}$$

$$\therefore OE = \frac{1}{2} \times KL = \frac{1}{2} \times 12 = 6$$

$$EM^2 = OM^2 - OE^2 \text{ (pythagoras theorem)}$$

$$EM^2 = (7.5)^2 - (6)^2$$

$$EM = \sqrt{20.25}$$

$$EM = 4.5 \text{ cm}$$

Alternative method

$$\angle KLM = 90^\circ \text{ (}\angle \text{ in a semicircle)}$$

$$ML^2 = 15^2 - 12^2 \text{ (pythagoras theorem)}$$

$$= 225 - 144$$

$$ML = \sqrt{81}$$

$$ML = 9$$

$$\text{But, } EM = \frac{1}{2} LM \text{ (Mid - point theorem)}$$

$$= \frac{1}{2} \times 9$$

$$= 4.5 \text{ cm}$$

Q26) Solution

Given: Area of rectangle ABCD = 72 cm^2 ,
width = 6 cm

Need to find area of similar rectangle KLMN,
width = 9 cm

Ratio ABCD: KLMN

$$\text{Scale factor } 6:9 = \frac{6}{9} = \frac{2}{3}$$

Area factor = (scale factor)²

$$\frac{72}{x} = \left(\frac{2}{3}\right)^2$$

$$\frac{72}{x} = \frac{4}{9} \text{ (Cross multiply)}$$

$$4x = 72 \times 9$$

$$x = \frac{648}{4} = 162$$

Alternative method

$$\frac{KLMN}{ABCD} = \left(\frac{9}{6}\right)^2$$

$$\frac{\text{Area of KLMN}}{72} = \frac{81}{36}$$

$$\therefore \text{Area of KLMN} = \frac{81 \times 72}{36} = 162 \text{ cm}^2$$

Q27) Solution

$$YD = \frac{1}{2} YD = \frac{12 \text{ cm}}{2} = 6 \text{ cm}$$

$$\sin \angle YXD = \frac{6}{9} = \frac{2}{3}$$

$$\therefore \angle YXD = \sin^{-1}\left(\frac{2}{3}\right)$$

$$= 41.81^\circ$$

$$\therefore \angle YXZ = 2\angle YXD \text{ (XD bisects } \angle YXZ)$$

$$= 2 \times 41.81^\circ$$

$$= 83.62^\circ$$

$$= 84^\circ$$

Q28) Solution

Since $\triangle TQU \parallel \triangle RSU$

$$\frac{TQ}{RS} = \frac{QU}{SU} = \frac{TU}{RU} = \frac{3}{6}$$

$$\frac{\text{Area of } \triangle TQU}{\text{Area of } \triangle RSU} = \left(\frac{TU}{RU}\right)^2$$

$$= \left(\frac{3}{6}\right)^2$$

$$= \frac{9}{36} = \frac{1}{4}$$

$\therefore$ Ratio of area of $\triangle TQU$: $\triangle RSU$ is $1:4$

Alternative method

Given $\triangle TQU \parallel \triangle RSU$

$\Rightarrow \triangle TQU \sim \triangle RSU$ and

$$\frac{TQ}{RS} = \frac{QU}{SU} = \frac{TU}{RU} = \frac{3}{6}$$

$$\frac{TU}{RU} = \frac{3}{6}$$

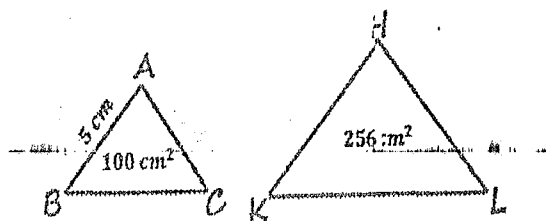
$$\therefore \text{Scale factor} = \frac{1}{2}$$

$$\begin{aligned} \text{Area factor} &= (\text{Scale factor})^2 \\ &= \left(\frac{1}{2}\right)^2 = \frac{1}{4} \end{aligned}$$

Hence, ratio of area of $\Delta TQU : \Delta RSU$ is 1:4

Q29) Solution

Sketch



Since $\Delta ABC \sim \Delta HKL$, then

$$\begin{aligned} \frac{\text{Area of } \Delta HKL}{\text{Area of } \Delta ABC} &= \left(\frac{HK}{AB}\right)^2 \\ \frac{256 \text{ cm}^2}{100 \text{ cm}^2} &= \left(\frac{HK}{5 \text{ cm}}\right)^2 \\ \frac{256 \text{ cm}^2}{100 \text{ cm}^2} &= \frac{HK^2}{25 \text{ cm}^2} \\ 25 \text{ cm}^2 \times \frac{256 \text{ cm}^2}{100 \text{ cm}^2} &= \frac{HK^2}{25 \text{ cm}^2} \times 25 \text{ cm}^2 \\ 64 &= HK^2 \\ HK &= \sqrt{64} \\ \therefore HK &= 8 \text{ cm} \end{aligned}$$

Topic 11

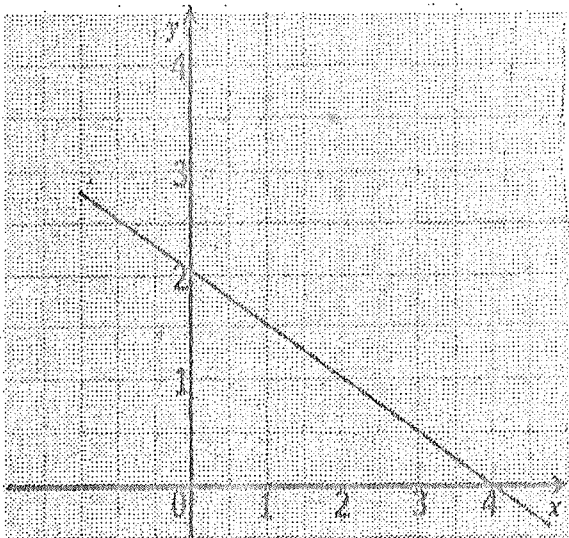
Coordinate Geometry (Equation of a straight line)

Reminder of some terms and formulae

- The horizontal x -axis and the vertical y -axis cross each other at their zero point called the **origin**.
- The coordinates of a point is presented as (x, y) . The order of the coordinates is important: the x -coordinate is given first, the y -coordinate second.
- A horizontal line through (a, k) will have the equation $y = k$ where k is some number.
- A vertical line through (k, a) will have the equation $x = k$ where k is some number.
- The gradient of a line is a measure of its steepness. The line $y = 3x + 2$ slopes up from left to right and has a **positive gradient**. The line $y = 1.5 - 3x$ slopes down from left to right and has a **negative gradient**.
- $\text{gradient} = (m) = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$
- $y = mx + c$ is the general form for the equation of the line where m is gradient and c is constant.
- The tangent of the angle between a straight line graph and the horizontal is the gradient of the line.
- The products of any pair of perpendicular lines is -1 .
- The distance between $M(x_1, y_1)$ and $N(x_2, y_2)$ is $MN = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

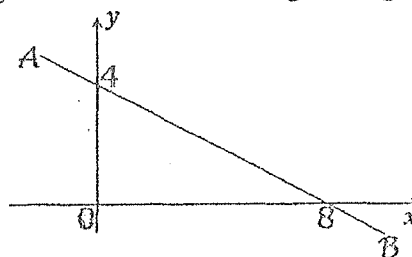
Questions on Coordinate Geometry Equation of a straight line

- Q1)** A straight line passing through the points $(0, 3)$ and $(2, 11)$. Write down
(i) its gradient
(ii) its equation in the form $y = mx + c$
- Q2)** A straight line passes through the points $(2, -3)$ and $(3, 5)$. Write the equation of this line in the form $y = mx + c$
- Q3)** Figure below shows a graph of a straight line AB



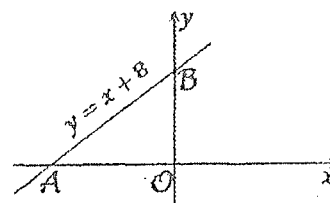
- (i) What is the gradient of the line AB?
(ii) Find the equation of the line AB
- Q4)** P is a point on the graph whose equation is $y = x^2 - 6x$. If the x -coordinate of P is 2. Calculate its y -coordinate.
- Q5)** A straight line passes through the point $(1, 6)$ and cuts the y -axis at 4, calculate its gradient.
- Q6)** The distance between two points $M(3, 5)$ and $N(8, a)$ is 13. Find the value of a .
- Q7)** A straight line passes through points $(-1, -2)$ and $(3, 4)$. Find the equation of the straight line in the form $y = mx + c$.
- Q8)** Find the equation of a straight line passing through the point $(0, 7)$ and parallel to the line $y = 2x + 5$.

- Q9)** Figure below shows a straight line graph



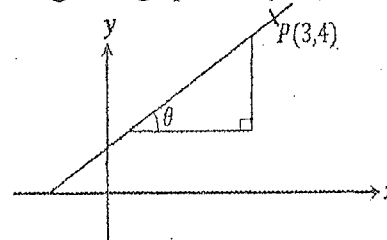
Find the equation of line AB and write it in the form $y = mx + c$

- Q10)** Calculate the distance between the points $A(-3, 1)$ and $B(3, 9)$
- Q11)** Given that $y = 2x - 3$ and $y = (b - 1)x + 5$ are graphs of two parallel lines, calculate the value of b .
- Q12)** Figure below shows a straight line graph $y = x + 8$ crossing the x and y axes at A and B respectively.



Calculate the distance between points A and B, leaving your answer correct to 2 decimal places.

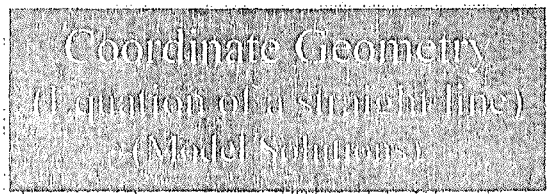
- Q13)** Find the gradient of a straight line whose equation is $\frac{y+2x}{4} = \frac{x}{3}$
- Q14)** Figure below shows a straight line passing through point $P(3, 4)$.



Given that $\tan \theta = \frac{2}{3}$, find the equation of the line in the form $y = mx + c$

- Q15)** A straight line which passes through $(3t, 7)$ and $(t, -5)$ has gradient 3. Find the equation of the line.

- Q16)** Find the distance between $L(-2,8)$ and $P(4,-1)$, giving your answer correct to 2 decimal places.
- Q17)** Find the gradient of a straight line whose equation is $y - 1 = mx$ and passes through $(-2,-5)$



Q1) Solution

- (i) Equation of the line is given by: $y = mx + c$
where m is the gradient, c is the y -intercept.

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{11 - 3}{2 - 0} = \frac{8}{2} = 4$$

$$\therefore m = 4$$

- (ii). Since $m = 4$

$$\Rightarrow y = 4x + c \dots \dots \dots (i)$$

Substituting $(0, 3)$ in the equation (i)

$$3 = 4(0) + c \dots \dots \dots (i)$$

$$3 - 0 = c$$

$$\therefore c = 3$$

$\therefore$ The equation of the line is $y = 4x + 3$

Q2) Solution

Equation of the line is given by: $y = mx + c$
where m is the gradient, c is the y -intercept.

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - (-3)}{3 - 2}$$

$$= \frac{8}{1} = 8$$

$$\therefore m = 8$$

$$\Rightarrow y = 8x + c \dots \dots \dots (i)$$

Substituting $(2, -3)$ in the equation (i)

$$-3 = 8(2) + c \dots \dots \dots (i)$$

$$-3 - 16 = c$$

$$\therefore c = -19$$

$\therefore$ The equation of the line is $y = 8x - 19$

Q3) Solution

- (i). Taking any two points on the line, e.g.

$$A(0, 2), B(4, 0)$$

$$\text{gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 - 2}{4 - 0}$$

$$= \frac{-2}{4} = \frac{-1}{2}$$

- (ii). Equation of the line is given by:

$y = mx + c$ where m is the gradient, c is the y -intercept.

$$\text{from (i), } m = \frac{-1}{2}$$

$$\text{from the graph, } c = 2$$

$\therefore$ The equation of the straight line AB is $y = \frac{-1}{2}x + 2$

Q4) Solution

Given the equation, $y = x^2 - 6x$

If $x = 2$, then

$$y = (2)^2 - 6(2)$$

$$= 4 - 12$$

$$= -8$$

$\therefore$ The y -coordinate is -8

Q5) Solution

Let $(x_1, y_1) = (1, 6); (x_2, y_2) = (x_2, 4)$

Since the line cuts the y -axis at 4, then

$$x_2 = 0,$$

$$\text{Hence } (x_2, 4) = (0, 4)$$

$$\text{gradient} = m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{6 - 4}{1 - 0}$$

$$= \frac{2}{1} = 2$$

$$\therefore \text{Gradient is } 2$$

Q6) Solution

Let $M(x_1, y_1) = (3, 5)$ and

$$N(x_2, y_2) = (8, a)$$

$$MN = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$13 = \sqrt{(8 - 3)^2 + (a - 5)^2}$$

$$13 = \sqrt{5^2 + a^2 - 10a + 25}$$

$$13 = \sqrt{25 + a^2 - 10a + 25}$$

$$13 = \sqrt{a^2 - 10a + 50}$$

$$\begin{aligned}
 13^2 &= (\sqrt{a^2 - 10a + 50})^2 \\
 169 &= a^2 - 10a + 50 \\
 \Rightarrow a^2 - 10a + 50 - 169 &= 0 \\
 a^2 - 10a - 119 &= 0 \\
 a^2 + 7a - 17a - 119 &= 0 \\
 a(a + 7) - 17(a + 7) &= 0 \\
 (a + 7)(a - 17) &= 0 \\
 \text{Either } a + 7 = 0 \Rightarrow a &= -7 \\
 \text{or } a - 17 = 0 \Rightarrow a &= 17 \\
 \therefore \text{The values of } a \text{ is } -7 \text{ or } 17
 \end{aligned}$$

Q7) Solution

Let $(x_1, y_1) = (-1, -2)$

$(x_2, y_2) = (3, 4)$

$$\begin{aligned}
 \text{gradient} = m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{4 - (-2)}{3 - (-1)} \\
 &= \frac{6}{4} = \frac{3}{2} \\
 y &= \frac{3}{2}x + c
 \end{aligned}$$

Substituting point $(3, 4)$ in the equation to find c , then,

$$4 = \frac{3}{2} \times 3 + c$$

$$4 = \frac{9}{2} + c$$

$$4 \times (2) = \frac{9}{2} \times (2) + c \times (2)$$

$$8 = 9 + 2c$$

$$2c = 8 - 9$$

$$2c = -1 \Rightarrow c = -\frac{1}{2}$$

Substitute $-\frac{1}{2}$ for c in the equation:

$$y = \frac{3}{2}x - \frac{1}{2}$$

$\therefore$ The equation of the straight line is $y = \frac{3}{2}x - \frac{1}{2}$

Q8) Solution

Let m_1 be gradient of the line passing through $(0, 7)$ and m_2 be the gradient of the line $y = 2x + 5$.

Since the lines are parallel, then their gradients should be equal.

$$\Rightarrow m_1 = m_2$$

Since $m_2 = 2$ then $m_1 = 2$

Then required equation $y = mx + c$, where m_1

is the gradient and c is a constant.

$$y = 2x + c \quad (\text{since } m_1 = 2)$$

At $(0, 7)$, $x = 0$ and $y = 7$.

Substituting them in $y = 2x + c$ to find c , then

$$7 = 2 \times 0 + c \Rightarrow c = 7$$

$\therefore$ The equation of the straight line passing through $(0, 7)$ is $y = 2x + 7$

Q9) Solution

When the line AB cuts the y -axis, the value of x is 0.

So the coordinate of A is $(0, 4)$.

Also, when line cuts x -axis, the value of y is 0.

So the B coordinate is $(8, 0)$

$$\begin{aligned}
 \therefore \text{gradient}(m) &= \frac{\text{change in } y}{\text{change in } x} \\
 &= \frac{0 - 4}{8 - 0} = \frac{-4}{8} \\
 &= -\frac{1}{2}
 \end{aligned}$$

Using $y = mx + c$, where m is the gradient and c is the y -intercept

$$m = -\frac{1}{2} \text{ and } c = 4$$

$$y = -\frac{1}{2}x + 4$$

$\therefore$ The equation of line AB is $y = -\frac{1}{2}x + 4$

Q10) Solution

Need to find the length of AB

For any two points $A(x_1, y_1)$ and $B(x_2, y_2)$,

Distance between A and B

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(3 - (-3))^2 + (9 - 1)^2}$$

$$AB = \sqrt{(3 + 3)^2 + (9 - 1)^2}$$

$$AB = \sqrt{(6)^2 + (8)^2}$$

$$AB = \sqrt{36 + 64}$$

$$AB = \sqrt{100}$$

$$AB = 10$$

$\therefore$ The distance between points

$A(-3, 1)$ and $B(3, 9)$ is 10 units

Q11) Solution

Given, $y = 2x - 3 \dots \dots \dots (i)$

$y = (b - 1)x + 5 \dots \dots \dots (ii)$

Gradient of (i) = 2

Gradient of (ii) = $(b - 1)$

Since the graphs of the two equations are parallel

Then, their gradients should be equal

$\Rightarrow b - 1 = 2$

$b = 2 + 1 = 3$

$\therefore$ The value of b is 3

Q12) Solution

Given $y = x + 8$

From the equation, y - intercept, $y = 8$

At $y = 8, x = 0$

$\therefore B = (0, 8)$

At point A, $y = 0$

Substitute 0 for y in the equation to find x :

$0 = x + 8 \Rightarrow x = -8$

$\therefore A = (-8, 0)$

$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$AB = \sqrt{(0 - (-8))^2 + (8 - 0)^2}$

$AB = \sqrt{8^2 + 8^2}$

$AB = \sqrt{64 + 64}$

$AB = \sqrt{128}$

$AB = 11.31$ (to 2 d.p)

$\therefore$ The distance between points A and B is 11.31 units

Q13) Solution

$\frac{y + 2x}{4} = \frac{x}{3}$

Cross-multiply to get

$3(y + 2x) = 4x$

$3y + 6x = 4x$

$3y = 4x - 6x$

$3y = -2x$

$y = -\frac{2}{3}x$

$\therefore$ The gradient of a straight line is $-\frac{2}{3}$

Q14) Solution

Given that $\theta = \frac{2}{3}$

Tan θ = gradient

$\therefore y = mx + c$ becomes $y = \frac{2}{3}x + c$

Substituting point $P(3, 4)$ in the equation

$4 = \frac{2}{3} \times 3 + c$

$c = 4 - 2 = 2$

$\therefore y = \frac{2}{3}x + 2$

Q15) Solution

Given coordinates on straight line

$(3t, 7)$ and $(t, -5)$

gradient = 3 = $\frac{\text{change in } y}{\text{change in } x}$

$\Rightarrow \frac{7 - (-5)}{3t - t} = 3$

$\frac{12}{2t} = 3$

$2t = \frac{12}{3}$

$t = \frac{4}{2} = 2$

Hence coordinates are $(6, 7), (2, -5)$ of the equation $y = mx + c$

Substituting $(6, 7)$ and gradient to get c

$7 = 3(6) + c$

$7 = 18 + c$

$c = 7 - 18$

$c = -11$

$\therefore$ The equation of the line is $y = 3x - 11$

Q16) Solution

Given $L(-2, 8)$ and $P(4, -1)$,

Distance between two points

$LP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(-2 - 4)^2 + (8 - (-1))^2}$

$= \sqrt{36 + 81}$

$= \sqrt{117}$

$= 10.82$

$\therefore$ Distance between L and P is 10.82 unit.

Q17) Solution

Given $y - 1 = mx$

$\Rightarrow y = mx + 1$

Since the line passes through $(-2, -5)$

$-5 = m(-2) + 1$

$\Rightarrow -5 = -2m + 1$

$\therefore -2m = -5 - 1$

$-2m = -6$

$m = 3 \Rightarrow \text{gradient} = 3$

Topic 12

Variation

Variation can be defined as the manner in which two or more quantities change relative to one another. Or can be defined as a function that relates the values of one variable to those of other variables.

Ratio

Is a comparison of two similar quantities.

Proportion

Refers to a statement of the fact that two given ratios are equal.

Definition of direct variation

When two variables are related in such a way that the ratio of their values always remains the same, the two variables are said to be in direct variation.

For example, a direct variation (D.V.) between two variables x and y that can be written as

$$y = kx, \text{ where } k \neq 0$$

This situation occurs when the ratio of two variables is constant

When, $y = kx$, we say that y varies directly with x

When, $z = kt$, we say that z varies directly with t

k is called the constant of variation

Two quantities are in *direct proportion* if they increase or decrease in the same ratio.

Definition of inverse (indirect) variation

When two quantities p and q are related by a formula of the form $pq = k$, we say that;

- p and q are inversely proportional or that p varies inversely as q .
- k is the constant of proportionality

We can restate the formula as $p = \frac{k}{q}$ or $p \propto \frac{1}{q}$

Two quantities are in *inverse proportion* if when one increases, the other decreases in the indirect ratio.

Joint variation

Joint variation shows a relationship where one variable depends on two or more others.

Partial variation

Partial variation as the word "*partial*" involves variation of different types happening together

Variation Questions

- Q1) If $p \propto q$, and $p = 8$ when $q = 2$, find the constant of proportionality.
- Q2) y varies directly as x and $y = 24$ when $x = 6$. find y when $x = 8$.
- Q3) (a) p varies directly as q^2 and $p = 6$, when $q = 3$. Find p when $q = 18$
 (b) Find the equation relating p and q in Q. 3(a).
- Q4) The length d cm of a vibrating string is inversely proportional to the square of the frequency of the vibration f cycles per second. Write down an equation for this.
- Q5) If p varies inversely as q^2 and $p = 4$ when $q = 6$, find p when $q = 8$.
- Q6) The electrical resistance (R) of a piece of wire with a circular cross section varies directly as the length (l) and inversely as the square of the radius (r). The resistance of a piece of wire 4.5 cm long and with a radius of 0.3 cm is 2 ohms. What length of wire with a radius of 0.2 cm will have a resistance of 10 ohms?
- Q7) The cost of making a shirt is partly constant and partly varies with the amount of time it takes to make it. If it takes 4 hours to make, it costs K120. If it takes 6 hours to make, it costs K140. Find the cost if it takes 3 hours.
- Q8) You are given that T is partly constant and partly varies as S . Suppose that when T is 29, $S = 12$ and when $T = 45$, $S = 20$. find the formula connecting T and S .
- Q9) If y varies directly as x^2 , and $y = 8$ when $x = 2$, find y when $x = 1$.

Q10) If y varies directly as x and z , and $y = 5$ when $x = 3$ and $z = 4$, then find y when $x = 2$ and $z = 3$.

Q11) A recipe for 6 cupcakes needs 1 cup of flour. The number of cupcakes you can make varies directly with the amount of flour. How many cupcakes can you make with 4 cups of flour?

Q12) P varies directly as t and inversely as the square of q . When $t = 1$ and $q = 2$, then $p = 3$. Find p when $q = \frac{1}{2}$ and $t = \frac{1}{3}$.

Q13) If p varies inversely as the square of q , and $p = 8$, when $q = 2$, find p when $q = 4$.

Q14) The number of people (N) who suffer from Malaria in a month is inversely proportional to the amount of insecticides (M) applied that month. When 5 litres of insecticide are applied, only 1 person suffers from Malaria. Find the equation connecting N and M .

Q15) A quantity b varies jointly with r and t , and $b = 108$ when $r = 3$ and $t = 6$. Find an equation which expresses b in terms of r and t .

Q16) A quantity q is the difference between two parts. The first part is constant and the second varies inversely as the square of p . If $q = 1$ then $p = 2$ and $q = 6$ when $p = 3$. Find the positive value of p when $q = 9$.

Q17) The cost (C) for an international call from Malawi to Europe, partly varies inversely as time, (t), and partly as the square of the time. A one minute call costs K120 and two-minute call costs K200. find the cost of a five-minute call.

Q18) Given that P varies as a product of q and r^2 , and that $P = 50$, when $q = 1$ and $r = 5$, find P when $q = 3$ and $r = 8$.

Q19) Given that $x \propto \frac{y}{z}$, when $x = 10$, $y = 2$ and $z = 4$. Find value of x when $y = 1$ and $z = 5$.

Q20) A quantity P varies directly as q and inversely as $(q^2 + 1)$. When $P = 1$, $q = 2$. Express P in terms of q only.

Q21) Given that $q \propto \sqrt{p}$ and $p = 4$ when $q = 3$, find the value of p when $q = 15$.

Q22) x varies directly as y and inversely as the square of n . if $x = 15$, $y = 24$ and $n = 4$. Calculate the value of n when $x = 8$ and $y = 20$.

Q23) A quantity T is proportional to m and the square of v . When $v = 3$ and $m = 5$, $T = 90$. Calculate the value of T when $m = 2$ and $v = 10$.

Q24) A variable y is partly constant and partly varies as x . When $x = 5$, $y = 4$ and when $x = 9$, $y = 8$. find y when $x = 1$.

Q25) Given that $V \propto rd$ and $V = 54$, when $r = 2$ and $d = 3$. Find V when $r = \frac{1}{2}$ and $d = 6$.

Q26) p varies directly as r and inversely as the square root of q . Given that $p = 4$ when $q = 9$ and $r = 1$, calculate q when $r = 2$ and $p = 6$.

Q27) Given that $x \propto \frac{y}{z^2}$ and $x = 12$ when $y = 2$ and $z = 1$. Find y when $x = 15$ and $z = 2$.

Variation (Model Solutions)

Q1) Solution

$$p \propto q$$

$$\Rightarrow p = kq$$

$$\text{So, } 8 = k \times 2$$

$$\frac{8}{2} = \frac{2k}{2}$$

$$\text{and } k = 4$$

$$\therefore \text{The proportionality equation is } p = 4q$$

Q2) Solution

$\frac{y}{x}$ is always the same,

$$\text{So } \frac{24}{6} = \frac{y}{8}$$

$$6y = 24 \times 8$$

$$\frac{6y}{6} = \frac{192}{6}$$

$$y = 32$$

Q3) (a) Solution

$$p = kq^2$$

$$\text{So, } k = \frac{p}{q^2} \text{ is constant}$$

$$\frac{6}{3^2} = \frac{p}{18}$$

$$p = \frac{6}{9} \times 18^2$$

$$= 216$$

(b) Solution

Since p varies directly as q^2 we know that $p = kq^2$.

We know that $p = 6$ when $q = 3$

$$\text{So } 6 = k \times 3^2$$

$$= 9k$$

$$k = \frac{6}{9} = \frac{2}{3}$$

$\therefore$ The proportionality equation is $p = \frac{2}{3}q^2$

Q4) Solution

$$d \propto \frac{1}{f^2}, \text{ so } d = \frac{k}{f^2} \text{ or } df^2 = k$$

Q5) Solution

In this case, if p varies inversely as q^2 , then pq^2 is constant.

$$\text{So, } p \times 8^2 = 4 \times 6^2$$

$$p = \frac{4 \times 6^2}{8^2}$$

$$p = \frac{4 \times 36}{64}$$

$$p = 2\frac{1}{4}$$

Q6) Solution

$$R \propto l \text{ and } R \propto \frac{1}{r^2}$$

$$\text{So, } R = k \cdot \frac{1}{r^2}$$

When $R = 2$ and $l = 4.5$, $r = 0.3$

$$\text{So, } 2 = \frac{k \times 4.5}{(0.3)^2}$$

$$2 = \frac{k \times 4.5}{0.09}$$

$$4.5k = 2 \times 0.09$$

$$k = \frac{0.18}{4.5}$$

$$k = \frac{1}{25}$$

$$\text{Hence, } R = \frac{l}{25r^2} \text{ and } l = 25Rr^2$$

When $R = 10$ and $r = 0.2$,

$$l = 25 \times 10 \times 0.2 \times 0.2$$

$$= 10 \text{ cm}$$

Q7) Solution

Let the constant part of the cost be a .

Part of the cost (C_1) varies with time (t),

$$\text{So, } (C_1) = kt$$

Hence total cost $C = a + kt$

For four hours, total cost is K120

$$120 = a + 4k$$

For six hours, total cost is K140

$$140 = a + 6k$$

There are two equations with two unknowns, which can be solved simultaneously.

$$140 = a + 6k$$

$$120 = a + 4k \quad (\text{Subtract})$$

$$20 = 2k$$

$$k = 10$$

If $120 = a + 4k$, when $k = 10$

$$\Rightarrow 120 = a + 4(10)$$

$$a = 120 - 40$$

$$a = 80$$

The equation is therefore

$$C = 80 + 10t$$

$$\text{When } t = 3 \text{ hrs, } C = 80 + 10 \times 3 = \text{K}110$$

Q8) Solution

From the information "when T is 29, S is 12," we get the equation:

$$29 = a + 12k$$

From the information "when T is 45, $S = 20$," we get the equation

$$45 = a + 20k$$

We now have two simultaneous equations:

$$29 = a + 12k \dots\dots\dots(i)$$

$$45 = a + 20k \dots\dots\dots(ii)$$

Subtract equation (i) from (ii), to give an equation in

$$29 = a + 12k$$

From the information "when T is 45, $S = 20$," we get the equation in k :

$$45 = a + 20k \dots\dots\dots(ii)$$

$$-29 = -a - 12k \dots\dots\dots(i)$$

$$16 = 8k$$

$$k = \frac{16}{8} = 2$$

To find the value of a , substitute 2 for k in equation (i)

$$29 = a + 12k \dots\dots\dots (i)$$

$$29 = a + 12(2)$$

$$a = 29 - 24$$

$$\therefore a = 5$$

Hence the required formula is $T = 5 + 2S$

Q9) Solution

Since this is direct variation, the formula is $y = kx^2$.

Given: data point $(x, y) = (2, 8)$

Need to find k .

Now,

$$y = kx^2$$

$$8 = k(2^2)$$

$$8 = 4k$$

$$2 = k$$

Rewriting the formula completely we have:

$$y = 2x^2, \text{ when } x = 1, y \text{ will be}$$

$$y = 2x^2$$

$$y = 2(1)^2$$

$$y = 2 \times 1$$

$$y = 2$$

Then the answer is when $x = 1, y = 2$

Q10) Solution

In this case, the formula is $y = kxz$

Data point given is $(x, y, z) = (3, 5, 4)$

Now, solving for k

$$y = kxz$$

$$\Rightarrow 5 = k(3)(4)$$

$$5 = 12k$$

$$\frac{12k}{12} = \frac{5}{12}$$

$$k = \frac{5}{12}$$

Now that we have the value of k , we can plug in the new values, and solve for the new value of y :

$$y = \frac{5}{12}xz$$

$$y = \left(\frac{5}{12}\right)(2)(3)$$

$$y = \left(\frac{5}{12}\right)(6)$$

$$y = \frac{5}{2}$$

$$\therefore y = \frac{5}{2}$$

Q11) Solution

Let x be the amount of flour

y be the number of cupcakes

since y varies directly with x , then $y = kx$

Need to find k when $x = 1$ and $y = 6$

$$y = kx$$

$$\Rightarrow 6 = k(1)$$

$$k = 6$$

Therefore, $y = 6x$ is the direct variation equation

When $x = 4, y = (6)(4) = 24$.

So you can make 24 cupcakes

Q12) Solution

In this case, the formula is

$$P \propto \frac{t}{q^2}$$

$$\Rightarrow P = k \cdot \frac{t}{q^2} \text{ Where } k \text{ is a constant.}$$

Substituting the values of P, t and q in the equation to find k .

$$3 = \frac{(k)(1)}{(2)^2}$$

$$3 = \frac{k}{4} \Rightarrow k = 12$$

When $q = 1/2$ and $t = 1/3$,

$$P = \frac{12 \times \frac{1}{3}}{\left(\frac{1}{2}\right)^2}$$

$$= \frac{4}{\frac{1}{4}}$$

$$= \frac{1}{\frac{1}{4}}$$

$$= 16$$

$$\therefore P = 16$$

Q13) Solution

$$p \propto \frac{1}{q^2}$$

$$\Rightarrow p = k \cdot \frac{1}{q^2}$$

$$p = \frac{k}{q^2}$$

Substituting, $p = 8, q = 2$ in the equation

$$p = \frac{k}{q^2}$$

$$8 = \frac{k}{(2)^2}$$

$$8 = \frac{k}{4}$$

$$\Rightarrow k = 32$$

Hence, the equation becomes $p = \frac{32}{q^2}$

We need to find p when $q = 4$

$$p = \frac{32}{(4)^2}$$

$$p = \frac{32}{16}$$

$$= 2$$

$$\therefore p = 2$$

Q14) Solution

$$N \propto \frac{1}{M}$$

Introducing the variation constant, k , then

$$N = \frac{k}{M}, \text{ where } M = 5, N = 1$$

Substitute them in the equation, $N = \frac{k}{M}$

$$1 = \frac{k}{5} \Rightarrow k = 5$$

Further, substituting 5 for k in the equation, $N = \frac{k}{M}$

$$N = \frac{5}{M}$$

$\therefore$ The equation connecting N and M is $N = \frac{5}{M}$

Q15) Solution

Given that b varies jointly with r and t

$$\Rightarrow b \propto rt$$

Introducing, the variation constant k , then we have;

$$b = krt$$

Substituting,

$$b = 108, r = 3 \text{ and } t = 6 \text{ into } b = krt;$$

Then,

$$108 = k(3)(6)$$

$$108 = 18k$$

$$\frac{18k}{18} = \frac{108}{18}$$

$$k = 6$$

Substituting 6 for k in the equation $b = krt$

$$\therefore b = 6rt$$

Q16) Solution

$$q \propto c - \frac{1}{p^2}$$

Introducing a variation constant k , then

$$q = c - \frac{k}{p^2}$$

If $q = 1$ then $p = 2$

Substituting, the numbers in $q = c - \frac{k}{p^2}$

$$1 = c - \frac{k}{(2)^2}$$

$$1 = c - \frac{k}{4}$$

Multiply each term by (4)

$$(4) \times 1 = (4) \times c - \frac{k}{4} \times (4)$$

$$4 = 4c - k$$

$$4c - k = 4 \dots\dots\dots(i)$$

If $q = 6$ then $p = 3$

$$6 = c - \frac{k}{(3)^2}$$

$$6 = c - \frac{k}{9}$$

Multiply each term by (9)

$$(9) \times 6 = (9) \times c - \frac{k}{9} \times (9)$$

$$54 = 9c - k$$

$$9c - k = 54 \dots\dots\dots(ii)$$

We have two equations as follows:

$$4c - k = 4 \dots\dots\dots(i)$$

$$9c - k = 54 \dots\dots\dots(ii)$$

(ii) - (i):

$$5c = 50$$

$$\Rightarrow c = 10$$

Substitute 10 for c in (i)

$$4 \times 10 - k = 4$$

$$40 - k = 4$$

$$-k = 4 - 40$$

$$-k = -36$$

Multiply both sides by (-1) to get

$$k = 36$$

Substituting, 10 for c and 36 for k in $q = c - \frac{k}{p^2}$

$$q = 10 - \frac{36}{p^2}$$

We need now to find p when $q = 9$

$$9 = 10 - \frac{36}{p^2}$$

$$\frac{36}{p^2} = 10 - 9$$

$$\frac{36}{p^2} = 1 \text{ (Cross-multiply)}$$

$$p^2 = 36$$

Taking the square root both sides:

$$p = \pm\sqrt{36}$$

$$= \pm 6$$

$\therefore$ The positive value of p is 6

Q17) Solution

Given C partly varies inversely as t and partly as the square of t

$$\Rightarrow C \propto \frac{1}{t} + t^2$$

Introducing the variation constant, i.e. m and k

$$\Rightarrow C = \frac{k}{t} + mt^2$$

One-minute call costs K120

$$\Rightarrow C = 120, \text{ when } t = 1$$

Substituting them in the equation

$$120 = \frac{k}{1} + m(1)^2$$

$$120 = k + m$$

$$k + m = 120 \dots\dots\dots(i)$$

Two-minute call costs K200

$$\Rightarrow C = 200, \text{ when } t = 2$$

Substituting them in the equation

$$200 = \frac{k}{2} + m(2)^2$$

$$200 = \frac{k}{2} + m(4)$$

$$200 \times (2) = (2) \times \frac{k}{2} + 4m \times (2)$$

$$\Rightarrow 400 = k + 8m$$

$$k + 8m = 400 \dots\dots\dots(ii)$$

We have two simultaneous equations as follows:

$$k + m = 120 \dots\dots\dots(i)$$

$$k + 8m = 400 \dots\dots\dots(ii)$$

$$(ii) - (i): 7m = 280 \Rightarrow m = 40$$

Substitute 40 for m in (i)

$$k + 40 = 120$$

$$k = 120 - 40$$

$$k = 80$$

Substitute 40 for m and 80 for k in the equation

$$\Rightarrow C = \frac{80}{t} + 40t^2$$

We need to find the cost (C) of a 5 minutes call i.e. to find C when $t = 5$

$$\Rightarrow C = \frac{80}{5} + 40(5)^2$$

$$C = 16 + 40(25)$$

$$= 16 + 1000$$

$$= 1016$$

$\therefore$ The cost of a 5 minute call is K1016

Q18) Solution

Given that P varies as a product of q and r^2

$$\Rightarrow P \propto qr^2$$

Introducing the variation constant, k , then

$$P = kqr^2$$

Substituting, $P = 50$, $q = 1$ and $r = 5$, in

$$P = kqr^2$$

$$\Rightarrow 50 = k(1)(5)^2$$

$$50 = 25k$$

$$\frac{25k}{25} = \frac{50}{25}$$

$$k = 2$$

Substitute 2 for k in $P = kqr^2$

$$\Rightarrow P = 2qr^2$$

Now, we need to find P when $q = 3$ and $r = 8$.

$$\Rightarrow P = 2(3)(8)^2 = 2 \times 3 \times 64 = 384$$

$$\therefore P = 384$$

Q19) Solution

Given $x \propto \frac{y}{z}$

Introducing the variation constant, k ;

$$x = \frac{ky}{z}, \text{ when } x = 10, y = 2 \text{ and } z = 4.$$

$$\Rightarrow 10 = \frac{k(2)}{(4)}$$

$$(4) \times 10 = \frac{(k)(2)}{(4)} \times (4)$$

$$40 = 2k$$

$$\frac{2k}{2} = \frac{40}{2}$$

$$k = 20$$

Substitute 20 for k in the equation:

$$\Rightarrow x = \frac{20y}{z}$$

Now, we need to find x when $y = 1$ and $z =$

$$x = \frac{20(1)}{5} = \frac{20}{5} = 4$$

$$\therefore x = 4$$

Q20) Solution

$$P \propto \frac{q}{(q^2 + 1)}$$

Introducing the variation constant, k

$$P = \frac{kq}{(q^2 + 1)}$$

Substituting $P = 1$ and $q = 2$ we get

$$1 = \frac{k(2)}{(2)^2 + 1}$$

$$1 = \frac{2k}{4 + 1}$$

$$1 = \frac{2k}{5} \quad (\text{Multiply both sides by 5})$$

$$2k = 5$$

$$\frac{2k}{2} = \frac{5}{2}$$

$$k = 2.5$$

Substitute 2.5 for k in the equation

$$P = \frac{2.5q}{q^2 + 1}$$

$$\therefore P \text{ in terms of } q \text{ only: } P = \frac{2.5q}{q^2 + 1}$$

Q21) Solution

Given that $q \propto \sqrt{p}$

Introducing the variation constant, k , then;

$$q = k\sqrt{p}$$

$$p = 4 \text{ when } q = 3$$

$$\Rightarrow 3 = k\sqrt{4}$$

$$3 = k \times 2$$

$$2k = 3 \quad (\text{Divide both sides by 2})$$

$$k = \frac{3}{2}$$

Substitute, $\frac{3}{2}$ for k in the equation:

$$q = \frac{3}{2}\sqrt{p}$$

Now, need to find p when $q = 15$

$$\Rightarrow 15 = \frac{3}{2}\sqrt{p} \quad (\text{Multiply both sides by 2})$$

$$30 = 3\sqrt{p}$$

$$\frac{3\sqrt{p}}{3} = \frac{30}{3}$$

$$\sqrt{p} = 10 \Rightarrow (\sqrt{p})^2 = (10)^2$$

$$p = 100$$

$\therefore$ The value of p is 100

Q22) Solution

Given that x varies directly as y and inversely as the square of n

$$\Rightarrow x \propto \frac{y}{n^2}$$

Introducing the variation constant, k , then;

$$x = \frac{ky}{n^2}$$

If $x = 15$, $y = 24$ and $n = 4$

Then,

$$15 = \frac{k(24)}{(4)^2}$$

$$15 = \frac{24k}{16} \quad (\text{Multiply both sides by 16})$$

$$15 \times 16 = 24k$$

$$24k = 240 \quad (\text{Divide both sides by 24})$$

$$k = 10$$

$$\therefore \text{The equation becomes } x = \frac{10y}{n^2}$$

Now, finding n when $x = 8$, $y = 20$

$$\Rightarrow 8 = \frac{10(20)}{n^2} \quad (\text{multiply both sides by } n)$$

$$8 \times n^2 = \frac{200}{n^2} \times n^2$$

$$8n^2 = 200$$

$$\frac{8n^2}{8} = \frac{200}{8}$$

$$n^2 = 25$$

$$n = \pm\sqrt{25}$$

$$= \pm 5$$

$\therefore$ The values of n are 5 or -5

Q23) Solution

Given that T is proportional to m and the square of v

$$\Rightarrow T \propto mv^2$$

Introducing the variation constant, k , then;

$$T = kmv^2$$

When $v = 3$ and $m = 5$, $T = 90$

Substituting them into $T = kmv^2$

$$90 = k(5)(3)^2$$

$$k \times 5 \times 9 = 90$$

$$45k = 90$$

$$\frac{45k}{45} = \frac{90}{45}$$

$$k = 2$$

Substitute 2 for k in $T = kmv^2$

$$T = 2mv^2$$

Now, need to find T when $m = 2$ and $v = 10$

$$\Rightarrow T = 2 \times (2) \times (10)^2$$

$$T = 4 \times 100 = 400$$

Hence, the value of T is 400.

Q24) Solution

Given that y is partly constant and partly varies as x

$$\text{i.e. } y \propto c + x$$

Introducing the variation constant, k , then;

$$y = c + kx \dots\dots\dots(\beta)$$

When, $x = 5, y = 4$. Substituting in (β) ;

$$4 = c + k(5)$$

$$c + 5k = 4 \dots\dots\dots(i)$$

When, $x = 9, y = 8$. Substituting in (β) ;

$$8 = c + k(9)$$

$$c + 9k = 8 \dots\dots\dots(ii)$$

We have therefore the following simultaneous equations

$$c + 5k = 4 \dots\dots\dots(i)$$

$$c + 9k = 8 \dots\dots\dots(ii)$$

$$(ii) - (i): 4k = 4$$

$$\Rightarrow k = 1$$

Substitute 1 for k in (i)

$$c + 5k = 4$$

$$\Rightarrow c + 5(1) = 4$$

$$c = 4 - 5$$

$$= -1$$

Substituting, (1) for k and (-1) for c in the equation:

$$y = c + kx$$

$$\Rightarrow y = -1 + (1)x$$

$$y = -1 + x$$

Now, need to find y when $x = 1$

$$y = -1 + 1 = 0$$

Hence, when $x = 1$, then $y = 0$

Q25) Solution

Given that $V \propto rd$ and $V = 54$, when $r = 2$ and $d = 3$

Need to find V when $r = 1/2$ and $d = 6$

$V = krd$, k is proportionality constant

Then;

$$V = krd \Rightarrow k = \frac{V}{rd}$$

$$V = 54, \text{ when } r = 2 \text{ and } d = 3$$

$$k = \frac{54}{2 \times 3}$$

$$k = 9$$

Substituting, 9 for k in the equation:

$$V = 9rd$$

Now, substituting $r = 1/2, d = 6$

$$V = 9(1/2)(6)$$

$$V = 27$$

Q26) Solution

$$p \propto \frac{r}{\sqrt{q}}$$

$$\Rightarrow p = \frac{kr}{\sqrt{q}}$$

Given that $p = 4$ when $q = 9$ and $r = 1$

$$4 = \frac{k(1)}{\sqrt{9}}$$

$$\frac{k(1)}{\sqrt{9}} = 4 \quad (\text{Multiply both sides by } \sqrt{9})$$

$$\sqrt{9} \times \frac{k(1)}{\sqrt{9}} = 4 \times \sqrt{9}$$

$$k = 4 \times \sqrt{9}$$

$$k = 4 \times 3$$

$$= 12$$

Need to find q when $r = 2$ and $p = 6$

$$\text{using } p = \frac{12r}{\sqrt{q}}$$

$$\Rightarrow 6 = \frac{12 \times 2}{\sqrt{q}}$$

$$\sqrt{q} \times 6 = \frac{12 \times 2}{\sqrt{q}} \times \sqrt{q}$$

$$6\sqrt{q} = 24$$

$$\frac{6\sqrt{q}}{6} = \frac{24}{6}$$

$$\sqrt{q} = 4$$

$$(\sqrt{q})^2 = (4)^2 \quad (\text{Squaring both sides})$$

$$q = 16$$

Q27) Solution

$$x \propto \frac{y}{z^2}$$

Introducing k as constant proportionality

$$\Rightarrow x = \frac{ky}{z^2}$$

$x = 12$ when $y = 2$ and $z = 1$

$$12 = \frac{k(2)}{(1)^2}$$

$$\therefore k = \frac{12}{2} = 6$$

$$\text{Hence, } x = \frac{6y}{z^2}$$

Now, to find y when $x = 15$ and $z = 2$

$$15 = \frac{6y}{(2)^2} = \frac{6y}{4}$$

$$\begin{aligned} 6y &= 60 \\ \therefore y &= 10 \end{aligned}$$

Topic 13

Linear inequalities & Linear Programming

Linear inequalities

An inequality is an algebraic statement which contains an inequality sign.

Note: Some of the terms and symbols that imply an inequality include:

- less than ($<$)
- greater than or more than ($>$)
- less than or equal to ($\leq$)
- greater than or equal to ($\geq$)

Realistic problems often involve many inequalities. For example: a manufacturing problem might produce inequalities resulting from production requirements, and also inequalities about cost requirement.

A collection of at least *two* inequalities is called a *system of inequalities*. Normally, we show an inequality on a graph by shading the region whose points do *not* satisfy the inequality. Shading of unwanted regions on graphs uses:

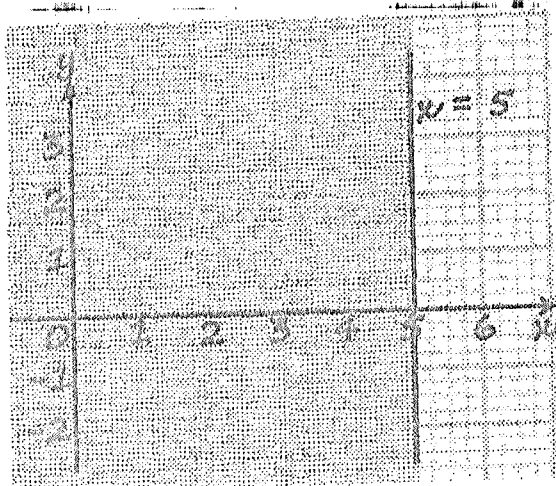
- dotted lines for $<$ or $>$
- solid lines for $\leq$ or $\geq$

Example

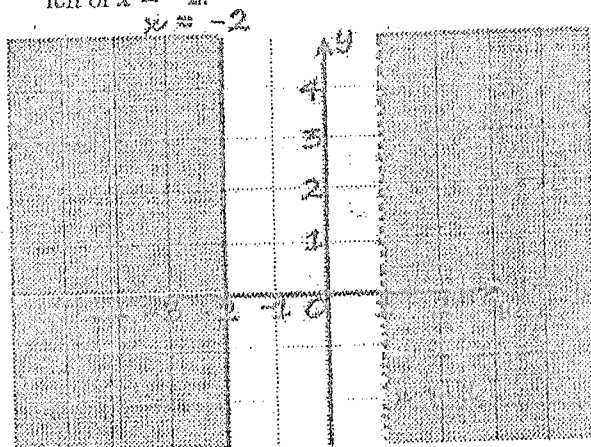
Draw graphs to show the following inequalities:

- (i). $x \geq 5$
- (ii). $-2 \leq x < 1$
- (iii). $y \geq 1$

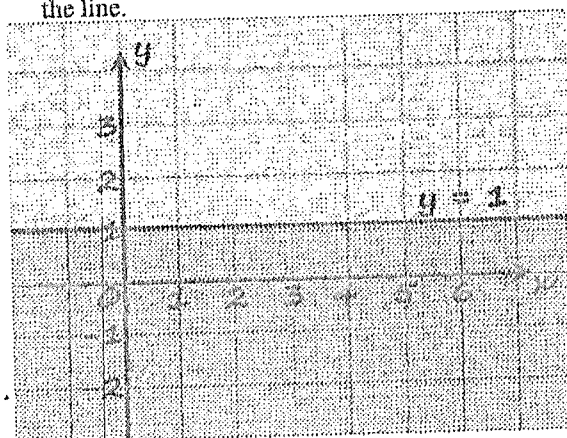
- (i). We first draw the solid line $x = 5$ as points on the line are included. We shade the region to the left of the line, as points with $x > 5$ are to the right.



- (ii). We draw solid line $x = -2$ (line is included) and broken line $x = 1$ (line not included). We shade the region to the right of $x = 1$ and to the left of $x = -2$.



- (iii). We first draw the solid line $y = 1$ as the points on the line are included. We shade the region below the line, as points with $y > 1$ are above the line.



Linear programming

Introduction

Linear programming and its applications

It is widely used in many mathematical models designed to solve problems in business, biology and economics involve finding the optimal value (either maximum or minimum) of the function, subject to certain restrictions and the problems often relate to obtaining maximum profits for given costs and production levels.

Linear programming

- ✓ **Definition:** It is a method used to find the *best* solution to problems that can be expressed in terms of linear equations or inequalities.

- ✓ Solutions are usually found by drawing graphs of inequalities and looking for optimal value (either maximum or minimum) values of the *objective function* that satisfy the required conditions. If an optimal value (either maximum or minimum) of the objective function exists, it will occur at one of the *corner points* of the feasible region.

- ✓ The inequalities may have:
 - one variable, for example $x \geq 1$
 - or two variables; for example, $x + y \leq 2$

Since the *solution set* of an inequality is usually *infinite*, the clearest and most meaningful description of the solution is often its graph.

The unshaded region on a graph is the region which satisfies all the conditions or *constraints* of the inequalities. This region is called the *feasible region* or the *accessible region* on a graph. Within the feasible region an expression can have various values.

In forming inequalities, we need to define the variables that represent the various quantities under consideration.

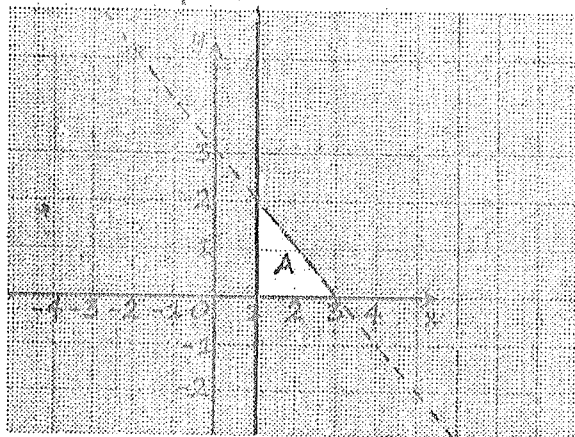
When solving linear programming problems, the following points are important:

- 1) Summarise the information in a table
- 2) Write down the objective function and all the necessary constraints.
- 3) Graph the feasible region.
- 4) Identify all the corner points.
- 5) Find the value of the *objective function* or *search line* at each corner point. The search line is of the form $ax + by = k$, where a, b and k .
- 6) For a bounded region, the solution is given by the corner point giving the optimal value of the objective function.
- 7) For the unbounded region, check that a solution actually exists. If it does it will occur at a corner point.

Questions on Linear Inequalities and Linear Programming

Q1) Draw a graph to show the region $x - y < 3$. Shade the unwanted region.

Q2) Figure below shows the unshaded region A on a Cartesian plane.



Write three inequalities that describe the unshaded region A.

Q3) Sketch the region represented by the following inequalities by shading the unwanted region.

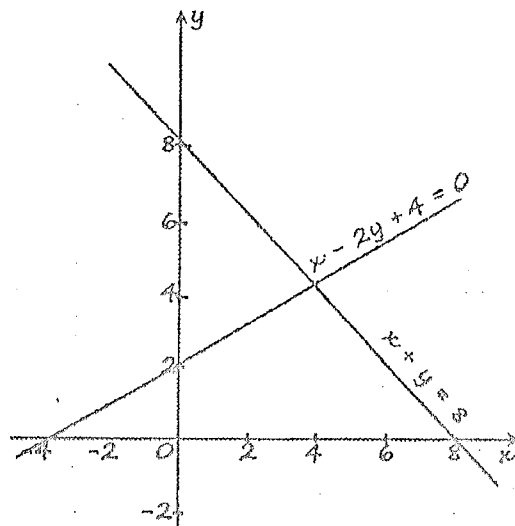
$$\begin{aligned} x &\geq -1 \\ y &\geq -4 \\ y &< 2x + 4 \\ y + 2x &\leq 6 \end{aligned}$$

Q4) On the same axes, sketch the graphs of the regions described by the following inequalities:

$$\begin{aligned} x &\geq 0 \\ y &= 0 \\ y &\leq 3x + 2 \\ y + 4x &< 8 \end{aligned}$$

Shade the unwanted region.

Q5) Figure below shows the graphs of $x + y = 8$ and $x - 2y + 4 = 0$.



Copy the figure on the graph paper provided and show the region bounded by the following inequalities.

$$\begin{aligned} x &\geq 0 \\ x + y &\leq 8 \\ x - 2y + 4 &\geq 0 \end{aligned}$$

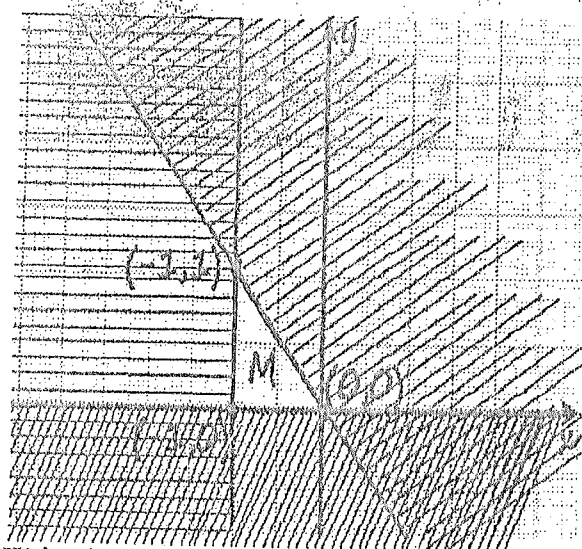
By shading the unwanted region

Q6) P is a set of points (x, y) which satisfies the three inequalities:

$$\begin{aligned} x &\geq 0 \\ x + y &\leq 4 \\ y &\geq x + 1 \end{aligned}$$

Show on a graph the region which satisfies the three inequalities.

- Q7) Figure below shows an unshaded region (M) bounded by three inequalities.



Write down the *three* inequalities

- Q8) A farmer had K5 000 to buy axes and hoes only. An axe costs K500 while a hoe costs K200. She decided to spend the money as follows:

- buy at least 3 axes and at least 5 hoes.
- Spend more on axes than hoes by not more than K1 000.

- (i). Taking x to represent number of axes and y to represent number of hoes to be bought, write down four inequalities in x and y that satisfies the above information.
- (ii). Using a scale of 2 cm to represent 2 units on the horizontal axis and 2 cm to represent 5 units on the vertical axis, draw a graph to show the region represented by the inequalities, shading the unwanted region.
- (iii). Use your graph to find the maximum number of hoes the farmer would buy if she bought 4 axes.

- Q9) A vendor decides to spend up to K600 to buy packets of sweets and biscuits. She does not want to buy more than 10 packets altogether but wants to buy at least 2 packets of each other. A packet of sweets costs K40, and that of biscuits costs K80. The profit on each packet of sweets is K10 and on each packet of biscuits is K15.

- (i). By taking x to represent the number of packets of sweets and y to represent the number of packets of biscuits, write down *four* inequalities that satisfy the above information.
- (ii). Using a scale of 2 cm to represent 2 units on both vertical and horizontal axes, draw graphs to show the region represented by the inequalities.
- (iii). Use your graph to find the number of packets of sweets and biscuits respectively for the vendor to get the maximum profit.

- Q10) Mala has K6000 to buy skirts and blouses: Skirts cost K600 each while blouses cost K300 each. She wants to spend at least K1 200 more on skirts than on blouses. She would like to buy at least 4 skirts and 2 blouses.

- (i). If x represents the number of skirts and y represents the number of blouses, write down four inequalities in x and y .
- (ii). Using a scale of 2 cm to represent 2 units on both axes, draw the graphs to show the region which represents the four inequalities.
- (iii). Using your graph, find the maximum number of skirts and blouses she can buy.

- Q11) A company hires two types of minibuses A and B to carry its employees. Type A carries 16 passengers and type B carries 8 passengers. The company requires at most 10 minibuses to carry a maximum of 128 employees every day.

- (i). If x represents number of the 16-passenger minibuses and y represents number of the 8-passenger minibuses, write down two inequalities in addition to $x \geq 0$ and $y \geq 0$.
- (ii). Using a scale of 2 cm to represent 2 units on both axes, draw graphs to show the region represented by the inequalities.
- (iii). Use your graph to find the number of minibuses of each type the company must hire to meet its target of transporting 128 employees at one using a minimum number of type A minibuses.

- Q12) Mrs Mlimi has a piece of land on which she would like to plant maize and groundnuts. The cost of planting maize is K300 per hectare while groundnuts cost K900 per hectare. Maize requires 3 labourers per hectare while groundnuts require 6 labourers per hectare. She hired at least 60 labourers and she has only K18 000 for planting costs.

- (i). Write down two inequalities in addition to $x \geq 0$ and $y \geq 0$. By taking x to represent the number of hectares of groundnuts and y to represent number of hectares of maize.
- (ii). Taking 2 cm to represent 5 units on the horizontal axis and 2 cm to represent 10 units on the vertical axis, draw graphs to show the region represented by the inequalities in x and y .
- (iii). Using your graph, find the maximum number of maize hectares if she plants 10 hectares of groundnuts.

Q13) A farmer gets a loan of K50 000 to buy sheep and goats only. Sheep cost K5 000 each while goats cost K2 000 each. She would like to spend at least K10 000 more on sheep than on goats.

- (i). If x represents the number of sheep and y represents the number of goats, write down two inequalities in x and y in addition to $x \geq 0$ and $y \geq 0$.
- (ii). Using a scale of 2 cm to represent 2 units on the horizontal axis and 2 cm to represent 5 units on the vertical axis, draw the region which represents the four inequalities.
- (iii). How many sheep and goats can the family buy to have maximum number of animals with the loan?

Linear Inequalities and Linear Programming (Model Solutions)

Q1) Solution

We find three points on the line $x - y = 3$, which can be rewritten as:

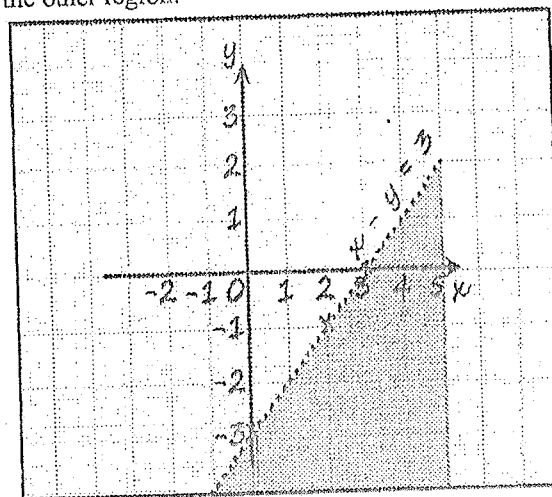
$y = x - 3$, from a table.

x	0	2	3
y	-3	-1	0

We plot the points

$(0, -3)$, $(2, -1)$ and $(3, 0)$ and draw $x - y = 3$ as broken line.

We test whether the point $(0, 0)$ is in the region we want. $x = 0$ and $y = 0$ gives $x - y = 0 < 3$, so we want the region containing $(0, 0)$ and we shade the other region.



Q2) Solution

- (i). The line $y = 0$ gives the inequality $y \geq 0$
- (ii). The line $x = 1$ gives the inequality $x \geq 1$
- (iii). For the line cutting the y -axis at $(0, 3)$ and x -axis at $(3, 0)$, we do the following:

$$\text{Gradient of line} = \frac{3-0}{0-3} = -1$$

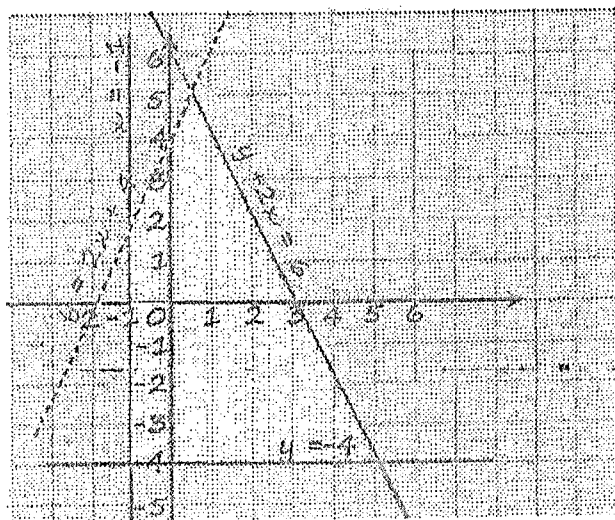
$$\text{Equation of the line} = y = mx + c$$

One of the points on this line is $(2, 1)$

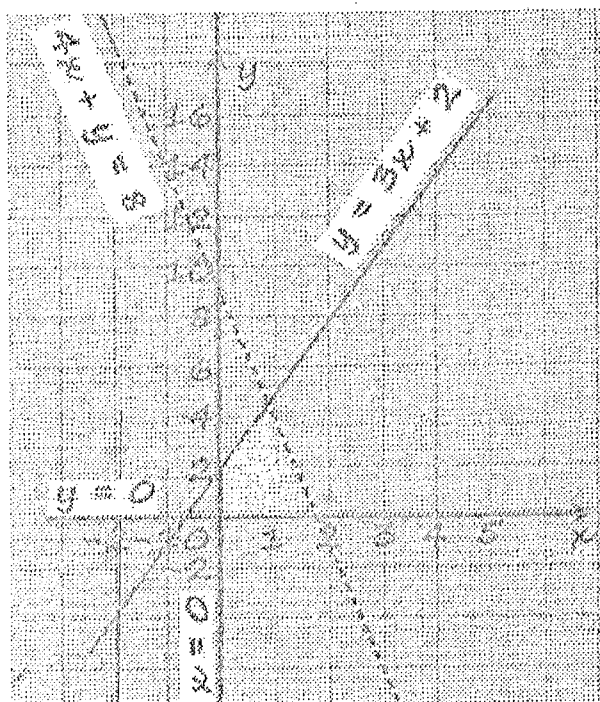
$$\therefore 1 = -(2) + c \Rightarrow c = 3$$

$\therefore$ equation of line is $y = -x + 3$
 Substituting one of the points, (0,0) in the equation
 $0 = -0 + 3$
 $0 = 3$
 The inequality is true for $0 < 3$
 $\therefore y < -x + 3$

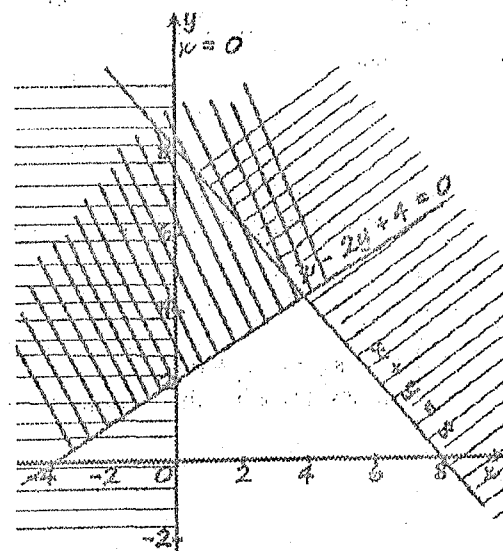
Q3) Solution



Q4) Solution



Q5) Solution

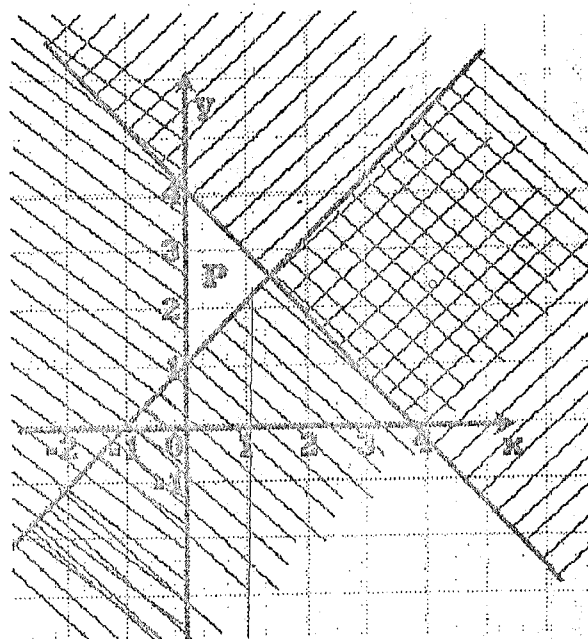


Q6) Solution

$x = 0$ is the vertical straight line passing through the origin

$x + y = 4$ is the equation of the straight line that crosses the x -axis at (4,0) and y -axis at (0,4)

$y = x + 1$ is the equation of straight line that crosses the x -axis at (-1,0) and the y -axis at (0,1).



Q7) Solution

Using the graph, since the region is above x - axis, we have

$y \geq 0$; Since the region is to the right of $x = -1$, we have $x \geq -1$

Consider the line that passes through the origin:

This is a straight line whose equation is of the form $y = mx + c$, where m is the gradient, c is the y - intercept.

$$\text{Gradient} = \frac{1-0}{-1-0} = \frac{1}{-1}$$

The equation becomes $y = -x + c$

Since the line passes through the origin, then y - intercept is 0

$$\therefore c = 0$$

$$y = -x + c \text{ becomes } y = -x$$

According to the shaded region, then, $y \leq -x$

$$\Rightarrow y + x \leq 0$$

$\therefore$ The *three* inequalities then become

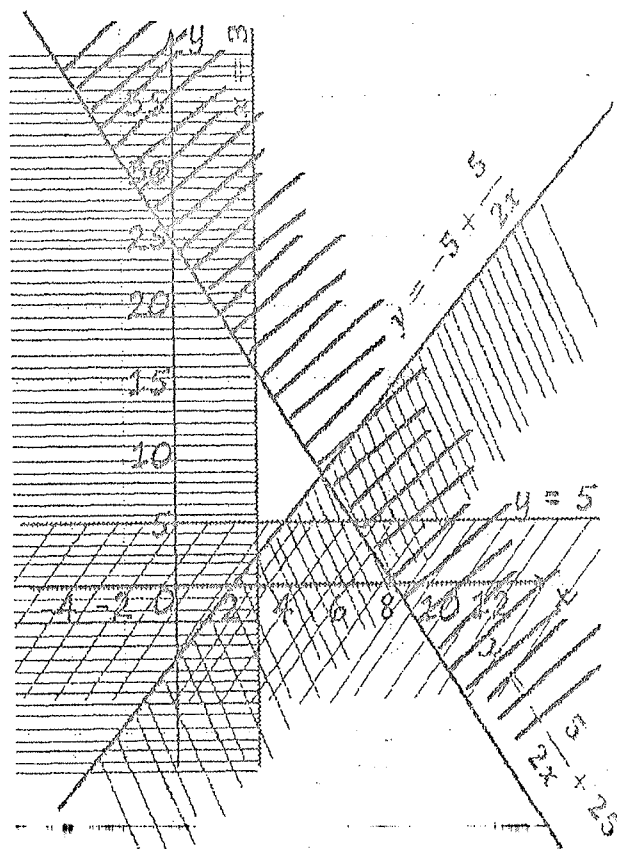
- (i). $x \geq 0$
- (ii). $x + y \leq 8$
- (iii). $x - 2y + 4 \geq 0$

Q8) Solution

(i). Inequalities formed are:

- a) at least 3 axes $\Rightarrow x \geq 3$
- b) at least 5 hoes $\Rightarrow y \geq 5$
- c) K5 000 for axes and hoes at K500 and K200 respectively
 $\Rightarrow 500x + 200y \leq 5000$
 $5x + 2y \leq 50$
- d) Spend more on axes than hoes by more than K1 000
 $\Rightarrow 500x + 200y \leq 1000$
 $5x + 2y \leq 10$

(ii). Graph



- (iii). From the graph the maximum number of hoes the farmer would buy if she bought 4 axes is 15 hoes.

Q9) Solution

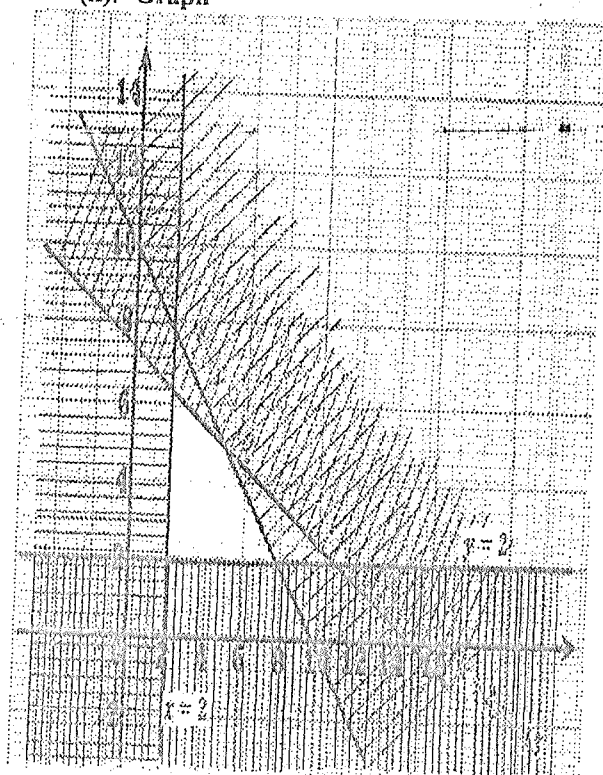
Given: x Sweets
 y Biscuits

Item	Number	Cost	Profit
Sweets	x	40	10
Biscuits	y	80	15
Total	10	K600	

(i). Inequalities

- $x \geq 2$ (i)
- $y \geq 2$ (ii)
- $x + y \leq 10$ (iii)
- $40x + 80y \leq 600$ (iv)
- $\Rightarrow x + 2y \leq 15$

(ii). Graph



(iii). Objective functions $10x + 15y$. From the graph we that coordinate

$$(2,2) \Rightarrow 10(2) + 15(2) = 50$$

$$(2,6) \Rightarrow 10(2) + 15(6) = 110$$

$$(5,5) \Rightarrow 10(5) + 15(5) = 125$$

$$(8,2) \Rightarrow 10(8) + 15(2) = 110$$

$\therefore$ The vendor should buy 5 packets of sweets and 5 packets of biscuits to make a maximum profit.

Q10) Solution

x represents number of skirts

y represents number of blouses

(i). The inequalities are:

$$x \geq 4 \quad (i)$$

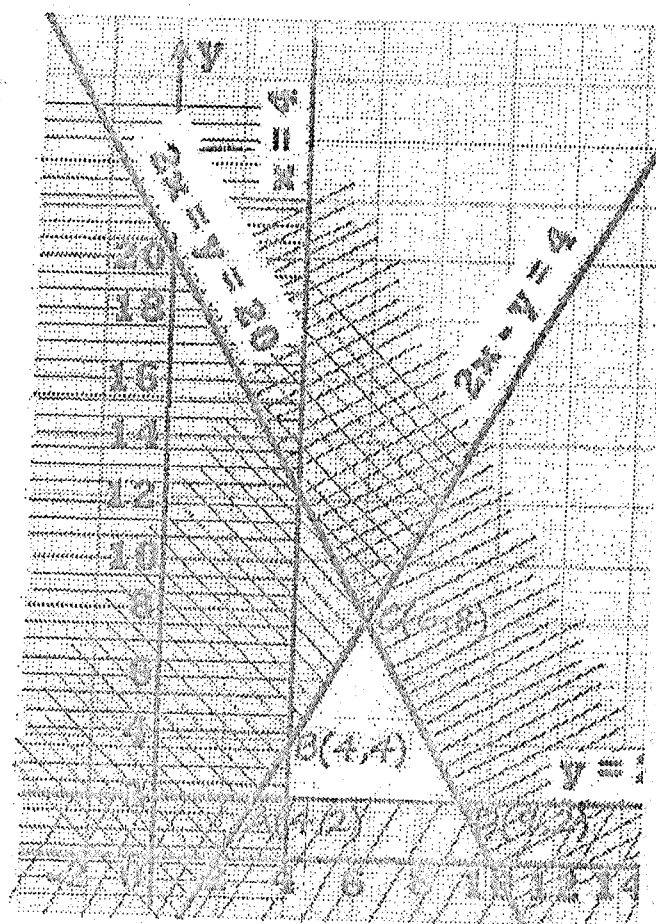
$$y \geq 2 \quad (ii)$$

$$600x + 300y \leq 6000 \text{ (divide by 300)}$$

$$2x + y \leq 1200 \quad (iii)$$

$$2x - y \geq 4 \quad (iv)$$

(ii). Graph



(iii). Corner points $(4,2), (4,4), (6,8), (9,2)$

Since she wants to spend at least K1200 more on skirts than blouses

	Skirts	Blouses	Total
$(4,2)$	$4 \times 600 = 2400$	$2 \times 300 = 600$	3000
$(4,4)$	$4 \times 600 = 2400$	$4 \times 300 = 1200$	3600
$(6,8)$	$6 \times 600 = 3600$	$8 \times 300 = 2400$	6000
$(9,2)$	$9 \times 600 = 5400$	$2 \times 300 = 600$	5400

The maximum number of skirts = 6 and maximum number of blouses = 8

Q11) Solution

(i).

- Since the company requires at most 10 minibuses, then $x + y \leq 10$
 - Total number of passengers that can board type A minibuses = $16x$
 - Total number of passengers that can board type B minibuses = $8y$
- Maximum number of employees to be transported = 128

$$\therefore 16x + 8y \leq 128 \text{ (dividing each term by 8)}$$

$$\Rightarrow 2x + y \leq 16$$

$\therefore$ The two other inequalities are:

a) $x + y \leq 10$

b) $2x + y \leq 16$

(ii). $x = 0$ is the y -axis

$y = 0$ is the x -axis

$x + y = 10$, crosses both the x -axis at (10, 0)

and the y -axis at (0, 10)

$2x + y = 16$, crosses the x -axis at (8, 0) and the y -axis at (0, 16)



(iii). Objective function = $16x + 8y$

From the graph, the coordinates bounding the wanted region are as follows:

(0, 0); (0, 10); (8, 0); (6, 4)

Substitute the coordinates in the objective function:

(0, 0): $16 \times 0 + 8 \times 0 = 0 + 0 = 0$

(0, 10): $16 \times 0 + 8 \times 10 = 0 + 80 = 80$

(8, 0): $16 \times 8 + 8 \times 0 = 128 + 0 = 128$

(6, 4): $16 \times 6 + 8 \times 4 = 96 + 32 = 128$

$\Rightarrow$ From above, point (6, 4) represent the number of minibuses that can transport 128 people with minimum number of type A minibuses.

$\therefore$ The company must hire 6 minibuses of type A and 4 minibuses of type B

Q12) Solution

(i).

• Total costs of planting maize = $K300y$

• Total costs of planting groundnuts = $K900x$

Since Mrs Mlimi has only K18 000 for planting costs, then

$$900x + 300y \leq 18\,000 \text{ (divide each term by 300)}$$

$$\Rightarrow 3x + y \leq 60$$

• Total number of labourers required for maize = $3y$

• Total number of labourers required for groundnuts = $6x$

Since Mrs Mlimi hired at least 60 labourers, then:

$$6x + 3y \geq 60 \text{ (divide each term by 3)}$$

$$2x + y \geq 20$$

$\therefore$ The other two inequalities are:

a) $3x + y \leq 60$

b) $2x + y \geq 20$

(ii). $x = 0$ is the y -axis

$y = 0$ is the x -axis

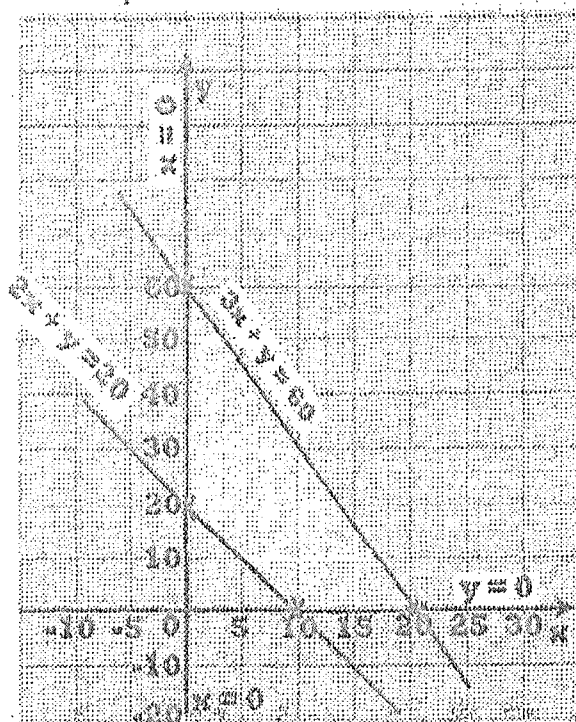
$3x + y = 60$ crosses the x -axis at (20, 0)

and y -axis at (0, 60)

$2x + y \geq 20$ crosses the x -axis at (10, 0)

and y -axis at (0, 20)

Graph



(iii). Need to find the maximum of y if $x = 10$.
From the graph, this is the y - coordinate of point M which is 30.

∴ The maximum number of maize hectares is 30.

Q13). Solution

(i). Each sheep cost K5 000 and each goat cost K2000

∴ Total amount of money to be spent on buying sheep = K5000x

Total amount of to be spent on buying on goats = K2000y

Since the farmer has an amount of K50 000 to be spent, then,

$5000x + 2000y \leq 50\,000$ (divide each term by 1000)

$$5x + 2y \leq 50$$

The farmer intends to spend at least K10 000 more on sheep than on goats, then

$5000x - 2000y \geq 10\,000$ (divide each term by 1000)

$$5x - 2y \geq 10$$

∴ The other two inequalities are as follows:

a) $5x + 2y \leq 50$

b) $5x - 2y \geq 10$

(ii). $x = 0$ is the y - axis

$y = 0$ is the x - axis

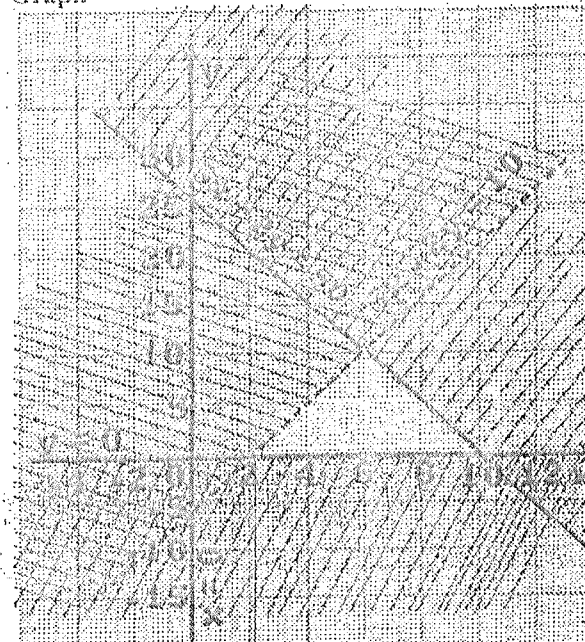
$$5x + 2y = 50$$

x	0	6	10
y	25	10	0

$$5x - 2y = 50$$

x	0	2	4
y	-5	0	5

Graph



(iii). Objective function = $x + y$

Need to maximise the objective function subject to:

$$x \geq 0;$$

$$y \geq 0;$$

$$5x + 2y \leq 50;$$

$$5x - 2y \geq 10$$

Coordinates connecting with the wanted region:

(2, 0); (10, 0); (6, 10)

Substitute the coordinates into the objective function:

Coordinate (x, y)	Objective function (x + y)
(2, 0)	$2 + 0 = 2$
(10, 0)	$10 + 0 = 10$
(6, 10)	$6 + 10 = 16$

Since 16 is the maximum, the farmer should buy 6 sheeps and 10 goats

Topic 14

Graphs of Cubic Functions

Graphs of cubic functions

A cubic function is in the form:

$y = ax^3 + bx^2 + cx + d$, where a, b, c and d are constants. Most graphs of cubic functions have two turning points. As in quadratic functions, we first determine the values of x and y before drawing graphs of cubic functions. To make the calculations easier, the function is split into suitable parts.

Graphical solutions of cubic functions

In general, when a graph of a cubic function is drawn, the solutions of the cubic equation is found by reading the values of the x -coordinate at the intersection of the curve and the line $y = 0$ or the x -axis.

Questions on Graphs of Cubic Functions

- Q1 (a)** Draw the graph of $y = x^3 + 2x^2 - 5x - 8$ for the values of x in the range $-4 \leq x \leq 3$.
- (b)** Use your graph to find:
- the solutions of $x^3 + 2x^2 - 5x - 8 = 0$.
 - the coordinates of the turning points.
- (c)** By drawing a suitable straight line, solve the equations:
- $x^3 + 2x^2 - 5x - 7 = 0$
 - $3 + 3x - 2x^2 - x^3 = 0$

- Q2 (a)** (i) Copy and Complete the table of values for the equation

x	-3	-2	-1	0	1	2	3
y	-15	0		0	-3		15

- (ii) Using a scale of 2 cm to represent 1 unit on x -axis and 2 cm to represent 5 units on y -axis, draw the graph of $y = x^3 - 4x$.
- (iii) Use your graph to solve the equation:

- Q3** Table below shows some of the values for the equation $y = x^3 - 2x^2 - 5x + 6$.

(i). Copy and complete the table.

Table 2: $y = x^3 - 2x^2 - 5x + 6$.

x	-3	-2	-1	0	1	2	3	4
y	-24		8		0	-4	0	18

- (ii). Using a scale of 2 cm to represent 1 unit on the x -axis and 2 cm to represent 5 units on the y -axis, draw the graph of $y = x^3 - 2x^2 - 5x + 6$.
- (iii). Use your graph to solve the equation $x^3 - 2x^2 - 5x + 4 = 0$

- Q4 (i)** Copy and complete the table of values for the equation $y = x(x^2 - x - 6)$.

x	-3	-2	-1	0	1	2	3	4
y	18	0		0	-6		0	24

- (ii) Using a scale of 2 cm to represent 1 unit on the y -axis, draw the graph of $y = x(x^2 - x - 6)$.

- (iii) Use your graph to solve the equation $x(x^2 - x - 6) + 5 = 0$

Graphs of Cubic Functions (Model Solutions)

Q1) Solution

- a) Make a table of values of x in the range $-4 \leq x \leq 3$. Work out the values of $x^3, 2x, -5x$, hence find the values of y .

x	-4	-3	-2	-1	0	1	2	3
x^3	-64	-27	-8	-1	0	1	8	27
$2x^2$	32	18	8	2	0	2	8	18
$-5x$	20	15	10	5	0	-5	-10	-15
-8	-8	-8	-8	-8	-8	-8	-8	-8
y	-20	-2	-2	-2	-8	-10	-2	22

The graph obtained from these values is as shown in Figure in the next column.

- c) (i) The curve $y = x^3 + 2x^2 - 5x - 8$ cuts the x -axis ($y = 0$) at three points. Therefore, there are three values of x for which $y = x^3 + 2x^2 - 5x - 8 = 0$. The solutions of $y = x^3 + 2x^2 - 5x - 8$ are:
 $x = -2.8$ or $x = -1.3$ or $x = 2.1$
- (ii) The coordinates of the turning points are:
 $(-2, 2)$ and $(1, -10)$.

- d) (i) The required straight line equation is obtained by eliminating terms involving x^3 and x^2 in the cubic function. This may be either by adding or subtracting corresponding terms of the cubic equation. A straight line equation is obtained by subtracting corresponding terms in the cubic equations.

$$y = x^3 + 2x^2 - 5x - 8$$

$$0 = x^3 + 2x^2 - 5x - 7$$

$$y = -1$$

The graph of $y = -1$ is then drawn. Read off the x -coordinates at the point of intersection of the curve and this straight line. The solutions of the equation

$$x^3 + 2x^2 - 5x - 7 = 0 \text{ are } x = -2.9 \text{ or } x = -1.2 \text{ or } x = 2.1.$$

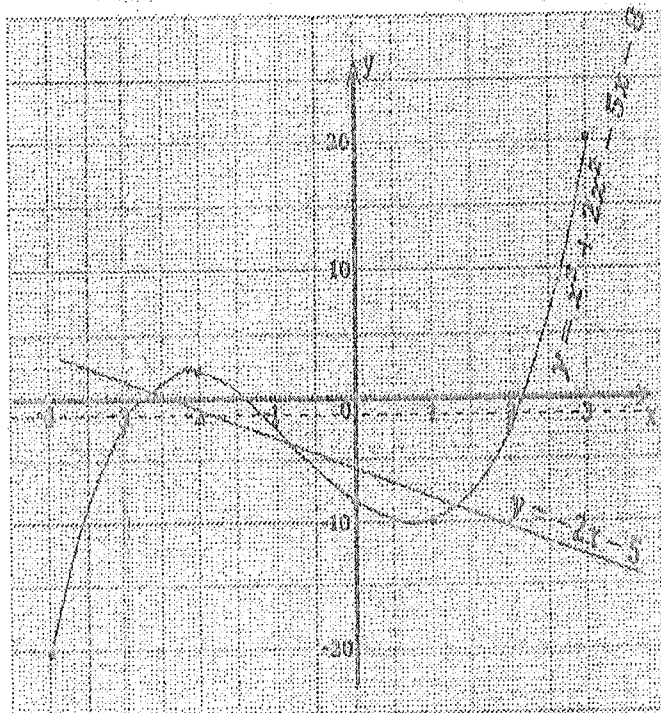
- (ii) The required straight line equation is obtained by adding corresponding terms.

$$y = x^3 + 2x^2 - 5x - 8$$

$$0 = -x^3 - 2x^2 + 3x + 3$$

$$y = -2x - 5$$

The graph of $y = -2x - 5$ is drawn as shown in Figure below. From the graph, the solutions of the equation $y = 3 + 3x - 2x^2 - x^3 = 0$, are $x = -2.7$ or $x = -0.8$ or $x = 1.5$.



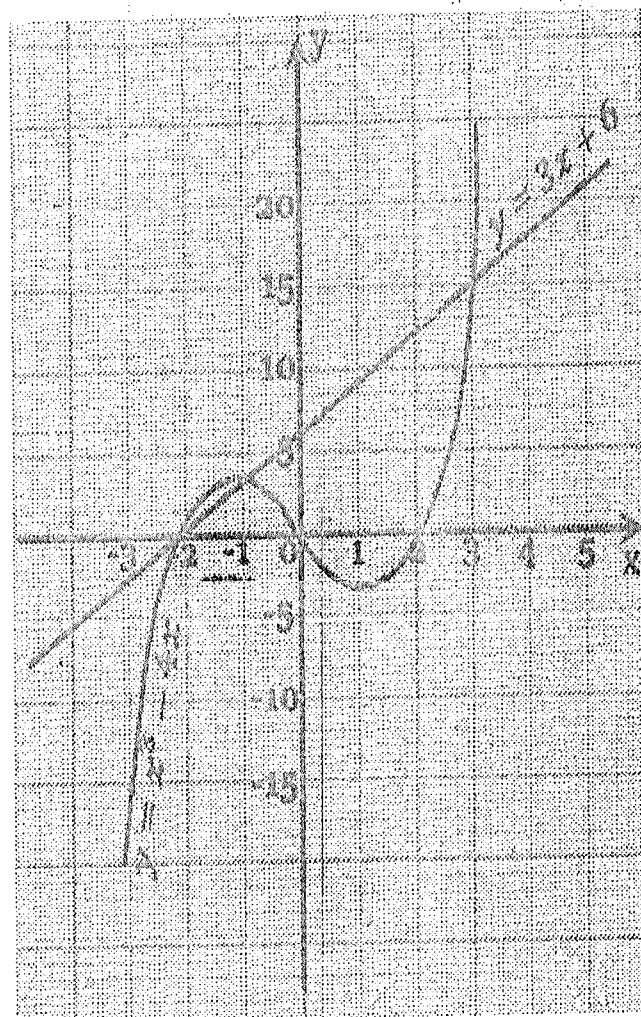
The graph of $y = x^3 + 2x^2 - 5x - 8$

Q2) Solution

- (a) (i) Copy and Complete the table of values for the equation

x	-3	-2	-1	0	1	2	3
y	-15	0	3	0	-3	0	15

Graph



To solve $x^3 - 7x - 6 = 0$

$$x^3 - 7x - 3x - 6 + 6 = 3x = 6$$

$$x^3 - 4x = 3x + 6$$

$$y = 3x + 6$$

Solution for $x^3 - 7x - 6 = 0$ is where the line $y = 3x + 6$ cut $y = x^3 - 4x$, from the graph

$$x = -2, -1 \text{ and } 3$$

Q3) Solution

(i)

x	-3	-2	-1	0	1	2	3	4
y	-24	0	8	6	0	-4	0	18

If $x = -2$, then

$$y = (-2)^3 - 2(-2)^2 - 5(-2) + 6$$

$$= -8 - 8 + 10 + 6$$

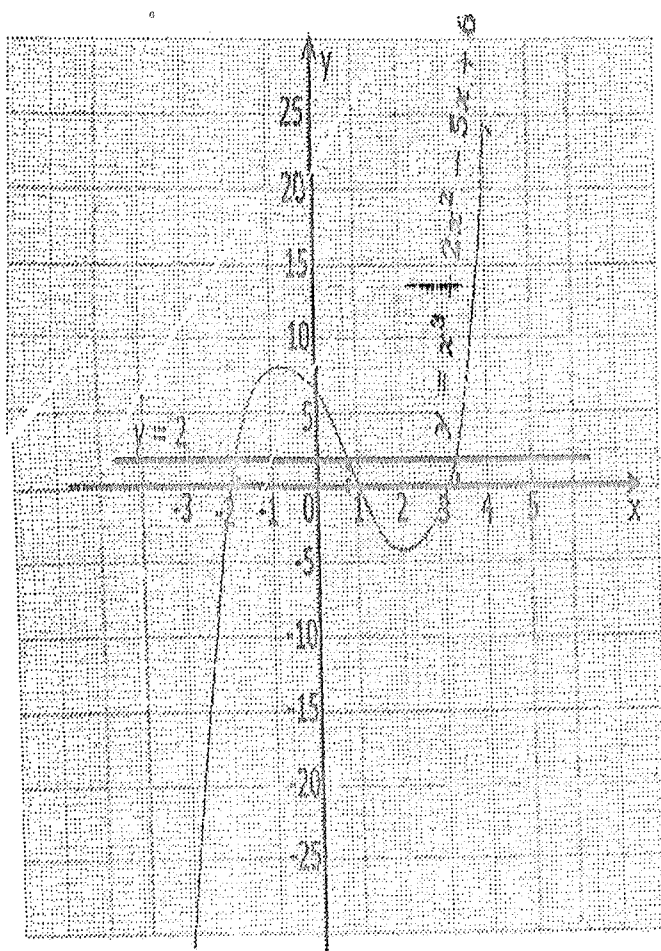
$$= 0$$

If $x = 0$, then $y = (0)^3 - 3(0)^2 - 5(0) + 6$

$$= 0 - 0 - 0 + 6$$

$$= 6$$

(ii) Graph



(iii) $x^3 - 2x^2 - 5x + 4 = 0$

$$x^3 - 2x^2 - 5x + 4 + 2 = 0 + 2$$

$$x^3 - 2x^2 - 5x + 6 = 2$$

$$y = 2$$

Draw the graph of $y = 2$ on the same axes where the graph of $y = x^3 - 2x^2 - 5x + 6$ is drawn. The

solutions lie where the equation of the equations of the two graphs intersect.

According to the graph, $x = -1.9$ or 0.7 or 3.2 .

Q4) Solution

(i). Completing the table of values.

x	-3	-2	-1	0	1	2	3	4
y	18	0	4	0	-6	-8	0	24

Given: $y = x(x^2 - x - 6)$

If $x = -1$, then

$$y = -1(-1)^2 - 1 - 6$$

$$= -1(1 + 1 - 6)$$

$$= -1(2 - 6)$$

$$= -1(-4)$$

$$= 4$$

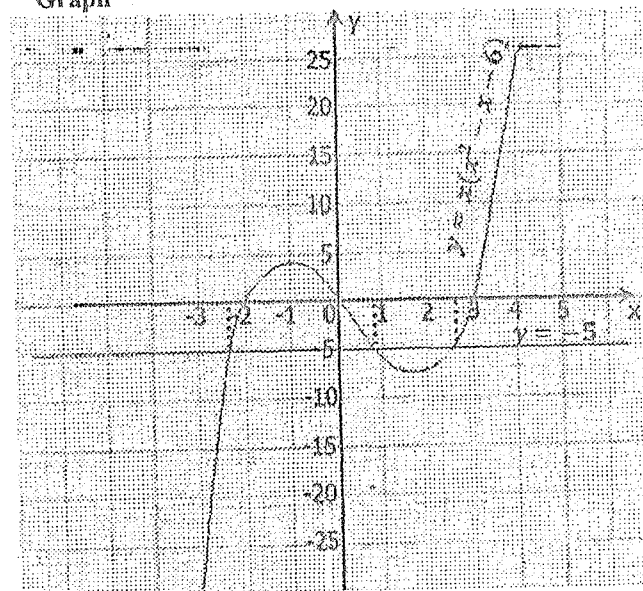
If $x = 2$, then $y = 2(2^2 - 2 - 6)$

$$= 2(4 - 2 - 6)$$

$$= 2(-4)$$

$$= -8$$

Graph



(ii). $x(x^2 - x - 6) + 5 = 0$

$$x(x^2 - x - 6) = -5$$

$$y = -5$$

Draw the graph of $y = -5$ on the same pair of axes where the graph of

$y = x(x^2 - x - 6)$ is drawn.

The solutions lie where the graphs of the two equations intersect

From the graph, $x = -2.4$ or 0.8 or 2.5

Topic 15

Statistics

Introduction:

In simple terms, the term statistics means a branch of mathematics which deals with collection, analysis and interpretation of data using statistical tools and finally come up with logical results.

The use of statistical methods is becoming increasingly important in all fields of everyday life. Indeed, it is almost impossible to name an activity which does not use statistics as an aid to influencing human behaviour.

Statistical methods for example are used in, accounting, economics, education, physics, engineering, biology, computer science, queuing theory and finance just to name a few.

Definitions of some important terms:

Measures of dispersion

Data can be summarised and compared by the use of averages, but an average on its own does *not* give sufficient information about the distribution of the scores.

The mean, median and mode are different ways of describing the average of a set of scores.

(a) The mean

This is a representative value that is found when sum of the given values in the data is divided by the number of values or items in the data set.

As a formula, the mean = $\frac{\text{sum of the scores}}{\text{number of scores}}$

For example,

In a test, the marks obtained out of 20 by 5 pupils were 12, 9, 16, 10 and 5

Calculate the mean mark for the 5 pupils.

$$\text{Mean mark} = \frac{12+9+16+10+5}{5} = \frac{52}{5} = 10.4$$

(b) The median

If the sample observations are arranged in order from smallest to largest, the median is defined as the middle observation if the number of observations is odd, and as the number-half way between the two middle observations if the number of observations is even

(c) The mode

The mode is defined as the observation in the sample which occurs most frequently, if there is such an observation.

(d) The range

The range is a way of describing the dispersion or spread of data. The range of a set of data is the largest value minus the smallest value. A *disadvantage* with the range is that it only uses two values and so it can be distorted by very high or very low values.

(e) The Interquartile range

The Interquartile range (IQR) of data is the difference between the upper quartile and the lower quartile of the data.

Note: Lower quartile is the score $\frac{1}{4}$ way along the distribution.

Upper quartile is the score $\frac{3}{4}$ way along the distribution.

(f) The variance

Variance is the average of squares of the deviations from the mean of a set of data.

It is also called the mean square deviation.

Variance can be obtained by dividing the sum of the squares of the deviations $\sum(x - \bar{x})^2$ by the total number of deviations ($\sum f$). Thus,

$$\text{Variance} = \frac{\sum f(x - \bar{x})^2}{(\sum f)}$$

(g) The standard deviation

The square root of the variance is called the standard deviation. This is the standard measure of the spread of a set of data from the mean and is denoted by s .

In general the given values are denoted as values of x , the mean as $\bar{x}$, and the total number of values (items) as n .

$$\text{Standard deviation} = \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$$

Questions on Statistics

- Q1)** Six athletes ran the 200 m race. The time they took, in seconds, were recorded as 21.9, 23.4, 23.5, 22.9, 22.5 and 22.3. Find their mean time
- Q2)** Five girls weigh 53kg, 58kg, 56kg, 59kg and x kg respectively. Their mean mass is 56 kg. Find the value of x .
- Q3)** Given that
 $x = a^2 - a$,
 $y = 2 - a$,
 $z = 7 + 5a - a^2$.
 Calculate the mean of x , y and z in terms of a .
- Q4)** The mean height of 12 boys is 152 cm. What is their total height?
- Q5)** Find the standard deviation of the following marks.
 4 6 7 10 13
- Q6)** Given the following data as prices of a certain product in kwacha:
 K4.44, K5.77, K6.08, K 8.33, 6.52, K7.18, 4.21, K5.16, K6.06, K3.64
 Using the above data:
 Find:
 i. the range
 ii. the median price
 iii. the mean
- Q7)** The variance of two temperature measurements, in degrees Celsius, 2 and $2a$ is 9. Calculate the possible values of a .
- Q8)** Calculate the standard deviation of the following numbers
 $x - 1, x, x + 1$
- Q9)** Find the standard deviation of each set of data whose deviations from the mean are given below:
 (a) -11, -6, -2, -1, 4, 5, 9, 10.
 (b) Use the table below:

Deviations(d)	-4	-2	-1	0	3	4	5
Frequency(f)	2	3	4	7	2	1	1

- (c) Explain your observation in terms of the results obtained from data in part (a) compared with the data in part in (b)

- Q10)** The table below shows frequency distribution of examination marks out of 100.

Marks Interval	<30	30 to 39	40 to 49	50 to 59	60 to 69	70 to 79	80 to 89	≥ 90
Frequency	0	3	9	8	9	4	2	0

- i. How many candidates took the examination?
 ii. What is the median interval?

- Q11)** The table below shows the marks scored by learners in a test.

Marks	10	11	12	13	14	15
Frequency	2	3	4	12	5	4

Calculate the mean mark

- Q12)** Given two sets of numbers 2, 3, 5, 7, 7, 8, 10 and x , 4, 6, 6, 8, 8 have equal mean, calculate the value of x .

Q13) The frequency table below shows the statistics recorded in a survey of the number of children in a family.

Number of children in a family	Frequency
1	5
2	7
3	6
4	2
5	6
6	1

- Draw a bar chart to illustrate the given information given in the table.
- Calculate the mean number of children per family

Q14) The table below shows scores obtained by a student in an end of term examination.

SUBJECT	STUDENT'S SCORE	TOTAL MARKS
Mathematics	16	20
Science	42	50
English	28	40
Bible knowledge	60	80

- Calculate the mean percentage mark
- When home economics mark is included, the mean percentage is 77. What was the student's score in home economics if the test marked out of 25?

Q15) Find the standard deviation for a set of data whose deviations from the mean are -2, -3, 4, 1.

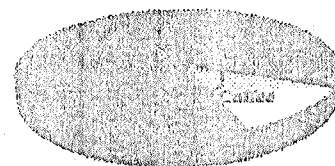
Q16) The ages of 10 students at a certain school are:

15, 12, 20, 18, 23, 16, 20, 24, 20, 15.

- Find (i) the modal age
(ii) the median age

Q17) The following are six marks in a test: 4, 9, 3, 4, m , 9. If the mean mark was 6.5, find m .

Q18) Figure below is a pie chart representing sales of three commodities; tobacco, tea and coffee



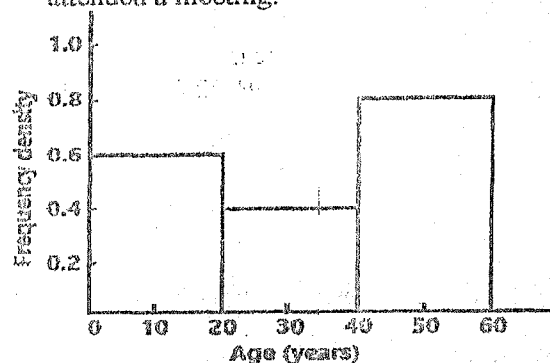
Express coffee sales as a percentage of tea total sales

Q19) Table below shows the distribution of the number of employees in 43 factories in a town.

Number of Employees	0-39	40-59	60-79	80-99
Number of Factories	5	15	13	10

Draw a histogram

Q20) In figure below, the histogram is representing the ages of the people who attended a meeting.



Calculate the number of people who were between 20 and 40 years old at the meeting.

Q21) Table below shows marks that student X and student Y got from tests. Student X sat for 5 tests while student Y sat for 4 tests. Student Y has mark q missing.

STUDENT X	55	70	80	30	65
STUDENT Y	67	60	q	53	

Given that the mean mark of student X is the same as the mean mark of student Y, calculate the value of q .

Q22) Table below shows tallies of students' marks in a test

MARK (m)	TALLY
$0 \leq M \leq 10$	
$10 < M \leq 20$	///
$20 < M \leq 30$	////
$30 < M \leq 40$	////
$40 < M \leq 50$	//// //
$50 < M \leq 60$	////
$60 < M \leq 70$	//// //
$70 < M \leq 80$	//// /
$80 < M \leq 90$	////
$90 < M \leq 100$	/

- (a) Find the modal class
 (b) If 50% of the students were to pass, find the passing mark for the test.

Q23) In a certain Form 1 class at Dowa Secondary School the ages of the pupils are given by the table below:

Age	12	13	14	15	16	17	18
No. of pupils	1	3	16	20	20	14	6

- (a) Draw a frequency polygon to illustrate this data.
 (b) Give the modal ages.
 (c) Calculate the mean age of the pupils below 15

Q24) The ages of five students at Mayeso Secondary School are.
 14, 16, 19, 20 and 21.
 Calculate the variance, show all your working.

Statistics Questions (Model Solutions)

Q1) Solution

The sum of their time is

$$21.9 + 23.4 + 23.5 + 22.9 + 22.5 + 22.3 = 136.5 \text{ seconds}$$

$$\therefore \text{The mean} = \frac{136.5}{6} = 22.75 \text{ seconds}$$

Q2) Solution

$$\text{Total mass} = 53 + 58 + 56 + 59 + x = 226 + x$$

$$\text{The mean} = \frac{226 + x}{5}$$

$$\therefore \frac{226 + x}{5} = 56 \quad (\text{Cross-multiplying we get})$$

$$226 + x = 280$$

$$x = 280 - 226$$

$$x = 54 \text{ kg}$$

Q3) Solution

$$\text{Given that } x = a^2 - a,$$

$$y = 2 - a,$$

$$z = 7 + 5a - a^2$$

$$\text{Mean} = \frac{x + y + z}{3}$$

$$= \frac{(a^2 - a) + (2 - a) + (7 + 5a - a^2)}{3}$$

$$= \frac{a^2 - a + 2 - a + 7 + 5a - a^2}{3}$$

$$= \frac{3a + 9}{3} = \frac{3(a + 3)}{3} = a + 3$$

Q4) Solution

$$\text{The mean} = \frac{\text{total height}}{\text{number of boys}}$$

$$\text{Therefore, } 152 = \frac{\text{total height}}{12}$$

$$\text{Total height} = 152 \times 12 = 1824 \text{ cm}$$

Q5) Solution

$$\text{The mean} = \frac{4 + 6 + 7 + 10 + 13}{5} = \frac{40}{5} = 8$$

Marks	Deviation from mean	Square of deviation
4	-4	16
6	-2	4
7	-1	1
10	2	4
13	5	25
Total	0	50

$$\text{Variance} = \frac{\sum f(x - \bar{x})^2}{(\sum f)} = \frac{50}{5} = 10$$

$$\begin{aligned}\therefore \text{The standard deviation} &= \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}} \\ &= \sqrt{10} \\ &= 3.2 \text{ (1 d.p.)}\end{aligned}$$

Q6) Solution

Firstly, arrange the prices in ascending order as follows

K3.64, K4.21, K4.44, K5.16, K5.77, K6.06, K6.08, K6.52, K7.18, 8.33

- The lowest price = K3.64
The maximum price = K8.33
 $\therefore \text{Range} = K8.33 - K3.64 = K4.69$
- The median price is the average of the 5th and 6th prices
 $= (K5.77 + K6.06) \div 2$
 $= K5.92$ to the nearest tambala
 $\therefore \text{The median price is K5.92}$

- The mean price
$$= \frac{\text{Total number of the prices}}{\text{Number of prices}}$$

$$\begin{aligned}&= \frac{K3.64 + K4.21 + 4.44 + K5.16 + 5.77 + K6.06 + K6.08 + K6.52 + K7.18 + K8.33}{10} \\ &= \frac{K57.39}{10} \\ &= 5.74 \text{ (to the nearest Tambala)} \\ &\therefore \text{The mean price} = K5.74\end{aligned}$$

Q7) Solution

x	frequency(f)	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
2	1	$1 - a$	$(a - 1)^2$	$(1 - a)^2$
2a	1	$a - 1$	$(1 - a)^2$	$(a - 1)^2$

$$\sum f = 2$$

$$\sum f(x - \bar{x})^2 = (1 - a)^2 + (a - 1)^2$$

$$\text{Variance} = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

$$9 = \frac{(1-a)^2 + (a-1)^2}{2}$$

$$18 = (1 - a)^2 + (a - 1)^2 \text{ (Both sides multiplied by 2)}$$

$$18 = 1 - 2a + a^2 + a^2 - 2a + 1$$

$$18 = 2a^2 - 4a + 2$$

$$2a^2 - 4a + 2 - 18 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

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$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$2a^2 - 4a - 16 = 0$$

$$(a + 2)(a - 4) = 0$$

$$\text{Either } (a + 2) = 0 \Rightarrow a = -2$$

$$\text{or } (a - 4) = 0 \Rightarrow a = 4$$

$\therefore \text{The possible values of } a \text{ are } -2 \text{ or } 4$

Q8) Solution

Given numbers; $x - 1, x, x + 1$

$$\text{Mean} = \frac{\text{Sum of the numbers}}{\text{number of terms}}$$

$$= \frac{x-1+x+x+1}{3}$$

$$= \frac{3x}{3}$$

$$\therefore \bar{x} = x$$

No. (y)	Freq. (f)	$y - \bar{x}$	$(y - \bar{x})^2$	$f(y - \bar{x})^2$
$x - 1$	1	-1	1	1
x	1	0	0	0
$x + 1$	1	1	1	1
	$\sum f = 3$			$\sum f(y - \bar{x})^2$

$$\text{Standard deviation} = \sqrt{\text{variance}}$$

$$\text{Variance} = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

$$\therefore \text{Standard deviation} = \sqrt{\frac{2}{3}}$$

$$= 0.82 \text{ (to 2 decimal places)}$$

Q9) Solutions

(a) Firstly, variance of the given data is found by:

Finding the squares of the deviations, then get their sum. Divide the sum by the number of these deviations.

In this case, the squares of deviations are: 121, 36, 4, 1, 16, 25, 81 and 100.

The sum of the squares of the deviations is:

$$121 + 36 + 4 + 1 + 16 + 25 + 81 + 100 = 384$$

The number of deviations is 8.

$$\text{The variance} = \frac{384}{8} = 48$$

$\therefore \text{The variance } s^2 \text{ of the data is } 48.$

$\therefore \text{standard deviation (s) is } \sqrt{48} = 6.928$

(b) The variance (s^2) is 5.35

$\therefore \text{standard deviation (s) is } \sqrt{5.35} = 2.313$

- (c) The standard deviation of the data in part (a) is greater than the standard deviation of the data in part (b). This means that the set of data in part (a) is more concentrated around the mean than the set of data in part (b)

Q10) Solution

- i. Number of candidates who took the examination
 $= 0 + 3 + 9 + 8 + 9 + 4 + 2 + 0$
 $= 35$
- ii. The median interval is 50 – 59

Q11) Solution

Mean mark =

$$\frac{(10 \times 2) + (11 \times 3) + (12 \times 4) + (13 \times 12) + (14 \times 5) + (15 \times 4)}{2 + 3 + 4 + 12 + 5 + 4}$$

$$= \frac{20 + 33 + 48 + 156 + 70 + 60}{30}$$

$$= \frac{387}{30}$$

$$= 12.9$$

Alternative method

Using frequency Table

Mark	Frequency	Total Marks
10	2	20
11	3	33
12	4	48
13	12	156
14	5	70
15	4	60
Total	30	387

$$\text{Mean} = \frac{387}{30} = 12.9$$

Q12) Solution

$$x + 4 + 6 + 6 + 8 + 8 = \frac{2 + 3 + 5 + 7 + 7 + 8 + 10}{6}$$

$$\frac{x + 32}{6} = \frac{42}{7} \quad (\text{Cross multiply})$$

$$7(x + 32) = 42 \times 6$$

$$7x + 224 = 252$$

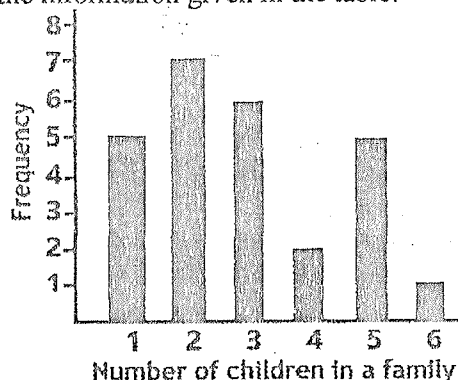
$$7x = 252 - 224$$

$$7x = 28$$

$$x = 4$$

Q13) Solution

- i. The following is a bar chart representing the information given in the table:



ii. Mean = $\frac{\text{total number of children}}{\text{number of families}}$

$$= \frac{(1 \times 5) + (2 \times 7) + (3 \times 6) + (4 \times 2) + (5 \times 6) + (6 \times 1)}{5 + 7 + 6 + 2 + 6 + 1}$$

$$= \frac{5 + 14 + 18 + 8 + 30 + 6}{27}$$

$$= \frac{81}{27}$$

$$= 3 \text{ children per family}$$

Q14) Solution

$$\% \text{ marks for Mathematics} = \frac{16 \times 100}{20} = 80\%$$

$$\% \text{ marks for Science} = \frac{42 \times 100}{50} = 84\%$$

$$\% \text{ marks for English} = \frac{28 \times 100}{40} = 70\%$$

$$\% \text{ marks for B/Knowledge} = \frac{60 \times 100}{80} = 75\%$$

i. Mean % mark = $\frac{80 + 84 + 70 + 75}{4} = 77.25\%$

- ii. Since the total mark for 4 subjects is 309

$$\text{Then } \frac{x + 309}{5} = 77$$

$$\Rightarrow x + 309 = 77 \times 5$$

$$x + 309 = 385$$

$$x = 385 - 309$$

$$x = 76$$

Let the student's score in home economics be k

$$\frac{k}{25} = \frac{76}{100} \quad (\text{Cross multiply})$$

$$k = \frac{25 \times 76}{100} = 19$$

Q15) Solution

In this case, given the deviations from the mean as follows: -2, -3, 4, 1.

$x - \bar{x}$	$(x - \bar{x})^2$	frequency (f)
-2	4	1
-3	9	1
4	16	1
1	1	1

This implies

$$\sum (x - \bar{x})^2 = 30$$

$$\sum f = 4$$

∴ Standard deviation (SD)

$$= \sqrt{\frac{\sum (x - \bar{x})^2}{\sum f}}$$

$$(SD) = \sqrt{30/4}$$

$$(SD) = 2.7$$

Hence the standard deviation is 2.7

Q16) Solution

- Modal age = 20
- Firstly, arrange the numbers as follows:
12, 15, 15, 16, 18, 20, 20, 20, 23, 24

$$\text{Median age} = \frac{18+20}{2} = 19$$

Q17) Solution

$$\text{Mean} = \frac{\text{sum of the marks}}{\text{number of marks}}$$

$$\Rightarrow 6.5 = \frac{4+9+3+4+m+9}{6}$$

$$\Rightarrow 6.5 = \frac{29+m}{6} \text{ (multiply both sides by 6)}$$

$$\Rightarrow 6 \times 6.5 = 29 + m$$

$$\Rightarrow 39 = 29 + m$$

$$\Rightarrow m = 39 - 29$$

$$\therefore m = 10$$

Q18) Solution

In this case, we need first to find the angle represented by coffee sales

Thus,

$$= 360 - (208^\circ + 107^\circ)$$

$$= 360^\circ - 315^\circ$$

$$= 45^\circ$$

∴ Coffee sales as a %

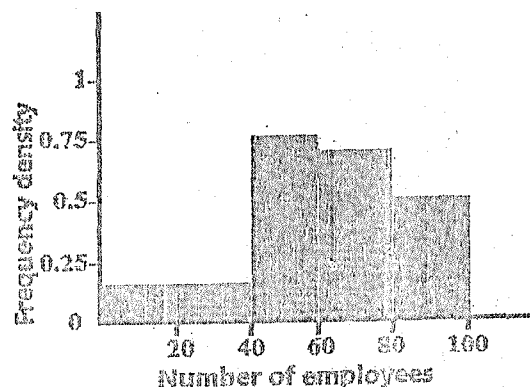
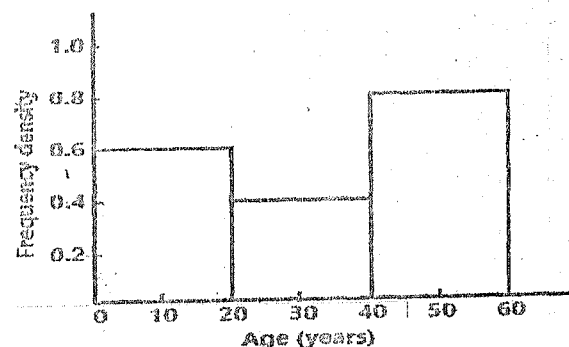
$$\Rightarrow 45^\circ / 360^\circ \times 100\%$$

$$= 1/8 \times 100\%$$

$$= 12.5\%$$

Q19) Solution

Number of employees	Frequency	F/Class interval	Width
0 - 39	5	5/40	0.125
40 - 59	15	15/20	0.75
60 - 79	13	13/20	0.65
80 - 99	10	10/20	0.5

**Q20) Solution**

Number of people between 20 and 40 years = Area of the middle rectangle

$$= (40 - 20) \times (0.4 - 0)$$

$$= 20 \times 0.4$$

$$= 8$$

Alternative method

Using the histogram above (area of the middle rectangle)

Number of people between 20 and 30 years
 $= 10 \times 0.4 = 4$

Number of people between 30 and 40 years
 $= 10 \times 0.4 = 4$

Therefore, total number of people
 $= 4 + 4 = 8$

Q21) Solution

Let the mean mark for the student X be $\bar{x}_1$ and the mean mark for student Y be $\bar{x}_2$.

$$\begin{aligned}\therefore \bar{x}_1 &= \frac{\text{sum of the marks of student X}}{\text{number of tests}} \\ &= \frac{55 + 70 + 80 + 30 + 65}{5} \\ &= \frac{300}{5} \\ &= 60\end{aligned}$$

Similarly,

$$\begin{aligned}\bar{x}_2 &= \frac{\text{sum of the marks of student Y}}{\text{number of tests}} \\ &= \frac{67 + 60 + q + 53}{4} \\ &= \frac{180 + q}{4}\end{aligned}$$

But $\bar{x}_1 = \bar{x}_2$ (given)

$$\Rightarrow 60 = \frac{180 + q}{4}$$

$$240 = 180 + q \quad (\text{both sides multiplied by 4})$$

$$q = 240 - 180$$

$$q = 60$$

$\therefore$ The value of q is 60

Q22) Solution

MARK (m)	TALLY
$0 < M \leq 10$	
$10 < M \leq 20$	///
$20 < M \leq 30$	///
$30 < M \leq 40$	///
$40 < M \leq 50$	//// ////
$50 < M \leq 60$	///
$60 < M \leq 70$	//// ////
$70 < M \leq 80$	////
$80 < M \leq 90$	///
$90 < M \leq 100$	///

i. The modal class is the class with more tallies (i.e. one with highest appearances)

With reference to the table, the class with more tallies is

$$40 < M \leq 50$$

ii. Total number of students

$$= 2 + 3 + 3 + 10 + 2 + 9 + 6 + 4 + 1 = 40$$

If 50% of the students were to pass, then

$$\Rightarrow \frac{50\%}{100\%} \times 40 = 20$$

$\therefore$ 20 students were to pass the test.

Need only the last 20 students

Using cumulative tallies from the bottom of the table going up we set

$$1 = 1$$

$$1 + 4 = 5$$

$$1 + 4 + 6 = 11$$

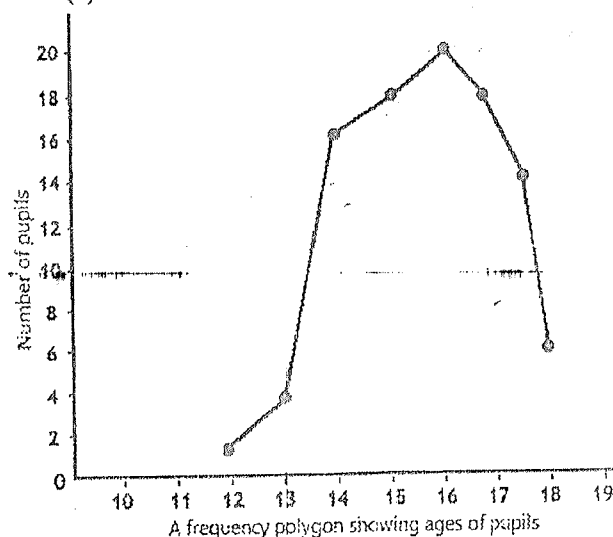
$$1 + 4 + 6 + 9 = 20$$

$\Rightarrow$ All students in the range $60 < M \leq 100$ were to pass the test

$\therefore$ The passing mark for the test should be greater than 60%

Q23) Solution

(a)



(b) The modal ages are 15 and 16

(c) Mean age of pupils below 15

$$\begin{aligned}
 &= \frac{(1 \times 12) + (3 \times 13) + 16 \times 14}{1 + 3 + 16} \\
 &= 275/20 \\
 &= 13 \text{ years 9 months}
 \end{aligned}$$

Q24) Solution

Given ages 14, 16, 19, 20, 21

$$\begin{aligned}
 \text{Mean} = \bar{x} &= \frac{\text{sum of the ages}}{\text{number of ages}} \\
 &= \frac{14 + 16 + 19 + 20 + 21}{5} \\
 &= \frac{90}{5} \\
 &= 18
 \end{aligned}$$

Age(x)	frequency (f)	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
14	1	-4	16	16
16	1	-2	4	4
19	1	1	1	1
20	1	2	4	4
21	1	3	9	9
	$\sum f = 5$			$\sum f(x - \bar{x})^2 = 34$

$$\begin{aligned}
 \text{Variance} &= \frac{\sum f(x - \bar{x})^2}{\sum f} \\
 &= \frac{34}{5} = 6.8
 \end{aligned}$$



Key Words and Definitions

(a) Matrix

A matrix is rectangular arrangement of numbers, letters or symbols that are called elements.

i.e. $\begin{pmatrix} 20 & 25 & 19 \\ 28 & 29 & 27 \end{pmatrix}$ and $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$

Note: we do not use commas in matrices. The elements are separated by space.

(b) Elements or components

The numbers or letters within the matrix are called elements.

In general, for a matrix $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$

a, b, c, d, e and f are elements or components

(c) The order of matrix

The elements of matrices are arranged in rows and columns. A matrix is described by giving the number of rows and columns. This is referred to as the order of a matrix. The order of a matrix is given as the number of rows $\times$ the number of columns.

Note that, the symbol $\times$ means by not multiplication. For example, the matrix shown above has 2 rows and 3 columns, hence its order is 2×3 .

The number of rows is always given first

(d) The matrix notation

Sometimes a matrix is represented by a bold, capital letter. For example,

$$A = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ 8 & 6 \\ 2 & 9 \end{pmatrix}$$

(e) **Types of matrices**

i. Row and column matrices

A matrix that has only one row is called a row matrix. For example, (1 2 3 4),

While, a column matrix is the one that has only one column. For example,

$$\begin{pmatrix} 4 \\ 6 \\ 7 \end{pmatrix}$$

ii. A square matrix

This is a matrix that has an equal number of rows and columns. For example,

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is a 2×2 matrix and

$B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is a 3×3 matrix

In general, if the order of a matrix is $n \times n$, then it's a square matrix

iii. An identity matrix

This is a square matrix that does not alter (change) another matrix after multiplication.

Examples of identity matrices are

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Such matrices are denoted (represented) by the letter I and for any matrix A of the same order as I , $AI = IA = A$

iv. A null or zero matrix

A null matrix has all its elements equal to

zero. For example, $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is 3×3

matrix. Any null matrix is denoted by letter O .

If matrix A is the same order as a null matrix O , then $A + O = A$ and $A - O = A$.

Also, $AO = O$. If $PQ = O$, then either P or Q or both are null matrices.

v. Equal matrices

These are matrices that are of the same order, and elements in corresponding positions are

equal. For instance, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} w & x \\ y & z \end{pmatrix}$

and $A = B$, then $a = w$, $b = x$, $c = y$ and $d = z$.

If $A = \begin{pmatrix} 4 & 3 \\ 1 & 6 \end{pmatrix}$, $B = \begin{pmatrix} 4 & x \\ y & 6 \end{pmatrix}$ and $A = B$, the $x = 3$ and $y = 1$.

Adding and subtracting matrices

In general, if $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$,

$$\begin{aligned} \text{Then } A + B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ &= \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{Similarly, } A - B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ &= \begin{pmatrix} a-e & b-f \\ c-g & d-h \end{pmatrix} \end{aligned}$$

Note: Matrices are added or subtracted only when they are of the same order.

Multiplying a matrix by a scalar

In general, if $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is multiplied by a scalar k , the $kM = k \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$

Transformations by matrix multiplication

A transformation can be described using a matrix that multiplies the position vector of a point.

Point P has coordinates (x, y) and its position vector, $\underline{p} = \begin{pmatrix} x \\ y \end{pmatrix}$.

If we have the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\text{Then } M\underline{p} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax & by \\ cx & dx \end{pmatrix}$$

We say that the matrix M transforms the point $P(x, y)$ to point $P'(x', y')$, where $x' = ax + by$ and $y' = cx + dy$

Questions on Matrices

$$C = \begin{pmatrix} 5 & a & 4 \\ 6 & 6 & 1 \\ 5 & 0 & e \end{pmatrix}$$

Find a, b, c, d and e so that $A + B = C$

Q1) State the order of the following matrices

- a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
 b) $\begin{pmatrix} 11 & 8 & 9 \\ 1 & 7 & 3 \\ 4 & 5 & 2 \end{pmatrix}$
 c) $\begin{pmatrix} 1 & 5 \end{pmatrix}$
 d) $\begin{pmatrix} 2 \\ 8 \\ 6 \end{pmatrix}$

Q2) Evaluate the products of the following matrices

- a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix}$
 b) $\begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
 c) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
 d) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Q3) If $P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, find a 2×2 matrix Q

such that $PQ = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Q4) $A = \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & 6 \end{pmatrix}$
 $B = \begin{pmatrix} -1 & 1 & 1 \\ 3 & 1 & -2 \end{pmatrix}$

- a) Find $A + B$
 b) Find $B + A$

What can you say about the sums $A + B$ and $B + A$?

Q5) $A = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ $B = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

Find: (a) $A + B$
 (b) $B + A$

Does $A + B = B + A$?

Q6) $A = \begin{pmatrix} 1 & 2 & c \\ 3 & 4 & 5 \\ 3 & d & 8 \end{pmatrix}$
 $B = \begin{pmatrix} 4 & 3 & -1 \\ 3 & 2 & b \\ 2 & 2 & -4 \end{pmatrix}$

Q7) $X = \begin{pmatrix} 1.2 & t-2 \\ q & s \\ 3 & r \end{pmatrix}$ $Y = \begin{pmatrix} 4.9 & 3 \\ 3.4 & s \\ 2 & 2.3 \end{pmatrix}$

$$Z = \begin{pmatrix} p & 2t \\ -2 & s \\ 5 & 2.3 \end{pmatrix}$$

Find p, q, r, s and t so that $X + Y = Z$

Q8) $A = \begin{pmatrix} 0 & 2 & 4 \\ 4 & -1 & 2 \end{pmatrix}$ $B = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{pmatrix}$

Find:

- a) $3A$
 b) $-2B$
 c) $3A - B$

Q9) $P = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$ $Q = \begin{pmatrix} 2 & -4 & -2 \end{pmatrix}$

Work out the following matrices

- a) $10P$
 b) $-\frac{1}{2}Q$
 c) $3P + 3Q$

Q10) If $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix}$
 $C = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$

Evaluate:

- a) A^2
 b) $B^2 - A^2$
 c) ABC

Q11) Find the unknown value in the matrix equation below.

a) $\begin{pmatrix} 2 & p & 3 \end{pmatrix} \begin{pmatrix} 3 \\ -5 \\ 7 \end{pmatrix} = (-3)$

b) $\begin{pmatrix} 1 & q \end{pmatrix} \begin{pmatrix} 3 \\ q \end{pmatrix} = (12)$

Q12) Work out the following:

$$2 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 3 \begin{pmatrix} 6 & 7 \\ 8 & 9 \end{pmatrix}$$

Q13) $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ $B = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 0 & 4 \end{pmatrix}$
 $C = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 1 & 2 \\ 4 & 0 & 1 \end{pmatrix}$

Work out, if possible

- a) AB
- b) BA
- c) BC

Q14) $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

Show that $IA = AI$

Q15) If $M = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$ is the matrix of transformation and $P = (6, 7)$, find the image of P under M .

Q16) The square ABCD has vertices $A(0,0)$, $B(1,0)$, $C(1,1)$ and $D(0,1)$

Find the image of ABCD under

transformation $M = \begin{pmatrix} 2 & 3 \\ -1 & 5 \end{pmatrix}$

Q17) A transformation T transforms points $P(0,0)$, $Q(2,1)$ and $R(4,1)$ to $P'(0,0)$, $Q'(3,2)$ and $R'(5,2)$ respectively. Find the matrix of the transformation T .

Q18) What is the point whose image is the point $(7, 9)$ when it is transformed with matrix $M = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$?

Q19) Find the image of $(3, 4)$ after a rotation of 30° . The matrix for a 30° rotation is $\begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{pmatrix}$

Q20) Find the image of the reflection of $(3, -2)$ in

- a) the x-axis
- b) the y-axis
- c) the line $y = x$

Q21) $A = \begin{pmatrix} 2 & 4 \\ 3 & 1 \\ 4 & 5 \end{pmatrix}$ $B = \begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 5 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 6 \end{pmatrix}$

Find: $A + B + C$

Q22) Let $P = \begin{pmatrix} 4 & -6 \\ -2 & 8 \end{pmatrix}$,

Find:

- a) Find $2P$
- b) Find P^2

c) Find Q when $P \times Q = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$

Q23) Given that $A = \begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix}$ and

$B = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$, find $A + 2B$

Q24) Calculate the values of a and b if

$\begin{pmatrix} 3a & 18 \\ 4 & 6 \end{pmatrix} - \begin{pmatrix} 2a & 2 \\ 2 & -6 \end{pmatrix} = \begin{pmatrix} 4 & 16 \\ 2 & b \end{pmatrix}$

Q25) Find the values of x and y in the following matrix equation.

$\begin{pmatrix} 6 & 3 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}$

Q26) Show that

$\begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 5 & -1 \end{bmatrix} + \begin{bmatrix} 8 & -12 \\ -8 & -1 \end{bmatrix}$

is zero matrix

Q27) Given that

$\begin{pmatrix} 6 & 8 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} w & -4 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Find the value of w .

Q28) Given that matrix $V = \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}$

and $W = \begin{pmatrix} -1 & 1 \\ 0 & 2 \end{pmatrix}$, Find $VW - V$

Q29) A and B are two matrices. If $A = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}$, find B given that $A^2 = A + B$

Q30) $A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}$, Find $2A - B^2$

Q31) Given that matrix $P = \begin{pmatrix} 7 & 5 \\ 2 & 4 \end{pmatrix}$ and $Q = \begin{pmatrix} 3 & 10 \\ 5 & 1 \end{pmatrix}$, find PQ

Q32) Given that $A = \begin{pmatrix} 2 & 1 \\ -3 & 4 \end{pmatrix}$ and

$B = \begin{pmatrix} 0 & -10 \\ 8 & 14 \end{pmatrix}$

Calculate $A + \frac{1}{2}B$

Q33) Given that $X = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$, find the value of $X^2 - 2X$

Q34) Given that matrix $M = \begin{pmatrix} 3 & 3 \\ -1 & -1 \end{pmatrix}$ and $M^2 = kM$ where k is a constant, calculate the value of k .

Q35) Given that $\begin{pmatrix} 3c & c \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 28 \\ 22 \end{pmatrix}$, find the value of c .

Q36) Given that $M = \begin{pmatrix} a & d \\ 1 & 2 \end{pmatrix}$ and

$N = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix}$, find MN .

Matrices (Model Solutions)

Q1) Solution

a) 2×2

b) 3×3

c) 1×2

d) 3×1

Q2) Solution

$$\begin{aligned} \text{a) } & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 4 + 0 \times 3 & 1 \times 1 + 0 \times 5 \\ 0 \times 4 + 1 \times 3 & 0 \times 1 + 1 \times 5 \end{pmatrix} \\ &= \begin{pmatrix} 4 + 0 & 1 + 0 \\ 0 + 3 & 0 + 5 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{b) } & \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \times 1 + 1 \times 0 & 4 \times 0 + 1 \times 1 \\ 3 \times 1 + 5 \times 0 & 3 \times 0 + 5 \times 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 + 0 & 0 + 1 \\ 3 + 0 & 0 + 5 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{c) } & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} a \times 1 + b \times 0 & a \times 0 + b \times 1 \\ c \times 1 + d \times 0 & c \times 0 + d \times 1 \end{pmatrix} \\ &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{d) } & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 \times a + 0 \times c & 1 \times b + 0 \times d \\ 0 \times a + 1 \times c & 0 \times b + 1 \times d \end{pmatrix} \\ &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \end{aligned}$$

Q3) Solution

Let $Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Then $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} a + 2c & b + 2d \\ 3a + 4c & 3b + 4d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Thus, $a + 2c = 0 \dots\dots\dots (i)$

and $3a + 4c = 0 \dots\dots\dots (ii)$

$2a + 4c = 0 \dots\dots\dots (iii)$

Multiply (i) by 2

Subtract (iii) from (ii) to get

$3a + 4c - 2a - 4c = 0$

$\therefore a = 0$ and $c = 0$

Also $b + 2d = 0 \dots\dots\dots (i)$

and $3b + 4d = 0 \dots\dots\dots (ii)$

Multiply (i) by 3

$2b + 4d = 0 \dots\dots\dots (iii)$

Subtract (iii) from (ii) we get

$3b + 4d - 2b - 4d = 0$

$\therefore b = 0$ and $d = 0$

$Q = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Q4) Solution

$$\begin{aligned} \text{a) } A + B &= \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & 6 \end{pmatrix} + \begin{pmatrix} -1 & 1 & 1 \\ 3 & 1 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 1 + (-1) & 2 + 1 & -3 + 1 \\ 4 + 3 & 5 + 1 & 6 + (-2) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 3 & -2 \\ 7 & 6 & 4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{b) } B + A &= \begin{pmatrix} -1 & 1 & 1 \\ 3 & 1 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & 6 \end{pmatrix} \\ &= \begin{pmatrix} -1 + 1 & 1 + 2 & 1 + (-3) \\ 3 + 4 & 1 + 5 & -2 + 6 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 3 & -2 \\ 7 & 6 & 4 \end{pmatrix} \end{aligned}$$

From the results it could be concluded that

$A + B = B + A$

Q5) Solution

$$\begin{aligned} \text{(a) } A + B &\Rightarrow \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 3 + (-1) \\ 1 + 3 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{(b) } B + A &= \begin{pmatrix} -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -1 + 3 \\ 3 + 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 4 \end{pmatrix} \end{aligned}$$

Yes, $A + B = B + A$

Q6) Solution

$A + B = C$ implies

$$\begin{pmatrix} 1 & 2 & c \\ 3 & 4 & 5 \\ 3 & d & 8 \end{pmatrix} + \begin{pmatrix} 4 & 3 & -1 \\ 3 & 2 & b \\ 2 & 2 & -4 \end{pmatrix} = \begin{pmatrix} 5 & a & 4 \\ 6 & 6 & 1 \\ 5 & 0 & e \end{pmatrix}$$

$$\begin{pmatrix} 1+4 & 2+3 & c+(-1) \\ 3+3 & 4+2 & 5+b \\ 3+2 & d+2 & 8+(-4) \end{pmatrix} = \begin{pmatrix} 5 & a & 4 \\ 6 & 6 & 1 \\ 5 & 0 & e \end{pmatrix}$$

$$\begin{pmatrix} 5 & 5 & c-1 \\ 6 & 6 & 5+b \\ 5 & d+2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & a & 4 \\ 6 & 6 & 1 \\ 5 & 0 & e \end{pmatrix}$$

Thus,

$$a = 5$$

$$5 + b = 1 \Rightarrow b = -4$$

$$c - 1 = 4 \Rightarrow c = 5$$

$$d + 2 = 0 \Rightarrow d = -2$$

$$e = 4$$

Hence, $a = 5$, $b = -4$, $c = 5$, $d = -2$

and $e = 4$

Check

$$\text{LHS} = \begin{pmatrix} 1+4 & 2+3 & c+(-1) \\ 3+3 & 4+2 & 5+b \\ 3+2 & d+2 & 8+(-4) \end{pmatrix}$$

$$= \begin{pmatrix} 1+4 & 2+3 & 5+(-1) \\ 3+3 & 4+2 & 5+(-4) \\ 3+2 & -2+2 & 8+(-4) \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 5 & 4 \\ 6 & 6 & 1 \\ 5 & 0 & 4 \end{pmatrix}$$

$$\text{RHS} = \begin{pmatrix} 5 & 5 & 4 \\ 6 & 6 & 1 \\ 5 & 0 & 4 \end{pmatrix}$$

Hence LHS = RHS proved

Q7) Solution

$X + Y = Z$ implies

$$\begin{pmatrix} 1.2 & t-2 \\ q & s \\ 3 & r \end{pmatrix} + \begin{pmatrix} 4.9 & 3 \\ 3.4 & s \\ 2 & 2.3 \end{pmatrix} = \begin{pmatrix} p & 2t \\ -2 & s \\ 5 & 2.3 \end{pmatrix}$$

$$\begin{pmatrix} 1.2+4.9 & t-2+3 \\ q+3.4 & s+s \\ 3+2 & r+2.3 \end{pmatrix} = \begin{pmatrix} p & 2t \\ -2 & s \\ 5 & 2.3 \end{pmatrix}$$

$$\begin{pmatrix} 6.1 & t+1 \\ q-2 & 2s \\ 8 & r+2.3 \end{pmatrix} = \begin{pmatrix} p & 2t \\ -2 & s \\ 5 & 2.3 \end{pmatrix}$$

Thus,

$$p = 1.2 + 4.9 = 6.1$$

$$q - 2 = -2 \Rightarrow q = -2 + 2 = 0$$

$$r + 2.3 = 2.3 \Rightarrow r = 2.3 - 2.3 = 0$$

$$2s = s \Rightarrow 2s - s = 0 \Rightarrow s = 0$$

$$2t = t + 1 \Rightarrow t = 1$$

Hence, $p = 6.1$, $q = 0$, $r = 0$, $s = 0$ and $t = 1$

Check

$$\text{LHS} = \begin{pmatrix} 6.1 & 1+1 \\ 0-2 & 2 \times 0 \\ 8 & 0+2.3 \end{pmatrix}$$

$$= \begin{pmatrix} 6.1 & 2 \\ -2 & 0 \\ 8 & 2.3 \end{pmatrix}$$

$$\text{RHS} = \begin{pmatrix} 6.1 & 2 \times 1 \\ -2 & 0 \\ 5 & 2.3 \end{pmatrix}$$

$$= \begin{pmatrix} 6.1 & 2 \\ -2 & 0 \\ 8 & 2.3 \end{pmatrix}$$

Hence LHS = RHS proved

Q8) Solution

$$3A = 3 \begin{pmatrix} 0 & 2 & 4 \\ 4 & -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \times 0 & 3 \times 2 & 3 \times 4 \\ 3 \times 4 & 3 \times (-1) & 3 \times 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 6 & 12 \\ 12 & -3 & 6 \end{pmatrix}$$

$$-2B = -2 \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \times 2 & -2 \times 1 & -2 \times 3 \\ -2 \times 0 & -2 \times (-2) & -2 \times 1 \end{pmatrix}$$

$$= \begin{pmatrix} -4 & -2 & -6 \\ 0 & 4 & -2 \end{pmatrix}$$

$$3A - B = 3 \begin{pmatrix} 0 & 2 & 4 \\ 4 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 6 & 12 \\ 4 & -3 & 6 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0-2 & 6-1 & 12-3 \\ 4-0 & -3-(-2) & 6-1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 5 & 9 \\ 4 & -1 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 5 & 9 \\ 12 & -1 & 5 \end{pmatrix}$$

Q9) Solution

$$\begin{aligned} \text{a) } 10P &= 10 \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \\ &= (10 \times 1 \quad 10 \times 2 \quad 10 \times 3) \\ &= (10 \quad 20 \quad 30) \end{aligned}$$

$$\begin{aligned} \text{b) } -\frac{1}{2}Q &= -\frac{1}{2} \begin{pmatrix} 2 & -4 & -2 \end{pmatrix} \\ &= (-\frac{1}{2} \times 2 \quad -\frac{1}{2} \times (-4) \quad -\frac{1}{2} \times (-2)) \\ &= (-1 \quad 2 \quad 1) \end{aligned}$$

$$\begin{aligned} \text{c) } 3P + 3Q &\text{ implies} \\ &= 3 \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} + 3 \begin{pmatrix} 2 & -4 & -2 \end{pmatrix} \\ &= (3 \quad 6 \quad 9) + (6 \quad -12 \quad -6) \\ &= (3+6 \quad 6+(-12) \quad 9+(-6)) \\ &= (9 \quad -6 \quad 3) \end{aligned}$$

Q10) Solution

$$\begin{aligned} \text{a) } A^2 &= AA \\ &= \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \times 2 + 1 \times 3 & 2 \times 1 + 1 \times 4 \\ 3 \times 2 + 4 \times 3 & 3 \times 1 + 4 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4+3 & 2+4 \\ 6+12 & 3+16 \end{pmatrix} \\ &= \begin{pmatrix} 7 & 6 \\ 18 & 19 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{b) } B^2 - A^2 &= \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} - \begin{pmatrix} 7 & 6 \\ 18 & 19 \end{pmatrix} \\ &= \begin{pmatrix} 11 & 35 \\ 14 & 46 \end{pmatrix} - \begin{pmatrix} 7 & 6 \\ 18 & 19 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 29 \\ -4 & 27 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{c) } ABC &= \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \\ &= \left[\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} \right] \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 16 \\ 11 & 39 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 36 & 20 \\ 89 & 50 \end{pmatrix} \end{aligned}$$

Alternative method

$$\begin{aligned} \text{d) } ABC &= \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \left[\begin{pmatrix} 1 & 5 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \right] \\ &= \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 11 & 6 \\ 14 & 8 \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} 36 & 20 \\ 89 & 50 \end{pmatrix}$$

Note: The order in which the matrices are grouped does not matter.

Q11) Solution

$$\begin{aligned} \text{a) } (2 \times 3 \quad p \times (-5) \quad 3 \times 7) &= -3 \\ 6 - 5p + 21 &= -3 \\ -5p + 27 &= -3 \\ p &= 6 \end{aligned}$$

$$\begin{aligned} \text{b) } (1 \times 3 \quad q \times q) &= 12 \\ 3 + q^2 &= 12 \\ q^2 &= \sqrt{9} \\ q &= \pm 3 \end{aligned}$$

Q12) Solution

$$\begin{aligned} &= \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} + \begin{pmatrix} 18 & 21 \\ 24 & 27 \end{pmatrix} \\ &= \begin{pmatrix} 2+18 & 4+21 \\ 6+24 & 8+27 \end{pmatrix} \\ &= \begin{pmatrix} 20 & 25 \\ 14 & 35 \end{pmatrix} \end{aligned}$$

Q13) Solution

$$\begin{aligned} \text{a) } AB &= \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 1 & 0 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 2 + 2 \times 1 & 1 \times 1 + 2 \times 0 & 1 \times 3 + 2 \times 4 \\ 2 \times 2 + 3 \times 1 & 2 \times 1 + 3 \times 0 & 2 \times 3 + 3 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 1 & 11 \\ 7 & 2 & 18 \end{pmatrix} \end{aligned}$$

$$\text{b) } BA = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} =$$

Impossible

Matrix B and A below cannot be multiplied together because the number of columns in B does not equal the number of rows in A. In this case, the multiplication of these two matrices is not defined.

$$\begin{aligned} \text{c) } BC &= \begin{pmatrix} 2 & 1 & 3 \\ 1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 4 \\ 3 & 1 & 2 \\ 4 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 5 & 13 \\ 17 & 2 & 8 \end{pmatrix} \end{aligned}$$

Q14) Solution

Firstly,

$$\begin{aligned}
 IA &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \\
 &= \begin{pmatrix} 1 \times a & 0 \times d & 0 \times g \\ 0 \times b & 1 \times e & 0 \times h \\ 0 \times c & 0 \times f & 1 \times i \end{pmatrix} \\
 &= \begin{pmatrix} a & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & i \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 AI &= \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} a \times 1 & b \times 0 & c \times 0 \\ d \times 0 & e \times 1 & f \times 0 \\ g \times 0 & h \times 0 & i \times 1 \end{pmatrix} \\
 &= \begin{pmatrix} a & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & i \end{pmatrix}
 \end{aligned}$$

Hence $IA = AI$ **Q15) Solution**

$$\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 6 \\ 7 \end{pmatrix} = \begin{pmatrix} 12 & 21 \\ 24 & 35 \end{pmatrix}$$

$$= \begin{pmatrix} 33 \\ 59 \end{pmatrix}$$

The image of P under transformation M is the point (33, 59).

Q16) Solution

$$\begin{pmatrix} 2 & 3 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 5 & 3 \\ 0 & -1 & 4 & 5 \end{pmatrix}$$

The image of A (0, 0) is A' (0, 0)

The image of B (1, 0) is B' (2, -1)

The image of C (1, 1) is C' (5, 4)

The image of D (0, 1) is D' (3, 5)

Q17) Solution

This is not a translation.

$$\text{Let } T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{Then, } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 2 & 2 \end{pmatrix}$$

$$2a + b = 3 \dots (i)$$

$$4a + b = 5 \dots (ii)$$

$$2c + d = 2 \dots (iii)$$

$$4c + d = 2 \dots (iv)$$

From (i) and (ii), by elimination we get

$$2a = 2$$

$$a = 1$$

From (iii) and (iv), by elimination we get

$$2c = 0$$

$$c = 0$$

Substituting in (iii) gives $d = 2$

Therefore, the required matrix is $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$

Note: It is a good idea to check your answer:

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 2 & 2 \end{pmatrix} \text{ as required.}$$

Q18) Solution

Let the point be (a, b)

$$\text{Then } \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\text{and } 2a + b = 7$$

$$3a + b = 9$$

We solve these simultaneous equations by elimination, and get $a = 2$ and $b = 3$.

Q19) Solution

$$\begin{pmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2.598 & -2 \\ 1.5 & 3.964 \end{pmatrix} \\
 = \begin{pmatrix} 0.598 \\ 4.964 \end{pmatrix}$$

The image of (3, 4) is (0.598, 4.964)

Q20) Solution

$$a) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \text{ so the image is } (3, 2).$$

$$b) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix} \text{ so the image is } (-3, -2).$$

$$c) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \text{ so the image is } (-2, 3).$$

Q21) Solution

$$A + B + C \Rightarrow$$

$$\begin{pmatrix} 2 & 4 \\ 3 & 1 \\ 4 & 5 \end{pmatrix} + \begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 5 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 2+3+1 & 4+2+3 \\ 3+1+2 & 1+1+3 \\ 4+5+4 & 5+3+6 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 9 \\ 6 & 5 \\ 13 & 14 \end{pmatrix}$$

Q22) Solution

$$\begin{aligned} \text{a) } 2 \begin{pmatrix} 4 & -6 \\ -2 & 8 \end{pmatrix} \\ = \begin{pmatrix} 2 \times 4 & 2 \times (-6) \\ 2 \times (-2) & 2 \times 8 \end{pmatrix} \\ = \begin{pmatrix} 8 & -12 \\ -4 & 16 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{b) } P^2 &= \begin{bmatrix} 4 & -6 \\ -2 & 8 \end{bmatrix} \times \begin{bmatrix} 4 & -6 \\ -2 & 8 \end{bmatrix} \\ &= \begin{bmatrix} (4 \times 4) + (-6 \times -2) & (4 \times -6) + (-6 \times 8) \\ (-2 \times 4) + (8 \times -2) & (-2 \times -6) + (8 \times 8) \end{bmatrix} \\ &= \begin{bmatrix} 28 & -72 \\ -24 & 76 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{c) Since } \begin{bmatrix} 4 & -6 \\ -2 & 8 \end{bmatrix} \times Q &= \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \text{ we have} \\ Q &= \begin{bmatrix} 4 & -6 \\ -2 & 8 \end{bmatrix}^{-1} \times \begin{bmatrix} 5 \\ 0 \end{bmatrix} \end{aligned}$$

Let's use our calculator to put P in

[A] and $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$ in [B].

$$\text{Then } [A]^{-1} \times [B] = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

Q23) Solution

$$\begin{aligned} A + 2B &= \begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix} + 2 \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 4 & 6 \\ -2 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 0+4 & 1+6 \\ 5+(-2) & 6+8 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 7 \\ 3 & 14 \end{pmatrix} \end{aligned}$$

$$\text{Hence } A + 2B = \begin{pmatrix} 4 & 7 \\ 3 & 14 \end{pmatrix}$$

Q24) Solution

$$\begin{pmatrix} 3a & 18 \\ 4 & 6 \end{pmatrix} - \begin{pmatrix} 2a & 2 \\ 2 & -6 \end{pmatrix} = \begin{pmatrix} 4 & 16 \\ 2 & b \end{pmatrix}$$

$$\begin{pmatrix} 3a-2a & 18-2 \\ 4-2 & 6-(-6) \end{pmatrix} = \begin{pmatrix} 4 & 16 \\ 2 & b \end{pmatrix}$$

$$\begin{pmatrix} a & 16 \\ 2 & 12 \end{pmatrix} = \begin{pmatrix} 4 & 16 \\ 2 & b \end{pmatrix}$$

Hence $a = 4$ and $b = 12$

Q25) Solution

$$\begin{pmatrix} 6 \times x & + & 3 \times y \\ 2 \times x & + & -3 \times y \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}$$

$$\begin{pmatrix} 6x + 3y \\ 2x - 3y \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}$$

Then

$$6x + 3y = 12 \dots \dots \dots \text{(i)}$$

$$\text{and } 2x - 3y = -4 \dots \dots \dots \text{(ii)}$$

Adding (i) to (ii) we get

$$8x = 8$$

$$x = 1$$

Substituting $x = 1$, into (i) we get

$$6(1) + 3y = 12$$

$$3y = 12 - 6$$

$$3y = 6$$

$$y = 2$$

Hence $x = 1$ and $y = 2$

Q26) Solution

$$\begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 5 & -1 \end{bmatrix} + \begin{bmatrix} 8 & -12 \\ -8 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 4 \times (-2) + 0 \times 5 & 4 \times 3 + 0 \times (-1) \\ 1 \times (-2) + 2 \times 5 & 1 \times 3 + 2 \times (-1) \end{bmatrix} + \begin{bmatrix} 8 & -12 \\ -8 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -8 & 12 \\ 8 & 1 \end{bmatrix} + \begin{bmatrix} 8 & -12 \\ -8 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -8+8 & 12-12 \\ 8-8 & 1-1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{Hence } \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 5 & -1 \end{bmatrix} + \begin{bmatrix} 8 & -12 \\ -8 & -1 \end{bmatrix}$$

gives

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ which is a zero matrix (proved)}$$

Q27) Solution

$$\begin{pmatrix} 6 & 8 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} w & -4 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 6 \times w + 8 \times (-1) & 6 \times (-4) + 8 \times 3 \\ 2 \times w + 3 \times (-1) & 2 \times (-4) + 3 \times 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 6w - 8 & 0 \\ 2w - 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore, $6w - 8 = 1 \Rightarrow 6w = 9$

$$w = \frac{9}{6} = \frac{3}{2}$$

And

$$2w - 3 = 0 \Rightarrow 2w = 3$$

$$w = \frac{3}{2}$$

$$\text{Therefore } w = \frac{3}{2}$$

Q28) Solution

$VW - V$

$$= \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \times (-1) + 5 \times 0 & 3 \times 1 + 5 \times 2 \\ -1 \times (-1) + 1 \times 0 & -1 \times 1 + 1 \times 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 13 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 - 3 & 13 - 5 \\ 1 - (-1) & 1 - 1 \end{pmatrix}$$

$$= \begin{pmatrix} -6 & 8 \\ 2 & 0 \end{pmatrix}$$

Q29) Solution

$$A^2 = A + B \Rightarrow$$

$$\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + B$$

$$\begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 1 + 1 \times 2 \\ 3 \times 1 + 2 \times 3 & 3 \times 1 + 2 \times 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + B$$

$$\begin{pmatrix} 4 & 5 \\ 9 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + B$$

$$\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + B = \begin{pmatrix} 4 & 5 \\ 9 & 7 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 & 5 \\ 9 & 7 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 - 1 & 5 - 1 \\ 9 - 3 & 7 - 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 4 \\ 6 & 5 \end{pmatrix}$$

Q30) Solution

$$2A - B^2 = 2 \begin{pmatrix} 6 & -2 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}^2$$

Thus

$$\begin{pmatrix} 6 & -2 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & -2 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 2 & -5 \\ -10 & 27 \end{pmatrix}$$

$$= \begin{pmatrix} 6 - 2 & -2 - (-5) \\ 2 - (-10) & 0 - 27 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 3 \\ 12 & -27 \end{pmatrix}$$

Q31) Solution

$$PQ = \begin{pmatrix} 7 & 5 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 3 & 10 \\ 5 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 7 \times 3 + 5 \times 5 & 7 \times 10 + 5 \times 1 \\ 2 \times 3 + 4 \times 5 & 2 \times 10 + 4 \times 1 \end{pmatrix}$$

$$= \begin{pmatrix} 31 & 75 \\ 26 & 24 \end{pmatrix}$$

Q32) Solution

$$A + \frac{1}{2}B = \begin{pmatrix} 2 & 1 \\ -3 & 4 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & -10 \\ 8 & 14 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 1 \\ -3 & 4 \end{pmatrix} + \begin{pmatrix} 0 & -5 \\ 4 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 2 + 0 & 1 + (-5) \\ -3 + 4 & 4 + 7 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -4 \\ 1 & 11 \end{pmatrix}$$

Q33) Solution

$$X^2 - 2X = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix} - 2 \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$$

$$X^2 - 2X = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} -2 & 0 \\ 4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} -2 & 0 \\ 4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 0 \\ -4 & -1 \end{pmatrix}$$

Q34) Solution

$$M^2 = kM \Rightarrow$$

$$\begin{pmatrix} 3 & 3 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 3 & 3 \\ -1 & -1 \end{pmatrix} = k \begin{pmatrix} 3 & 3 \\ -1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 6 & 6 \\ -2 & -2 \end{pmatrix} = \begin{pmatrix} k3 & k3 \\ -k1 & -k1 \end{pmatrix}$$

$$\text{Thus, } k3 = 6$$

$$k = 2$$

$$\text{Or } -k1 = -2$$

$$k = 2$$

Hence the value of k is 2

Arithmetic Progressions

In mathematics, an arithmetic progression (AP) or arithmetic sequence is a sequence of numbers such that the difference between any two successive members of the sequence is a constant. For example, the sequence 3, 5, 7, 9, 11, 13, ... is an arithmetic progression with a common difference of 2. An arithmetic progression is therefore a sequence in which each term is formed by adding a constant to the one before it.

The n^{th} term
The n^{th} term is a term at any given position n .
There is a formula for the n^{th} term of a sequence. Look at the sequence below:
1, 4, 7, 10, 13, 17, 20

1 st term	2 nd term	3 rd term	4 th term
1	4	7	10
1 + 3	1 + 3 + 3	1 + 3 + 3 + 3	1 + 3 + 3 + 3 + 3
1 + 3	1 + 2 (3)	a + 2d	a + 3d

If a is the first term and d is the common difference of a sequence,
the second term is: $a + d$
the third term is $a + 2d$
the fourth term is $a + 3d$
For the 3rd term we add $2d$. For the 4th term we add $3d$ and so on.
To get the n^{th} term we add $(n - 1)$ times d to the first term
Therefore the n^{th} term is $a_n = a + (n - 1)d$ or $a_n = a + d(n - 1)$

Geometric Progressions

An arithmetic progression is given by $a, (a + d), (a + 2d), (a + 3d), \dots$ where a = the first term, d = the common difference

We can use this formula to find any term of an arithmetic progression without having to write out all the terms.

1 st term	2 nd term	3 rd term	4 th term	5 th term
3	6	12	24	48
3	3 × 2	3 × 2 × 2	3 × 2 × 2 × 2	3 × 2 × 2 × 2 × 2
a	ar	ar ²	ar ³	ar ⁴

A sequence where each term is multiplied by a constant ratio to give the next term is called a geometrical progression (GP). The ratio is called the *common ratio* of the sequence.
If we know the first term a , and the common ratio, r , then we can find any term. The second term is ar , the third term is ar^2 , the fourth term is ar^3 , and so on.

We can generalise this as the n^{th} term of a GP written as ar^{n-1}

Note: A Geometric Progression (GP) is given by $a, ar, ar^2, ar^3, \dots$ where a = the first term, r = the common ratio

Topic 17: PROGRESSIONS AND SERIES

Q35) Solution

$$\begin{pmatrix} 3c \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 22 \\ 28 \end{pmatrix}$$

$$\begin{pmatrix} 3c \times 4 + c \times 2 \\ 28 \end{pmatrix} = \begin{pmatrix} 5 \times 4 + 1 \times 2 \\ 22 \end{pmatrix}$$

$$\begin{pmatrix} 12c + 2c \\ 28 \end{pmatrix} = \begin{pmatrix} 20 + 2 \\ 22 \end{pmatrix}$$

$$\begin{pmatrix} 14c \\ 28 \end{pmatrix} = \begin{pmatrix} 22 \\ 22 \end{pmatrix}$$

Therefore, $14c = 28$
 $c = 2$

Q36) Solution

$$MN = \begin{pmatrix} a & d \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} a \times 5 + d \times 1 & a \times 2 + d \times 3 \\ 1 \times 5 + 2 \times 1 & 1 \times 2 + 2 \times 3 \end{pmatrix} = \begin{pmatrix} 5a + d & 2a + 3d \\ 7 & 8 \end{pmatrix}$$

Introduction

Progressions are very important in our daily lives for they are used in life counting, economics, education and many others. Geometric series for example are used throughout mathematics, and they have important applications in physics, engineering, biology, economics, computer science, queuing theory, and finance.

Key words

a) Sequence

A collection of numbers arranged according to a definite rule e.g. 6, 11, 16, 21 and 18, 15, 12, 9. In the first sequence, terms are formed by adding 5 and subtracting 3 in the second sequence respectively to find the next term.

b) Series

A successive addition of terms in a sequence e.g. $6 + 11 + 16 + 21$ forms a series of numbers.

d) Common ratio

Consider the sequence 1, 3, 9, 27, 81... In this sequence there is no common difference, each term is formed from the preceding term by multiplying it with a constant 3. The constant term is called the common ratio. The common ratio is denoted r.

difference is denoted by d

This is the amount of decrease or increase between any two consecutive terms in a sequence or series. For example, in the first sequence and series above the common difference is 5. If the common difference is negative, terms in a sequence will be decreasing and if the common difference is positive, terms will be increasing in a sequence. For instance, the sequence 18, 15, 12, 9 has a negative common difference of -5 and the terms are decreasing. The common difference is denoted by d

c) Common Difference

$$S_n = \begin{cases} \frac{a(r^n - 1)}{r - 1} & (\text{if } r > 1) \\ \frac{a(1 - r^n)}{1 - r} & (\text{if } r < 1) \end{cases}$$

where a = the first term
 r = common ratio
 n = number of terms

Questions on Arithmetic Progressions (AP)

- Q1) Find the 26th term of the arithmetic progression beginning 1, 7, 13, ...
- Q2) Find the number of terms in the arithmetic progression: -8, -3, 2, ..., 47.
- Q3) The fifth term of an arithmetic progression is 26 and twelfth term is 75. Find the eight term.
- Q4) The first three terms of an arithmetic progression are p , $2p - 1$, $2p + 4$.
 (i) Find p and hence write down the values of the first three terms.
 (ii) Find the 10th term.
- Q5) Suppose we have a sequence 1, 6, 11, 16, 21, ..., 501. How many terms do we have in the sequence.
- Q6) Find the sum of the first eight terms of the arithmetic progression 3, 7, 11, ...
- Q7) Find the number of terms and the sum of the following series
 a) $2 + 5 + 8 + \dots + 299$
 b) $205 + 199 + 193 + \dots - 11$
- Q8) The sum of the first 6 terms of an arithmetic progression is 57, and the tenth term is 29. Find the first 3 terms.
- Q9) The first term of an arithmetic sequence is equal to 6 and the common difference is equal to 3. Find a formula for the n^{th} term and the value of the 50th term
- Q10) An arithmetic sequence has its 5th term equal to 22 and its 15th term equal to 62. Find its 100th term.
- Q11) Find $4 + 7 + 10 + 13 + 16 + \dots$ up to 20 terms
- Q12) Find $6 + 9 + 12 + \dots + 30$
- Q13) The sum of the fourth and twelfth term of an arithmetic progression is 20. What is the sum of the first 15 terms of the arithmetic progression?
- Q14) What is the sum of all 3 digit numbers that leave a remainder of 2 when divided by 3?
- Q15) Find the sum of the 10 terms of the arithmetic progression if $a_1 = 5$ and $d = 4$.
- Q16) Find the sum of terms in the sequence; $1 + 2 + 3 + \dots + 100$
- Q17) The n^{th} term of a sequence is given by the formula $n^2 + 5$. Find the first two terms and the thirteen terms.
- Q18) The 4th term and 10th terms of arithmetic sequences are 8 and 50, respectively. Find the 1st term and the common difference.
- Q19) In an AP, the sum of the first 20 terms is 400 and the 8th term is three times the 3rd term.
 Find:
 (a) The first term, a
 (b) The common difference, d .
- Q20) A woman had a basket of eggs for sale. She had 4 customers.
 The 1st customer bought half her eggs and half an egg
 The 2nd customer bought half of those remaining and half an egg
 The 3rd customer bought half of those remaining and half an egg

The 4th customer bought half of those remaining and half an egg
The woman was left with 8 eggs and did not break any when trading.

- How many eggs did she start with?
- How many eggs did each customer buy?

Q21) The sum of n terms of an arithmetic progression is $5n^2 - 11n$.

Calculate:

- the first term.
- the common difference

Q22) The n th term of an arithmetic progression is $(3n - 2)$. Calculate the sum of the first four terms.

Q23) The sum of n terms of an arithmetic progression is $\frac{5n^2 + 3n}{2}$. Calculate the first two terms of the arithmetic progression.

Arithmetic Progressions (Model solutions)

Q1) Solution:

In this case, $a = 1$, $d = 6$ and $n = 26$

$$n^{\text{th}} \text{ term} = a + d(n - 1)$$

$$26^{\text{th}} \text{ term} = 1 + 6(26 - 1)$$

$$= 1 + 6 \times 25$$

$$= 151$$

Q2) Solution:

In this case, $a = -8$, $d = 5$ and the

n^{th} term is 47

$$n^{\text{th}} \text{ term} = a + d(n - 1)$$

$$47 = -8 + 5(n - 1)$$

$$47 = -8 + 5n - 5$$

$$47 = -13 + 5n$$

$$5n = 60$$

$$n = 12$$

So, there are 12 terms

Q3) Solution:

$$n^{\text{th}} \text{ term} = a + d(n - 1)$$

$$5^{\text{th}} \text{ term} = a + d(5 - 1)$$

$$26 = a + 4d \quad \dots \quad (\alpha)$$

$$12^{\text{th}} \text{ term} = a + d(12 - 1)$$

$$75 = a + 11d \quad \dots \quad (\beta)$$

Solving (α) and (β) simultaneously

$$a + 11d = 75$$

$$\text{Subtract } a + 4d = 26$$

$$7d = 49$$

$$d = 7$$

Substitute $d = 7$ into the equation

$$a + 4d = 26 \text{ to get}$$

$$a + 4(7) = 26$$

$$a = 26 - 28$$

$$a = -2$$

To find the eight term:

$$n^{\text{th}} \text{ term} = a + d(n - 1)$$

$$8^{\text{th}} \text{ term} = a + d(8 - 1)$$

$$= -2 + 7 \times 7$$

$$= -2 + 49 = 47$$

The 8th term is 47

Q4) Solution:

(i) Common difference can be found by subtracting 1st term from 2nd term or 2nd term from 3rd term as shown below:

$$d = (2p - 1) - t \text{ or } (2p + 4) - (2p - 1)$$

$$= 2p - 1 - t \text{ or } 2p + 4 - 2p + 1$$

$$= p - 1 \text{ or } 5$$

Hence $d = 5$ and $p - 1 = 5$ (the common difference is constant)

Therefore, $p = 6$

Q9) Solution:

Use the value of the common difference $d = 3$ and the first term $a_1 = 6$ in the formula for the n^{th} term given above

$$a_n = a_1 + (n - 1)d$$

$$= 6 - 3(n - 1)$$

$$= 3n + 3$$

The 50th term is found by setting $n = 50$ in the above formula.

$$a_{50} = 3(50) + 3 = 153$$

Q10) Solution:

We use the n^{th} term formula for the 5th and 15th terms to write

$$a_5 = a_1 + (5 - 1)d = 22 \quad \text{... (i)}$$

$$a_{15} = a_1 + (15 - 1)d = 62 \quad \text{... (ii)}$$

We obtain a system of 2 linear equations where the unknown are a_1 and d . Subtract the right and left term of the two equations to obtain

$$62 - 22 = 14d - 4d$$

Solving for d we get $d = 4$

Now use the value of d in one of the equations to find a_1 .

$$a_1 + (5 - 1)4 = 22$$

Solving for a_1 to obtain

$$a_1 = 6$$

Now that we have calculated a_1 and d we use them in the n^{th} term formula to find the 100th term.

$$a_{100} = 6 + 4(100 - 1) = 402$$

Q11) Solution:

$$a = 4, d = 7 - 4 = 3$$

Sum of first 20 terms,

$$S_{20} = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{20}{2} [(2 \times 4) + (20 - 1)3]$$

$$= 10[8 + (19 \times 3)]$$

$$= 10[8 + 57]$$

$$= 650$$

Q12) Solution:

$$a = 6$$

$$l = 30$$

$$d = 9 - 6 = 3$$

$$n = \frac{d}{(l-a)} + 1$$

$$= \frac{3}{(30-6)} + 1$$

$$= \frac{24}{3} + 1$$

$$= 8 + 1 = 9$$

$$\text{Sum, } S = \frac{n}{2} [a+l]$$

$$= \frac{9}{2} [6+30]$$

$$= \frac{9}{2} \times 36$$

$$= 9 \times 18$$

$$= 162$$

Q13) Solution:

The sum of the 4th and 12th term = 20.
Let a_1 be the first term, a_4 be the 4th term, and a_{12} be the 12th term

$$\text{Then } a_4 + a_{12} = 20$$

a_4 can be expressed as $a_1 + 3d$

Similarly, a_{12} can be expressed as $a_1 + 11d$

Then $a_4 + a_{12} = 20$ can be expressed as

$$a_1 + 3d + a_1 + 11d = 20$$

$$\Rightarrow 2a_1 + 14d = 20$$

$$\Rightarrow a_1 + 7d = 10$$

$$\Rightarrow a_8 = 10$$

The sum of the first 15 terms

$$= \frac{n}{2} (a + l) = \frac{15}{2} (a_1 + a_{15})$$

In an arithmetic progression

$$a_1 + a_{15} = a_1 + a_1 + 14d$$

$$= 2a_1 + 14d$$

$$\begin{aligned} 1^{\text{st}} \text{ term} &= 6 \\ 2^{\text{nd}} \text{ term} &= 2p - 1 = 12 - 1 = 11 \\ 3^{\text{rd}} \text{ term} &= 2p + 4 = 12 + 4 = 16 \end{aligned}$$

Q5) Solution:

$$a = 1, d = 5 \text{ nth term} = 501$$

$$n^{\text{th}} \text{ term} = a + d(n-1)$$

$$501 = 1 + 5(n-1)$$

$$501 = 1 + 5n - 5$$

$$501 = -4 + 5n$$

$$5n = 501 + 4$$

$$5n = 505$$

$$n = 101$$

Hence, there are 101 terms

$$\text{Or Use this formula: } n = \frac{a_n - a}{d} + 1$$

Thus,

$$n = \frac{501 - 1}{5} + 1$$

$$= 101 \text{ terms}$$

Q6) Solution:

First term, $a = 3$, common difference, $d = 4$

$$S_n = \frac{1}{2} n [2a + (n-1)d]$$

$$S_8 = \frac{1}{2} \times 8 [6 + (7 \times 4)]$$

$$= \frac{1}{2} \times 8 \times 34$$

$$= 136$$

Q7) Solution:

(a) $a = 2$, $d = 3$, $n^{\text{th}} \text{ term} = 299$

$$n^{\text{th}} \text{ term} = a + d(n-1)$$

$$299 = 2 + 3(n-1)$$

$$299 = 2 + 3n - 3$$

$$\text{So, } 3n = 300$$

$$n = 100$$

We know the first term and the last term.

$$S_n = \frac{1}{2} n (a + l)$$

$$S_{100} = \frac{1}{2} \times 100 (2 + 299)$$

$$= 15050$$

b) $a = 205$, $d = -6$, $n^{\text{th}} \text{ term} = -11$

$$n^{\text{th}} \text{ term} = a + d(n-1)$$

$$-11 = 205 + (-6)(n-1)$$

$$-11 = 205 - 6n + 6$$

$$6n = 222$$

$$n = 37$$

$$S_n = \frac{1}{2} n [2a + (n-1)d]$$

$$S_{37} = \frac{1}{2} \times 37 (205 - 11)$$

$$= \frac{1}{2} \times 37 (194)$$

$$= 3589$$

Q8) Solution

$$S_n = \frac{1}{2} n [2a + (n-1)d]$$

The sum of the first 6 terms is 57, so

$$S_6 = 57 = \frac{1}{2} \times 6 [2a + (6-1)d]$$

$$57 = 3(2a + 5d)$$

$$57 = 6a + 15d \dots (i)$$

(i)

$$\text{The tenth term} = 29$$

$$n^{\text{th}} \text{ term} = a + d(n-1)$$

$$29 = a + d(10-1)$$

$$29 = a + 9d$$

(ii)

Solving equations (i) and (ii) simultaneously we

get

$$6a + 15d = 57 \dots (i)$$

$$a + 9d = 29 \dots (ii)$$

Multiplying equation (ii) by 6 throughout we get

$$6(a + 9d = 29) \Rightarrow 6a + 54d = 174$$

(ii)

Subtracting (ii) from (i) we get

$$6a + 54d = 174$$

$$(6a + 15d = 57)$$

$$39d = 117$$

$$d = 3$$

Substituting $d = 3$ into (ii) we get,

$$a + 9d = 29$$

$$\Rightarrow a + 9(3) = 29$$

$$a = 29 - 27 = 2$$

$$\text{Hence, } a = 2, d = 3$$

$$1^{\text{st}} \text{ term} = 2$$

$$2^{\text{nd}} \text{ term} = 2 + 3 = 5$$

$$3^{\text{rd}} \text{ term} = 2 + 3 + 3 = 8$$

$$= 2(a_1 + 7d)$$

$$= 2(a_8)$$

Therefore, the sum of the first 15 terms

$$= \frac{15}{2} (2 \times a_8)$$

$$= \frac{15}{2} (2 \times 10)$$

$$= 150$$

Q14) Solution:

The smallest 3 digit number that will leave a remainder of 2 when divided by 3 is 101. The next number that will leave a remainder of 2 when divided by 3 is 104, 107,.....

The largest 3 digit number that will leave a remainder of 2 when divided by 3 is 998.

So, the given series is an AP with the first term being 101 and the last term being 998 and the common difference being 3.

Sum of an AP

$$= \left(\frac{\text{First term} + \text{last Term}}{2} \right) \times \text{Number of terms}$$

We know that in an AP, the nth term

$$a_n = a_1 + (n - 1) \times d$$

In this case, therefore,

$$998 = 101 + (n - 1) \times 3$$

$$\text{i.e., } 897 = (n - 1) \times 3$$

$$\text{Therefore, } n - 1 = 299$$

$$\text{or } n = 300$$

Sum of the AP will therefore, be

$$\frac{101 + 998}{2} \times 300 = 164,850$$

Q15) Solution:

$$S_n = \frac{n}{2} [2a_1 + (n - 1)d]$$

$$= \frac{10}{2} [2 \times 5 + (10 - 1)4]$$

$$= 5 \times 46$$

$$= 230$$

Q16) Solution:

Since, $a_1 = 1$, $d = 1$, $n = 100$,

$a_{100} = 100$. The sum is:

$$S_n = \frac{n}{2} (a + l)$$

$$= \frac{100}{2} (1 + 100)$$

$$= 5050$$

Q17) The n^{th} term is $n^2 + 5$

When $n = 1$, the term is $1^2 + 5 = 6$

When $n = 2$, the term is $2^2 + 5 = 9$

When $n = 13$, the term is $13^2 + 5 = 174$

Q18) Solution:

The 4th term = $a + 3d = 8$... (i)

The 10th term = $a + 9d = 50$.. (ii)

Subtracting (i) from (ii) gives

$$6d = 42$$

Therefore, $d = 7$

But, $a + 3d = 8$

Thus, $a + 3 \times 7 = 8$

$$a + 21 = 8$$

$$a = -13$$

The 1st term is -13 and the common difference is 7.

Q19) Solution:

(a) Using $S_n = \frac{1}{2} n \{2a + (n - 1)d\}$,

$$S_{20} = \frac{1}{2} \times 20 \{2a + 19d\}$$

$$= 10(2a + 19d)$$

The sum of the first 20 terms is 400.

$$\Rightarrow 10(2a + 19d) = 400$$

$$\therefore 2a + 19d = 40 \dots (i)$$

the 8th term = $a + 7d$

the 3rd term = $a + 2d$

The 8th term is 3 times the 3rd term

$$\Rightarrow a + 7d = 3(a + 2d)$$

$$a + 7d = 3a + 6d$$

$$\therefore d = 2a \dots (ii)$$

Substitute $2a$ for d in (i) we get

$$2a + 19(2a) = 40$$

$$2a + 38a = 40$$

$$40a = 40$$

$$a = 1$$

- (b) Substitute 1 for a in (ii) to get $d = 2$. The first term of the AP is 1 and the common difference is 2.

Q20) (a) 143

- (b) 1st customer: 72 ($71\frac{1}{2} + \frac{1}{2}$) leaving 71
 2nd customer: 36 ($35\frac{1}{2} + \frac{1}{2}$) leaving 35
 3rd customer: 18 ($17\frac{1}{2} + \frac{1}{2}$) leaving 17
 4th customer: 9 ($8\frac{1}{2} + \frac{1}{2}$) leaving 8

Q21) Solution:

- (a) S_n for an AP is $S_n = 5n^2 - 11n$

Since we need to find first term; a_1

For a_1 ; $n = 1$

$$\begin{aligned}\therefore a_1 &= 5(1)^2 - 11(1) \\ &= 5 - 11 \\ &= -6\end{aligned}$$

$$\begin{aligned}(b) S_2 &= 5(2)^2 - 11(2) \\ &= 20 - 22 \\ &= -2\end{aligned}$$

$$\begin{aligned}2^{\text{nd}} \text{ term} &= -2 - (-6) \\ &= 4\end{aligned}$$

$$\therefore d = 4 - (-6) = 10$$

Q22) Solution:

In this case, n^{th} term $= (3n - 2)$

$$1^{\text{st}} \text{ term} = 3 \times 1 - 2 = 3 - 2 = 1$$

$$4^{\text{th}} \text{ term} = 3 \times 4 - 2 = 12 - 2 = 10$$

$$S_n = \frac{n}{2}(a + l)$$

$$S_n = \frac{4}{2}(1 + 10)$$

$$\begin{aligned}S_n &= 2 \times 11 \\ &= 22\end{aligned}$$

$\therefore$ The sum of the first 4 terms is 22

The other alternative method would be

$$n^{\text{th}} \text{ term} = (3n - 2)$$

$$1^{\text{st}} \text{ term} = 3 \times 1 - 2 = 3 - 2 = 1$$

$$2^{\text{nd}} \text{ term} = 3 \times 2 - 2 = 6 - 2 = 4$$

$$3^{\text{rd}} \text{ term} = 3 \times 3 - 2 = 9 - 2 = 7$$

$$4^{\text{th}} \text{ term} = 3 \times 4 - 2 = 12 - 2 = 10$$

$$\text{Sum of first 4 terms} = 1 + 4 + 7 + 10 = 22$$

$\therefore$ The sum of the first 4 terms is 22

Q23) Solution

In this case, we denote the sum of an arithmetic progression as S_n

When $n = 1$, S_1 = the first term

Then, it follows:

$$S_n = \frac{5n^2 + 3n}{2}$$

a_2 = the second term

$$\begin{aligned}a_1 &= S_1 = \frac{5(1)^2 + 3(1)}{2} \\ &= \frac{5 \times 1 + 3 \times 1}{2} \\ &= \frac{8}{2} = 4\end{aligned}$$

$$\therefore S_n = 4$$

$$\begin{aligned}S_2 &= \frac{5(2)^2 + 3(2)}{2} \\ &= \frac{5 \times 4 + 3 \times 2}{2} \\ &= \frac{20 + 6}{2} = \frac{26}{2} = 13\end{aligned}$$

From the previous steps

Since $a_1 = S_1$

Then it $\Rightarrow S_1 + a_2 = 13$

$$4 + a_2 = 13$$

$$a_2 = 13 - 4 = 9$$

$$\therefore a_2 = 9$$

Hence the first two terms of an arithmetic progression are 4 and 9

Questions on Geometric Progressions

Q1) The sum of the first n terms of a geometric progression, GP is $2^{n+1} - 1$.

Calculate:

- i. the first term of the GP
- ii. the common ratio of the GP.

Q2) The sum of the first two terms of a geometric progression is 4. If the first term is 3, find the common ratio.

Q3) The first three terms of a GP are

$$x + 1, x^2 - 1 \text{ and } (x^2 - 1)(2x - 4)$$

Calculate the value of x .

Q4) Find the sum of the first 12 terms of the following GP.

$$\frac{1}{2187}, \frac{1}{729}, \frac{1}{243}, \dots$$

Give your answer correct to decimal 2 places.

Q5) In a geometric progression, the sum of the second and third terms is 9, If the seventh term is eight times the fourth, find:

- the first term
- the common ratio

Q6) A geometric progression has 6 terms. If its first term is 3 and the last term is 96, calculate the common ratio.

Q7) (a) Find the common ratio for the GP

$$2, 8, 32, \dots$$

(b) Find the 8th term of the GP

$$5, 10, 20, 40, \dots$$

Q8) Find the common ratio and the tenth term of the geometric series $2 + 3 + \dots + \dots$

Q9) The second term of a geometric series is 8 and the fifth term is 64. Find the first term and the common ratio.

Q10) Suppose that after finishing secondary school (Form 4) Likuni Girls Secondary School, Omega who just turned 17, deposits K1000.00 in the bank and leaves it untouchable until at the age of 60. Assume that the bank pay 5% interest annually. If no other withdrawals or deposits were made.

- How much would be in her bank account when she is
 - 23 years old
 - 25 years old
- How much money would be in the bank account at the time she retires?

Q11) Pemphero is a form two student at a boarding school and at their school they write weekly Mathematics test. Term one had 11 weeks. She got a score in the first test. Her score increased by 4% every week test that she

wrote in the term. Assume that she wrote all the tests in this term.

Calculate:

- Pemphero's score in the second and third term.
- Pemphero's score in the 10th test.
- If grade were given to each student using their last week's score in the term as the final grade for the term. The grades were awarded using the table below

Score	Grade
40 - 50	D
51 - 60	C
61 - 75	B
75 - 100	A

- What grade did Pemphero get at the end of the term?

Q12) Find the 10th term in the series
2, 4, 8, 16, ...

Q13) Find 5th term in the series 5, 15, 45, ...

Q14) If the ratio of the sum of the first 6 terms of a GP to the sum of the first 3 terms of the GP is 9, what is the common ratio of the GP?

Q15) Find $4 + 12 + 36 + \dots$ up to 6 terms

Q16) Find $1 + \frac{1}{2} + \frac{1}{4} + \dots$ up to 5 terms

Q17) A G.P has first term 3, and the common ratio 2. Find the sum of the first ten terms.

Q18) A GP has first term 8, and the common ratio 0.5. Find the sum of the first six terms.

Q19) What is the largest number of terms of the geometrical progression $8 + 24 + 72 + \dots$ that will give a total less than 3 000 000?

Q20) A business lady saves from her profits as follows.

She saves K3 the first week, K6 the second week, K12 the third week, K24 the fourth week, and so on. If she manages to keep on saving under this system.

- How much would she save in the tenth week?

- ii. What would be the total amount of her savings in the first ten weeks?

Q21) A geometric progression has positive terms. The sum of the first six terms is nine times the sum of the first three terms. The seventh term is 320.

Find:

- the common ratio.
- the first term.
- the sum of the first five terms.

Q22) A geometric progression has a positive common ratio, and the sixth term is $7\frac{9}{32}$. The sum of the first two terms and the sum of the third and fourth terms are in the ratio 4:9. Find the sum of the first six terms.

Q23) How many terms are there in the geometric progression 2, 4, 8, ..., 128?

Q24) How many terms in the geometric progression, 1.1, 1.21, 1.331, ... will be needed so that the sum of the first n terms is greater than 20?

Q25) Given the first two terms in a geometric progression as 5 and $\frac{1}{2}$, what is the 7th term?

Q26) Given the first two terms of a geometric progression as 5 and $\frac{1}{2}$, what is the sum of the first 7 terms?

Q27) The 5th and 8th terms of a geometric sequence are 6 and 48 respectively. Find the first term and the common ratio of this sequence.

Geometric Progressions (GP) (Model Solutions)

Q1) Solution

Given $S_n = 2^{n+1} - 1$ of a GP

(i) To find the first term of the GP

Let the values of n be 1, 2, 3 and 4

When;

$$n = 1, \text{ then } S_1 = 2^{1+1} - 1 = 4 - 1 = 3$$

$$n = 2, \text{ then } S_2 = 2^{2+1} - 1 = 8 - 1 = 7, \text{ } n^{\text{th}} \text{ term } (7 - 3) = 4$$

$$n = 3, \text{ then } S_3 = 2^{3+1} - 1 = 16 - 1 = 15, \text{ } n^{\text{th}} \text{ term } (15 - 7) = 8$$

$$n = 4, \text{ then } S_4 = 2^{4+1} - 1 = 32 - 1 = 31, \text{ } n^{\text{th}} \text{ term } (31 - 15) = 16$$

Therefore 1st term is 3

Alternative method

No. of terms (n)	Sum of terms $(2^{n+1}) - 1$	n^{th} term
1	$2^{1+1} - 1 = 4 - 1 = 3$	3
2	$2^{2+1} - 1 = 8 - 1 = 7$	$(7 - 3) = 4$
3	$2^{3+1} - 1 = 16 - 1 = 15$	$(15 - 7) = 8$
4	$2^{4+1} - 1 = 32 - 1 = 31$	$(31 - 15) = 16$

$\therefore$ The first term is 3

(ii) Common ratio $= \frac{8}{4} = \frac{16}{8} = 2$

Q2) Solution

In this case, $a + ar = 4$, where $a =$ first term and $ar =$ second term.

Now, since $a = 3$

$$a + ar = 4$$

$$\Rightarrow 3 + 3r = 4$$

$$3r = 4 - 3$$

$$3r = 1$$

$$r = \frac{1}{3}$$

Hence, the common ratio is $\frac{1}{3}$

Q3) Solution

In this case, common ratio

$$= x^2 - 1/x + 1$$

$$= (x-1)(x+1)/x+1$$

$$= (x-1)$$

$$\text{Also, } \frac{(x^2-1)(2x-4)}{x^2-1}$$

$$\frac{(x-1)(x+1)(2x-4)}{(x-1)(x+1)}$$

$$= (2x-4)$$

Therefore, equating the common ratios

$$(x-1) = (2x-4)$$

$$x-2x = -4+1$$

$$-x = -3$$

$$x = 3$$

Hence the value of $x = 3$ **Q4) Solution**

$$a = 1/2187, r = 1/729 \div 1/2187$$

$$= 1/729 \times 2187/1 = 3$$

$$\text{and } n = 12$$

$$\text{Since, } r > 3, \text{ we use } S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{12} = \frac{1/2187(3^{12} - 1)}{3 - 1}$$

$$S_{12} = \frac{1/2187(531441 - 1)}{2}$$

$$S_{12} = \frac{1/2187(531440)}{2}$$

$$S_{12} = 243/2 \text{ (to 2 d.p.)}$$

$$S_{12} = 121.50$$

Q5) Solution

$$T_2 = ar$$

$$T_3 = ar^2$$

$$T_7 = ar^6$$

$$T_4 = ar^3$$

$$\Rightarrow ar + ar^2 = 9 \dots\dots\dots (i)$$

$$\text{and } ar^6 = 8ar^3 \dots\dots\dots (ii)$$

From (ii)

$$\frac{ar^6}{ar^3} = \frac{8ar^3}{ar^3}$$

$$r^3 = 8$$

$$r^3 = 2^3$$

 $\therefore$ common ratio is 2Substituting $r = 2$ in (i)

$$a(2) + a(2)^2 = 9$$

$$2a + 4a = 9$$

$$\frac{6a}{6} = \frac{9}{6}$$

$$a = 1\frac{1}{2}$$

Q6) Solution

$$\text{Here } a = 3 \dots\dots\dots (i)$$

$$\text{and } ar^5 = 96 \dots\dots\dots (ii)$$

$$\text{Common ratio } ar^5/a = 96/3$$

$$r^5 = 32 \text{ (But } 32 = 2^5)$$

$$r^5 = 2^5$$

$$\text{Hence } r = 2$$

Alternative methodIn a GP, $T_n = ar^{n-1}$, a = first term r = common ratio n = number of terms

$$\text{Hence, } T_{12} = 96, a = 3$$

$$r = ?, n = 6$$

$$T_n = 3r^{6-1} = 96$$

$$3r^5 = 96$$

$$r^5 = \frac{96}{3}$$

$$r^5 = 32 \Rightarrow \sqrt[5]{r^5} = \sqrt[5]{32} \Rightarrow r = 2$$

Q7) (a) The ratio between any term and the term before it is

$$\frac{8}{2} = \frac{32}{8} = 4$$

Common ratio is 4

$$(b) \text{ Common ratio } = \frac{10}{5} = 2, a = 5$$

$$n^{\text{th}} \text{ term} = ar^{n-1}$$

$$8^{\text{th}} \text{ term} = 5 \times 2^{8-1}$$

$$= 5 \times 128$$

$$= 640$$

Q8) Here the first term is $a = 2$, and the second term $ar = 3$

The common ratio, $r = \frac{ar}{a} = \frac{3}{2} = 1.5$

The n^{th} term $= ar^{n-1}$

The 10^{th} term $= 2 \times 1.5^{10-1}$
 $= 2 \times 1.5^9$
 $= 76.89$ (to 4 s.f.)

Q9) Here $ar = 8$ (i)

and $ar^4 = 64$ (ii)

Common ratio $\frac{ar^4}{ar} = \frac{64}{8}$

$$r^3 = 8 \Rightarrow r^3 = 2^3$$

$$r = 2$$

Substituting $r = 2$ into (i) we get

$$2a = 8$$

and so $a = 4$

The first term is 4 and the common ratio is 2.

Q10) (a) (i) $1000 \times 1.05^3 = \text{K}1157.62$

(ii) $1000 \times 1.05^5 = \text{K}1276.28$

(b) $1000 \times 1.05^{40} = \text{K}7039.98$

Q11) (i) 1^{st} score $= 40$,

2^{nd} score $= 40 \times 1.04 = 41.6$

3^{rd} score $= 40 \times 1.04^2 = 43.264$

(ii) 10^{th} score $= 40 \times 1.04^9 = 56.93$

(iii) The term had 11 weeks so her final score was the 11^{th} score

$$= 40 \times 1.04^{10} = 59.20$$

(iv) She got grade C

Q12) $a = 2$, $r = \frac{4}{2} = 2$, $n = 10$

10^{th} term, $a_{10} = ar^{n-1}$

$$= 2 \times 2^{10-1}$$

$$= 2 \times 2^9$$

$$= 2 \times 512 = 1024$$

Q13) $a = 5$, $r = \frac{15}{5} = 3$, $n = 5$

5^{th} term, $a_5 = ar^{n-1}$

$$= 5 \times 3^{5-1}$$

$$= 5 \times 3^4$$

$$= 5 \times 81$$

$$= 405$$

Q14) The sum of the first n terms of a GP is given

by $\frac{a(r^n - 1)}{r - 1}$, where ' a ' is the first term of the GP,

' r ' is the common ratio and ' n ' is the number of terms in the GP.

Therefore, the sum of the first 6 terms of the

GP will be equal to $\frac{ar^6 - 1}{r - 1}$

and sum of the first 3 terms of the GP will

be equal to $\frac{ar^3 - 1}{r - 1}$

The ratio of the sum of the first 6 terms: sum of first 3 terms $= 9:1$, i.e.

$$\frac{ar^6 - 1}{r - 1}$$

$$= 9$$

$$\frac{ar^3 - 1}{r - 1}$$

$$\Rightarrow \frac{r^6 - 1}{r^3 - 1} = \frac{(r^3 + 1)(r^3 - 1)}{r^3 - 1} = 9$$

$$\Rightarrow (r^3 + 1) = 9$$

$$\Rightarrow r^3 = 8$$

$$\Rightarrow r = 2$$

Q15) Finding $4 + 12 + 36 + \dots$ up to 6 terms

$$a = 4, r = \frac{12}{4} = 3, n = 6$$

Here $r > 1$. Hence,

$$S_6 = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{4(3^6 - 1)}{3 - 1}$$

$$= \frac{4(729 - 1)}{2}$$

$$= \frac{4 \times 728}{2}$$

$$= 2 \times 728 = 1456$$

Q16) Finding $1 + \frac{1}{2} + \frac{1}{4} + \dots$ up to 5 terms

$$a = 1, \quad r = \left(\frac{1}{2}\right) = \frac{1}{2}, \quad n = 5$$

Here $r < 1$.

$$\begin{aligned} \text{Hence, } S_5 &= \frac{a(1-r^n)}{1-r} \\ &= \frac{1[1-(\frac{1}{2})^5]}{(1-\frac{1}{2})} \\ &= \frac{1[1-(\frac{1}{32})]}{(\frac{1}{2})} \end{aligned}$$

$$= \frac{(\frac{31}{32})}{(\frac{1}{2})} = \frac{31}{16} = 1\frac{15}{16}$$

Q17) $a = 3, r = 2$ and $n = 10$

Since $r > 1$, we use

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ S_{10} &= \frac{3(2^{10} - 1)}{2 - 1} \\ &= \frac{3(1024 - 1)}{1} \\ &= 3 \times 1023 \\ &= 3069 \end{aligned}$$

Q18) $a = 8, r = 0.5$ and $n = 6$

Since $r < 1$, we use

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} \\ &= \frac{8(1-(\frac{1}{2})^6)}{1-\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} S_6 &= 8 \times 2 \times \frac{63}{64} \\ &= \frac{63}{4} \\ &= 15\frac{3}{4} \end{aligned}$$

Q19) $a = 8, r = 3$ and $S_n = 3\,000\,000$. What is n ?

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{8(3^n - 1)}{3 - 1} \end{aligned}$$

Hence $4(3^n - 1) < 3\,000\,000$ (divide by 4 each side)

$$3^n - 1 < 750\,000$$

$$3^n < 750\,001$$

Taking logarithms (base 10) of both sides

$$\log 3^n < \log 750\,001$$

$$n \log 3 < \log 750\,001$$

$$n < \frac{\log 750\,001}{\log 3}$$

$$n < \frac{5.8751}{0.4771}$$

$$n < 12.31 \quad (\text{to 4 s.f.})$$

Therefore, the number of terms required is 12.

Note: This solution shows how quickly the terms of a GP grow when the common ratio is more than 1. For instance, twelve terms have a sum of almost three million, yet the first term is 8 and the ratio is only 3.

Q20) (i) The series is

$$3 + 6 + 12 + 24 + \dots$$

In this case, the first term, a , is 3, common ratio, r , is 2

Using the formula

$$n^{\text{th}} \text{ term} = ar^{n-1} \text{ we have;}$$

$$\begin{aligned} 10^{\text{th}} \text{ term} &= 3 \times 2^9 \\ &= 3 \times 512 \\ &= 1536 \end{aligned}$$

In the tenth week she would save K1536

(ii) Using the formula

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ S_{10} &= \frac{3(2^{10} - 1)}{2 - 1} \\ &= \frac{3(1024 - 1)}{1} \\ &= 3 \times 1023 \\ &= 3069 \end{aligned}$$

The total amount of the savings would be K3069

Q21) (i) Sum of the first six terms = 9 (sum of first 3 terms)

$$\begin{aligned} \frac{a(r^6 - 1)}{r - 1} &= \frac{9a(r^3 - 1)}{r - 1} \\ (r^6 - 1) &= 9(r^3 - 1) \end{aligned}$$

$$r^6 - 9r^3 + 8 = 0$$

$$(r^3 - 8)(r^2 - 1) = 0$$

$$\text{Either } r^3 = 8 \text{ or } r^2 = 1$$

$$\text{Hence } r = 2 \text{ and } r = 1$$

$r = 2$ and $r = 1$ is rejected as the GP is increasing

$$\begin{aligned} \text{(ii) Seventh term} &= ar^{n-1} = 5 \times 2^{7-1} = 320 \\ a2^6 &= 5 \times 2^6 \\ a &= 5 \end{aligned}$$

So the first term is 5

(iii) Sum of the first five terms

$$\begin{aligned} &= \frac{a(1-r^n)}{1-r} \\ &= \frac{5(1-2^5)}{1-2} \\ &= \frac{5(1-32)}{-1} \\ &= 155 \end{aligned}$$

Q22) $\frac{(\text{First term} + \text{second term})}{(\text{third term} + \text{fourth term})} = \frac{4}{9}$

$$\begin{aligned} \frac{a+ar}{ar^2+ar^3} &= \frac{4}{9} \\ \frac{a(1+r)}{ar^2(1+r)} &= \frac{4}{9} \\ \frac{1}{r^2} &= \frac{4}{9} \\ r^2 &= \frac{9}{4} \\ r &= \frac{3}{2} \end{aligned}$$

(Since $a \neq 0$, $r \neq 1$ but greater than 1)

Then,

$$\begin{aligned} \text{Sixth term} &= 1 \times \frac{3^5}{2} \\ a &= 1 \end{aligned}$$

Therefore, sum of the first six terms

$$= \frac{1\left(\left(\frac{3}{2}\right)^6 - 1\right)}{\left(\frac{3}{2} - 1\right)} = 13.1875$$

Q23) Solution

In this sequence $a = 2$ and $r = 2$. We also know that the n^{th} term is 128. But the formula for the n^{th} term is ar^{n-1} .

$$\text{So } 128 = 2 \times 2^{n-1}$$

$$64 = 2^{n-1}$$

$$2^6 = 2^{n-1}$$

$$6 = n-1$$

$$n = 7.$$

So there are 7 terms in this geometric progression.

Q24) Solution

The sequence is a geometric progression with $a = 1$ and $r = 1.1$.

We want to find the smallest value of n such that $S_n > 20$.

Now,

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

So

$$\begin{aligned} \frac{1 \times (1-1.1^n)}{1-1.1} &> 20 \\ \frac{1-1.1^n}{-0.1} &> 20 \\ 1.1^n - 1 &> 2 \\ 1.1^n &> 3 \end{aligned}$$

If we now take logarithms of both sides, we get

$$\begin{aligned} \log 1.1^n &> \log 3 \\ n \log 1.1 &> \log 3 \end{aligned}$$

And as $\log 1.1 > 0$ we obtain

$$n > \log 3 / \log 1.1 = 11.5267...$$

And therefore the smallest whole number value of n is 12.

Q25) $a = 5$, $r = \frac{1}{10}$

Then

$$\begin{aligned} a^7 &= 5 \times \left(\frac{1}{10}\right)^{7-1} \\ &= \frac{5}{1000000} \\ &= 0.000005 \end{aligned}$$

Q26) We know that $a = 5$ and $r = \frac{1}{10}$. Then

$$\begin{aligned} S_7 &= \frac{a(1-r^n)}{(1-r)} \\ &= \frac{5(1-\frac{1}{10}^7)}{(1-\frac{1}{10})} \\ &= \frac{5(1-\frac{1}{10^7})}{(\frac{9}{10})} \\ &= 5.55555 \end{aligned}$$

Q27) The 5th term is $ar^4 = 6$... (i)

The 8th term is $ar^7 = 48$... (ii)

Dividing (ii) by (i) gives $\frac{ar^7}{ar^4} = \frac{48}{6}$

$$\begin{aligned} \text{Therefore, } r^3 &= 8 \\ r &= 2 \end{aligned}$$

$$r^3 = 8 \rightarrow r^3 = 2^3$$

$$\text{Therefore, } r = 2$$

$$\text{But, } ar^4 = 6$$

$$a \times 2^4 = 6$$

$$\therefore a = \frac{6}{16}$$

Topic 18

Travel Graphs

Speed-time graphs

In speed-time graphs, speed is represented on the vertical axis and time on the horizontal axis. If the motion is in a specific direction then velocity should be represented on the vertical axis.

Definitions of terms and formulae associated with travel graphs

- ✓ **Velocity:** Displacement per unit time or *Speed* in particular direction
- ✓ **Acceleration:** Increase in velocity per unit time, SI unit is m/s^2 or ms^{-2}

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{change in time}} = \frac{v - u}{t}$$

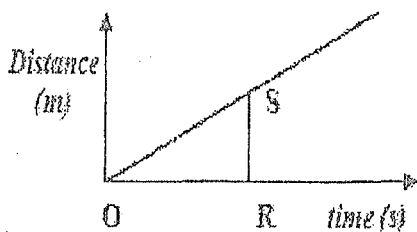
where u = initial velocity,

v = Final velocity and t = time taken

Note: *acceleration* is also obtained by finding the gradient of the graph

- ✓ **Deceleration:** Decrease of velocity per unit time. Acceleration in the opposite direction SI unit is m/s^2 or ms^{-2}
- ✓ **Speed** = $\frac{\text{distance}}{\text{time}}$
- ✓ **Average speed** = $\frac{\text{total distance}}{\text{total time}}$

The gradient (or slope) of the distance-time graph



$$\text{Gradient} = \frac{\text{vertical side}}{\text{horizontal side}} = \frac{RS}{RO}$$

RS represents change in the distance travelled and RO represents the time taken.

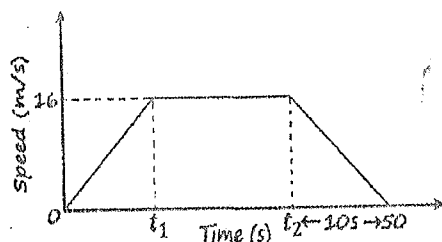
Questions on Travel Graphs

- Q1) The velocity of car changes from 2m/s to 14m/s in 4 seconds. Find the acceleration of car.
- Q2) What is the speed of a racing car in metres per second if the car covers 360 km in 2 hours?
- Q3) A car moving along a straight road ABC as shown in figure below maintains an average speed of 90km/h between points A and B and 36km/h between points B and C.



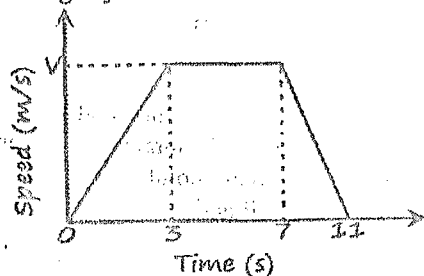
Calculate the:

- a) Total time taken in seconds by the car between points A and C.
 - b) Average speed in metres per second of the car between points A and B
- Q4) At the top of a hill, a cyclist's velocity was 5 m/s . At the bottom of the hill, the cyclist's velocity was 10 m/s . If it took the cyclist 2 seconds to reach the bottom of the hill, calculate the average acceleration of the cyclist.
- Q5) Suppose a cyclist was climbing a hill and his velocity at the bottom of the hill was 24 m/s . After 4 seconds he had reached at the top and his velocity was 10 m/s , calculate his average acceleration.
- Q6) A train accelerates uniformly from rest and reaches a speed of 20 m/s in 5 seconds. It then continues moving at the same speed for 3 seconds after which it accelerates constantly for 4 seconds and reaches a speed of 40 m/s . Using a scale of 2 cm to represent 10 m/s on the vertical axis and 2 cm to represent 2 seconds on the horizontal axis, draw a speed-time graph to represent the motion of the train.
- Q7) Figure below is the speed time graph of a car. The car starts from rest and accelerates at 2 m/s^2 for t_1 seconds its speed reaching 16 m/s . It then maintains this speed for some time after which it decelerates uniformly for 10 seconds and stops.



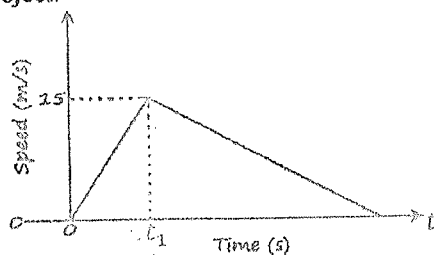
Calculate the distance that the car travelled from time t_1 seconds to the time t_2 when it started decelerating.

- Q8) Figure below shows the speed-time graph of a moving object.



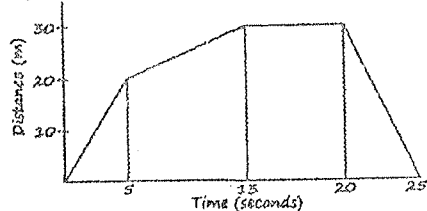
If the total distance travelled by the object was 90 m, calculate the speed V of the object.

- Q9) Figure below is a speed-time of a moving object.



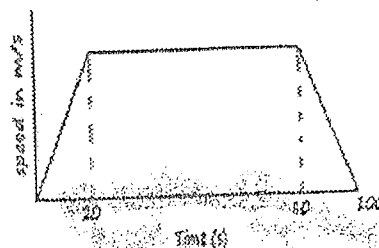
Given that acceleration for the first t_1 seconds is 0.3 ms^{-1} , calculate the value of t_1 .

- Q10) Figure below shows a distance/time graph of an object.



Calculate the speed of the object between $t = 5$ and $t = 13$.

- Q11) Figure below shows a speed time graph of a car.



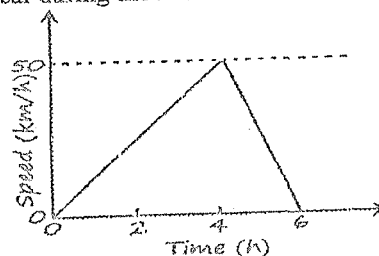
Given that the total distance travelled is 3200 m, calculate the deceleration during the last 20 seconds.

- Q12) Table below shows the speed of a train recorded every 10 seconds.

Table				
SPEED (m/s)	200	400	600	800
TIME (s)	10	20	30	40

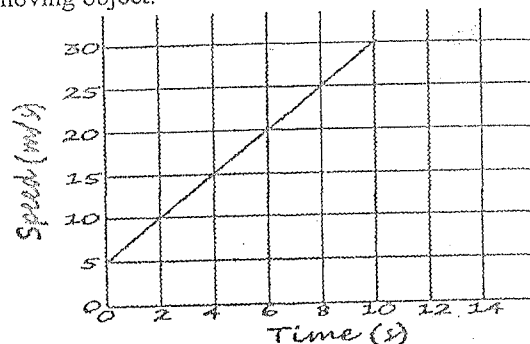
Using a scale of 2 cm to represent 100 m/s on the vertical axis and 2 cm to represent 10 seconds on the horizontal axis, draw a speed-time graph and use it to calculate the acceleration of the train.

- Q13) Figure below shows a speed-time graph for a car during the first 6 hours.



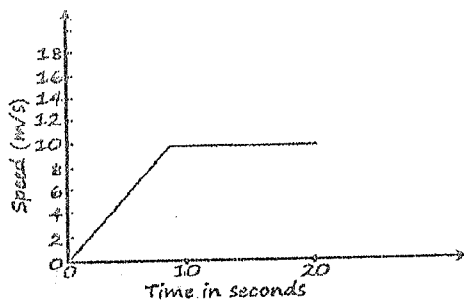
Calculate the total distance travelled during the 6 hours.

- Q14) Figure below shows the speed time graph of a moving object.

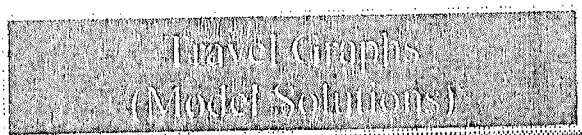


Use the graph to find the acceleration of the object.

Q15) Figure below shows a speed-time graph for a particle during the first 20 seconds of its motion.



Calculate the particle's average speed during the 20 seconds.



Q1) Solution

$$a = \frac{v - u}{t}, u = 2 \text{ m/s}, v = 14 \text{ m/s and } t = 4$$

$$\text{Therefore, } a = \frac{14 - 2}{4 \text{ s}} \text{ m/s}$$

$$a = \frac{12 \text{ m/s}}{4} \\ = 3 \text{ m/s}^2$$

Q2) Solution

$$\text{Speed} = \frac{\text{total distance}}{\text{total time}} \\ = \frac{360 \text{ km}}{2 \text{ hrs}} \\ = 180 \text{ km/hr}$$

(Changing km to m)

$$= \frac{180 \times 1000 \text{ m}}{1 \times 60 \times 60 \text{ s}} \\ = 50 \text{ m/s}$$

Q3) Solution

$$a) \text{ Average speed} = \frac{\text{total distance}}{\text{total time}}$$

$$\text{Total time between A and B} = \frac{\text{total distance}}{\text{average speed}}$$

$$\frac{1.5 \text{ h}}{90} = \frac{1.5}{90} \times 60 \times 60 = 60 \text{ s}$$

$$\text{Total time taken between B and C} = \frac{\text{total distance}}{\text{average speed}}$$

$$= \frac{2.4}{36} \times 60 \times 60 = 240 \text{ s}$$

$$\text{Total time between A and C} = 60 \text{ s} + 240 \text{ s} = 300 \text{ s}$$

$$b) \text{ Average speed} = \frac{\text{total distance}}{\text{total time}}$$

$$= \frac{(1.5 + 2.4) \times 1000 \text{ m}}{300 \text{ s}} \\ = 13 \text{ m/s}$$

Q4) Solution

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{change in time}} = \frac{v - u}{t} \\ = \frac{(10 \text{ m} - 5) \text{ m/s}}{2 \text{ s}} \\ = \frac{5 \text{ m/s}}{2 \text{ s}} \\ = \frac{5 \text{ m}}{\text{s}} \times \frac{1}{2 \text{ s}} \\ = \frac{2.5 \text{ m}}{\text{s}^2} \\ = 2.5 \text{ m/s}^2$$

Note: An acceleration of 2.5 m/s^2 means the velocity increases by 2.5 m/s every second.

In this case,

At the end of the 1st second, the velocity of the cyclist will be:

$$5 + 2.5 = 7.5 \text{ m/s}$$

At the end of the 2nd second, the velocity of the cyclist will be:

$$7.5 + 2.5 = 10 \text{ m/s}$$

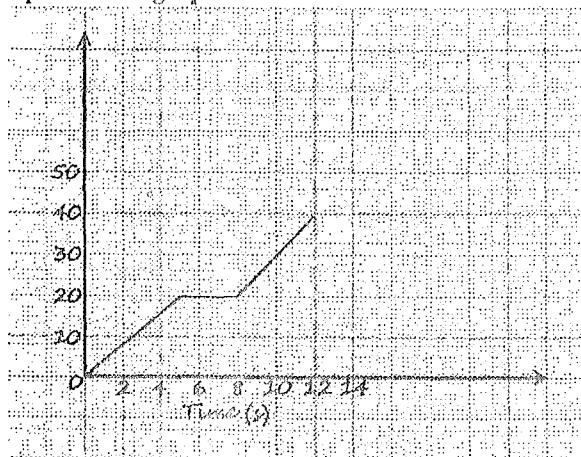
Q5) Solution

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{change in time}} \\ = \frac{(10 - 24) \text{ m/s}}{4 \text{ s}} \\ = \frac{-14 \text{ m/s}}{4 \text{ s}} \\ = -3.5 \text{ m/s}^2$$

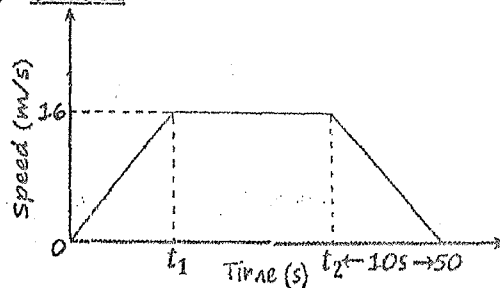
The significance of the negative sign in the calculated value is that, the velocity is decreasing by 3.5 m/s every second.

Q6) Solution

Speed-Time graph for a Train



Q7) Solution



Distance travelled = area under graph

Required to find time between t_1 and t_2

$$\text{Time from 0 to } t_1 = \frac{16 \text{ m/s}}{2 \text{ m/s}^2} = 8 \text{ seconds}$$

$$\begin{aligned} \text{Time between } t_1 \text{ and } t_2 &= 50 - (8 + 10) \\ &= 32 \text{ seconds} \end{aligned}$$

$$\begin{aligned} \therefore \text{Distance between } t_1 \text{ and } t_2 &= \frac{16 \text{ m}}{s} \times 32 \text{ s} \\ &= 512 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Distance travelled from time } t_1 \text{ to time } t_2 &= \\ &= 512 \text{ m} \end{aligned}$$

Q8) Solution

Total distance = Area under graph

$$\text{Area} = \frac{1}{2} (\text{sum of // sides}) \times h$$

$$= \frac{1}{2} (4 + 11) \times h$$

$$= \frac{15}{2} h$$

$$\text{But } \frac{15}{2} h = 90$$

$$\therefore h = \frac{90}{15} \times 2 = 12$$

$$\therefore \text{Speed, } V = 12 \text{ m/s}$$

Q9) Solution

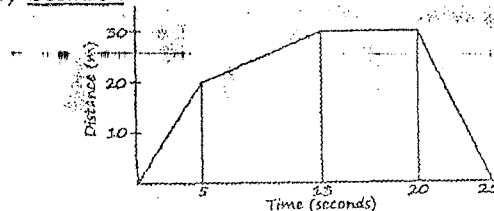
$$\text{Acceleration} = \frac{\text{Change in speed}}{\text{Change in time}}$$

$$0.3 \text{ m/s/s} = \frac{(15 - 0) \text{ m/s}}{t_1 - 0}$$

$$0.3 \text{ m/s/s} = \frac{15 \text{ m/s}}{t_1}$$

$$\begin{aligned} \Rightarrow t_1 &= \frac{15 \text{ m/s}}{0.3 \text{ m/s/s}} \\ &= 50 \text{ s} \end{aligned}$$

Q10) Solution



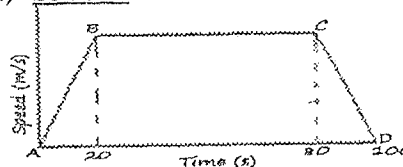
On a distance/time graph, speed is the gradient

$$\text{Speed} = \text{gradient} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\begin{aligned} \Rightarrow \text{Speed} &= \frac{30 \text{ m} - 20 \text{ m}}{13 \text{ s} - 5 \text{ s}} \\ &= \frac{10 \text{ m}}{8 \text{ s}} \\ &= 1.25 \text{ m/s} \end{aligned}$$

$\therefore$ The speed of the object between $t = 5$ and $t = 13$ is 1.25 m/s

Q11) Solution



Distance travelled = Area under the speed-time graph
= Area of the trapezium ABCD

$$= \frac{1}{2} (BC + AD) h$$

$$3200 = \frac{1}{2} [(80 - 20) + (100 - 0)] h$$

$$3200 = \frac{1}{2} (60 + 100) h$$

$$3200 = \frac{160}{2} h$$

$$80h = 3200$$

$$\frac{80h}{80} = \frac{3200}{80}$$

$$\therefore h = 40 \text{ m/s}$$

Deceleration = gradient of CD

$$= \frac{0 - 40 \text{ m/s}}{100 \text{ s} - 80 \text{ s}}$$

$$= \frac{-40 \text{ m/s}}{20 \text{ s}}$$

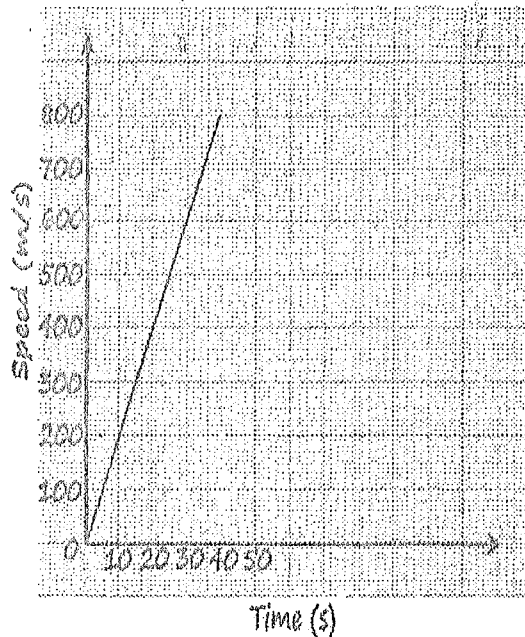
$$= -2 \text{ m/s}^2$$

$\therefore$ The deceleration during the last 20 seconds is 2 m/s^2

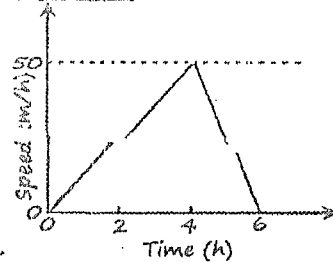
Note: the (-) sign is usually ignored in the presentation of the last stage.

Q12) Solution

Speed-Time graph for a Train



Q13) Solution



On a speed-time graph the total distance is the area under the graph

Since the graph is triangular in shape, then
Area = $\frac{1}{2} \times \text{base} \times \text{height}$ (area of the Δ)

$$\therefore \text{Distance} = \frac{1}{2} \times 6h \times 50 \text{ km/h}$$

$$= 150 \text{ km}$$

$\therefore$ The distance travelled during the 6 hours is 150 km.

Q14) Solution

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

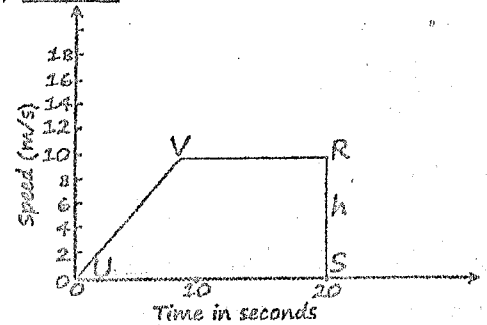
By reading off values from the graph

$$\text{acceleration} = \frac{(30 - 5) \text{ m/s}}{(12 - 0) \text{ s}}$$

$$= \frac{25 \text{ m/s}}{12 \text{ s}}$$

$$\therefore \text{acceleration} = 2.8 \text{ m/s/s.}$$

Q15) Solution



$$\text{Average speed} = \frac{\text{Distance}}{\text{Time taken}}$$

In this case, distance = Area of trapezium UVRS

$$= \frac{1}{2} (VR + US)h$$

$$= \frac{1}{2} (10 + 20)10$$

$$= \frac{1}{2} \times 30 \times 10$$

$$= 150$$

But, time taken = 20 seconds

$$\therefore \text{Average speed} = \frac{150 \text{ m}}{20 \text{ s}} = 7 \frac{1}{2} \text{ m/s}$$

The average speed during the 20 seconds is $7 \frac{1}{2} \text{ m/s}$

Note: On a speed-time graph, distance is the area under the graph

Topic 19

POLYNOMIALS

Definition of a polynomial

A polynomial expression is made up of two or more terms where each consist of a coefficient and a variable(s) raised to some non-negative power which must be a whole number. The power can be zero.

For example, linear expressions such as $2x + 4$, quadratic expressions such as $3x^2 - 7 + 6$ and cubic expressions such as $x^3 + 12x + 6$ are called polynomials but $3x^2 - x^{-3} + 3$ and $\frac{4}{x^3} + \frac{7}{x^4}$ are not polynomials. A polynomial in x is any function built up from powers of x , where x is a variable. The degree of the polynomial is the highest power of x in the expression.

For instance,

$2x + 4$ is a polynomial of degree 1,

$3x^2 - 7x + 2$ is a polynomial of degree 2,

$2x^3 - 7x^2 + 2x + 9$ is a polynomial of degree 3.

Note: Most questions under this chapter will be about polynomial of degree 3.

The remainder theorem

The remainder theorem states that:

If $f(x)$ is a polynomial in x and $f(a) = k$, then k is the remainder when $f(x)$ is divided by $x - a$.

The factor theorem

The factor theorem states that:

If $f(x)$ is a polynomial in x and $f(a) = 0$, then $x - a$ is a factor of $f(x)$.

Roots of polynomial equations

If a is a solution of a polynomial $f(x) = 0$, then a is said to be a root of the equation and $x - a$ is a factor of the polynomial.

Questions on Polynomials

- Q1) Divide $x^3 + 6x^2 - x - 30$ by $x + 2$
 Q2) Divide $x^3 + 7x^2 - 5x + 9$ by $x + 3$
 Q3) Find the quotient and remainder when $x^3 - 3x^2 + 4x + 1$ is divided by $x - 2$
 Q4) Divide $2x^4 + x^3 + 3x^2 + 7$ by $x + 2$
 Q5) Divide $4x^3 - 8x^2 - x + 2$ by $(2x - 1)$
 Q6) Find the remainder when, $x^4 + 4x^3 + 6x^2 + 5x - 2$ is divided by $x + 3$
 Q7) When $2x^4 + x^3 + bx^2 + 5$ is divided by -2 , the remainder is 33. Find the value of b .
 Q8) When the expression, $x^3 + ax^2 - 3x + b$ is divided by $x + 1$ and $x - 2$, the remainders are 7 and 3 respectively. Find the values of a and b .
 Q9) Use the remainder theorem to find the remainder when $f(x) = x^3 - 3x^2 + 6x + 5$ is divided by $(x - 2)$
 Q10) Find the remainder when $f(x) = x^3 + 2x^2 - 5x - 6$ is divided by $x + 3$
 Q11) Use the remainder theorem to prove that $(x - y)$ is a factor of a polynomial $x^2(y - 2) + y^2(2 - x)$
 Q12) When the polynomial $ax^2 + bx + c$ is divided by $(x + 1)$ and $(x + 3)$ gives a remainder of 2 in each case. Find the polynomial.

- Q13) Given that $(x - 2)$ is a factor of $x^3 - 6x^2 - 11x + p$, find p .
- Q14) When a polynomial, $x^3 + px^2 - 3x + 4$ is divided by $x + 2$, the remainder is 14. Calculate the value of p .
- Q15) If $(x + 1)$ is a factor of $x^3 - 5x^2 + 2x + 8$
- Find the other two factors, hence
 - Find the roots of the equation $x^3 - 5x^2 + 2x + 8 = 0$
- Q16) Find the remainder when $t^3 + 2t^2 - 3(t + 1)$ is divided by $(t + 1)$
- Q17) Find the remainder when $2x^3 - 13x^2 - 8x + 12$ is divided by $2x - 1$
- Q18) Given that $(3x^2 + x - 6)(x + 1)$ and $3x^3 + kx^2 + hx - 6$ are identical, find the value of k and h .
- Q19) Given that $(4x^2 - 9)(Bx + C)$ is identical to $16x^3 + 24x^2 - 36x - 54$, calculate the values of B and C given that they are positive.
- Q20) Given that the roots of a quadratic equation $y = x^2 + ax + b$ are -3 and 2 , find the values of a and b in the equation.
- Q21) If $(y + 2)$ and $(y - 3)$ are factors of $2y^3 + by^2 + cy - 6$, find the values of b and c .
- Q22) Given that the roots of quadratic equation in y are -5 or 3 . Formulate the equation in the form $ay^2 + by + c = 0$
- Q23) Given that $(x + 2)$ is a factor of $2x^2 - 3x^2 - 11x + 6$, solve the equation $2x^2 - 3x^2 - 11x + 6 = 0$
- Q24) Show that $k + 3$ is a factor of $k^3 + 3k^2 - 4k - 12$
- Q25) Given that $(x + 1)$ and $(x - 3)$ are two factors of the polynomial $ax^2 + bx - 6$, calculate the values of a and b .
- Q26) Given that $(x - 2)$ is a factor of $x^3 - 6x^2 + 11x + p$, find the p .
- Q27) The polynomials $x^3 - x^2 + x + 9$ and $x^3 - 4x + 3$ leave the same remainder when divided by $(x - a)$. Calculate the values of a .
- Q28) Find the values of A and B for which $(x - Bx + 8)(x + A) \equiv x^3 - 5x^2 + 2x + 8$.

Q29) Show that $(2x - 1)$ is a factor of $8x^3 + 20x^2 - 2x - 5$

Q30) Given that $(x + 3)(x + 1)^2 \equiv Ax^3 + Bx^2 + Cx + D$, find the value of C .

Q31) Find the remainder when $2x^3 - 13x^2 - 8x + 12$ is divided by $2x - 1$

Q32) The remainder when $x(x + b)(x - 2b)$ is divided by $x - b$ is -16 . Find the value of b

Polynomials (Model Solutions)

Q1) Solution

$$\begin{array}{r}
 x^2 + 4x - 9 \\
 x + 2 \overline{) x^3 + 6x^2 - x - 30} \\
 \underline{x^2 + 2x^2} \\
 4x^2 - x \\
 \underline{4x^2 - 8x} \\
 -9x - 30 \\
 \underline{-9x - 18} \\
 -12
 \end{array}$$

No further steps can be done since there are no terms in x . Therefore, $x^3 + 6x^2 - x - 30$ divided by $x + 2$ gives $x^2 + 4x - 9$ remainder -12

Q2) Solution

$$\begin{array}{r}
 x^2 + 4x - 17 \\
 x + 3 \overline{) x^3 + 7x^2 - 5x + 9} \\
 \underline{x^2 + 3x^2} \\
 4x^2 - 5x \\
 \underline{4x^2 - 12x} \\
 -17x - 9 \\
 \underline{-17x - 51} \\
 60
 \end{array}$$

Since $(x + 3)$ cannot be divided into 60 , the division stops here. The quotient is $x^2 + 4x - 17$ and the remainder is 60 .

Q3) Solution

$$\begin{array}{r}
 x^2 - x + 2 \\
 x - 2 \sqrt{x^3 - 3x^2 + 4x + 1} \\
 \underline{x^3 + 2x^2} \\
 -x^2 + 4x \\
 \underline{x^2 + 2x} \\
 +2x - 1 \\
 \underline{-2x - 4} \\
 5
 \end{array}$$

The quotient is $x^2 - x + 2$ and the remainder is 5.

Q4) Solution

$$\begin{array}{r}
 2x^3 - 3x^2 + 3x - 6 \\
 x + 2 \sqrt{2x^4 - x^3 + x^2 + 0x + 7} \\
 \underline{2x^4 + 4x^3} \\
 -3x^3 - 3x^2 \\
 \underline{-(-3x^3 - 6x^2)} \\
 3x^2 + 0x \\
 \underline{-(3x^2 + 6x)} \\
 -6x + 7 \\
 \underline{-(-6x - 12)} \\
 19
 \end{array}$$

Therefore, $2x^4 + x^3 + 3x^2 + 7$ divided by $x + 2$ gives $2x^3 - 3x^2 + 3x - 6$ as quotient and the remainder is 19.

Q5) Solution

$$\begin{array}{r}
 2x^2 - 3x - 2 \\
 2x - 1 \sqrt{4x^3 - 8x^2 - x + 2} \\
 \underline{4x^3 - 2x^2} \\
 -6x^2 - x \\
 \underline{-6x^2 - 8x} \\
 -4x + 2 \\
 \underline{-4x + 2} \\
 0
 \end{array}$$

$4x^3 - 8x^2 - x + 2$ divided by $(2x - 1)$ gives $2x^2 - 3x - 2$ as the quotient

Since there is no remainder, the divisor is said to be a factor of the polynomial

Q6) Solution

In this case, $x = -3$

$$\begin{aligned}
 f(-3) &= (-3)^4 + 4(-3)^3 + 6(-3)^2 + \\
 &\quad 5(-3) - 2 \\
 &= 81 - 108 + 54 - 15 - 2 \\
 &= 135 - 125 \\
 &= 10
 \end{aligned}$$

Therefore, the remainder is 10.

Q7) Solution

In this case, $a = 2$ and $f(2) = 33$.

$$\begin{aligned}
 \therefore f(2) &= 2(2)^4 + (2)^3 + b(2)^2 + 5 = 33 \\
 2 \times 16 + 8 + 4b + 5 &= 33 \\
 32 + 8 + 4b + 5 &= 33 \\
 4b &= -12 \\
 b &= -3
 \end{aligned}$$

Q8) Solution

In this case, when $x + 1$, then $x = -1$

And when $x - 2$, then $x = 2$

Thus

When $x = -1$

$$f(-1) = (-1)^3 + a(-1)^2 - 3(-1) + b = 7 \quad (i)$$

When $x = 2$

$$f(2) = (2)^3 + a(2)^2 - 3(2) + b = 3 \quad (ii)$$

$$-1 + a + 3 + b = 7 \quad (i)$$

$$8 + 4a - 6 + b = 3 \quad (ii)$$

$$a + b = 5 \quad (i)$$

$$-(4a + b = 1) \quad (ii)$$

Subtracting (ii) and (i) we get

$$-3a = 4$$

$$a = -\frac{4}{3}$$

Substituting $a = -\frac{4}{3}$ into (i) to find (b)

$$\left(-\frac{4}{3}\right) + b = 5$$

$$b = 5 + \frac{4}{3} = \frac{19}{3}$$

$$\text{Thus, } a = -\frac{4}{3} \text{ and } b = \frac{19}{3}$$

Q9) Solution

The remainder is $f(2)$.

$$\begin{aligned}\text{If } x = 2, & x^3 - 3x^2 + 6x + 5 \\ &= 2^3 - 3(2)^2 + 6(2) + 5 \\ &= 8 - 12 + 12 + 5 \\ &= 13\end{aligned}$$

Note: You can check this by dividing; you will find that:

$$\begin{aligned}& x^3 - 3x^2 + 6x + 5 \\ &= (x - 2)((x^2 - x + 4) + 13\end{aligned}$$

Q10) Solution

If $x + 3 = 0, x = -3$

$$\begin{aligned}f(-3) &= (-3)^3 + 2(-3)^2 - 5(-3) - 6 \\ &= -27 + 18 + 15 - 6 \\ &= 0\end{aligned}$$

The remainder is 0.

Q11) Solution

In this case, if $(x - y)$ is a factor then substituting $x = -y$ into the polynomial will give 0

Thus,

$$\begin{aligned}f(-y) &= x^2(y - 2) + y^2(2 - x) = 0 \\ y^2(y - 2) + y^2(2 - y) &= 0 \\ y^3 - 2y^2 + 2y^2 - y^3 &= 0\end{aligned}$$

Rearranging like terms

$$y^3 - y^3 - 2y^2 + 2y^2 = 0$$

Gives $LHS = RHS \Rightarrow 0 = 0$

Hence, $(x - y)$ is a factor of a polynomial

$$x^2(y - 2) + y^2(2 - x)$$

Q12) Solution

Since $(x + 1)$ and $(x + 3)$ gives a remainder of 2 in each case

Then,

$$(x + 1)(x + 3) + \text{remainder} = ax^2 + bx + c$$

Expanding the LHS we get

$$\begin{aligned}x(x + 3) + 1(x + 3) + 2 &= ax^2 + bx + c \\ \Rightarrow x^2 + 3x + x + 3 + 2 &= ax^2 + bx + c \\ \Rightarrow x^2 + 4x + 5 &= ax^2 + bx + c\end{aligned}$$

Hence, $(x^2 + 4x + 5)$ is the required polynomial

and $a = 1, b = 4$ and $c = 5$

Q13) Solution

Given that $(x - 2)$ is a factor of $x^3 - 6x^2 - 11x + p$

Then, $\Rightarrow x - 2 = 0 \therefore x = 2$

$$\begin{aligned}f(2) &= 2^3 - 6(2)^2 + 11(2) + p \\ 8 - 24 + 22 + p &= 0 \\ 6 + p &= 0 \\ p &= -6\end{aligned}$$

Q14) Solution

In this case, $x = -2$ and $f(-2) = 14$

Thus,

$$\begin{aligned}f(-2) &= (-2)^3 + p(-2)^2 - 3(-2) + 4 = 14 \\ -8 + 4p + 6 + 4 &= 14 \\ 4p &= 14 - 2 \\ 4p &= 12 \\ p &= 3\end{aligned}$$

Q15) Solution

(i) Using trial method, by trying factors of the constant term which is 8

$$\begin{aligned}f(2) &= 2^3 - 5(2)^2 + 2(2) + 8 \\ &= 8 - 20 + 4 + 8 \\ &= 8 + 8 + 4 - 20 = 20 - 20 \\ &= 0\end{aligned}$$

Therefore, $x - 2$ is a factor of $x^3 - 5x^2 + 2x + 8$.

$$\begin{aligned}\text{When } f(4) &= (4)^3 - 5(4)^2 + 2(4) + 8 \\ &= 64 - 80 + 8 + 8 \\ &= 64 + 8 + 8 - 80 = 80 - 80 \\ &= 0\end{aligned}$$

Therefore, $x - 4$ is again another factor of $x^3 - 5x^2 + 2x + 8$.

(ii) Finding the roots of the equation

$$x^3 - 5x^2 + 2x + 8 = 0$$

8 has factors: $\pm 1, \pm 2, \pm 4, \pm 8$,

$$f(1) = (1)^3 - 5(1)^2 + 2(1) + 8 = 6$$

So $(x - 1)$ is not a factor

$$f(2) = (2)^3 - 5(2)^2 + 2(2) + 8 = 0$$

So $(x - 2)$ is a factor

$$f(4) = (4)^3 - 5(4)^2 + 2(4) + 8 = 0$$

So $(x - 4)$ is a factor

$$f(8) = (8)^3 - 5(8)^2 + 2(8) + 8 = 216$$

So $(x - 8)$ is not a factor

Hence, the factors are $x = -1$, $x = 2$ and $x = 4$

Alternative method

- (i) We need to divide $(x^3 - 5x^2 + 2x + 8)$ by $(x + 1)$ to find the quotient

$$\begin{array}{r} x^2 - 6x + 8 \\ x + 1 \overline{) x^3 - 5x^2 + 2x + 8} \\ \underline{-(x^3 + x^2)} \\ -6x^2 + 2x \\ \underline{-(-6x^2 - 6x)} \\ 4x + 8 \\ \underline{-(4x + 4)} \\ 0 \end{array}$$

Hence, $x^2 - 6x + 8 \Rightarrow (x - 2)(x - 4)$
 $\therefore$ The factors are $x - 2$ and $x - 4$

- (ii) $x^3 - 5x^2 + 2x + 8 = 0$
 $(x + 1)(x^2 - 6x + 8) = 0$
 $(x + 1)(x - 2)(x - 4) = 0$
 Either $(x + 1) = 0 \Rightarrow x = -1$
 or $(x - 2) = 0 \Rightarrow x = 2$
 or $(x - 4) = 0 \Rightarrow x = 4$
 Hence $x = -1$ or $x = 2$ or $x = 4$

Q16) Solution

Firstly,

$$t^3 + 2t^2 - 3(t + 1) = t^3 + 2t^2 - 3t - 3$$

$$\text{If } (t + 1) = 0, t = -1$$

$$\begin{aligned} f(-1) &= (-1)^3 + 2(-1)^2 - 3(-1) - 3 \\ &= -1 + 2 + 3 - 3 \\ &= 1 \end{aligned}$$

Therefore, the remainder is 1

Q17) Solution

If $2x - 1 = 0$, then $x = \frac{1}{2}$

$$\begin{aligned} f\left(\frac{1}{2}\right) &= 2\left(\frac{1}{2}\right)^3 - 13\left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 12 \\ &= 2\left(\frac{1}{8}\right) - 13\left(\frac{1}{4}\right) - 4 + 12 \\ &= \frac{1}{4} - \frac{13}{4} - 4 + 12 \\ &= \frac{20}{4} \\ &= 5 \end{aligned}$$

Q18) Solution

Expanding LHS we get

$$\begin{aligned} (3x^2 + x - 6)(x + 1) \\ &= 3x^3 + 3x^2 + x^2 + x - 6x - 6 \\ &= 3x^3 + 4x^2 - 5x - 6 \end{aligned}$$

But

$$\begin{aligned} (3x^2 + x - 6)(x + 1) &\equiv 3x^3 + kx^2 + hx - 6 \\ \Rightarrow 3x^3 + 4x^2 - 5x - 6 &\equiv 3x^3 + kx^2 + hx - 6 \end{aligned}$$

So equating the coefficients gives

$$k = 4 \text{ and } h = -5$$

Q19) Solution

Expanding the LHS we get

$$\begin{aligned} (4x^2 - 9)(Bx + C) \\ &= 4Bx^3 + 4Cx^2 - 9Bx - 9C \end{aligned}$$

But

$$(4x^2 - 9)(Bx + C) = 16x^3 + 24x^2 - 36x - 54$$

So equating the coefficients we get

$$4Bx^3 = 16x^3 \text{ Or } -9Bx = -36x$$

Thus, $B = 4$

$$\text{And } 4Cx^2 = 24x^2 \text{ Or } -9C = -54$$

Hence, $C = 6$

Therefore, $B = 4$ and $C = 6$

Q20) Solution

Since -3 is a root, then $(x + 3)$ is a factor
 Similarly, $(x - 2)$ is another factor for the equation.

Therefore, the equation is

$$\begin{aligned} (x + 3)(x - 2) &= x^2 + ax + b \\ x^2 - 2x + 3x - 6 &= x^2 + ax + b \\ x^2 + x - 6 &= x^2 + ax + b \end{aligned}$$

So equating the coefficients we get

$$a = 1 \text{ and } b = -6$$

Q21) Solution

Since $(y + 2)$ and $(y - 3)$ are factors of
 $2y^3 + by^2 + cy - 6$

In this case, $y = -2 \Rightarrow$

$$f(-2) = 2(-2)^3 + b(-2)^2 + c(-2) - 6$$

$$= -16 + 4b - 2c - 6$$

$$= 4b - 2c - 22 \dots \quad (\alpha)$$

And $y = 3 \Rightarrow$

$$f(3) = 2(3)^3 + b(3)^2 + c(3) - 6$$

$$= 54 + 9b + 3c - 6$$

$$= 9b + 3c + 48 \dots \quad (\beta)$$

We have two equations as follows:

$$4b - 2c = 22 \dots \quad (\alpha)$$

$$9b + 3c = -48 \dots \quad (\beta)$$

$$(\alpha) \times 9 \text{ and } (\beta) \times 4$$

$$36b - 18c = 198 \dots \quad (@)$$

$$36b + 12c = -192 \dots \quad (\textcircled{C})$$

Subtracting $(\textcircled{C})$ from $(@)$ we get

$$-30c = 390$$

$$c = -13$$

Substituting $c = -13$ into (i) to get

$$4b - 2(-13) = 22 \dots \quad (\alpha)$$

$$4b + 26 = 22$$

$$4b = -4$$

$$b = -1$$

Hence $b = -1$ and $c = -13$

Q22) Solution:

Since -5 is a root, $(y + 5)$ is a factor.

Similarly, $(y - 3)$ is another factor of the equation

Therefore, the equation is:

$$(y + 5)(y - 3) = 0$$

$$y(y - 3) + 5(y - 3) = 0$$

$$y^2 - 3y + 5y - 15 = 0$$

$$y^2 + 2y - 15 = 0$$

Q23) Solution:

Given $x + 2$ is a factor of $2x^2 - 3x^2 - 11x + 6$. Using long division to find the quadratic factor:

$$\begin{array}{r} 2x^2 - 7x + 3 \\ x + 2 \overline{) 2x^2 - 3x^2 - 11x + 6} \\ \underline{(2x^2 + 4x^2)} \\ - (7x^2 - 11x) \\ \underline{-(7x^2 - 14x)} \\ 3x + 6 \\ \underline{-(3x + 6)} \\ 0 \end{array}$$

$$\therefore 2x^2 - 3x^2 - 11x + 16 = (x + 2)(2x^2 - 7x + 3)$$

$$\text{But } 2x^2 - 7x + 3 = 2x^2 - x - 6x + 3$$

$$= x(2x - 1) - 3(2x - 1)$$

$$= (x - 3)(2x - 1)$$

$$\therefore 2x^2 - 7x + 3 = (x + 2)(x - 3)(2x - 1)$$

$$\Rightarrow (x + 2)(x - 3)(2x - 1) = 0$$

Either $(x + 2) = 0 \Rightarrow x = -2$

or $(x - 3) = 0 \Rightarrow x = 3$

$$2x - 1 = 0 \Rightarrow x = \frac{1}{2}$$

$\therefore$ The values of x are $-2, 3$ or $\frac{1}{2}$

Q24) Solution:

If $k + 3$ is a factor then substituting $k = -3$ into the polynomial will give 0

$$f(-3) = (-3)^3 + 3(-3)^2 - 4(-3) - 12 = 0$$

$$= -27 + 27 + 12 - 12 = 0$$

$$= 0 = 0$$

Hence $k + 3$ is a factor of $k^3 + 3k^2 - 4k - 12$

Q25) Solution:

Since $(x + 1)$ and $(x - 3)$ are two factors of the polynomial

Then, $(x + 1) \Rightarrow x = -1$,

$$\text{Thus, } f(-1) = a(-1)^2 + b(-1) - 6$$

$$= a - b - 6$$

$$\Rightarrow a - b = 6 \dots \quad (i)$$

Also, $(x - 3) \Rightarrow x = 3$,

$$\text{Thus, } f(3) = a(3)^2 + b(3) - 6$$

$$= 9a + 3b - 6$$

$$\frac{27}{3}a + \frac{3}{3}b = \frac{6}{3}$$

$$\Rightarrow 9a + b = 2 \dots \quad (ii)$$

Adding (i) to (ii) we get

$$a - b = 6 \dots \quad (i)$$

$$+ (9a + b = 2) \dots \quad (ii)$$

$$8a = 8$$

$$a = 1$$

Substituting $a = 1$ into (ii) to get

$$9(1) + b = 2$$

$$b = 2 - 9$$

$$b = -7$$

Hence $a = 1$ and $b = -7$

Q26) Solution:

If $(x - 2)$ is a factor then substituting $x = 2$ into the polynomial will give 0

$$\begin{aligned} f(2) &= (2)^3 - 6(2)^2 + 11(2) + p = 0 \\ 8 - 24 + 22 + p &= 0 \\ p &= -6 \end{aligned}$$

Q27) Solution

In this case, $(x - a) \Rightarrow x = a$
And let $f(a) = y$

$$\therefore f(a) = a^3 - a^2 + a + 9 = y$$

$$\text{Also } f(a) = a^3 - 4a + 3 = y$$

Equating the two equations we get

$$\begin{aligned} a^3 - a^2 + a + 9 &= a^3 - 4a + 3 \\ a^3 - a^3 - a^2 + a + 4a &= 3 - 9 \\ -a^2 + 5a &= -6 \end{aligned}$$

Factorising we get

$$-a(a - 5) = -6$$

Either $a = 6$ or $a = -1$

Therefore the values of a are

$$a = 6 \text{ or } a = -1$$

Q28) Solution

Expanding the LHS we get:

$$\begin{aligned} (x - Bx + 8)(x + A) \\ = (x^3 - Ax^2 + Bx^2 + ABx + 8x + 8A) \\ = x^3 - (A + B)x^2 + (AB + 8)x + 8A \end{aligned}$$

Equating with the RHS we have:

$$\begin{aligned} x^3 - (A + B)x^2 + (AB + 8)x + 8A \\ \equiv x^3 - 5x + 2x + 8 \end{aligned}$$

Comparing the coefficients:

$$A + B = -5 \dots \dots (i)$$

$$AB + 8 = 2 \dots \dots (ii)$$

$$8A = 8 \dots \dots (iii)$$

From (iii) $A = 1$

Substitute 1 for A in (i) we have

$$1 + B = -5$$

Therefore, $B = -6$

Q29) Solution

If $2x - 1$ is a factor then substituting

$x = \frac{1}{2}$ into the polynomial will give 0

$$\begin{aligned} f\left(\frac{1}{2}\right) &= 8\left(\frac{1}{2}\right)^3 + 20\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right) - 5 \\ &= 0 \end{aligned}$$

$$= 1 + 5 - 1 - 5 = 0$$

$$= 0 = 0$$

Hence $(2x - 1)$ is a factor of

$$8x^3 + 20x^2 - 2x - 5$$

Q30) Solution

Expanding the LHS we get:

$$\begin{aligned} (x + 3)(x + 1)^2 \\ = (x + 3)(x + 1)(x + 1) \\ = [(x + 3)(x + 1)](x + 1) \\ = (x^2 + x + 3x + 3)(x + 1) \\ = x^3 + x^2 + x^2 + x + 3x^2 + 3x + 3x + 3 \\ = x^3 + 5x^2 + 7x + 3 \end{aligned}$$

Equating the RHS we get:

$$x^3 + 5x^2 + 7x + 3 \equiv Ax^3 + Bx^2 + Cx + D$$

Comparing the coefficients:

$$Ax^3 = x^3 \Rightarrow A = 1$$

$$Bx^2 = 5x^2 \Rightarrow B = 5$$

$$Cx = 7x \Rightarrow C = 7$$

$$\text{And } D = 3$$

Hence the value of $C = 7$

Q31) Solution

Using the remainder theorem

$$\text{Let } 2x - 1 = 0 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2}$$

Substituting $x = \frac{1}{2}$ in the polynomial

$$\begin{aligned} 2\left(\frac{1}{2}\right)^3 - 13\left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 12 \\ = 2 \times \frac{1}{8} - 13 \times \frac{1}{4} - 8 \times \frac{1}{2} + 12 \\ = \frac{1}{4} - \frac{13}{4} + 8 \\ = \frac{1 - 13 + 32}{4} \\ = \frac{20}{4} \\ = 5 \end{aligned}$$

$\therefore$ The remainder is 5

Q32) Solution

$$(x - b) \Rightarrow x = b,$$

$$\text{Thus, } f(b) = b(b + b)(b - 2b) = -16$$

$$(b^2 + b^2)(b - 2b) = -16$$

$$2b^2(-b) = -16$$

$$-2b^3 = -16$$

$$b^3 = 8$$

$$b^3 = 2^3$$

Hence $b = 2$

Topic 20 Probability

Meaning of probability

Probability is a measure of the likelihood of a *required outcome* happening. In other words, it can be defined as numerical representation of a chance. It is usually given as a fraction.

$$\text{probability} = \frac{\text{number of required outcomes}}{\text{number of possible outcomes}}$$

Experimental probability

It is the probability of an event that can be determined from past experiences. For example, in an experiment of throwing a coin, each throw is called a *trial* and each result is called an *outcome* or an *event*.

$$\text{experimental probability} = \frac{\text{number of favourable outcomes}}{\text{total number of trials}}$$

Possibility space

This is a list of all possible outcomes of a trial

Certainty

The probability of a certain event happening is 1.

The probability of an impossible event is 0.

$$0 \leq P \leq 1 \begin{cases} \text{The maximum probability can be 1} \\ \text{The minimum probability can be 0} \end{cases}$$

The sum of all probabilities of all outcomes = 1

$$P(\text{event}) + P(\text{not event}) = 1$$

Mutually exclusive events

Two events are said to be *mutually exclusive*, if the occurrence of one event prevents the other event from occurring.

The addition law

When A and B are mutually exclusive events, then, $P(A \text{ or } B) = P(A) + P(B)$.

A or B means that A occurs, or B occurs or both A and B occur.

When A and B are not mutually exclusive, there are some elements common to both A and B.

$$\begin{aligned} \text{Thus, } P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B) \end{aligned}$$

Independents events

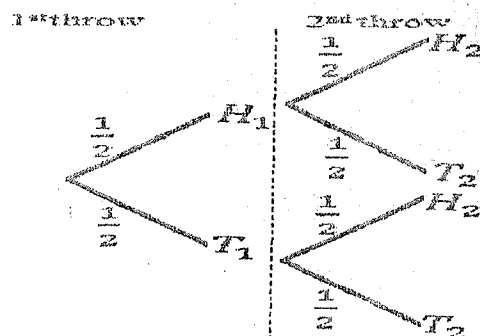
Two events are independent if the occurrence of one does not affect the occurrence of another. Independent events can occur at the same time. For example, when a coin and a die are thrown at the same time, the outcome of the coin does not affect the outcome of the die.

Multiplication law

If A and B are independent events, then $P(A \text{ and } B) = P(A) \times P(B)$

Tree diagrams

A useful way of sorting out probability problems that involve more one event is to draw a *probability tree*. Consider a fair coin tossed twice. The outcomes can be represented on a tree diagram as shown in the figure below.



Note:

- (i). The outcomes are written at the end of the branch and their probabilities are written along the branch.
- (ii). The sum of the probabilities at any branches from the same point is 1, no matter how many outcomes there are for

one trial. This is the most important feature of the probability tree diagram.

(iii) In general, we multiply the probabilities when we follow a path along the branches of the probability tree.

(iv) In general, we add the probabilities when following several paths of the same outcomes; this is done when there are several similar outcomes.

Questions on Probability

Q1) A fair coin is tossed 20 times and heads appears 13 times. What is the probability of getting heads with this coin?

Q2) A die numbered 1 to 6 is thrown 90 times. Each time the die is thrown, the number on the top is recorded as the score. The results are shown in the following table.

Score	1	2	3	4	5	6
Frequency	12	14	17	19	15	13

Find the experimental probability of getting:

- a) a 2.
- b) an odd number.
- c) a prime number

Q3) A card is picked from the pack of cards. What are the following probabilities?

- a) $P(\text{picking an ace})$
- b) $P(\text{picking a red card})$

Q4) It is known that out of every 1000 new cars, 50 develop a mechanical fault in the first 3 months. What is the probability of buying a car that will develop a mechanical fault within 3 months?

Q5) A market trader has 100 oranges for sale. 4 of them are bad. What is the probability that an orange chosen at random is good?

Q6) A die is thrown. Find the probability that the outcome is divisible by 3.

Q7) Three coins are thrown. What is the probability of getting 2 heads and 1 tail?

Q8) What is the probability of 5 heads if I toss 5 coins?

Q9) A bag contains 4 red and 5 blue beads. Two beads are taken out of it. What is the probability that they are both blue

- a) if the first bead is put back?
- b) if the first bead is not replaced?

Q10) Bag A contains 3 blue marbles and 2 green marbles. Bag B contains 5 blue marbles and 1 green marble. A marble is drawn from each bag at random. What is the probability that they are both blue?

Q11) The letters M, A, L, A, W, I are written on separate identical cards. The cards are placed upside down. If one card is picked at random, what is the probability that it has the letter A?

Q12) A fair die is rolled once. What is the probability of obtaining

- a) A prime number?
- b) Perfect square?

Q13) A bookshelf contains 10 Mathematics books, 9 History books and 11 books on sports. All books have the same dimensions. If Chimwenwe selects a book at random, what is the probability that it is a book on sports?

Q14) A box contains 5 red balls, 8 white balls and 7 black balls. If one ball is selected at random, calculate the probability that it is white or black.

Q15) In a plastic bag there are x blue pens, 6 black pens and 4 red pens. If the probability of picking a red pen is $\frac{1}{5}$, calculate the number of blue pens.

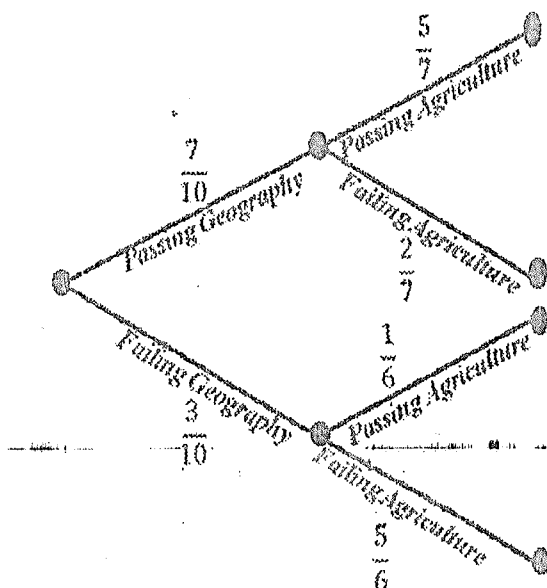
Q16) A die and a coin are rolled and tossed respectively

- a) Draw a probability space table.
- b) Use the probability space table to find the probability of obtaining an even number on the die and a tail on the coin

Q17) A family has 3 children born at different times. Assuming it is likely to have a baby boy or a baby girl,

- draw a tree diagram to show the probabilities of having a boy or a girl on each of the three births.
- calculate the probability that the family has 2 boys and 1 girl in any order of births.

Q18) Figure below is a tree diagram illustrating the probability of a student passing Agriculture and Geography in an examination. The probability of passing Geography in the examination is $\frac{7}{10}$ and the probability of passing Agriculture after one has passed Geography is $\frac{5}{7}$. The probability of passing Agriculture after one failed Geography is $\frac{1}{6}$.



Calculate the probability of a student passing Agriculture.

Q19) Coin A is tossed followed by B. The probability that coin A shows head is $\frac{1}{2}$ while the probability that coin B shows head $\frac{1}{4}$. Using a tree diagram, calculate the probability that both coins A and B show tails.

Q20) A coin is tossed three times. Calculate the probability that the first and the third tosses give heads as outcomes, regardless of what the outcome of the second toss is.

Q21) Two fair dice, each with faces numbered 0, 1, 2, 3, 4, 5 are rolled at the same time. Table below shows the possible sums of the numbers on the dice.

	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	6
2	2	3	4	5	6	7
3	3	4	5	6	7	8
4	4	5	6	7	8	9
5	5	6	7	8	9	10

Calculate the probability of getting:

- a sum less than 4
- a sum which is a multiple of 5

Q22) The probability of a bus arriving early at a depot is $\frac{1}{10}$ and arriving late is $\frac{3}{5}$. If 400 buses are expected at the depot during the day, calculate the number of buses that are likely to arrive at the depot in time.

Q23) Table below shows the distribution of ages of learners in a Form 2 class.

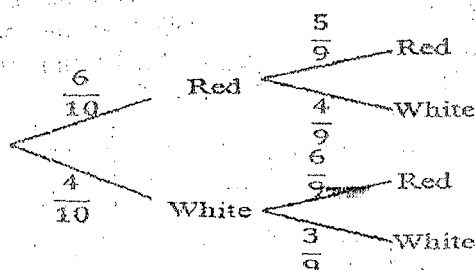
Age	14	15	16	17	18	19
Number of learners	2	10	8	4	9	3

What is the probability of picking at random a learner of 18 years of age?

Q24) A set of three cards numbered 1, 2 and 3 respectively, are placed in a bag. Draw a tree diagram and use it to answer the questions below;

- how many two-digit numbers can be formed by selecting with replacement two cards with different digits in any order.
- the probability that any two-digit number formed with different digits is even.

- Q25)** Figure below is a tree diagram which shows the probability of picking two balls one at a time without replacement from a bag containing 6 red and 4 white balls.



Use the tree diagram to calculate the probability of picking two balls of different colours, leaving your answer in its simplest form.

- Q26)** A bag contains beans, groundnuts and maize seeds. The probability of getting at random a bean seed is $\frac{1}{5}$, a groundnut seed is $\frac{x}{15}$ and a maize seed is $\frac{1}{3}$. Find the value represented by x .
- Q27)** In Ziweto village, families are allowed to keep three types of animals: chickens, goats and pigs. The probability that a family will keep a chicken and a pig is $\frac{1}{2}$, $\frac{1}{5}$ and $\frac{1}{10}$ respectively.
- Draw a tree diagram to illustrate this information.
 - Calculate the probability that a family will keep only one type of animal.
- Q28)** A box has 5 cards marked 1, 2, 3, 4 and 5, another box has 2 cards marked A and B. If a card is picked randomly from each box at the same time, calculate the probability of picking a card marked 3 from one box and a card marked B from the other box.
- Q29)** In a class 30 pupils like English and 17 like science. All 46 students say they like at least one of these subjects. What is the probability that a pupil chosen at random likes exactly one subject?

Probability (Model Solutions)

Q1) Solution

Out of 20 trials there are 13 heads.

$$P(\text{heads}) = \frac{\text{number of heads recorded}}{\text{number of trials}} = \frac{13}{20}$$

In 20 throws of a fair coin, 10 are expected. Thus, the probability of getting heads is expected to be $\frac{10}{20} = \frac{1}{2}$. But this is not the case from the experiment. An experiment does not always give what is expected.

Q2) Solutions

- In this experiment 2 has been recorded 14 times out of 90 times.
 $\therefore P(2) = \frac{14}{90} = \frac{7}{45}$
- The odd numbers between 1 and 6 inclusive are 1, 3 and 5.
 1 has been recorded 12 times;
 3 has been recorded 17 times and
 5 has been recorded 15 times.
 In total, all of them have been recorded
 $= 12 + 17 + 15 = 44 \text{ times}$
 $\therefore P(\text{odd number}) = \frac{44}{90} = \frac{22}{45}$
- The prime numbers between 1 and 6 inclusive are 2, 3 and 5 which have been recorded $14 + 17 + 15 = 46 \text{ times}$.
 $\therefore P(\text{prime number}) = \frac{46}{90} = \frac{23}{45}$

Q3) Solutions

- There are 52 cards in a pack, 4 of which are aces.
 $P(\text{picking an ace}) = \frac{4}{52} = \frac{1}{13}$
- There are 26 red cards in a pack.

$$P(\text{picking a red card}) = \frac{26}{52} = \frac{1}{2}$$

Q4) Solution

Number of cars developing a faults = 50

Number of cars altogether = 1 000

$$\therefore P(\text{buying a faulty car}) = \frac{50}{1\,000} = \frac{1}{20}$$

Q5) Solution

4 out 100 oranges are bad,

Thus, 96 out of 100 oranges are good.

$$\therefore P(\text{a good orange}) = \frac{96}{100} = \frac{24}{25}$$

or

$$P(\text{bad orange}) = \frac{4}{100} = \frac{1}{25}$$

$$P(\text{good orange}) = 1 - \frac{1}{25} = \frac{24}{25}$$

Q6) Solution

There are 6 possible outcomes:

1; 2; 3; 4; 5; 6

Two of these are divisible by 3: 3 and 6

$$\therefore P(\text{no. divisible by 3}) = \frac{2}{6} = \frac{1}{3}$$

Q7) Solution

$$\xi = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Let the Q be the probability representing 2 heads and 1 tail

$$\text{Then } Q = \{HHT, HTH, THH\}$$

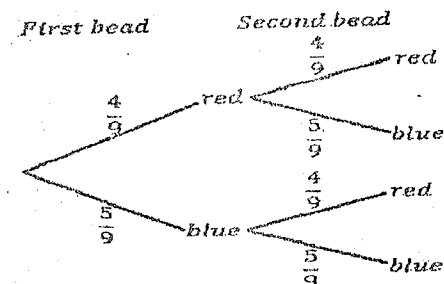
$$\therefore P(Q) = \frac{n(Q)}{n(\xi)} = \frac{3}{8}$$

Q8) Solution

$$P(H H H H H) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Q9) Solution

a) If the first bead is replaced, tree diagram looks like this on the other column:



From the tree diagram, $P(\text{blue, blue})$

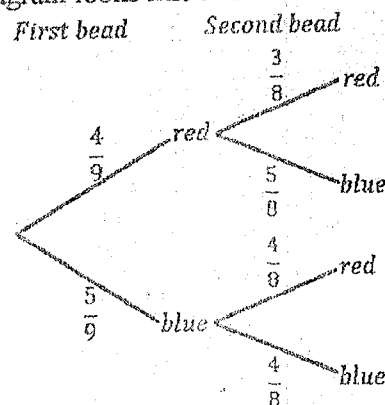
$$= \frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$$

b) When the bead is not replaced the number of beads in the second group of branches has dropped to 8.

That is, if the first bead removed was red, there will be 3 red beads and 5 blue left.

Similarly, if the first bead removed was blue, then there will be 4 blue and 4 red beads.

$\therefore$ The probabilities of the second branches will be different. The tree diagram looks like this:



$$\therefore P(\text{blue, blue}) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$$

Q10) Solution

Both events are independent,

$$\therefore P(\text{blue marble from A}) = \frac{3}{5}$$

$$P(\text{blue marble from B}) = \frac{5}{6}$$

$$P(\text{both are blue}) = \frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$$

Q11) Solution

The letters M, A, L, A, W, I are 6 in total

Number of A's = 2

$\therefore P(\text{picking an A})$

$$= \frac{\text{Number of A}}{\text{Total no. of letters}}$$

$$= \frac{2}{6}$$

$$= \frac{1}{3}$$

Q12) Solution

- a) When a fair die is rolled once, the sample space (S) is as follows;
1, 2, 3, 4, 5 and 6

In this case, prime numbers (P) are 2, 3 and 5

$$\therefore P(\text{prime number}) = \frac{n(P)}{n(S)}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

Note: (Prime numbers are numbers which have got exactly two factors, the number itself and 1)

- b) Total number of possible outcomes = 6
No. of favourable outcomes = 2
(i.e. getting 1 or 4)

$$\therefore P(\text{perfect square}) = \frac{2}{6} = \frac{1}{3}$$

Q13) Solution

Let's denote total number of books to be T_b
and number of books on sports be s

$$\text{Then, } T_b = 10 + 9 + 11$$

$$= 30$$

and, $s = 11$

$$\therefore P(\text{books on sports}) = \frac{s}{T_b}$$

$$= \frac{11}{30}$$

Hence, the probability that Chimwenwe selects a book on sports is $\frac{11}{30}$

Q14) Solution

In this case, let R denote red ball, W denote white ball, B denote black ball and N denote total number of balls

$$N = 5 + 8 + 7 = 20$$

We need to find $P(W \text{ or } B) = P(W \cup B)$

$$\text{Thus, } P(W) = \frac{n(W)}{n(N)} = \frac{8}{20}$$

$$\text{and } P(B) = \frac{n(B)}{n(N)} = \frac{7}{20}$$

$$\therefore P(W \cup B) = \frac{8}{20} + \frac{7}{20}$$

$$= \frac{15}{20}$$

$$= \frac{3}{4}$$

$\therefore$ The probability that the ball selected is white or black is $\frac{3}{4}$

Q15) Solution

Given: number of blue pens = x

number of black pens = 6

number of red pens = 4

Let O be an outcome of picking a red pen,
 N be the number of pens

$$N = x + 6 + 4$$

$$= x + 10$$

$$P(O) = \frac{n(\text{red pens})}{\text{Total number of pens}}$$

$$= \frac{4}{x+10}$$

Since $P(O) = \frac{1}{5}$ (Given)

$$\Rightarrow \frac{4}{x+10} = \frac{1}{5}$$

$$x + 10 = 20$$

$$x = 20 - 10$$

$$x = 10$$

$\therefore$ The number of blue pens is 10

Q16) Solution

- a) Probability space table

		Die					
		1	2	3	4	5	6
Coin	H	(H,1)	(H,2)	(H,3)	(H,4)	(H,5)	(H,6)
	T	(T,1)	(T,2)	(T,3)	(T,4)	(T,5)	(T,6)

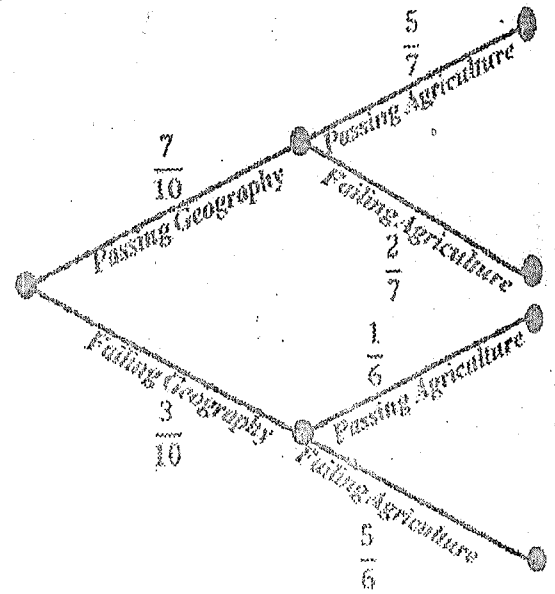
Where H is head, T is tail and 1, 2, 3, 4, 5, 6 are the six faces of a die.

Q18) Solution

- b) Let E be the required event
 From the table, the required events,
 E is represented by:
 $(T, 2); (T, 4); (T, 6)$.
 Furthermore, denoting N as the total
 number of outcomes and n be
 number of required events.
 $\therefore N = 12$ and $n = 3$

$$P(E) = \frac{n}{N} = \frac{3}{12} = \frac{1}{4}$$

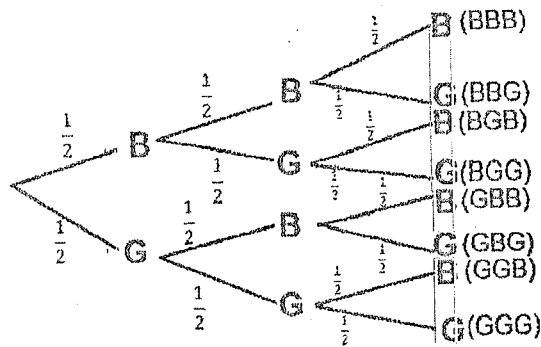
$\therefore$ The probability of obtaining an
 even number on the die and a tail on
 the coin is $\frac{1}{4}$



Q17) Solutions

- (i). Denoting an event of having a baby
 boy as B
 and G as an event of having a baby
 girl.
 Since there are equal chances of
 having a baby boy or baby girl
 Then, $P(B) = \frac{1}{2}$, $P(G) = \frac{1}{2}$

The tree diagram will look as the
 one on the other column:



$$\begin{aligned} \text{(ii). } P(2B \text{ and } 1G) &= P(BBG) + P(BGB) + P(GBB) \\ &= (\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}) + (\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}) + (\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}) \\ &= (\frac{1}{8} + \frac{1}{8} + \frac{1}{8}) \\ &= \frac{3}{8} \end{aligned}$$

$\therefore$ The probability that the family
 has 2 boys and 1 girl in any order is
 $\frac{3}{8}$

Denoting P_A as passing of Agriculture
 P_G as passing of Geography
 F_G as failing of Geography

Using the tree diagram we get;

$$\begin{aligned} P &= (P_A) = P(P_G \text{ and } P_A) + P(F_G \text{ and } P_A) \\ &= \left(\frac{7}{10} \times \frac{5}{7}\right) + \left(\frac{3}{10} \times \frac{1}{6}\right) \\ &= \frac{1}{2} + \frac{1}{20} \\ &= \frac{10 + 1}{20} \\ &= \frac{11}{20} \end{aligned}$$

$\therefore$ The probability of passing Agriculture is $\frac{11}{20}$

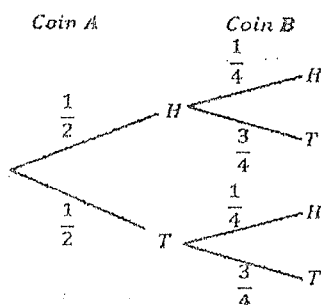
Q19) Solution

In this case, let's denote
 head by H
 and tail by T

$$\begin{aligned} \text{On coin A, } P(H) &= \frac{1}{2} \\ \Rightarrow P(T) &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{On coin B, } P(H) &= \frac{1}{4} \\ \Rightarrow P(T) &= 1 - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

The tree diagram will like this:



$$P(TT) = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

∴ The probability that both coin A and B shows tails is $\frac{3}{8}$

Q20) Solution

Let H represent head and T represent tail.

The sample space for the three tosses is shown below;

$$\xi = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

∴ The probability that the 1st and the 3rd tosses give heads is;

$$\begin{aligned} &P(HHH) \text{ and } P(HTH) \\ &\Rightarrow P(HHH) + P(HTH) \\ &= \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = \frac{1}{4} \end{aligned}$$

∴ The probability that the 1st and the 3rd tosses give heads is $\frac{1}{4}$

Alternatively, you can use a tree diagram.

Q21) Solution

Denoting N as the number of possible outcomes

	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	6
2	2	3	4	5	6	7
3	3	4	5	6	7	8
4	4	5	6	7	8	9
5	5	6	7	8	9	10

∴ counting from the table, $N = 36$

Further, let's denote n as the number of events,

E_1 to represent sum less than 4

Then, E_1 consist of,

$$(0+0) = 0, (0+1) = 1$$

$$(0+2) = 2, (0+3) = 3$$

$$(0+1) = 1, (1+1) = 2$$

$$(1+2) = 3, (0+2) = 2,$$

$$(1+2) = 3, (2+1) = 3$$

$$\therefore 0, 1, 2, 3, 1, 2, 3, 2, 3, 3, 3$$

Total of elements in E_1 is 10

$$\text{hence } P(E_1) = \frac{n}{N} = \frac{10}{36} = \frac{5}{18}$$

∴ The probability of getting a sum less than 4 is $\frac{5}{18}$

Q22) Solution

In this case,

Let O_1 be an outcome of arriving early

O_2 be an outcome of arriving late

and O_3 be an outcome of arriving in time

$$\text{From the statement, } P(O_1) = \frac{1}{10}$$

$$P(O_2) = \frac{3}{5}$$

Since the total probabilities must add to 1

$$\Rightarrow P(O_1) + P(O_2) + P(O_3) = 1$$

$$\frac{1}{10} + \frac{3}{5} + P(O_3) = 1$$

$$P(O_3) = 1 - \frac{1}{10} - \frac{3}{5}$$

$$P(O_3) = \frac{10-1-6}{10}$$

$$P(O_3) = \frac{3}{10}$$

But, the total number of buses = 400

∴ The number of buses likely to arrive in time

$$= \frac{3}{10} \times 400 = 120$$

Q23) Solution

Age	14	15	16	17	18	19
Number of learners	2	10	8	4	9	3

Let's denote total number of learners as N and n to be number of learners with 18 years of age.

In reference to the table

$$N = 2 + 10 + 8 + 4 + 9 + 3 = 36$$

$$n = 9$$

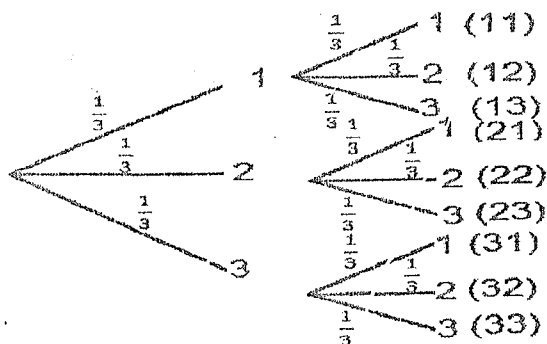
Further, let E be an event of picking a learner of 18 years of age:

$$\therefore P(E) = \frac{n}{N} = \frac{9}{36} = \frac{1}{4}$$

Hence, the probability of picking at random a learner of 18 years of age is $\frac{1}{4}$

Q24) Solution

- a) This problem would better be solved using a tree diagram shown below



From tree diagram above, the two digits number with different digits, in any order are:

12, 13, 21, 23, 31, 32

$\therefore$ There are 6 numbers

- b) Let N be the total number of outcomes and n be the number of events needed.

In this case, the required events E are: 12, 32

$\therefore n = 2$ and $N = 9$

$$P(E) = \frac{n}{N} = \frac{2}{9}$$

Hence, the probability that any two digits number formed with different digits is even is $\frac{2}{9}$

Q25) Solution

Denoting R to represent red and W to represent white

The balls picked can be in either order RW or WR

Then, probability of picking two balls of different colours:

$$\Rightarrow P(RW) = \frac{6}{10} \times \frac{4}{9} = \frac{4}{15}$$

$$\text{and } P(WR) = \frac{4}{9} \times \frac{6}{10} = \frac{4}{15}$$

$$\therefore P(RW) \text{ or } P(WR) = \frac{4}{15} + \frac{4}{15} = \frac{8}{15}$$

Q26) Solution

$$\text{Given: } P(\text{bean}) = \frac{1}{5}$$

$$P(\text{groundnuts}) = \frac{x}{15}$$

$$P(\text{maize}) = \frac{1}{3}$$

Since total probability is equal to 1

$$\Rightarrow \frac{1}{5} + \frac{x}{15} + \frac{1}{3} = 1$$

$$\frac{3 + x + 5}{15} = 1$$

Multiply both sides by 15 to get

$$x + 8 = 15$$

$$x = 15 - 8$$

$$x = 7$$

Q27) Solution

- (i). From the statement above,

$$P(\text{chickens kept}) = \frac{1}{2}$$

Then,

$$P(\text{no chickens}) = 1 - \frac{1}{2} = \frac{1}{2}$$

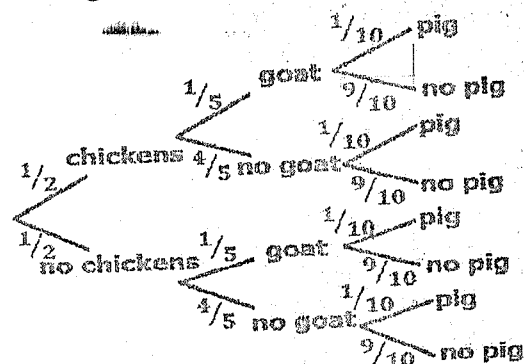
$$P(\text{goats kept}) = \frac{1}{5}$$

$$\text{Then, } P(\text{no goats}) = 1 - \frac{1}{5} = \frac{4}{5}$$

$$P(\text{pigs kept}) = \frac{1}{10}$$

$$\text{Then, } P(\text{no pigs}) = 1 - \frac{1}{10} = \frac{9}{10}$$

A tree diagram will be as follows:



Probability of keeping one type of animal
 $P(\text{chickens}) \text{ or } P(\text{goats}) \text{ or } P(\text{pigs})$

$$\begin{aligned}
 P(\text{chickens only}) &= \frac{1}{2} \times \frac{4}{5} \times \frac{9}{10} \\
 &= \frac{36}{100}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{goats only}) &= \frac{1}{2} \times \frac{1}{5} \times \frac{9}{10} \\
 &= \frac{9}{100}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{pigs only}) &= \frac{1}{2} \times \frac{4}{5} \times \frac{1}{10} \\
 &= \frac{4}{100}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{chickens or pig or goat only}) &= \frac{36}{100} + \frac{4}{100} + \frac{9}{100} \\
 &= \frac{49}{100}
 \end{aligned}$$

Hence, the probability that a family will keep only one type of animal is $\frac{49}{100}$

Q28) Solution

In this case, the probability spaces for boxes are as follows:

First box = {1, 2, 3, 4, 5},

Second box = {A, B}

$$P(\text{picking a card marked 3}) = \frac{1}{5}$$

$$P(\text{picking a card marked B}) = \frac{1}{2}$$

Since these events are independent of each other

$$\text{Then, } P(3 \text{ and } B) = P(3) \times P(B)$$

$$\Rightarrow P(3 \cap B) = \frac{1}{5} \times \frac{1}{2} = \frac{1}{10}$$

Hence, the probability of picking a card marked 3 from one box and a card marked B from the other box is $\frac{1}{10}$.

Q29) Solution

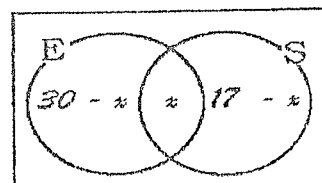
Let's denote E to represent English
 S to represent science
 and $n(E \cap S) = x$

$$\text{Given: } n(E) = 30,$$

$$n(S) = 17$$

$$n(\xi) = 46$$

We draw a Venn diagram



To find x

$$30 - x + x + 17 - x = 46$$

$$x = 1$$

The numbers are English only 29, both subjects 1, science only 16.

$$P(\text{exactly one subject}) = \frac{29 + 16}{46} = \frac{45}{46}$$

Topic 21

Three-Dimensional Figures

Introduction:

Some of the common regular three-dimensional figures are cubes, cuboids, cylinders, pyramids, prisms, cones and spheres. These figures are also called *solids*. Solids may have faces, edges and vertices. These faces, edges and vertices determine the geometrical properties of a solid. Solids with flat faces are called *polyhedral* (singular-*polyhedron*). Polyhedron means *many faces*. A regular polyhedron consists of faces that are made of regular polygons.

Useful Formulae

Mensuration formulae

Shape	Area	Perimeter
Square length l	l^2	$4l$
Rectangle length l , breadth b	lb	$2(l + b)$
Triangle base b , height h	$\frac{1}{2}(bh)$	
Parallelogram base l , height h	bh	
Trapezium height h parallels a and b	$\frac{1}{2}(a + b)h$	
Circle radius r	π^2	$2\pi r$

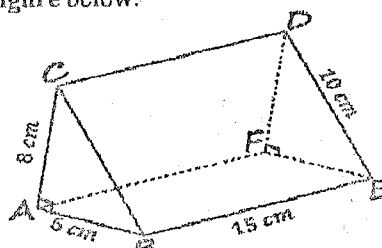
Shape

Shape	Volume	Surface area
Cube Length of side l	l^3	$6l^2$
Cuboid length l , breadth b , height h	lbh	$2(lb + lh + bh)$
Prism height h base area A	Ah	
Cylinder radius r , height h	$\pi r^2 h$	curved surface area $= 2\pi rh$ total surface area $= 2\pi r(r + h)$

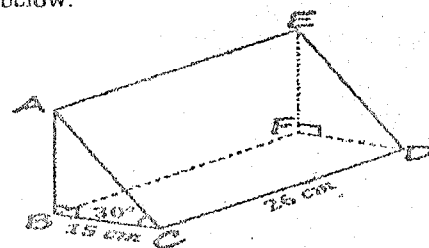
Pyramid height h base area A	$\frac{1}{3} Ah$	
Cone radius r , height h slant height l	$\frac{1}{3} \pi r^2 h$	curved surface area $= \pi rl$ total surface area $= \pi r(r + l)$
Sphere radius r	$\frac{4}{3} \pi r^3$	$4\pi r^2$

Questions on Three-Dimensional Figures

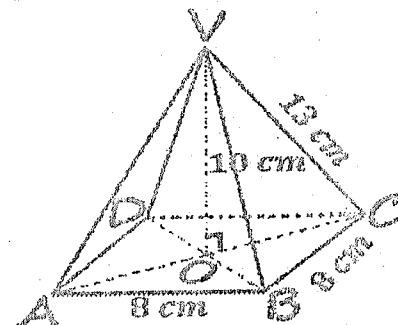
- Q1) Find the surface area of a cube whose length is 6 cm.
 Q2) Find the surface area of a cuboid measuring 6 cm by 4 cm by 3 cm.
 Q3) Find the surface area of prism ABCDEF in Figure below.



- Q4) Find the surface area of the prism in Figure below.



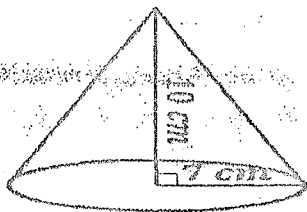
- Q5) A right pyramid has a square base of side 8 cm. If its height is 10 cm and the slant height is 13 cm, find the surface area.



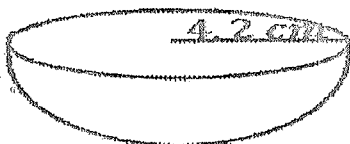
Q6) Find the surface area of a cone with a base radius 2.8 cm and the slant height 7.4 cm .

Q7) Find the volume of the cone in Figure below.

Use $\pi = \frac{22}{7}$

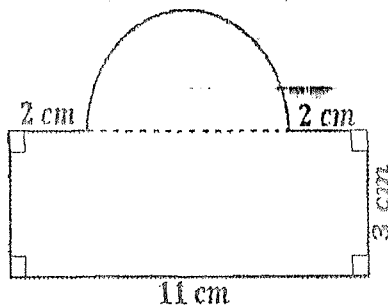


Q8) The Figure below is a hemisphere of radius 4.2 cm



Find its volume. Take $\pi = 3.142$.

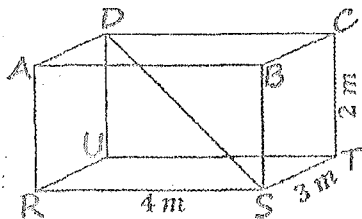
Q9) Figure below is made up of a rectangle and a semi-circle.



Calculate

- the perimeter of the figure
- the total area of the figure. (Take $\pi = \frac{22}{7}$)

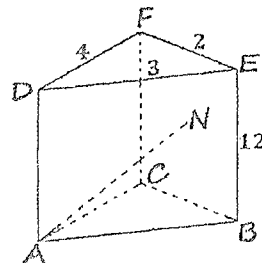
Q10) Figure below shows a solid cuboid of length 4 m , width 3 m and height 2 m .



Calculate

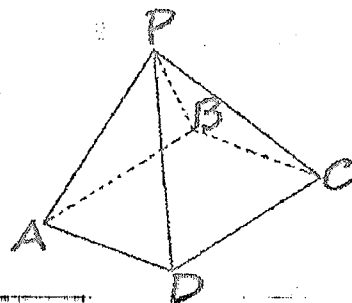
- the length of DS , correct to three decimal places.
- the angle DSC to the nearest degree.

Q11) Figure below is a triangular prism with $DE = 3 \text{ cm}$, $EF = 2 \text{ cm}$, $DF = 4 \text{ cm}$ and $BE = 12 \text{ cm}$.



If N is the point where diagonals of the face $FEBC$ meet, find the angle which NA makes with the plane ABC .

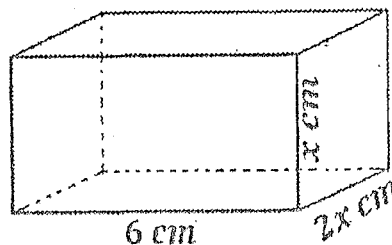
Q12) Figure below shows a right pyramid with vertex P and $PA = PB = PC = PD = 13 \text{ cm}$. $ABCD$ is a rectangular base with $AB = 8 \text{ cm}$ and $AD = 6 \text{ cm}$.



Calculate

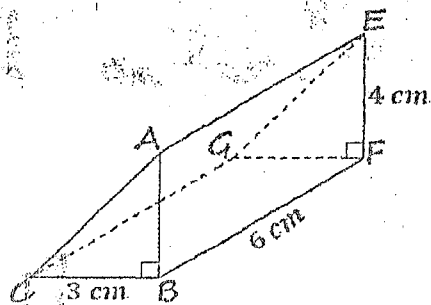
- the height of the pyramid.
- the height between the base and an edge.

Q13) Figure below shows a rectangular box with an open top. The box measures 6 cm long, $2x \text{ cm}$ wide and $x \text{ cm}$ high.



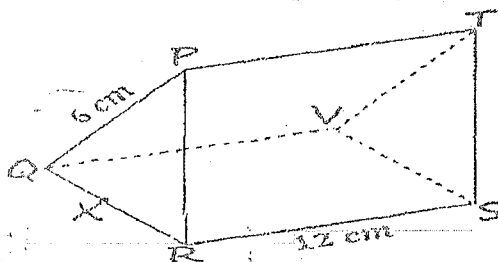
Given that the total outer surface area of the box is 108 cm^2 . Form an equation in x and show that it simplifies to $x^2 + 6x - 27 = 0$.

- Q14) Figure below shows a triangular prism in which $BC = 3 \text{ cm}$, $EF = 4 \text{ cm}$, $BF = 6 \text{ cm}$ and angle $ABC = 90^\circ$.



Calculate the volume of the prism.

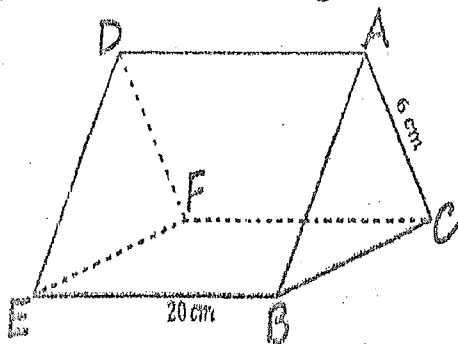
- Q15) Figure below is a diagram with a cross section of an equilateral triangle PQR. PRST, PQVT and QRSV are rectangles, such that $RS = 12 \text{ cm}$, $QP = 6 \text{ cm}$ and X is a midpoint of QR.



Calculate

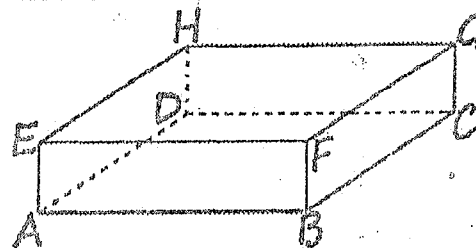
- the length of TX giving your answer correct to 2 decimal places.
- the angle between TX and the base QRSV giving your answer correct to 2 decimal places.

- Q16) Figure below shows a wooden prism with triangular cross-section ABC. $BE = 20 \text{ cm}$, $AB = BC = AC = 6 \text{ cm}$, ACFD, BCFD and ADEB are rectangles.



Calculate angle DEC

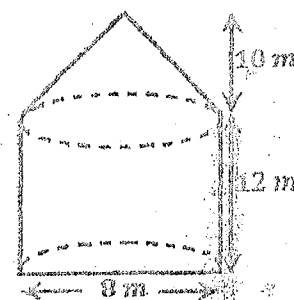
- Q17) Figure below shows a water tank with rectangular faces. $AB = 8 \text{ m}$, $BC = 5 \text{ m}$ and $CG = 3 \text{ m}$.



Calculate:

- the length of AG leaving your answer in its simplest surd form.
- the angle which AG makes with the plane ABCD.

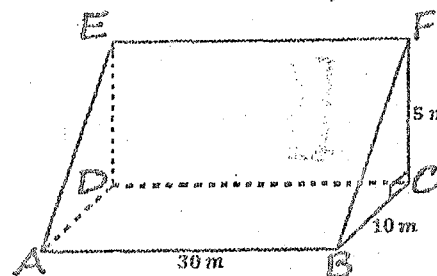
- Q18) Figure below shows a cone on top of a cylinder. The height of the cone is 10 m and that of a cylinder is 12 m . The diameter of both cone and the cylinder is 8 m .



Calculate the total volume of the shape to 2 decimal places.

(Volume of cone $= \frac{1}{3} \pi r^2 h$, take $\pi \approx 3.14$).

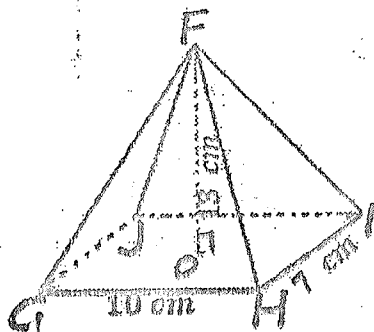
- Q19) Figure below shows a prism ABCDEF. $AB = 30 \text{ m}$, $BC = 10 \text{ m}$, $CF = 5 \text{ m}$ and angle $BCF = 90^\circ$.



Calculate

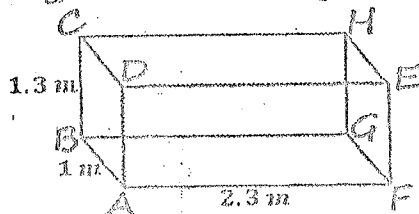
- the volume of the prism.
- angle FAC to the nearest degree.

- Q20) Figure below shows a rectangular based right pyramid FGHIJ, $GH = 10\text{ cm}$ and $HI = 7\text{ cm}$.



If the height of the pyramid is 13 cm, calculate the angle between FHI and the base.

- Q21) Figure below shows a cuboid which is 2.3 m long, 1 m wide and 1.3 m high.



Calculate:

- the length of EB,
- the angle between EB and the base of the cuboid. (Giving your answer correct to 1 decimal place in both cases).

Three-Dimensional Figures (Model Solutions)

- Q1) Solution

$$\begin{aligned}\text{Surface area} &= 6l^2\text{ cm}^2 \\ &= 6 \times 6 \times 6\text{ cm}^2 \\ &= 216\text{ cm}^2\end{aligned}$$

- Q2) Solution

$$\begin{aligned}\text{Surface area} &= 2(lb + lh + bh) \\ &= 2(6 \times 4 + 6 \times 3 + 4 \times 3) \\ &= 2(24 + 18 + 12) \\ &= 2(54) \\ &= 108\text{ cm}^2\end{aligned}$$

- Q3) Solution

$$\text{Area of } \triangle ABC = \text{Area of } \triangle DEF$$

$$\begin{aligned}&= \frac{1}{2} \times 6 \times 8 \\ &= 24\text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } BCDE &= 10 \times 15\text{ m}^2 \\ &= 150\text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } ACDF &= 8 \times 15\text{ m}^2 \\ &= 120\text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } ABEF &= 6 \times 15\text{ m}^2 \\ &= 90\text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Total surface area} &= (2 \times 24) + 150 + 120 + 90\text{ m}^2 \\ &= 408\text{ m}^2\end{aligned}$$

- Q4) Solution

$$\tan 30^\circ = \frac{AB}{15}$$

$$\begin{aligned}AB &= 15 \times \tan 30^\circ \\ &= 8.66\end{aligned}$$

$$\text{Area of } \triangle ABC = \text{Area of } \triangle EFD$$

$$\begin{aligned}&= \frac{1}{2} \times 8.66\text{ cm} \times 15\text{ cm} \\ &= 64.95\text{ cm}^2\end{aligned}$$

$$\cos 30^\circ = \frac{15\text{ cm}}{AC}$$

$$\therefore AC = \frac{15}{\cos 30^\circ}$$

$$\begin{aligned}&= \frac{15}{0.866} \\ &= 17.3\text{ cm}\end{aligned}$$

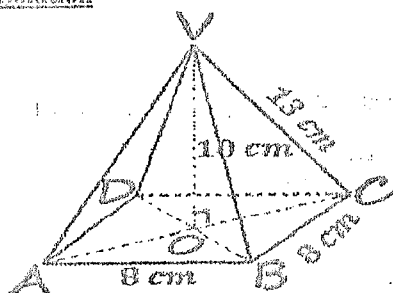
$$\begin{aligned}\text{Area of } ACDE &= 17.3\text{ cm} \times 26\text{ cm} \\ &= 449.8\text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } BCDF &= 15\text{ cm} \times 26\text{ cm} \\ &= 390\text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } ABFE &= 8.66\text{ cm} \times 26\text{ cm} \\ &= 225.2\text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Surface area of the prism} &= (2 \times 64.95) + 449.5 + 390 + 225.2 \\ &= 1195\text{ cm}^2\end{aligned}$$

Q5) Solution

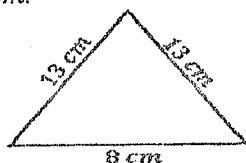


A pyramid in which the vertex is vertically above the centre of the base is right pyramid. All the slant edges of a right pyramid are equal.

Area of the base = 8×8

$$= 64 \text{ cm}^2$$

Each slanting face is an isosceles triangle of side 8 cm by 13 cm.



Using the formula,

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Where } s = \frac{13+13+8}{2} = 17$$

$\therefore$ The area of the triangular face

$$= \sqrt{17(17-13)(17-13)(17-8)}$$

$$= \sqrt{(17 \times 4 \times 4 \times 9) \text{ cm}^2}$$

$$= 49.5 \text{ cm}^2$$

There are four identical triangular faces.

$$\therefore \text{Area of triangular faces} = 4 \times 49.5 \text{ cm}^2$$

$$= 198 \text{ cm}^2$$

$$\text{Total surface area} = 64 + 198 \text{ cm}^2$$

$$= 262 \text{ cm}^2$$

Q6) Solution

$$\text{Surface area} = \pi r l + \pi r^2$$

$$= \frac{22}{7} \times 2.8 \times 7.4 + \frac{22}{7} \times 2.8 \times 2.8$$

$$= 89.75 \text{ cm}^2$$

Q7) Solution

$$\text{The base area} = \frac{22}{7} \times 7 \times 7$$

$$= 22 \times 7$$

$$= 154 \text{ cm}^2$$

$$\text{Height} = 10 \text{ cm}$$

$$\text{Volume} = 154 \times 10$$

$$= 1540 \text{ cm}^3$$

Q8) Solution

$$\text{Volume of a sphere} = \frac{4}{3} \times \pi \times r^3$$

$$\text{Volume of a hemisphere} = \frac{1}{2} \times \frac{4}{3} \times \pi \times r^3$$

$$\text{In this case, } V = \frac{1}{2} \times \frac{4}{3} \times 3.142 \times 4.2^3$$

$$= 155.19 \text{ cm}^3$$

Q9) Solution

Perimeter = circumference of the semi-circle

$$+ (11 + 3 \times 2 + 2 \times 2) \text{ cm}$$

$$= \left(\frac{1}{2} \times \frac{22}{7} \times \frac{7}{1} \right) + 21 \text{ cm}$$

$$= 11 \text{ cm} + 21 \text{ cm}$$

$$= 32 \text{ cm}$$

Total area of the figure = area of the semi-circle + area of the rectangle.

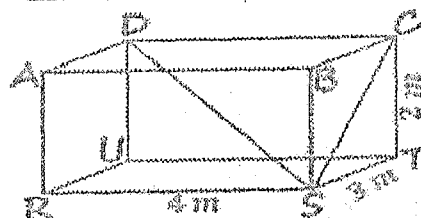
$$= \left(\frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \right) \text{ cm}^2 + (11 \times 3) \text{ cm}^2$$

$$= \frac{77}{4} \text{ cm}^2 + 33 \text{ cm}^2$$

$$= 19.25 \text{ cm}^2 + 33 \text{ cm}^2$$

$$= 52.25 \text{ cm}^2$$

Q10) Solution



(i). In $\triangle BCS$,

$$BC^2 + BS^2 = CS^2 \text{ (Pythagoras)}$$

$$3^2 + 2^2 = CS^2$$

$$CS^2 = 13$$

$$CS = \sqrt{13}$$

$$CS = 3.606 \text{ cm (from square root tables)}$$

$$CS = 3.61 \text{ (to 3 s.f.)}$$

(ii). In $\triangle CDS$,

$$DS^2 = DC^2 + CS^2 \text{ (Pythagoras)}$$

$$DS^2 = 4^2 + 3.612^2$$

$$DS^2 = 16 + 13.03$$

$$= 29.03$$

$$DS = \sqrt{29.03}$$

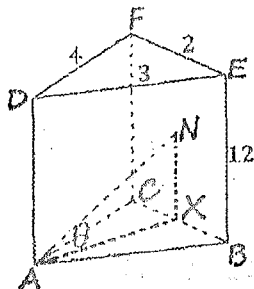
$$= 5.41 \text{ (to 3 s.f.)}$$

$$\sin \angle DSC = \frac{DC}{DS} = \frac{4}{5.41} = 0.7394$$

$$= 47.6779$$

$$= 48 \text{ (to the nearest degree)}$$

Q11) Solution



Draw line from N to BC bisecting BC at X.

$$\Rightarrow NX = \frac{1}{2}EB = 6$$

Join AX

$$\Rightarrow \angle AXB = 90^\circ$$

$$AX^2 = AB^2 - BX^2 \quad (\text{Pythagoras})$$

$$AX^2 = 3^2 - 1^2$$

$$AX = \sqrt{8}$$

$$AX = 2.8284$$

$$AX = 2.83$$

$$\tan \theta = \frac{6}{2.83} = 2.12$$

$$\theta = 65^\circ$$

Since diagonals of a rectangle bisect each other, then

$$AO = OC = \frac{10}{2} = 5 \text{ cm}$$

$$AO = 5 \text{ cm}$$

In ΔAPO :

$$\angle POA = 90^\circ$$

$$\therefore PO^2 = PA^2 - AO^2$$

$$= (13 \text{ cm})^2 - (5 \text{ cm})^2$$

$$= 169 \text{ cm}^2 - 25 \text{ cm}^2$$

$$= 144 \text{ cm}^2$$

$$PO = \sqrt{144 \text{ cm}^2}$$

$$PO = 12 \text{ cm}$$

$\therefore$ The height of the pyramid is 12 cm

(ii). We need to find angle PAO:

$$\sin \angle PAO = \frac{PO}{PA}$$

$$\sin \angle PAO = \frac{12 \text{ cm}}{13 \text{ cm}}$$

$$\sin \angle PAO = \frac{12}{13}$$

$$\Rightarrow \angle PAO = \sin^{-1} \left(\frac{12}{13} \right)$$

$$67^\circ \text{ (to the nearest degree)}$$

$\therefore$ The angle between the base and an edge is 67°

Q12) Solution

Given: A right pyramid with

$$PA = PB = PD = 13 \text{ cm}$$

A rectangular base ABCD with

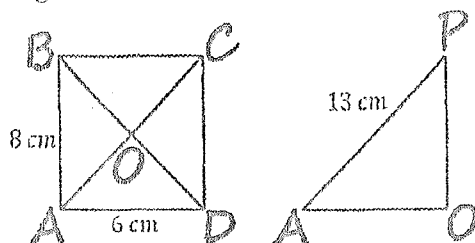
$$AB = 8 \text{ cm and } AD = 6 \text{ cm}$$

Required to find:

- (i). the height of the pyramid.
- (ii). the height between the base and an edge

(i). Join AC and BD intersecting at O

The following can be taken out from the diagram.



Since ABCD is a rectangle, then

$$\angle ADC = 90^\circ$$

$$CD = AB = 8 \text{ cm}$$

$$AC^2 = AD^2 + CD^2 \quad (\text{Pythagoras})$$

$$= (6 \text{ cm})^2 + (8 \text{ cm})^2$$

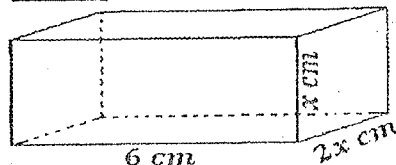
$$= 36 \text{ cm}^2 + 64 \text{ cm}^2$$

$$= 100 \text{ cm}^2$$

$$\therefore AC = \sqrt{100 \text{ cm}^2}$$

$$= 10 \text{ cm}$$

Q13) Solution



There are *two* outer rectangular surfaces measuring 6 cm by x cm (i.e.

ABCD and GEFH)

$$\text{Total area} = 2(lb)$$

$$= 2(6 \text{ cm} \times x \text{ cm})$$

$$= 12x \text{ cm}^2$$

There is *one* outer rectangular surface measuring 6 cm by 2x cm (i.e. AHFD)

$$\text{Area} = lb$$

$$= 6 \text{ cm} \times 2x \text{ cm}$$

$$= 12x \text{ cm}^2$$

There are *two* outer rectangular surfaces measuring

2x cm by x cm (i.e. CEFD and BGHA)

$$\text{Total area} = 2(lb)$$

$$= 2(2x \text{ cm} \times x \text{ cm})$$

$$= 4x^2 \text{ cm}^2$$

Total outer surface area of the box

$$= 12x \text{ cm}^2 + 12x \text{ cm}^2 + 4x^2 \text{ cm}^2$$

$$108 \text{ cm}^2 = 12x \text{ cm}^2 + 12x \text{ cm}^2 + 4x^2 \text{ cm}^2$$

$$108 \text{ cm}^2 = 24x \text{ cm}^2 + 4x^2 \text{ cm}^2$$

$$108 = 24x + 4x^2$$

$$4x^2 + 24x = 108 \text{ (Dividing each term by 4)}$$

$$x^2 + 6x = 27$$

$$\therefore x^2 + 6x - 27 = 0 \text{ as required.}$$

Q14) Solution

Volume = cross sectional area $\times$ length

Cross sectional area = Area of $\triangle ABC$

$$= \frac{1}{2}bh$$

$$= \frac{1}{2} \times 3 \text{ cm} \times 4 \text{ cm}$$

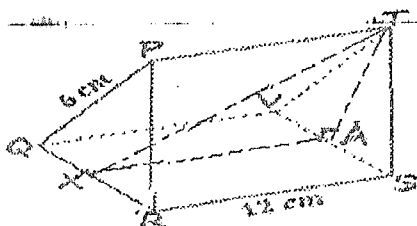
$$= 6 \text{ cm}^2$$

Length of the prism = 6 cm

$$\therefore \text{Volume} = 6 \text{ cm}^2 \times 6 \text{ cm}$$

$$= 36 \text{ cm}^3$$

Q15) Solution



Construction: Join TX, XA, TA and PX.

(i). Need to find the length of TX

Now, QR = 6 cm ($\triangle PQR$ is equilateral)

$$\therefore RX = \frac{6 \text{ cm}}{2} \text{ (X is a middle point of QR)}$$

$$RX = 3 \text{ cm}$$

In $\triangle PXR$:

$$XP^2 = PR^2 - RX^2 \text{ (Pythagoras)}$$

$$= 6^2 - 3^2$$

$$= 36 - 9$$

$$= 27$$

$$XP = \sqrt{27} = TA \text{ (XP = TA)}$$

In $\triangle TXA$:

$$TX^2 = XA^2 + TA^2 \text{ (Pythagoras)}$$

$$= 12^2 + (\sqrt{27})^2$$

$$= 144 + 27$$

$$TX = \sqrt{171}$$

$$TX = 13.08 \text{ (to d.p.)}$$

(ii). Need to calculate $\angle TXA$:

$$\cos \angle TXA = \frac{AX}{TX}$$

$$\cos \angle TXA = \frac{12}{13.08}$$

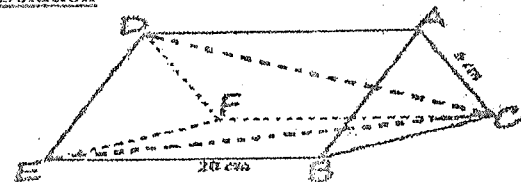
$$\angle TXA = \cos^{-1}\left(\frac{12}{13.08}\right)$$

$$= \cos^{-1}(0.9174)$$

$$= 23.45^\circ \text{ (to 2 d.p.)}$$

$\therefore$ The angle between TX and the base QRSV is 23.45° .

Q16) Solution



Given: The wooden prism with triangular section above

To calculate: $\angle DCE$

Construction: Join DC and EC.

In $\triangle DAC$,

$$DC^2 = DA^2 + AC^2 \text{ (Pythagoras thrm.)}$$

$$= 20^2 + 6^2$$

$$= 400 + 36$$

$$= 436$$

$$DC = \sqrt{436}$$

Since the rectangular faces of the prism are equal, then their diagonals are equal.

$$\therefore EC = DC = \sqrt{436}$$

In $\triangle DEC$,

Using cosine rule, we have

$$DE^2 = DC^2 + EC^2 - 2DC \times EC \times \cos \angle DCE$$

$$\Rightarrow 6^2 = (\sqrt{436})^2 + (\sqrt{436})^2 - 2(\sqrt{436}) \times$$

$$(\sqrt{436}) \times \cos \angle DCE$$

$$36 = 436 + 436 - 2 \times 436 \times \cos \angle DCE$$

$$\text{(Since } \sqrt{436} \times \sqrt{436} = 436)$$

$$36 = 436 + 436 - 872 \times \cos \angle DCE$$

$$36 = 872 - 872 \times \cos \angle DCE$$

$$-872 \times \cos \angle DCE = 36 - 872$$

$$-872 \times \cos \angle DCE = -836$$

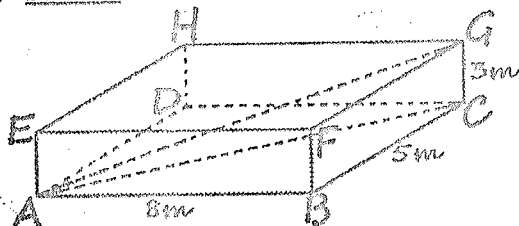
$$\cos \angle DCE = \frac{-836}{-872}$$

$$\cos \angle DCE = 0.9587$$

$$\therefore \angle DCE = \cos^{-1}(0.9587)$$

$$= 17^\circ \text{ (to the nearest degree)}$$

Q17) Solution



Given: water tank with rectangular faces,
 $AB = 8\text{ m}$, $BC = 5\text{ m}$ and $CG = 3\text{ m}$

Required to find:

- the length of AG leaving your answer in its simplest surd form.
- the angle which AG makes with the plane $ABCD$.

(i). In $\triangle ABC$:

$$\begin{aligned} AC^2 &= AB^2 + BC^2 \quad (\text{Pythagoras Thrm}) \\ &= 8^2 + 5^2 \\ &= 64 + 25 \\ &= 89 \\ AC &= \sqrt{89} \end{aligned}$$

In $\triangle ACG$:

$$\begin{aligned} AG^2 &= AC^2 + CG^2 \quad (\text{Pythagoras Thrm}) \\ &= (\sqrt{89})^2 + 3^2 \\ &= 89 + 9 \\ &= 98 \\ AG &= \sqrt{98} \\ AG &= \sqrt{2 \times 49} \\ AG &= 7\sqrt{2} \end{aligned}$$

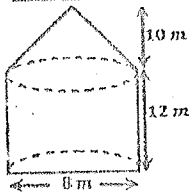
$\therefore$ The length of AG is $7\sqrt{2}\text{ m}$

(ii). Need to find angle CAG .

$$\begin{aligned} \text{Now, } \tan \angle CAG &= \frac{CG}{AC} \\ \tan \angle CAG &= \frac{3}{\sqrt{89}} \\ \angle CAG &= \tan^{-1} \left(\frac{3}{\sqrt{89}} \right) \\ &= 18^\circ \quad (\text{to the nearest degree}) \end{aligned}$$

$\therefore$ The $\angle$ which AG makes with the plane $ABCD$ is 18°

Q18) Solution



Volume of the shape = Volume of the cylinder + Volume of the cone

Now, Volume of the cylinder = $\pi r^2 h$, where r is radius and h is the height of the cylinder.

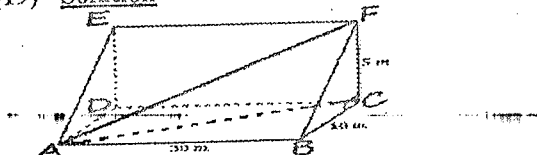
$$\begin{aligned} &= 3.14 \times \left(\frac{8\text{m}}{2} \right)^2 \times 12\text{m} \\ &= 3.14 \times (4\text{m})^2 \times 12\text{m} \\ &= 3.14 \times 16\text{m}^2 \times 12\text{m} \\ &= 602.88\text{ m}^3 \end{aligned}$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$\begin{aligned} &= \frac{1}{3} \times 3.14 \times \left(\frac{8\text{m}}{2} \right)^2 \times 10\text{m} \\ &= \frac{1}{3} \times 3.14 \times (4\text{m})^2 \times 10\text{m} \\ &= \frac{1}{3} \times 3.14 \times 16\text{m}^2 \times 10\text{m} \\ &= \frac{502.4}{3}\text{ m}^3 \end{aligned}$$

$$\begin{aligned} \therefore \text{Volume of the shape} &= 602.88\text{ m}^3 + \frac{502.4}{3}\text{ m}^3 \\ &= 770.35\text{ m}^3 \end{aligned}$$

Q19) Solution



- Volume of the prism = Cross-sectional area $\times$ length
 Now, Cross-sectional area = Area of $\triangle FBC$

$$\begin{aligned} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times 10\text{ m} \times 5\text{ m} \\ &= 25\text{ m}^2 \end{aligned}$$

Length of the prism = 30 m

$$\therefore \text{Volume of the prism} = 25\text{ m}^2 \times 30\text{ m} = 750\text{ m}^3$$

- Angle FAC to the nearest degree.

Construction: Join FA and AC

$$\text{Now, } \tan \angle FAC = \frac{FC}{AC}$$

$$\tan \angle FAC = \frac{5\text{ m}}{AC}$$

$$\begin{aligned} \text{But } AC^2 &= AB^2 + BC^2 \\ &= 30^2 + 10^2 \\ &= 900 + 100 \\ &= 1000 \end{aligned}$$

$$\therefore AC = \sqrt{1000} = 10\text{m}$$

$$\text{Hence } \tan \angle FAC = \frac{5\text{ m}}{10\text{ m}}$$

$$\tan \angle FAC = \frac{1}{2}$$

$$\angle FAC = \tan^{-1} \left(\frac{1}{2} \right)$$

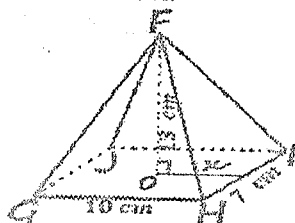
$$= 27^\circ \text{ (to the nearest degree)}$$

$\therefore$ The value of $\angle FAC$ is 27°

Q20) Solution

To find angle between FHI and GHJ

Construct perpendicular bisector from O to line HI and label it X.



Since O is centre

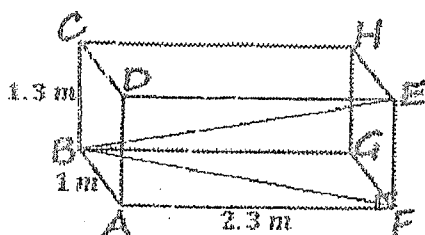
$$\Rightarrow OX = \frac{1}{2} GH = OX = 5$$

$$\tan \theta = \frac{13}{5}$$

$$\theta = \tan^{-1} \left(\frac{13}{5} \right)$$

$$\therefore \theta = 69^\circ \text{ (to the nearest degree)}$$

Q21) Solution



Join BF and find its length

$$BF^2 = AF^2 + BA^2$$

$$= (2.3)^2 + (1)^2$$

$$= 5.29 + 1 = 6.29$$

$$BF = \sqrt{6.29}$$

$$= 2.5$$

To find BE, Join BE

$$BE^2 = BF^2 + EF^2$$

$$= (2.5)^2 + (1.3)^2$$

$$BE = 2.8 \text{ m}$$

$$\text{angle } EBF = \tan \theta = \frac{1.3}{2.5} = 0.52$$

$$\text{angle } EBF = 27.5^\circ$$

Topic 22

Simultaneous Equations

Introduction

To begin with, an equation is a mathematical expression with an equal sign in the middle (i.e. it has the left hand side-LHS, and right hand side-RHS. In this case, the LHS has to always balance with the RHS, thus; $LHS = RHS$.

On the other hand, *simultaneous equations* are equations where the given values of say, x and y both satisfy the given equations when solved at the same time (*simultaneously*). Note, to satisfy is to make the equation true. Simultaneous equations are normally given as a pair; to solve such a pair either use:

- use graphical method,
- elimination method and
- Substitution method

Simultaneous *linear* and *quadratic* equations can be solved by substitution. It is easier to make one unknown as the subject of the linear equation and then substitute the expression in the quadratic equation. The resulting quadratic equation with one unknown can then be solved by factorization.

In most case, there are two solutions for the unknown. Substitute one solution at a time in the linear equation to find the value of the other unknown. Ensure it is clear in your answer which value of x pairs with which value of y .

Question 1

Simultaneous Equations

Q1) Solve the simultaneous equations below

$$x^2 + xy = 4$$

$$y - x = 2$$

Q2) Solve the simultaneous equations below.

$$x - 2y = 1$$

$$x^2 + y^2 = 29$$

Q3) Solve each of the following simultaneous equations

a) $y - x = 2$, $y^2 - x^2 = 12$.

b) $x + y = 6$, $x^2 - 2xy + y^2 = 16$.

Q4) Solve the following simultaneous equations

$$2h^2 - mh = 24$$

$$m + h = 6$$

Q5) Solve the simultaneous equations

$$x + y - 3 = x - 2y = 7$$

Q6) Solve the simultaneous equations:

$$x^2 - y - 5 = 0$$

$$\frac{1}{2}y - x = -1$$

Q7) Solve the simultaneous equations.

$$y = x^2$$

$$y = 5x - 6$$

Q8) Solve the simultaneous equations.

$$xy = -9$$

$$y = x + 6$$

Q9) Solve the simultaneous equations.

$$m - n = 5$$

$$m^2 - n^2 = 35$$

Q10) Solve the simultaneous equations.

$$y = x - 2$$

$$xy + 1 = 0$$

Q11) Solve the simultaneous equations.

$$x - 3y - 1 = 0$$

$$x^2 - 8y^2 - y + 5 = 0$$

Q12) Solve the simultaneous equations.

$$x - y = 0$$

$$xy = 9$$

Simultaneous Equations (Model Solutions)

Q1) Solution

$$x^2 + xy = 4 \dots \dots \dots (i)$$

$$y - x = 2 \dots \dots \dots (ii)$$

Rewrite equation (ii) as $y = x + 2$ and then substitute $(x + 2)$ for y in equation (i). Thus,

$$x^2 + x(x + 2) = 4$$

$$x^2 + x^2 + 2x = 4$$

$$2x^2 + 2x - 4 = 0$$

$$\frac{2x^2}{2} + \frac{2x}{2} - \frac{4}{2} = \frac{0}{2}$$

$$x^2 + x - 2 = 0$$

$$(x - 1)(x + 2) = 0$$

$$\text{Either } (x - 1) = 0 \Rightarrow x = 1$$

$$\text{or } (x + 2) = 0 \Rightarrow x = -2$$

When we substitute for x in equation (ii), we get that when:

$$x = 1 \text{ then } y = 3$$

$$\text{and when } x = -2 \text{ then } y = 0.$$

If the graphs of the equations $x^2 + xy = 4$ and $y - x = 2$ are plotted, the points $(1, 3)$ and $(-2, 0)$ are points of intersection.

Note that x can also be made the subject of the linear equation.

Q2) Solution

$$x - 2y = 1 \dots \dots \dots (i)$$

$$x^2 + y^2 = 29 \dots \dots \dots (ii)$$

$$\text{From equation (i), } x = 1 + 2y \dots \dots (iii)$$

When we substitute for x in (ii) we get,

$$(1 + 2y)^2 + y^2 = 29$$

$$1 + 4y + 4y^2 + y^2 = 29$$

$$5y^2 + 4y - 28 = 0$$

$$(5y + 14)(y - 2) = 0$$

$$\therefore \text{ Either } (5y + 14) = 0 \Rightarrow y = -\frac{14}{5}$$

$$\text{or } (y - 2) = 0 \Rightarrow y = 2$$

$$\text{Thus, } y = 2 \text{ or } -\frac{14}{5}$$

$$\text{From (iii), when } y = 2 \text{ then } x = 5$$

$$\text{and when } y = -\frac{14}{5}, x = -\frac{23}{5}$$

Q3) Solution

$$a) \quad y - x = 2 \dots \dots \dots (i)$$

$$y^2 - x^2 = 12 \dots \dots \dots (ii)$$

The quadratic equation is the difference of two squares. Thus,

$$(y - x)(y + x) = 12.$$

$$\text{But, } y - x = 2$$

$$\therefore 2(y + x) = 12$$

$$\Rightarrow 2y + 2x = 12 \text{ (Divide each term by 2)}$$

$$y + x = 6 \dots \dots \dots (iii)$$

Adding (i) and (iii) gives:

$$y - x + y + x = 2 + 6$$

$$2y = 8$$

$$y = 4$$

$$\text{Therefore, } y = 4 \text{ and } x = 2$$

This is a special case where there is only one pair of solutions.

$$b) \quad x + y = 6 \dots \dots \dots (i)$$

$$x^2 - 2xy + y^2 = 16 \dots \dots \dots (ii)$$

The quadratic equation can be written as
 $(x - y)^2 = 16$ or $x - y = \pm 4$

Adding $x + y = 6$ and $x - y = 4$ gives
 $2x = 10$. When $x = 5$, $y = 1$.

Also adding $x + y = 6$ and $x - y = -4$
 gives
 $2x = 2$. When $x = 1$, $y = 5$

Q4)

Solution

$$2h^2 - mh = 24 \dots\dots\dots (i)$$

$$m + h = 6 \dots\dots\dots (ii)$$

From (ii) make $m = 6 - h$

Substitute $m = 6 - h$ in (i)

$$2h^2 - (6 - h)h = 24$$

$$2h^2 - 6h + h^2 = 24$$

$$3h^2 - 6h = 24$$

Divide both sides by 3 to get

$$h^2 - 2h = 8$$

Subtract 8 from both sides

$$h^2 - 2h - 8 = 0$$

Using factor method,

$$h^2 - 2h - 8 = (h - 4)(h + 2) = 0$$

$$\therefore h = 4 \text{ or } h = -2$$

Substitute into (ii) to find the values of m

$$\text{When } h = 4$$

$$m + 4 = 6$$

$$m = 2$$

$$\text{When } h = -2$$

$$m - 2 = 6$$

$$\therefore m = 8$$

Q5)

Solution

$$x + y - 3 = x - 2y \dots\dots (i)$$

$$x - 2y = 7 \dots\dots\dots (ii)$$

Simplify (i)

$$x - x + y + 2y = 3$$

$$\therefore 3y = 3$$

$$\therefore y = 1$$

Substitute into (ii) to find x

$$x - 2 \times 1 = 7$$

$$x - 2 = 7$$

$$x = 7 + 2$$

$$\therefore x = 9$$

Q6) Solution

$$x^2 - y - 5 = 0 \dots\dots (i)$$

$$\frac{1}{2}y - x = -1 \dots\dots (ii)$$

Make y the subject in (ii)

$$y = 2x - 2$$

Substitute $2x - 2$ for y in (i)

$$x^2 - (2x - 2) - 5 = 0$$

$$x^2 - 2x + 2 - 5 = 0$$

$$x^2 - 2x - 3 = 0$$

$$x^2 + x - 3x - 3 = 0$$

$$x(x + 1) - 3(x + 1) = 0$$

$$(x + 1)(x - 3) = 0$$

$$\text{Either } (x + 1) = 0 \Rightarrow x = -1$$

$$\text{or } (x - 3) = 0 \Rightarrow x = 3$$

Substitute -1 for x in (i)

$$(-1)^2 - y - 5 = 0$$

$$1 - y - 5 = 0$$

$$-y - 4 = 0$$

$$-y = 4$$

$$y = -4 \text{ (Divide both sides by } -1)$$

Substitute 3 for x in (i) again

$$3^2 - y - 5 = 0$$

$$9 - y - 5 = 0$$

$$4 - y = 0$$

$$-y = -4 \text{ (Divide both sides by } -1)$$

$$y = 4$$

$$\therefore x = -1, y = -4$$

$$x = 3, y = 4$$

Q7)

Solution

$$y = x^2 \dots\dots\dots (i)$$

$$y = 5x - 6 \dots\dots\dots (ii)$$

Substitute x^2 for y in (ii)

$$x^2 = 5x - 6$$

$$x^2 - 5x + 6 = 0$$

$$x^2 - 2x - 3x + 6 = 0$$

$$x(x - 2) - 3(x - 2) = 0$$

$$(x - 2)(x - 3) = 0$$

$$\text{Either } (x - 2) = 0 \Rightarrow x = 2$$

$$\text{or } (x - 3) = 0 \Rightarrow x = 3$$

If $x = 2$, then substituting 2 for x in (i):

$$y = 2^2$$

$$= 4$$

If $x = 3$, then substituting 3 for x in (i) again:

$$y = 3^2$$

$$= 9$$

The solutions of the simultaneous equations are as follows:

Either $x = 2; y = 4$

or $x = 3; y = 9$

Q8) Solution

$$xy = -9 \dots \dots \dots (i)$$

$$y = x + 6 \dots \dots \dots (ii)$$

Substitute $x + 6$ for y in (i)

$$x(x + 6) = -9$$

$$x^2 + 6x = -9$$

$$x^2 + 6x + 9 = 0$$

$$(x + 3)(x + 3) = 0 \quad (\text{Factorization})$$

$$\text{Either } (x + 3) = 0 \Rightarrow x = -3$$

$$\text{or } (x + 3) = 0 \Rightarrow x = -3$$

Substitute -3 for x in (ii)

$$y = -3 + 6$$

$$y = 3 \text{ (twice)}$$

$$\therefore x = -3 \text{ (twice)}; y = 3 \text{ (twice)}$$

Q9) Solution

$$m - n = 5$$

$$m^2 - n^2 = 35$$

Make m the subject in (i):

$$m = 5 + n$$

Substituting $(5 + n)$ for m in (ii):

$$(5 + n)^2 - n^2 = 35$$

$$(5 + n)(5 + n) - n^2 = 35$$

$$25 + 10n + n^2 - n^2 = 35$$

$$10n = 35 - 25$$

$$10n = 10$$

(Divide both sides by 10)

$$n = 1$$

Substitute 1 for n in (i):

$$m - 1 = 5$$

$$m = 6$$

$$\therefore m = 6; n = 1$$

Q10) Solution

$$y = x - 2 \dots \dots \dots (i)$$

$$xy + 1 = 0 \dots \dots \dots (ii)$$

Substituting $x - 2$ for y in (ii):

$$x(x - 2) + 1 = 0$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)(x - 1) = 0$$

$$\Rightarrow x - 1 = 0$$

$$\Rightarrow x = 1 \text{ (twice)}$$

Substituting 1 for x in (i):

$$y = 1 - 2$$

$$= -1$$

$$\therefore x = 1 \text{ (twice)} \text{ and } y = -1 \text{ (twice)}$$

Q11) Solution

$$x - 3y - 1 = 0 \dots \dots \dots (i)$$

$$x^2 - 8y^2 - y + 5 = 0 \dots \dots \dots (ii)$$

Make x the subject in (i)

$$x = 3y + 1$$

Substitute $3y + 1$ for x in (ii)

$$(3y + 1)^2 - 8y^2 - y + 5 = 0$$

$$9y^2 + 6y + 1 - 8y^2 - y + 5 = 0$$

$$y^2 + 5y + 6 = 0$$

$$(y + 2)(y + 3) = 0 \quad (\text{Factorization})$$

$$\text{Either } (y + 2) = 0 \Rightarrow y = -2$$

$$\text{or } (y + 3) = 0 \Rightarrow y = -3$$

Substitute -2 for y in (i):

$$x - 3(-2) - 1 = 0$$

$$x + 6 - 1 = 0$$

$$x + 5 = 0$$

$$x = -5$$

Substitute -3 for y in (i) again:

$$x - 3(-3) - 1 = 0$$

$$x + 9 - 1 = 0$$

$$x + 8 = 0$$

$$x = -8$$

$$\therefore y = -2; x = -5$$

$$y = -3; x = -8$$

Q12) Solution

$$x - y = 0 \dots \dots \dots (i)$$

$$xy = 9 \dots \dots \dots (ii)$$

Making x subject of formula in (i)

$$x = y$$

Substituting x in (ii)

$$x \times y = 9$$

$$y^2 = 9$$

$$y = \sqrt{9}$$

$$y = \pm 3$$

Substituting $y = 3$ in (i)

$$x - 3 = 0$$

$$\therefore x = 3$$

and substituting $y = -3$ in (i) gives $x = -3$

Thus, when $y = 3; x = 3$

and when $y = -3; x = -3$

